ALGEBRA II · COMMON CORE HIGH SCHOOL MATH

Question Collection

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

85 question sets • 2555 questions

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Write complex numbers in the form a + bi

MAIN	Algebra II main question	**HSN-CN.A.1**
BACKWARD 1	Simplify square roots of positive numbers	**HSN-RN.A.2**
BACKWARD 2	No real number squares to a negative number	**8.EE.A.2**
BACKWARD 3	Divide each term of a sum by a number	**6.EE.A.3**
FORWARD 1	Use equality of complex numbers to find unknown real values	**HSN-CN.A.1**
FORWARD 2	Represent complex numbers as points in the complex plane	**HSN-CN.B.4**

SET 1 • HSN-CN.A.1

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write $\sqrt{(-49)}$ in the form bi, where b is a real number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q1/5

Write $\sqrt{(-18)}$ in the form bi with b in simplest radical form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q2/5

Write $6 - \sqrt{(-20)}$ in the form a + bi. Then state its real part and its imaginary part (each a real number).

a + bi: _____

Real part: _____

Imaginary part: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q3/5

Write $(-4 + \sqrt{(-75)}) / 2$ in the form a + bi. Then state its real part and its imaginary part (each a real number).

a + bi: _____

Real part: _____

Imaginary part: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q4/5

Write $(10 + \sqrt{(-64)}) / 4$ in the form a + bi. Then state its real part and its imaginary part (each a real number).

a + bi: _____

Real part: _____

Imaginary part: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 1

Simplify square roots of positive numbers • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

Simplify $\sqrt{50}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q1/5

Simplify $\sqrt{72}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q2/5

Which is √12 in simplest radical form?

A. 4√3

B. 3√2

C. 6√2

D. 2√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q3**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q3/5

True or false?

$\sqrt{(9 + 16)} = \sqrt{9} + \sqrt{16}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q4/5

Simplify $\sqrt{28}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 2

No real number squares to a negative number • Grade 8 • Expressions & Equations • 8.EE.A.2

Find all real numbers x with $x^2 = 81$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

True or false?

$(-3)^2 = -9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

Is there a real number x with $x^2 = -4$?

A. No: only the number 0 has a square that is negative, and $0^2 = 0$, not -4.

B. Yes: $x = -2$ works, because the square of a negative number is negative: $(-2)^2 = -4$.

C. Yes: $x = 2i$ is a real number whose square is -4.

D. No: the square of every real number is 0 or positive, so it cannot equal -4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

Which number has a square equal to 36?

A. 18

B. $\sqrt{-36}$

C. -6

D. -18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

Evaluate $(-7)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 3

Divide each term of a sum by a number • Grade 6 • Expressions & Equations • 6.EE.A.3

Write $(8 + 6\sqrt{2}) / 2$ as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q1/5

Write $(-6 + 4\sqrt{5}) / 2$ as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q2/5

True or false?

$(10 + \sqrt{7}) / 5 = 2 + \sqrt{7}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q3/5

(12 − 9√3) / 3 equals

A. 4 − 3√3

B. 4 − 9√3

C. 4 − 3√1

D. 12 − 3√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q4/5

Write $(-4 + 2\sqrt{3}) / (-2)$ as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • HSN-CN.A.1

FORWARD 1

Use equality of complex numbers to find unknown real values • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

x and y are real numbers and $(x - 3) + 2yi = 5 - 8i$. Find x and y.

$x =$ _____

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q1/5

a and b are real numbers and $3a + (b + 1)i = -12 + 4i$. Find a and b.

$a =$ _____

$b =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q2/5

True or false?

If x and y are real and x + yi = 7, then y = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q3/5

p and q are real and $(p + q) + (p - q)i = 10 + 2i$. What is p?

A. 6

B. 5

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q4/5

m is a real number. For what value of m is $(m^2 - 4) + (m + 3)i$ a real number? What real number is it then?

m = _____

The number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSN-CN.A.1

FORWARD 2

Represent complex numbers as points in the complex plane • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4

In the complex plane the horizontal axis is real and the vertical axis is imaginary. Give the point for $-3 + 5i$.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q1/5

Which complex number is represented by the point $(4, -2)$ in the complex plane?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q2/5

Find the distance from 0 to the point 6 + 8i in the complex plane.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q3/5

True or false?

Every real number lies on the horizontal axis of the complex plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q4/5

Which number lies on the vertical axis of the complex plane?

A. -7

B. $7 + i$

C. $-7i$

D. $1 - i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q5/5

Add and subtract complex numbers

MAIN	Algebra II main question	**HSN-CN.A.2**
BACKWARD 1	Combine like terms	**7.EE.A.1**
BACKWARD 2	Add and subtract integers and decimals	**7.NS.A.1**
BACKWARD 3	Write square roots of negative numbers with i	**HSN-CN.A.1**
FORWARD 1	Find real parameters that make a sum real or imaginary	**HSN-CN.A.2**
FORWARD 2	Add complex numbers as vectors in the complex plane	**HSN-CN.B.5**

SET 2 • HSN-CN.A.2

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Simplify (3 + 5i) + (−7 + 2i). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q1/5

Simplify $(6 - 4i) - (-2 + 9i)$. Write the answer in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q2/5

Simplify $(-1 + \sqrt{(-16)}) + (5 - \sqrt{(-9)})$. Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q3/5

$z_1 = 2.5 - i$, $z_2 = -1.5 + 3i$ and $z_3 = 4i$. Find $z_1 - z_2 + z_3$ in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q4/5

Find the complex number w for which $(4 - 7i) + w = -2 + 3i$.

$w =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 1

Combine like terms • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify $(3 + 5x) + (-7 + 2x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q1/5

Simplify $(6 - 4x) - (-2 + 9x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q2/5

$(2 - 3y) - (5 - y)$ equals

A. $-3 - 2y$

B. $7 - 4y$

C. $-3 + 2y$

D. $-3 - 4y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q3/5

True or false?

$(4 + 2t) + (1 - 6t) = 5 + 4t$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q4/5

Simplify $(0.5 - x) - (1.5 + 2x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 2

Add and subtract integers and decimals • Grade 7 • The Number System • 7.NS.A.1

Evaluate $6 - (-2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q1/5

Evaluate $-4 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q2/5

$-1 - 3 + 4$ equals

A. 8

B. 0

C. -8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$2.5 - (-1.5) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q4/5

Evaluate $-2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 3

Write square roots of negative numbers with i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write $\sqrt{(-16)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q1/5

Write $\sqrt{(-9)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q2/5

$\sqrt{(-25)}$ equals

A. -5

B. $25i$

C. $-5i$

D. $5i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-B3-Q3**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q3/5

True or false?

$\sqrt{(-1)} = i$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q4/5

Write $-\sqrt{(-36)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q5/5

SET 2 • HSN-CN.A.2

FORWARD 1

Find real parameters that make a sum real or imaginary • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

k is a real number. For what value of k is $(3 + ki) + (k - 5i)$ a real number? Give that real number.

k = _____

The sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q1/5

k is real. For what value of k is
$(2k - 6) + 4i - (k + 1 - i)$ a pure imaginary number (real part 0)?

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q2/5

True or false?

The sum of a complex number and its conjugate is always 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q3/5

a is real and $(a + 3i) - (2 + ai)$ is real. What is a?

A. 0

B. 2

C. 3

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q4/5

a and b are real numbers. Why is the sum of a + bi and a − bi always a real number?

A. The real parts a and a cancel, leaving 2bi, which is real.

B. The imaginary parts bi and −bi add to 0, leaving 2a, which is real.

C. Any sum of two complex numbers is real.

D. It is real only when b = 0, because otherwise the sum has an i term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSN-CN.A.2

FORWARD 2

Add complex numbers as vectors in the complex plane • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5

In the complex plane (real axis horizontal, imaginary axis vertical), the number a + bi is the point (a, b). Which point represents the sum (2 + i) + (1 + 3i)?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q1/5

The points 0, 2 + i, 1 + 3i and their sum are vertices of a quadrilateral in the complex plane. What kind of quadrilateral is it?

A. a parallelogram

B. a kite that is not a parallelogram

C. a trapezoid that is not a parallelogram

D. a square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q2/5

True or false?

Subtracting w from z gives the vector from the point w to the point z.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q3/5

What is the distance between the points $5 + 2i$ and $1 - i$ in the complex plane?

A. 25

B. $\sqrt{13}$

C. 5

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q4**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q4/5

Find the midpoint of the segment joining $6 - 2i$ and $-2 + 4i$, as a complex number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q5/5

Multiply complex numbers and simplify powers of i

MAIN	Algebra II main question	**HSN-CN.A.2**
BACKWARD 1	Multiply binomials	**HSA-APR.A.1**
BACKWARD 2	Use $i^2 = -1$ and the cycle of powers of i	**HSN-CN.A.1**
BACKWARD 3	Divide whole numbers to find a remainder	**4.NBT.B.6**
FORWARD 1	Find a real parameter that makes a product real	**HSN-CN.A.2**
FORWARD 2	Divide complex numbers using conjugates	**HSN-CN.A.3**

SET 3 • HSN-CN.A.2

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply $(3 + 2i)(4 - 5i)$. Write the product in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q1/5

Simplify $(-2 + i)^2$. Write the answer in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q2/5

Multiply $(5 - 3i)(5 + 3i)$. Write the product in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q3/5

Simplify $(1 + i)^3$. Write the answer in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q4/5

Simplify $i^{23} \cdot (4 - i)$. Write the answer in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 1

Multiply binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $(3 + 2x)(4 - 5x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q1/5

Expand $(-2 + x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q2/5

$(5 - 3x)(5 + 3x)$ equals

A. $25 + 9x^2$

B. $10 - 9x^2$

C. $25 - 9x^2$

D. $25 - 30x - 9x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B1-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

$(1 + x)^2 = 1 + x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q4/5

Multiply $x(4 - x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 2

Use $i^2 = -1$ and the cycle of powers of i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Simplify i^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q1/5

Simplify i^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q2/5

Simplify i^4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q3/5

True or false?

$-10i^2 = -10$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q4/5

Which equals 1?

A. i^8

B. i^9

C. i^3

D. i^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 3

Divide whole numbers to find a remainder • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Find the remainder when 23 is divided by 4.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q1/5

Find the remainder when 50 is divided by 4.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q2/5

What is the remainder when 36 is divided by 4?

A. 0

B. 4

C. 1

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q3/5

True or false?

The remainder when 101 is divided by 4 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q4/5

Write 39 as 4q + r with 0 ≤ r < 4.

q = _____

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q5/5

SET 3 • HSN-CN.A.2

FORWARD 1

Find a real parameter that makes a product real • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

a is a real number. For what value of a is $(a + 2i)(3 - i)$ a real number? Give the product.

a = _____

The product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q1/5

b is real. For what value of b is (2 + bi)(1 + i) pure imaginary (real part 0)?

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q2/5

True or false?

The product of two non-real complex numbers can be a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q3/5

For real c, $(c + i)^2$ is real when

A. $c = 0$

B. $c = 1$

C. $c = -1$

D. never

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q4/5

a and b are real numbers. Which work shows that $(a + bi)(a - bi)$ is always a real number?

A. $(a + bi)(a - bi) = a^2 - abi + abi - b^2i^2 = a^2 - b^2(-1) = a^2 + b^2$, which is real.

B. $(a + bi)(a - bi) = a^2 - b^2$, which is real.

C. $(3 + 2i)(3 - 2i) = 13$ is real, so every such product is real.

D. $(a + bi)(a - bi) = a^2 - b^2i^2$, and $a^2 - b^2i^2$ is real because it has no i in it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSN-CN.A.2

FORWARD 2

Divide complex numbers using conjugates • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

Write (3 + 4i) / (1 + 2i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q1/5

Write $10 / (3 - i)$ in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q2/5

True or false?

To divide by $2 - 5i$, multiply numerator and denominator by $2 - 5i$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q3/5

(2 + i) / i equals

A. $-1 + 2i$

B. $2 + i$

C. $1 - 2i$

D. $1 + 2i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q4**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q4/5

c and d are real and not both 0. Why does multiplying by the conjugate of the denominator c + di always give a real denominator?

A. $(c + di)(c - di) = c^2 - d^2$, a real number (the middle terms cancel).

B. The conjugate $c - di$ is itself a real number, so any product with it is real.

C. Multiplying by any complex number makes the denominator real.

D. $(c + di)(c - di) = c^2 + d^2$, a real number (positive because c and d are not both 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q5/5

Solve quadratic equations with complex solutions

MAIN	Algebra II main question	**HSN-CN.C.7**
BACKWARD 1	Use the quadratic formula and the discriminant	**HSA-REI.B.4.b**
BACKWARD 2	Write square roots of negative numbers with i	**HSN-CN.A.1**
BACKWARD 3	Solve equations of the form $(x - h)^2 = k$	**HSA-REI.B.4.b**
FORWARD 1	Find a quadratic from one complex solution	**HSN-CN.C.7**
FORWARD 2	Classify solutions for a family of quadratics	**HSN-CN.C.7**

SET 4 • HSN-CN.C.7

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^2 + 49 = 0$ over the complex numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q1/5

Solve $x^2 - 4x + 13 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q2/5

Solve $2x^2 + 2x + 5 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q3/5

Solve $(x + 3)^2 = -20$ over the complex numbers. Give exact solutions in the form a + bi.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q4/5

Solve $3x^2 - 6x + 7 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 1

Use the quadratic formula and the discriminant • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Find the discriminant $b^2 - 4ac$ of $x^2 - 4x + 13 = 0$.

Discriminant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q1/5

Find the discriminant of $2x^2 + 2x + 5 = 0$.

Discriminant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q2/5

True or false?

If the discriminant is negative, the quadratic equation has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q3/5

In $3x^2 - 6x + 7 = 0$, the values of a, b and c are

A. 3, −6, −7

B. 3, 6, 7

C. −3, −6, 7

D. 3, −6, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q4/5

Which is the quadratic formula for the solutions of $ax^2 + bx + c = 0$ $(a \neq 0)$?

A. $x = -b \pm \sqrt{(b^2 - 4ac)}/(2a)$

B. $x = (-b \pm \sqrt{(b^2 - 4ac)})/(2a)$

C. $x = (-b \pm \sqrt{(b^2 + 4ac)})/(2a)$

D. $x = (b \pm \sqrt{(b^2 - 4ac)})/(2a)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 2

Write square roots of negative numbers with i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write $\sqrt{(-36)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q1/5

Write $\sqrt{(-48)}$ in simplest form with i.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q2/5

$\sqrt{(-20)}$ equals

A. $2\sqrt{5}\,i$

B. $4\sqrt{5}\,i$

C. $-2\sqrt{5}$

D. $10i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{(-49)} = -7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q4/5

Write $(-2 + \sqrt{(-36)}) / 4$ in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 3

Solve equations of the form $(x - h)^2 = k$ • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $(x - 2)^2 = 9$ over the real numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q1/5

Solve $(x + 1)^2 = 5$ over the real numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q2/5

True or false?

Solving $x^2 = 16$ gives only $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q3/5

The solutions of $(x + 4)^2 = 25$ are

A. 5 and −5

B. −1 and 9

C. 1 and −9

D. 21 and −29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q4/5

Solve $x^2 = 50$ over the real numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q5/5

SET 4 • HSN-CN.C.7

FORWARD 1

Find a quadratic from one complex solution • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

$3 - 2i$ is a solution of $x^2 + bx + c = 0$, where b and c are real. Find b and c.

b = _____

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q1/5

$-1 + 4i$ is a solution of a quadratic equation $x^2 + bx + c = 0$ with real b and c. Find b and c.

b = _____

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q2/5

True or false?

If 2 + i is a solution of a quadratic with real coefficients, then 2 − i is also a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q3/5

Which quadratic with real coefficients has 5i as a solution?

A. $x^2 + 25 = 0$

B. $x^2 + 5 = 0$

C. $x^2 - 25 = 0$

D. $x^2 - 10x + 25 = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q4/5

Why does $x^2 - (2 + i)x = 0$ have the solution $2 + i$ but not the solution $2 - i$?

A. Substituting $2 - i$ gives $(2 - i)^2 - (2 + i)(2 - i) = 0$, so $2 - i$ is a solution after all.

B. Non-real solutions of every quadratic equation come in conjugate pairs, so $2 - i$ must also be a solution, along with $2 + i$.

C. Its coefficient $2 + i$ is not real, so its solutions need not come in conjugate pairs; factoring $x(x - (2 + i)) = 0$ gives only 0 and $2 + i$.

D. A quadratic equation can have only one non-real solution, so once $2 + i$ is found there is no room for $2 - i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSN-CN.C.7

FORWARD 2

Classify solutions for a family of quadratics • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

For which real values of k does $x^2 + kx + 9 = 0$ have two non-real solutions?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q1/5

Consider $x^2 - 4x + c = 0$ with real c. (a) For which real values of c are the solutions non-real? (b) Solve the equation when $c = 8$.

(a) _____

(b) $x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q2/5

True or false?

$x^2 + kx - 1 = 0$ has non-real solutions for some real k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q3/5

$x^2 + 6x + m = 0$ has exactly one (repeated) real solution when $m =$

A. 3

B. 36

C. -9

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q4/5

For $c > 4$, which gives the solutions of $x^2 - 4x + c = 0$ in terms of c?

A. $x = 2 \pm i\sqrt{(4 - c)}$

B. $x = -2 \pm i\sqrt{(c - 4)}$

C. $x = 2 \pm \sqrt{(c - 4)}$

D. $x = 2 \pm i\sqrt{(c - 4)}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q5/5

Factor polynomials over the complex numbers

MAIN	Algebra II main question	**HSN-CN.C.8**
BACKWARD 1	Factor a difference of squares	**HSA-SSE.A.2**
BACKWARD 2	Multiply complex conjugates	**HSN-CN.A.2**
BACKWARD 3	Solve quadratics with complex solutions	**HSN-CN.C.7**
FORWARD 1	Factor cubic and quartic polynomials over the complex numbers	**HSN-CN.C.8**
FORWARD 2	Write polynomials with real coefficients from complex zeros	**HSN-CN.C.9**

SET 5 • HSN-CN.C.8

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^2 + 25$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q1/5

Factor $x^2 + 7$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q2/5

Factor $4x^2 + 9$ into two linear factors with complex coefficients, each of the form $(2x + ki)$ with k real.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q3/5

Factor $x^4 - 16$ completely into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q4/5

Factor $x^2 - 6x + 13$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 1

Factor a difference of squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 - 49$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q1/5

Factor $9x^2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q2/5

$x^4 - 1$ equals

A. $(x - 1)^4$

B. $(x^2 - 1)^2$

C. $(x^2 - 1)(x^2 + 1)$

D. $(x^2 + 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B1-Q3**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q3/5

True or false?

$x^2 + 9 = (x + 3)(x - 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q4/5

Factor $x^2 - 5$ using radicals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 2

Multiply complex conjugates • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply $(x + 5i)(x - 5i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q1/5

Multiply $(2x + 3i)(2x - 3i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q2/5

True or false?

$(3i)^2 = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q3/5

$(x - (1 + i))(x - (1 - i))$ equals

A. $x^2 + 2x + 2$

B. $x^2 - 2x + 2$

C. $x^2 + 2$

D. $x^2 - 2x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q4/5

Simplify $(i\sqrt{7})^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 3

Solve quadratics with complex solutions • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^2 + 25 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q1/5

Solve $x^2 - 6x + 13 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q2/5

True or false?

If r is a zero of p(x), then x − r is a factor of p(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q3/5

The zeros of $x^2 + 4$ are

A. $2i$ and $-2i$

B. $4i$ and $-4i$

C. -4

D. 2 and -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q4/5

Solve $x^2 + 2x + 5 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • HSN-CN.C.8

FORWARD 1

Factor cubic and quartic polynomials over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^3 + 8$ completely into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q1/5

Factor $x^3 + 9x$ completely into linear factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q2/5

Factor $x^4 + 5x^2 + 4$ completely into linear factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q3/5

True or false?

$x^3 - 1 = (x - 1)(x^2 + x + 1)$, and $x^2 + x + 1$ has two non-real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q4**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q4/5

Which is a linear factor of $x^3 - 27$ over the complex numbers?

A. $x + 3i$

B. $x + 3$

C. $x - 3i$

D. $x - (-3/2 + (3\sqrt{3}/2)i)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSN-CN.C.8

FORWARD 2

Write polynomials with real coefficients from complex zeros • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

Write a polynomial with real coefficients, leading coefficient 1 and least degree, whose zeros include 2i and 1 − i. Give it in expanded form. Use ^ for exponents.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q1/5

What is the least degree of a polynomial with real coefficients that has zeros 3, 4i and 2 + i?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q2/5

True or false?

A polynomial with real coefficients can have exactly one non-real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q3/5

A cubic with real coefficients has zeros 5 and $-i$. Its third zero is

A. -5

B. $5i$

C. 1

D. i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q4/5

Why must a polynomial with real coefficients and degree 3 have at least one real zero?

A. The Fundamental Theorem of Algebra says that every zero of a polynomial with real coefficients is real, so all 3 zeros are real.

B. It has 3 zeros counted with multiplicity, and its non-real zeros come in conjugate pairs, so at most 2 are non-real and at least 1 is real.

C. Every polynomial has at least one real zero, whatever its degree.

D. Every cubic with real coefficients can be factored over the integers, so at least one of its 3 zeros is a whole number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q5/5

SET 6 • HSN-CN.C.9 • THE COMPLEX NUMBER SYSTEM

Use the Fundamental Theorem of Algebra

MAIN	Algebra II main question	**HSN-CN.C.9**
BACKWARD 1	Find the degree of a polynomial	**HSA-SSE.A.1.a**
BACKWARD 2	Find zeros from factored form	**HSA-SSE.B.3.a**
BACKWARD 3	Solve quadratics with complex solutions	**HSN-CN.C.7**
FORWARD 1	Relate the complex zeros to the real x-intercepts of a graph	**HSA-APR.B.3**
FORWARD 2	Prove existence statements about real zeros	**HSN-CN.C.9**

SET 6 • HSN-CN.C.9

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

How many complex zeros, counted with multiplicity, does $p(x) = 3x^5 - x^2 + 7$ have? (Use the Fundamental Theorem of Algebra.)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q1/5

$p(x) = (x - 2)^3(x^2 + 9)$. List every complex zero with its multiplicity.

Zero with multiplicity 3: _____

Zeros with multiplicity 1 (both): _____

Sum of the multiplicities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-M-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q2/5

Which argument shows that every quadratic $p(x) = ax^2 + bx + c$ (real a, b, c, $a \neq 0$) has exactly two complex zeros, counted with multiplicity?

A. $x^2 - 5x + 6 = (x - 2)(x - 3)$ has two zeros, so every quadratic has two zeros.

B. By the quadratic formula the zeros are $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$. If $b^2 - 4ac < 0$ the square root does not exist, so there are no zeros; otherwise there are two, so a quadratic never has more than two zeros.

C. The graph of every quadratic is a parabola, and a parabola that opens up or down always crosses the x-axis in exactly two places, so every quadratic $ax^2 + bx + c$ has exactly two zeros.

D. The quadratic formula $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$ gives every zero: two real zeros if $b^2 - 4ac > 0$, one zero of multiplicity 2 if $b^2 - 4ac = 0$, and two non-real conjugate zeros if $b^2 - 4ac < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-M-Q3**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q3/5

Find every complex zero of $p(x) = x^4 - 81$. How many are there, and does the count agree with the Fundamental Theorem of Algebra for degree 4?

All zeros: $x =$ _____

Number of zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q4/5

A student says $p(x) = x^3 + x + 1$ has no zeros because it does not factor with integers. Which statement is correct?

A. The student is right: a polynomial that does not factor over the integers has no zeros.

B. It has exactly 3 real zeros, because the Fundamental Theorem of Algebra says a polynomial of degree 3 has 3 zeros, and a polynomial with real coefficients has only real zeros.

C. It has 3 complex zeros counted with multiplicity, and at least one is real, because non-real zeros of a polynomial with real coefficients come in conjugate pairs.

D. It has 3 complex zeros counted with multiplicity, and all three could be non-real, because a polynomial that does not factor over the integers need not cross the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-M-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 1

Find the degree of a polynomial • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

What is the degree of $3x^5 - x^2 + 7$?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q1/5

What is the degree of $(x - 2)^3(x^2 + 9)$ when multiplied out?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q2/5

The degree of $x(x + 1)(x^2 - 4)$ is

A. 1

B. 4

C. 2

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-B1-Q3**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q3/5

True or false?

The degree of $4x^3 + 2x^4$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q4/5

What is the leading coefficient of $-2x^3 + 5x - 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 2

Find zeros from factored form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a

Find the zeros of $(x - 2)(x + 5)$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q1/5

$p(x) = (x - 1)^3$. Find the zero of p and its multiplicity.

Zero: _____

Multiplicity: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q2/5

True or false?

A product is 0 only when one of its factors is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q3/5

The zeros of $x(x + 4)$ are

A. 0 and 4

B. 0 and -4

C. -4 only

D. 4 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q4/5

Find the zeros of $(2x - 3)(x + 1)$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 3

Solve quadratics with complex solutions • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^2 + 9 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q1/5

Solve $x^2 + 2x + 2 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q2/5

True or false?

$x^2 + 1 = 0$ has no complex solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q3/5

The solutions of $x^2 - 2x + 5 = 0$ are

A. $1 \pm 4i$

B. $-1 \pm 2i$

C. $1 \pm 2i$

D. $2 \pm i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q4/5

Solve $x^2 = -81$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • HSN-CN.C.9

FORWARD 1

Relate the complex zeros to the real x-intercepts of a graph • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

A polynomial of degree 4 with real coefficients has a graph that crosses the x-axis at exactly two points and touches it nowhere else. Each crossing is a simple zero. How many non-real zeros does it have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q1/5

$p(x) = x^3 - 4x^2 + 5x - 2 = (x - 1)^2(x - 2)$. Give the multiplicity of each zero.

Multiplicity of the zero 1: _____

Multiplicity of the zero 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q2/5

True or false?

A polynomial of degree 6 whose graph never meets the x-axis has 6 non-real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q3/5

A degree-5 polynomial with real coefficients has exactly one real zero, of multiplicity 1. How many non-real zeros does it have?

A. 4

B. 5

C. 2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q4/5

Can a degree-4 polynomial with real coefficients have exactly three real zeros, each of multiplicity 1, and no other real zeros?

A. No: the fourth zero (counted with multiplicity) would be a single non-real zero, but non-real zeros come in conjugate pairs.

B. No: a degree-4 polynomial with real coefficients must always have exactly four distinct real zeros.

C. Yes: a degree-4 polynomial can have any number of zeros from 0 to 4.

D. Yes: for example $(x - 1)(x - 2)(x - 3)(x - i)$ has degree 4, real coefficients, and exactly three real zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSN-CN.C.9

FORWARD 2

Prove existence statements about real zeros • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

Which argument proves that every polynomial of odd degree with real coefficients has at least one real zero?

A. By the Fundamental Theorem of Algebra it has n zeros counted with multiplicity, n odd. Non-real zeros come one at a time, so there is an odd number of them, and an odd number of non-real zeros leaves at least one zero that is real.

B. By the Fundamental Theorem of Algebra it has n zeros counted with multiplicity, n odd. Non-real zeros come in conjugate pairs, so there is an even number of them; odd minus even is odd, so at least one zero is real.

C. The Fundamental Theorem of Algebra says that every polynomial of degree $n \geq 1$ with real coefficients has at least one real zero; a polynomial of odd degree has degree at least 1, so it has at least one real zero as well.

D. $x^3 - 1$ has the real zero 1, so every polynomial of odd degree has a real zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q1**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q1/5

A degree-4 polynomial with real coefficients has the zeros 1 + i and 3. Which describes all of its zeros, counted with multiplicity?

A. 1 + i, 1 − i, and 3 only (a degree-4 polynomial has 3 zeros here).

B. 1 + i, 3, and two more non-real zeros that need not include 1 − i.

C. 1 + i, 1 − i, 3, and one more real zero r (r may equal 3, making 3 a double zero).

D. 1 + i, 1 − i, 3, and −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q2/5

True or false?

A degree-4 polynomial with real coefficients must have at least one real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q3/5

A polynomial with real coefficients has degree 7. The possible numbers of real zeros (with multiplicity) are

A. 1, 3, 5, 7

B. 7 only

C. 0, 2, 4, 6

D. 0 to 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q4/5

Which statement about the Fundamental Theorem of Algebra is correct?

A. It guarantees that a polynomial of degree $n \geq 1$ has n complex zeros counted with multiplicity, but gives no method to find them.

B. It says that every zero of a polynomial with real coefficients is real, so a polynomial of degree n has n real zeros counted with multiplicity.

C. It gives a formula, like the quadratic formula, that finds the exact zeros of every polynomial of any degree.

D. It applies only to polynomials that factor over the integers, so it says nothing about other polynomials.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q5/5

Interpret the parts of polynomial, rational, and exponential expressions in context

MAIN	Algebra II main question	**HSA-SSE.A.1.a**
BACKWARD 1	Name terms, factors, and coefficients	**HSA-SSE.A.1.a**
BACKWARD 2	Translate between a percent change and a multiplying factor	**7.RP.A.3**
BACKWARD 3	Interpret a quotient as an amount per unit	**6.RP.A.2**
FORWARD 1	Build an expression whose parts have given meanings	**HSA-SSE.A.1.a**
FORWARD 2	Rewrite an exponential expression to reveal a different rate	**HSF-IF.C.8.b**

SET 7 • HSA-SSE.A.1.a

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A town's population is modeled by $P(t) = 12{,}500(0.97)^t$, where t is the number of years after 2020. Which interpretation of 12,500, 0.97, and $1 - 0.97$ is correct?

A. 12,500 is the population in 2020; 0.97 means the population decreases by 97% per year; $1 - 0.97 = 0.03$ is the population left after one year.

B. 12,500 is the population in 2020; 0.97 is the yearly factor (each year the population is 97% of the year before); $1 - 0.97 = 0.03$ means a 3% decrease per year.

C. 12,500 is the number of people lost per year; 0.97 is the yearly factor; $1 - 0.97 = 0.03$ means a 3% decrease per year.

D. 12,500 is the population after one year; 0.97 is the yearly factor; $1 - 0.97 = 0.03$ means a 3% increase per year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q1/5

The average cost per item of making x items is C(x) = (4,000 + 6x)/x dollars. Which interpretation of 4,000, 6x, and the division by x is correct?

A. 4,000 is a fixed cost in dollars; 6x is the variable cost; dividing by x gives the total cost of x items.

B. 4,000 is the cost of each item; 6x is the number of items made; dividing by x gives the total cost of all items.

C. 4,000 is a fixed cost in dollars; 6x is the variable cost ($6 per item for x items); dividing the total cost by x gives the cost per item.

D. 4,000 is a fixed cost in dollars; 6x is the profit on x items ($6 per item); dividing by x gives the profit per item.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q2/5

An open box is made by cutting squares of side x cm from the corners of a 20 cm by 20 cm sheet and folding up the sides. Its volume is $V(x) = x(20 - 2x)^2$ cm^3. Which interpretation of x, $20 - 2x$, and $(20 - 2x)^2$ is correct?

A. x is the side of the base; $20 - 2x$ is the height; $(20 - 2x)^2$ is the volume in cm^3.

B. x is the height; $20 - 2x$ is the side of the base (20 cm minus one cut); $(20 - 2x)^2$ is the perimeter of the base.

C. x is the height; $20 - 2x$ is the area of one side; $(20 - 2x)^2$ is the surface area of the box.

D. x is the height of the box; $20 - 2x$ is the side length of the square base (20 cm minus two cuts); $(20 - 2x)^2$ is the base area in cm^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q3/5

The amount of a medicine in the body t hours after a dose is $A(t) = 200(1/2)^{t/6}$ mg. Which interpretation of 200, 1/2, and t/6 is correct?

A. 200 mg is the dose at t = 0; 1/2 means the amount is halved repeatedly; t/6 counts 6-hour periods, so the amount halves every 6 hours.

B. 200 mg is the dose at t = 0; 1/2 means the amount is halved; t/6 means the amount halves every 1/6 hour.

C. 200 mg is the amount left after 6 hours; 1/2 means the amount is halved; t/6 counts 6-hour periods.

D. 200 mg is the dose at t = 0; 1/2 means half of the dose is removed every hour; t/6 is the number of doses taken.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q4/5

Maya keeps $1,500 in an account that earns 4% interest compounded once a year and also puts $200 per year in a jar (no interest). Her total after n years is $T(n) = 1500(1.04)^n + 200n$. Which interpretation of $1500(1.04)^n$, 1.04, and 200n is correct?

A. $1500(1.04)^n$ is the account balance after n years; 1.04 means 104% interest per year; 200n is the cash in the jar after n years.

B. $1500(1.04)^n$ is the account balance after n years; 1.04 is the yearly growth factor (4% interest added each year); 200n is the cash in the jar after n years.

C. $1500(1.04)^n$ is the account balance after n years; 1.04 is the yearly growth factor; 200n is the interest earned by the jar.

D. $1500(1.04)^n$ is the interest earned after n years; 1.04 is the yearly growth factor; 200n is the cash in the jar after n years.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 1

Name terms, factors, and coefficients • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

Which lists the terms of 4,000 + 6x?

A. 4,000 and 6

B. 4,000, 6, and x

C. 4,000 and 6x

D. 6 and x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q1/5

What is the coefficient of x in $7x - 3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q2/5

In $5x(x - 2)$, which is a factor?

A. $5x - 2$

B. $x - 10$

C. $x - 2$

D. -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q3/5

True or false?

In 3 + 2x, the number 3 is a factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q4/5

Which lists the factors of $200(0.5)^t$?

A. 0.5 and t

B. $200 + (0.5)^t$

C. 200 and 0.5t

D. 200 and $(0.5)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 2

Translate between a percent change and a multiplying factor • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A price is multiplied by 0.97. By what percent does it change? Give a signed percent (negative for a decrease).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q1/5

What factor multiplies an amount that increases by 4%?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q2/5

Multiplying by 1.25 is the same as

A. increasing by 2.5%

B. increasing by 25%

C. increasing by 125%

D. decreasing by 25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q3/5

True or false?

Multiplying by 0.8 decreases an amount by 80%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q4/5

An amount is multiplied by 0.85. By what percent does it decrease? (Hint: find 1 − 0.85.)

Percent decrease: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 3

Interpret a quotient as an amount per unit • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

A total cost of $600 is shared equally by 30 people. What does 600/30 = 20 represent?

A. The total cost, $20

B. The number of people per dollar

C. The cost per person, $20

D. The number of people, 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q1/5

A car travels 150 miles in 3 hours. What does 150/3 = 50 represent?

A. The average speed, 50 miles per hour

B. The time per mile, 50 hours per mile

C. The total distance, 50 miles

D. The number of hours, 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q2/5

If T is a total cost for n items, then T/n is

A. the profit

B. the number of items

C. the cost per item

D. the total cost

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q3/5

True or false?

Dividing a total by the number of items gives an average per item.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q4/5

Making x items costs a total of 4,000 + 6x dollars. What does (4,000 + 6x)/x equal when x = 100, and what does it mean?

A. $46: the cost of making the 100th item alone

B. $40: the fixed cost per item when 100 items are made

C. $4,600: the total cost of making 100 items

D. $46: the average cost per item when 100 items are made

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 1

Build an expression whose parts have given meanings • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A $24,000 car loses 15% of its value each year. Write an expression for its value in dollars t years after purchase.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q1/5

A gym charges a $90 joining fee and $35 per month. Write an expression for the average cost per month, in dollars, over m months.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q2/5

True or false?

An expression for a quantity that triples every 5 years, starting at 40, is $40(3)^{5t}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q3/5

A bacteria count starts at 500 and doubles every 3 hours. Which expression gives the count after h hours?

A. $2(500)^{h/3}$

B. $500(2)^{3h}$

C. $500(3)^{h/2}$

D. $500(2)^{h/3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q4/5

A gym charges a $90 joining fee and $35 per month, so the average cost per month over m months is 35 + 90/m dollars. What does the part 90/m represent?

A. The monthly fee, which stays the same $35 for every m.

B. The joining fee spread over the m months; it shrinks as m grows.

C. The total amount paid for the gym after m months of membership.

D. The joining fee, charged again in full each month.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 2

Rewrite an exponential expression to reveal a different rate • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

Rewrite $12{,}500(0.97)^{t}$ (t in years) to show the monthly factor, rounded to 5 decimal places.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q1/5

$A(t) = 200(1/2)^{t/6}$. Write A as $200 \cdot b^t$. Give b to 4 decimal places, and the hourly percent decrease to the nearest hundredth of a percent.

b = _____

Hourly percent decrease: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q2/5

True or false?

$1.04^{n} = (1.04^{1/12})^{12n}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q3/5

Which expression shows that $500(2)^{t/3}$ grows by about 26% per time unit?

A. $1.26(500)^{t}$

B. $500(1.26)^{3t}$

C. $500(1.26)^{t}$

D. $500(2.26)^{t}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q4/5

$A(t) = 200(1/2)^{t/6}$ is rewritten as $A(t) = 200(0.8909)^{t}$ (rounded). Why does an exact rewriting such as $200((1/2)^{1/6})^{t}$ give the same value as $200(1/2)^{t/6}$ for every t?

A. By the power of a power rule, $((1/2)^{1/6})^{t} = (1/2)^{t/6}$, so the two are equal for every t; only the displayed rate differs.

B. Rewriting an exponential always changes its values a little, so the two expressions are only close for each t.

C. They give the same value only at t = 0, where both equal 200; for other t the base 0.8909 makes them differ.

D. They are equal only when t is a multiple of 6, because the exponent t/6 is then a whole number and no rounding is involved.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q5/5

SET 8 • HSA-SSE.A.1.b • SEEING STRUCTURE IN EXPRESSIONS

Interpret complicated expressions by viewing parts as single entities

MAIN	Algebra II main question	**HSA-SSE.A.1.b**
BACKWARD 1	Squares are never negative	**8.EE.A.2**
BACKWARD 2	Powers of numbers greater than 1 and between 0 and 1	**8.EE.A.1**
BACKWARD 3	A fraction gets smaller as its denominator grows	**3.NF.A.3.d**
FORWARD 1	Find the set of all values of an expression on a domain	**HSA-SSE.A.1.b**
FORWARD 2	Use single-entity reasoning to find horizontal asymptotes	**HSF-IF.C.7.d**

SET 8 • HSA-SSE.A.1.b

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

$P(r) = 5000(1 + r)^{10}$ with $r > 0$ is the value of an investment. Which reasoning shows, without calculating, that $P(r) > 5000$?

A. $(1 + r)^{10} = 1 + r^{10}$ by distributing the exponent, and $1 + r^{10}$ is greater than 1 for $r > 0$, so $P(r) > 5000$.

B. $1 + r > 1$, so $(1 + r)^{10}$ is a product of ten numbers greater than 1 and is greater than 1; 5000 times a number greater than 1 is greater than 5000.

C. For $r = 0.05$, $P = 8{,}144.47 > 5000$, so $P(r) > 5000$ for every $r > 0$.

D. The exponent 10 is greater than 1, so any positive number raised to the 10th power is greater than 1, and $P(r) > 5000$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q1/5

The height of a ball is $h(t) = 80 - 16(t - 2)^2$ feet after t seconds. Without expanding, use the fact that $(t - 2)^2 \geq 0$ to find the greatest height and when it occurs.

Greatest height (ft): _____

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q2/5

$C(x) = 10 + 40/(x + 1)$ is a cost in dollars for $x \geq 0$. Which reasoning shows that $C(x)$ is always more than 10 and never more than 50?

A. $40/(x + 1)$ is at least 40 because $x + 1 \geq 1$, so $C(x)$ is at least $10 + 40 = 50$ for every $x \geq 0$.

B. $40/(x + 1)$ can be 0, so $C(x)$ can equal 10; it is at most 50 because $10 + 40 = 50$.

C. $C(0) = 50$ and $C(100) \approx 10.4$, and both are between 10 and 50, so every value of $C(x)$ is between 10 and 50.

D. $x + 1 \geq 1$, so $40/(x + 1)$ is positive and at most 40; adding 10 gives more than 10 and at most 50 (equal to 50 at $x = 0$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q3/5

The temperature of a drink is $T(t) = 70 - 30(0.8)^t$ °F after t minutes ($t \geq 0$). Which reasoning explains why $T(t)$ is always below 70 and gets closer to 70 as t grows?

A. $30(0.8)^t$ is always positive, so $T < 70$; $(0.8)^t$ shrinks toward 0 as t grows, so T approaches 70.

B. $(0.8)^t$ grows as t grows, so the amount $30(0.8)^t$ grows, and T rises toward 70 from below without reaching it.

C. $30(0.8)^t$ becomes exactly 0 after enough minutes, and from then on T equals 70 and stops changing.

D. $T(0) = 40$ and $T(10) \approx 66.8$ are both below 70 and rising, so T stays below 70 and approaches it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q4/5

$f(x) = 3/(x^2 + 1)$. Which gives the greatest value of $f(x)$ with a correct reason?

A. 4, at $x = 1$, because $x^2 + 1$ is smallest when $x = 1$.

B. 3, at $x = 0$: $x^2 + 1 \geq 1$, so $3/(x^2 + 1) \leq 3$, with equality only when $x = 0$.

C. 3, at $x = 0$, because $f(1) = 1.5$ and $f(2) = 0.6$ are smaller.

D. There is no greatest value, because $x^2 + 1$ can be as large as we like.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 1

Squares are never negative • Grade 8 • Expressions & Equations • 8.EE.A.2

True or false?

$(t - 2)^2 \geq 0$ for every real t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q1/5

What is the smallest value of $(x + 5)^2$, and for which x does it occur?

Smallest value: _____

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q2/5

Which expression can be negative for some real x?

A. $x^2 + 1$

B. $x - 1$

C. x^2

D. $(x - 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q3/5

What is the smallest value of $x^2 + 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q4/5

True or false?

$-(x - 3)^2$ is never positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 2

Powers of numbers greater than 1 and between 0 and 1 • Grade 8 • Expressions & Equations • 8.EE.A.1

Is 1.05^{10} greater than or less than 1?

A. Less than 1

B. Equal to 1

C. Greater than 1

D. It cannot be decided without a calculator

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q1/5

Is 0.8^5 greater than or less than 1?

A. It cannot be decided without a calculator

B. Greater than 1

C. Less than 1

D. Equal to 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q2/5

As n increases, 0.5^n

A. becomes negative

B. stays 0.5

C. gets closer to 0

D. grows without bound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q3/5

True or false?

A negative number raised to any whole-number power is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q4/5

Which order is from least to greatest?

A. 0.9^5, 0.9^2, 0.9

B. 0.9^2, 0.9^5, 0.9

C. 0.9^2, 0.9, 0.9^5

D. 0.9, 0.9^2, 0.9^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 3

A fraction gets smaller as its denominator grows • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d

Which is greater, 40/2 or 40/5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q1/5

True or false?

If $x \geq 0$, then $40/(x + 1) \leq 40$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q2/5

As x grows, 6/(x + 3)

A. increases

B. becomes negative

C. decreases toward 0

D. stays 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q3/5

What is the greatest value of $12/(x + 4)$ for $x \geq 0$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q4/5

True or false?

8/10 is greater than 8/9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 1

Find the set of all values of an expression on a domain • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

Find every value that $h(t) = 80 - 16(t - 2)^2$ takes for $0 \le t \le 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q1/5

Find every value of $f(x) = 3/(x^2 + 1)$ for real x. Write the answer as a compound inequality in y.

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q2/5

True or false?

For $x \geq 0$, $C(x) = 10 + 40/(x + 1)$ takes every value in $10 < C \leq 50$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q3/5

For $t \geq 0$, which set contains every value of $70 - 30(0.8)^t$?

A. $T \geq 70$

B. $40 \leq T < 70$

C. $40 < T \leq 70$

D. $0 < T < 70$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q4/5

Find every value of $g(x) = 5 - 2/(x^2 + 4)$ for real x. Write the answer as a compound inequality in y.

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 2

Use single-entity reasoning to find horizontal asymptotes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Why does $y = 10 + 40/(x + 1)$ have the horizontal asymptote $y = 10$?

A. The constant term of any rational expression, here 10, is always its horizontal asymptote.

B. As x grows without bound, $40/(x + 1)$ gets closer and closer to 0, so y gets closer to 10.

C. When $x = -1$ the fraction is undefined, so $y = 10$ there.

D. At $x = 0$, $y = 10 + 40 = 50$, and 50 is the asymptote shifted up by the 40 in the numerator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q1/5

Rewrite (2x + 7)/(x + 3) as 2 + 1/(x + 3) and state its horizontal asymptote.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q2/5

True or false?

$y = 3/(x^2 + 1)$ has the horizontal asymptote $y = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q3/5

The horizontal asymptote of $y = 5 - 2/(x - 4)$ is

A. $y = -2$

B. $y = 5$

C. $y = 4$

D. $y = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q4/5

Does the graph of $y = 10 + 40/(x + 1)$ ever reach its horizontal asymptote $y = 10$?

A. Yes: for very large x the fraction is 0, so y equals 10.

B. No: the graph stays far from $y = 10$ for every x.

C. Yes: at $x = 0$, where $y = 10$.

D. No: $40/(x + 1)$ is never 0, so y is never exactly 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q5/5

Factor higher-degree expressions by recognizing differences of squares and sums or differences of cubes

MAIN	Algebra II main question	**HSA-SSE.A.2**
BACKWARD 1	Take out a greatest common factor	**6.EE.A.3**
BACKWARD 2	Recognize perfect squares and perfect cubes	**8.EE.A.2**
BACKWARD 3	Multiply to check a factoring pattern	**HSA-APR.A.1**
FORWARD 1	Factor expressions whose parts are themselves expressions	**HSA-SSE.A.2**
FORWARD 2	Complete the factorization over the complex numbers	**HSN-CN.C.8**

SET 9 • HSA-SSE.A.2

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^4 - 81$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q1/5

Factor $8x^3 + 27$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q2/5

Factor $x^6 - 64$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q3/5

Factor $16x^4 - y^4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q4/5

Factor $2x^5 - 32x$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 1

Take out a greatest common factor • Grade 6 • Expressions & Equations • 6.EE.A.3

Factor 6x + 18.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q1/5

Factor $2x^5 - 32x$ by taking out the greatest common factor only.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q2/5

The greatest common factor of $12x^3$ and $18x$ is

A. $6x^3$

B. $6x$

C. $3x$

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$3x^2 + 9x = 3x(x + 9)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q4/5

Factor $5y^2 - 20y$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 2

Recognize perfect squares and perfect cubes • Grade 8 • Expressions & Equations • 8.EE.A.2

Write $16x^4$ as a square.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q1/5

Write $8x^3$ as a cube.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q2/5

Which is a perfect cube?

A. 32

B. 16

C. 36

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q3/5

True or false?

x^6 is both a perfect square and a perfect cube.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q4/5

Find the cube root of 27.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 3

Multiply to check a factoring pattern • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $(a + b)(a^2 - ab + b^2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q1/5

Multiply $(x - 2)(x^2 + 2x + 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q2/5

True or false?

$(x + 3)(x^2 - 6x + 9) = x^3 + 27.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q3/5

$(x^2 - 9)(x^2 + 9)$ equals

A. $x^4 - 18x^2 - 81$

B. $x^4 + 81$

C. $(x - 3)^4$

D. $x^4 - 81$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-B3-Q4**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q4/5

Multiply $(2x + 1)(4x^2 - 2x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • HSA-SSE.A.2

FORWARD 1

Factor expressions whose parts are themselves expressions • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $(x + 1)^2 - 9$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q1/5

Factor $(2x - 1)^3 + 8$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q2/5

True or false?

$a^6 - b^6$ can be factored first as a difference of squares or first as a difference of cubes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q3/5

$(x - y)^2 - (x + y)^2$ equals

A. 0

B. $4xy$

C. $-4xy$

D. $-2y^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q4/5

Factor $x^2(x + 2) - 4(x + 2)$ completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSA-SSE.A.2

FORWARD 2

Complete the factorization over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^4 - 81$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q1/5

Factor $16x^4 - 1$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q2/5

True or false?

Over the complex numbers, $x^3 + 8$ has only one linear factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q3/5

Which polynomial is fully factored over the integers but not over the complex numbers?

A. $x^2 + 9$

B. $x - 9$

C. $x^2 - 9$

D. $(x - 3)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q4/5

$2x^5 - 32x = 2x(x^2 + 4)(x + 2)(x - 2)$ over the integers. Complete the factorization over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q5/5

Rewrite expressions in quadratic form and factor

MAIN	Algebra II main question	**HSA-SSE.A.2**
BACKWARD 1	Factor quadratic trinomials	**HSA-SSE.A.2**
BACKWARD 2	Write powers as squares of other powers	**8.EE.A.1**
BACKWARD 3	Treat a repeated part as a single variable	**HSA-SSE.A.1.b**
FORWARD 1	Solve equations in quadratic form over the complex numbers	**HSN-CN.C.7**
FORWARD 2	Solve exponential equations in quadratic form	**HSF-LE.A.4**

SET 10 • HSA-SSE.A.2

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^4 - 5x^2 + 4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q1/5

Factor $x^4 + 3x^2 - 10$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q2/5

Factor $9^x - 4 \cdot 3^x + 3$ by writing it in quadratic form. Use ^ for exponents.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q3/5

Factor $x^6 - 9x^3 + 8$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q4/5

Factor $2x^4 - 7x^2 - 4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 1

Factor quadratic trinomials • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $u^2 - 5u + 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q1/5

Factor $u^2 + 3u - 10$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q2/5

$2u^2 - 7u - 4$ equals

A. $(2u + 1)(u - 4)$

B. $(u - 2)(2u + 2)$

C. $(2u + 4)(u - 1)$

D. $(2u - 1)(u + 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q3/5

True or false?

$u^2 - 9u + 8 = (u + 1)(u - 8)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q4/5

Factor $u^2 - 4u + 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 2

Write powers as squares of other powers • Grade 8 • Expressions & Equations • 8.EE.A.1

Write x^4 as a power of x^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q1/5

Write 9^x as a power of 3^x. Use ^ for exponents.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q2/5

x^6 equals

A. $(x^4)^2$

B. $(x^3)^2$

C. $x^3 \cdot x^3 \cdot x^3$

D. $(x^2)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q3/5

True or false?

$4^x = (2^x)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q4/5

Write 25^x as a power of 5^x. Use ^ for exponents.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 3

Treat a repeated part as a single variable • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

In $x^4 - 5x^2 + 4$, what single variable u turns the expression into $u^2 - 5u + 4$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q1/5

Rewrite $2(x - 1)^2 + 3(x - 1) - 5$ using $u = x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q2/5

True or false?

In $4^x - 2^x - 6$, the substitution $u = 2^x$ gives $u^2 - u - 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q3/5

Which substitution writes $x^6 - 9x^3 + 8$ as a quadratic?

A. $u = x^2$

B. $u = x^3$

C. $u = x^6$

D. $u = 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q4/5

After factoring $(u - 1)(u - 8)$ with $u = x^3$, write the result in terms of x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q5/5

SET 10 • HSA-SSE.A.2

FORWARD 1

Solve equations in quadratic form over the complex numbers • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^4 + 3x^2 - 4 = 0$ over the complex numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q1/5

Solve $x^4 - 5x^2 + 4 = 0$ over the complex numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q2/5

True or false?

$x^4 + 10x^2 + 9 = 0$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q3/5

How many complex solutions, with multiplicity, does $x^4 - 13x^2 + 36 = 0$ have?

A. 1

B. 3

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q4/5

Solve $x^4 + x^2 - 12 = 0$ over the complex numbers. Give exact solutions.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSA-SSE.A.2

FORWARD 2

Solve exponential equations in quadratic form • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $9^x - 5 \cdot 3^x + 6 = 0$. Give both solutions as decimals rounded to 3 decimal places.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q1/5

Solve $4^x - 6 \cdot 2^x + 8 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q2/5

True or false?

$4^x + 2^x - 2 = 0$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q3/5

The solutions of $25^x - 26 \cdot 5^x + 25 = 0$ are

A. 0 and 2

B. 1 and 25

C. 0 and 1

D. 1 and 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q4**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q4/5

Solve $e^{2t} - 7e^{t} + 10 = 0$. Give both solutions as decimals rounded to 3 decimal places.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q5/5

SET 11 • HSA-SSE.B.4 • SEEING STRUCTURE IN EXPRESSIONS

Derive the formula for the sum of a finite geometric series

MAIN	Algebra II main question	**HSA-SSE.B.4**
BACKWARD 1	Write terms of a geometric sequence	**HSF-LE.A.2**
BACKWARD 2	Distribute a factor over a sum	**6.EE.A.3**
BACKWARD 3	Use the product rule for exponents	**8.EE.A.1**
FORWARD 1	Prove the sum formula from a polynomial identity	**HSA-APR.C.4**
FORWARD 2	Describe how partial sums behave when the ratio is between -1 a...	**HSA-SSE.B.4**

SET 11 • HSA-SSE.B.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

$S_n = a + ar + ar^2 + \ldots + ar^{n-1}$ with $r \neq 1$. In the derivation of $S_n = a(1 - r^n)/(1 - r)$, which equation results from subtracting rS_n from S_n?

A. $S_n - rS_n = a - ar^n$

B. $S_n - rS_n = ar - ar^n$

C. $S_n - rS_n = a - ar$

D. $S_n - rS_n = a - ar^{n-1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q1/5

$S = 3 + 6 + 12 + \ldots + 3 \cdot 2^9$. With the multiply-and-subtract method, $2S - S$ equals which expression, and what is S?

A. $3 \cdot 2^9 - 3$, so $S = 1533$

B. $3 \cdot 2^{10} + 3$, so $S = 3075$

C. $3 \cdot 2^{10} - 6$, so $S = 3066$

D. $3 \cdot 2^{10} - 3$, so $S = 3069$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q2/5

Why does the derivation of $S_n = a(1 - r^n)/(1 - r)$ fail when $r = 1$, and what is S_n then?

A. When $r = 1$ the terms a, ar, ar^2, ... grow without bound, so S_n does not exist and no formula for it is possible.

B. Its last step divides by $1 - r$, which is 0 when $r = 1$. Every term then equals a, so $S_n = na$.

C. rS_n cannot be formed when $r = 1$, so $S_n = a$.

D. The formula gives 0/0, which means $S_n = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q3/5

$T = 1 + x + x^2 + \ldots + x^n$ ($n + 1$ terms, $x \neq 1$).
Multiplying by x and subtracting T gives which formula?
(Check: $x = 2$, $n = 3$ gives $1 + 2 + 4 + 8 = 15$.)

A. $T = (x^n - 1)/(x - 1)$

B. $T = (x^{n+1} - x)/(x - 1)$

C. $T = (x^{n+1} - 1)/(x - 1)$

D. $T = (x^{n+1} - 1)/(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q4/5

$T = 100 + 100(1.05) + 100(1.05)^2 + \ldots + 100(1.05)^{19}$ (20 yearly amounts). Using the multiply-and-subtract method with $1.05T - T$, which equation and total are correct? A calculator may be used.

A. $0.05T = 100(1.05^{19} - 1)$, so $T \approx \$3{,}053.90$

B. $1.05T = 100(1.05^{20} - 1)$, so $T \approx \$157.46$

C. $0.05T = 100(1.05^{21} - 1)$, so $T \approx \$3{,}571.93$

D. $0.05T = 100(1.05^{20} - 1)$, so $T \approx \$3{,}306.60$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 1

Write terms of a geometric sequence • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the first four terms of the geometric sequence with first term 3 and ratio 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q1/5

What is the 10th term of 3, 6, 12, ...?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q2/5

The nth term of a geometric sequence with first term a and ratio r is

A. $a + (n - 1)r$

B. $a \cdot n \cdot r$

C. ar^n

D. ar^{n-1}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-B1-Q3**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q3/5

True or false?

100, 105, 110.25, ... is arithmetic with common difference 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q4/5

Find the ratio of 81, 27, 9, 3, ...

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 2

Distribute a factor over a sum • Grade 6 • Expressions & Equations • 6.EE.A.3

Distribute: 2(3 + 6 + 12).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q1/5

Distribute: r(a + ar + ar²). Use ^ for exponents.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q2/5

True or false?

$x(1 + x + x^2) = x + x^2 + x^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q3/5

1.05(100 + 105) equals

A. 105 + 105

B. 100 + 110.25

C. 105 + 110.25

D. 205

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q4/5

Subtract: $(6 + 12 + 24 + 48) - (3 + 6 + 12 + 24)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 3

Use the product rule for exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Simplify $r \cdot r^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q1/5

Simplify $ar^{n-1} \cdot r$.

A. a^2r^n

B. ar^{n-2}

C. ar^n

D. ar^{n-1}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q2/5

$2 \cdot 2^9$ equals

A. 2^{10}

B. 4^{10}

C. 4^9

D. 2^9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q3/5

True or false?

$x \cdot x^n = x^{2n}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q4/5

Evaluate 2^{10}.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q5/5

SET 11 • HSA-SSE.B.4

FORWARD 1

Prove the sum formula from a polynomial identity • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Which argument proves the identity $x^n - 1 = (x - 1)(x^{n-1} + x^{n-2} + \ldots + x + 1)$?

A. Divide both sides by $x - 1$ to get $(x^n - 1)/(x - 1) = x^{n-1} + \ldots + x + 1$; this is the geometric sum formula, so the identity is true.

B. x times the sum is $x^n + x^{n-1} + \ldots + x$; subtracting the sum $x^{n-1} + \ldots + x + 1$ cancels every term except x^n and -1, leaving $x^n - 1$.

C. x times the sum is $x^n + x^{n-1} + \ldots + x$; subtracting the sum $x^{n-1} + \ldots + x + 1$ cancels every term except x^n and 1, leaving $x^n + 1$.

D. For $x = 2$ and $n = 3$: $7 = 1 \cdot 7$, so the identity holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q1/5

Use the identity $x^n - 1 = (x - 1)(x^{n-1} + \ldots + 1)$ with $x = 3$ to write $1 + 3 + 9 + \ldots + 3^5$ as $(3^m - 1)/(3 - 1)$. Give m and the value of the sum.

m = _____

Sum = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q2/5

True or false?

The identity shows that $x - 1$ is a factor of $x^n - 1$ for every whole number $n \geq 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q3/5

Which equals $1 + 2 + 4 + \ldots + 2^9$?

A. 2^{10}

B. $2^{10} - 1$

C. $2^9 - 1$

D. $(2^9 - 1)/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q4/5

The identity $x^n - 1 = (x - 1)(x^{n-1} + \ldots + x + 1)$ holds for every x. How does it give $a + ar + \ldots + ar^{n-1} = a(1 - r^n)/(1 - r)$ for $r \neq 1$?

A. Put $x = a$ and divide both sides by r: the sum is $(a^n - 1)/(a - 1)$, which is the same as $a(1 - r^n)/(1 - r)$ after simplifying.

B. Put $x = 1$: both sides of the identity are 0, so the identity is checked at that value, and therefore the sum formula holds for every ratio $r \neq 1$.

C. Put $x = r$ and add a to both sides of the identity: the sum is $r^n - 1 + a$, which equals $a(1 - r^n)/(1 - r)$ for $r \neq 1$.

D. Put $x = r$ and multiply by a: $a(r^n - 1) = a(r - 1)(1 + r + \ldots + r^{n-1})$; divide by $r - 1$ ($r \neq 1$), then multiply top and bottom by -1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSA-SSE.B.4

FORWARD 2

Describe how partial sums behave when the ratio is between −1 and 1 • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

For the series 8 + 4 + 2 + 1 + ..., find S_5 and S_{10} (the sums of the first 5 and first 10 terms), to 4 decimal places.

S_5 = _____

S_{10} = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q1/5

$S_n = 16(1 - 0.5^n)$ is the sum of the first n terms of $8 + 4 + 2 + \ldots$. Why is every partial sum less than 16, yet as close to 16 as you like for large n?

A. 0.5^n reaches 0 when $n = 16$, so the partial sums are less than 16 until the 16th term and equal 16 after it.

B. 0.5^n is always positive, so $1 - 0.5^n < 1$ and $S_n < 16$; 0.5^n shrinks toward 0 as n grows, so S_n gets as close to 16 as you like.

C. 0.5^n grows as n grows, so S_n approaches 16 from above.

D. $S_{10} \approx 15.98$ is less than 16 and very close to it, so every partial sum is less than 16 and as close as you like.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q2/5

True or false?

For $r = 2$, the partial sums of $a + ar + ar^2 + \ldots$ $(a > 0)$ stay below a fixed number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q3/5

For $10 - 5 + 2.5 - 1.25 + \ldots$, the partial sums get close to

A. 20/3

B. 15

C. 5

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q4/5

A ball falls 10 m, then rebounds to 60% of each previous height, so the total distance after n bounces is $10 + 2 \cdot 6(1 - 0.6^n)/0.4$ meters. What total distance (in m) can it approach but never reach?

Distance (m): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q5/5

Use the finite geometric series formula to solve problems

MAIN	Algebra II main question	**HSA-SSE.B.4**
BACKWARD 1	Identify the first term, ratio, and number of terms	**HSF-LE.A.2**
BACKWARD 2	Change a percent to a growth factor	**7.RP.A.3**
BACKWARD 3	Round results of calculations correctly	**HSN-Q.A.3**
FORWARD 1	Find the number of terms with logarithms	**HSF-LE.A.4**
FORWARD 2	Compare two payment plans with the series formula	**HSA-SSE.B.4**

SET 12 • HSA-SSE.B.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Find the sum of the first 12 terms of 5 + 15 + 45 + ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q1/5

Leah deposits $250 at the end of each month into an account that earns 0.5% interest per month. The balance just after the 36th deposit is $250 + 250(1.005) + \ldots + 250(1.005)^{35}$. Find it to the nearest cent.

Balance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q2/5

A $180,000 loan is repaid with 360 equal monthly payments P at a monthly interest rate of 0.4%. P satisfies
$180{,}000 = P/1.004 + P/1.004^2 + \ldots + P/1.004^{360}$. Use the geometric series formula to find P to the nearest cent.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q3/5

A ball is dropped from 10 m and each rebound reaches 60% of the previous height. Find the total vertical distance it has traveled when it hits the ground for the 6th time, to the nearest hundredth of a meter.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q4/5

The sum of the first n terms of 2 + 6 + 18 + ... is $S_n = 3^n - 1$. What is the least number of terms whose sum is greater than 10,000? Give that number and the sums of the first 8 and the first 9 terms.

Least number of terms: _____

Sum of the first 8 terms: _____

Sum of the first 9 terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 1

Identify the first term, ratio, and number of terms • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

For the geometric series 5 + 15 + 45 + ... with 12 terms, give the first term a, the ratio r, and the number of terms n.

a = _____

r = _____

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q1/5

For $250 + 250(1.005) + \ldots + 250(1.005)^{35}$, give the first term a, the ratio r, and the number of terms n.

a = _____

r = _____

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q2/5

For 6 + 3.6 + 2.16 + 1.296 + 0.7776, r is

A. 0.4

B. 1.67

C. 0.6

D. 2.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q3/5

True or false?

A list with exponents 0 through 35 has 35 terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q4/5

For $P/1.004 + P/1.004^2 + \ldots + P/1.004^{360}$, which gives the first term a and the common ratio r?

A. $a = P/1.004,\ r = 1/1.004$

B. $a = 1/1.004,\ r = P$

C. $a = P,\ r = 1.004$

D. $a = P/1.004,\ r = 1.004$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 2

Change a percent to a growth factor • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

What factor adds 0.5% interest?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q1/5

A height is 60% of the one before. What is the ratio?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q2/5

0.4% per month as a factor is

A. 1.04

B. 1.004

C. 0.004

D. 1.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q3/5

True or false?

Interest of 12% per year is a factor of 0.12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q4/5

A loss of 15% per year is what factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 3

Round results of calculations correctly • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3

Round 9834.0262 to the nearest cent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q1/5

Round 37.6672 to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q2/5

True or false?

Rounding 1.005 to 1.01 before raising it to the 36th power gives the same answer to the nearest cent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q3/5

1.005^{36} is closest to

A. 1.1967

B. 1.18

C. 1.2

D. 1.036

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q4/5

Evaluate 3^{12} exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q5/5

SET 12 • HSA-SSE.B.4

FORWARD 1

Find the number of terms with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

How many end-of-month deposits of $250 at 0.5% per month are needed for the balance to reach $20,000? Use $250(1.005^n - 1)/0.005 \geq 20{,}000$ and give the smallest whole number n.

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q1/5

The sum of the first n terms of $2 + 6 + 18 + \ldots$ is $3^n - 1$. Solve $3^n - 1 = 1{,}000{,}000$ for n (to 2 decimal places), then give the least whole number of terms whose sum is at least 1,000,000.

n (2 decimal places) = _____

Least number of terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q2/5

True or false?

For a ratio greater than 1, more terms always give a larger sum when $a > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q3/5

The sum $5(2^n - 1)$ first exceeds 5,000 when n =

A. 9

B. 11

C. 10

D. 1000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q4/5

The sum of a geometric series with positive terms first reaches a target when n ≈ 12.58. Why must n be rounded up to 13 rather than simply rounded to the nearest whole number?

A. Rounding down to 12 is safer: fewer terms always give a sum that still reaches the target.

B. Answers must always be rounded to the nearest whole number, and 12.58 rounds to 13 that way.

C. Any whole number near 12.58 gives about the same sum, so 12 and 13 are both acceptable answers, and the choice is a matter of style.

D. The sum increases with n and must reach the target; 12 terms give a sum below it, so round up to 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-F1-Q5**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSA-SSE.B.4

FORWARD 2

Compare two payment plans with the series formula • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Plan A pays $1,000 per month for 24 months. Plan B pays $500 the first month and 8% more each month after that, for 24 months. Find Plan B's total and how much more than Plan A it pays, to the nearest cent.

Plan B total ($): _____

Difference ($): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q1/5

Plan A pays $1,000 per month. Plan B pays $500 the first month and 8% more each month after that. After how many months does Plan B's total first exceed Plan A's total for the same number of months?

Number of months: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q2/5

True or false?

A geometric plan with ratio greater than 1 eventually pays more in total than any plan with a fixed monthly amount.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q3/5

Plan C pays $2,000 then 5% less each month. Its total over 24 months is closest to

A. $40,000

B. $24,000

C. $48,000

D. $28,320

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q4/5

Plan A pays $1,000 per month for 24 months; Plan B pays $500 the first month and 8% more each month after that. Comparing the plans only by their 24-month totals assumes which of the following?

A. That both plans pay exactly $24,000 in total over the 24 months.

B. That Plan B pays the same amount every month, just like Plan A.

C. That money received later is worth the same as money now (no interest or inflation).

D. That the 8% increase in Plan B is applied only once, in month 2, and not again after that.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-F2-Q5**

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q5/5

Add and subtract polynomials of higher degree

MAIN	Algebra II main question	**HSA-APR.A.1**
BACKWARD 1	Combine like terms with the same power	**7.EE.A.1**
BACKWARD 2	Subtract a whole expression	**6.EE.A.3**
BACKWARD 3	Write polynomials in standard form	**HSA-SSE.A.1.a**
FORWARD 1	Reason about the degree of a sum or difference	**HSA-APR.A.1**
FORWARD 2	Add and subtract rational expressions	**HSA-APR.D.7**

SET 13 • HSA-APR.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Simplify $(4x^3 - 2x^2 + 7) + (-x^3 + 5x - 9)$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q1/5

Simplify $(6x^4 - x^3 + 2x) - (2x^4 + 3x^3 - x^2 + 2x - 1)$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q2/5

A company's revenue is $R(x) = -0.01x^3 + 3x^2 + 40x$ dollars and its cost is $C(x) = 0.02x^2 + 15x + 900$ dollars for x items. Write the profit $P(x) = R(x) - C(x)$ in standard form.

$P(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q3/5

Subtract $(x^5 + 2x^4 - 3x^3 - 7)$ from $(x^5 - 3x^3 + x)$. Combine like terms and write the result in standard form; then give its degree.

Result: _____

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q4/5

$p(x) = 5x^3 + x^2 - 6$. Find the polynomial $q(x)$ for which $p(x) + q(x) = 2x^3 - x + 4$.

$q(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 1

Combine like terms with the same power • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify $4x^3 - x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q1/5

Simplify $-2x^2 + 5x - 3x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q2/5

True or false?

$x^3 + x^2 = x^5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q3/5

Which pair are like terms?

A. $4x^2$ and $-x^2$

B. $4x^2$ and $4x$

C. x^3 and $3x$

D. x and 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q4/5

Simplify $7 - 9 + 2x - 2x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 2

Subtract a whole expression • Grade 6 • Expressions & Equations • 6.EE.A.3

Rewrite $-(2x^4 + 3x^3 - x^2 + 2x - 1)$ without parentheses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q1/5

Simplify $(5 - x) - (3 - 4x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q2/5

True or false?

$a - (b - c) = a - b - c$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q3/5

$-(x^2 - 7)$ equals

A. $x^2 + 7$

B. $-x^2 - 7$

C. $-x^2 + 7$

D. $x^2 - 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q4/5

Simplify $10 - (x + 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 3

Write polynomials in standard form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

Write $7 + 5x - 2x^3 + x^2$ in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q1/5

What is the leading coefficient of $3 - 4x^2 + x^4$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q2/5

The degree of $-2x^4 + x + 7$ is

A. 7

B. 1

C. 4

D. -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q3/5

True or false?

The standard form of $2x - x^3$ is $2x - x^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q4/5

What is the constant term of $6x^3 - 2x + 9$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q5/5

SET 13 • HSA-APR.A.1

FORWARD 1

Reason about the degree of a sum or difference • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

$p(x)$ and $q(x)$ both have degree 4. Which statement about $p(x) + q(x)$ is true?

A. Its degree is 4 or less, or it is the zero polynomial: $(x^4 + x) + (-x^4 + 1) = x + 1$.

B. Its degree is always exactly 4, because adding never changes the highest power.

C. Its degree is always 4 or more, because adding can only raise the degree.

D. Its degree is always 8, because adding polynomials adds their degrees: $4 + 4 = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q1/5

For which value of k does
$(kx^3 + 2x - 1) - (5x^3 - x^2 + 3)$ have degree 2?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q2/5

True or false?

The difference of two polynomials is always a polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q3/5

p has degree 5 and q has degree 3. The degree of $p - q$ is

A. 5

B. 2

C. 3

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q4/5

Why is the sum of two polynomials always a polynomial?

A. The sum always has a higher degree than either polynomial, and anything with a higher degree is a polynomial.

B. Adding combines like terms, and each term is still a real number times a whole-number power of x.

C. Adding powers of x gives a power of x, as $x^2 + x^3 = x^5$ shows, so each term of the sum is still a power of x.

D. Any expression that contains x, such as $1/x$ or $\sqrt{x}$, is a polynomial, so every sum of such expressions is a polynomial too.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSA-APR.A.1

FORWARD 2

Add and subtract rational expressions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Write 3/x + 2/(x + 1) as one fraction with denominator x(x + 1). Give the numerator and the excluded values.

Numerator: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q1/5

Write $\frac{1}{(x - 2)} - \frac{1}{(x + 2)}$ as one fraction with denominator $(x - 2)(x + 2)$. Give the numerator and the excluded values.

Numerator: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q2/5

True or false?

Polynomials are closed under division.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q3/5

x/2 + x/3 equals

A. 5x/5

B. x/5

C. 5x/6

D. 2x/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q4/5

How is adding rational expressions like adding fractions of integers?

A. In both, the numerators are multiplied together and the denominators are added together.

B. In both, the numerators are added and the denominators are added.

C. Rational expressions never need a common denominator, unlike fractions of integers, which always do.

D. Both are rewritten over a common denominator and then the numerators are added; polynomials play the role of integers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q5**

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q5/5

Multiply polynomials of higher degree

MAIN	Algebra II main question	**HSA-APR.A.1**
BACKWARD 1	Distribute a monomial over a polynomial	**HSA-APR.A.1**
BACKWARD 2	Multiply powers with the same base	**8.EE.A.1**
BACKWARD 3	Multiply two binomials	**HSA-APR.A.1**
FORWARD 1	Find a coefficient of a product without full expansion	**HSA-APR.A.1**
FORWARD 2	Expand powers of binomials with the Binomial Theorem	**HSA-APR.C.5**

SET 14 • HSA-APR.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $(x^2 + 3x - 2)(2x - 5)$. Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q1/5

Expand $(x - 2)^3$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q2/5

Multiply $(x^2 - x + 1)(x^2 + x + 1)$. Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q3/5

Multiply $(2x + 1)(x - 3)(x + 4)$. Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q4/5

A box has length $x + 2$, width $x - 1$, and height $2x + 3$ (in cm). Write its volume as a polynomial in standard form.

$V(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 1

Distribute a monomial over a polynomial • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $2x(x^2 + 3x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q1/5

Multiply $-5(x^2 + 3x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q2/5

True or false?

$x^2(x + 4) = x^3 + 4x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q3/5

$-3x(x - 1)$ equals

A. $3x^2 - 3x$

B. $-3x^2 + 3x$

C. $-3x^2 - 3x$

D. $-3x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-B1-Q4**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q4/5

Multiply $x(x^2 - x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 2

Multiply powers with the same base • Grade 8 • Expressions & Equations • 8.EE.A.1

Simplify $x^2 \cdot x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q1/5

Simplify $2x \cdot 3x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q2/5

True or false?

$x^2 \cdot x^2 = x^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q3/5

$-x \cdot x^3$ equals

A. x^4

B. $-x^3$

C. $-x^4$

D. $-x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q4/5

Simplify $(x^2)(-4x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 3

Multiply two binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $(x - 3)(x + 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q1/5

Multiply $(x + 2)(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q2/5

$(x - 2)^2$ equals

A. $x^2 + 4$

B. $x^2 - 4$

C. $x^2 - 4x + 4$

D. $x^2 - 2x + 4$

True or false?

$(2x + 1)(x - 3) = 2x^2 - 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q4/5

Multiply $(2x + 3)(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • HSA-APR.A.1

FORWARD 1

Find a coefficient of a product without full expansion • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Find the coefficient of x^2 in $(x^2 + 3x - 2)(4x^2 - x + 5)$ without expanding everything.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q1/5

For which k does $(x + k)(x^2 - 3x + 1)$ have no x^2 term?

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q2/5

True or false?

The constant term of a product of polynomials is the product of their constant terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q3/5

The leading coefficient of $(2x^3 - 1)(5x^2 + x)$ is

A. 7

B. 5

C. 2

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q4/5

Find the coefficient of x in $(x - 1)(x - 2)(x - 3)$.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSA-APR.A.1

FORWARD 2

Expand powers of binomials with the Binomial Theorem • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Expand $(x + 1)^4$ using Pascal's Triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q1/5

Expand $(x - 2)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q2/5

True or false?

The coefficients of $(x + y)^5$ are 1, 5, 10, 5, 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q3/5

The coefficient of x^2y^2 in $(x + y)^4$ is

A. 2

B. 6

C. 4

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q4/5

Why are the coefficients of $(x + y)^n$ symmetric (the same read forward and backward)?

A. $C(n, k) = C(n, n - k)$: choosing which k of the n factors give x is the same as choosing which $n - k$ give y.

B. x and y always have equal values in the expansion, so the coefficients of matching terms are the same.

C. Every coefficient of $(x + y)^n$ equals 1.

D. The middle coefficient is always the largest one, so the list of coefficients must be symmetric.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F2-Q5**

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q5/5

Find remainders with the Remainder Theorem

MAIN	Algebra II main question	**HSA-APR.B.2**
BACKWARD 1	Evaluate a polynomial function	**HSF-IF.A.2**
BACKWARD 2	Powers of negative numbers	**7.NS.A.2**
BACKWARD 3	Dividend = divisor × quotient + remainder	**4.NBT.B.6**
FORWARD 1	Find unknown coefficients from given remainders	**HSA-APR.B.2**
FORWARD 2	Prove the Remainder Theorem	**HSA-APR.B.2**

SET 15 • HSA-APR.B.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Use the Remainder Theorem to find the remainder when $p(x) = x^3 - 4x^2 + 2x - 5$ is divided by $x - 3$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q1/5

Use the Remainder Theorem to find the remainder when $p(x) = 2x^4 + x^3 - 3x + 7$ is divided by $x + 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q2/5

Use the Remainder Theorem to find the remainder when $p(x) = x^{100} + 3$ is divided by $x + 1$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q3/5

Use the Remainder Theorem to find the remainder when $p(x) = 4x^3 - 6x^2 + x - 1$ is divided by $x - 1/2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q4/5

Use the Remainder Theorem to find the remainder when $p(x) = x^4 - 5x^2 + 4x$ is divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 1

Evaluate a polynomial function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

$p(x) = x^3 - 4x^2 + 2x - 5$. Find $p(1)$.

$p(1) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q1/5

$p(x) = 2x^2 - x + 3$. Find $p(-2)$.

$p(-2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q2/5

True or false?

If $p(x) = x^2 + 1$, then $p(-3) = -8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q3/5

If $p(x) = x^3 - x$, then $p(2) =$

A. 4

B. 6

C. 8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q4/5

$p(x) = 4x - 1$. Find $p(1/2)$.

$p(1/2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 2

Powers of negative numbers • Grade 7 • The Number System • 7.NS.A.2

Evaluate $(-2)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q1/5

Evaluate $(-2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q2/5

$(-1)^{100}$ equals

A. -100

B. -1

C. 100

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q3/5

True or false?

$-2^4 = (-2)^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q4/5

Evaluate $2(-2)^4 + (-2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 3

Dividend = divisor × quotient + remainder • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Write $47 = 5 \times q + r$ with $0 \leq r < 5$.

q = _____

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q1/5

A division gives quotient 12 and remainder 3 with divisor 7. What was the dividend?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q2/5

True or false?

If a division leaves remainder 0, the divisor is a factor of the dividend.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q3/5

$x^3 - 1 = (x - 1)(x^2 + x + 1) + 0$. The remainder on division by $x - 1$ is

A. 0

B. 1

C. $x^2 + x + 1$

D. -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q4/5

$p(x) = (x - 3)(x^2 - x - 1) - 8$. What is the remainder when $p(x)$ is divided by $x - 3$?

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • HSA-APR.B.2

FORWARD 1

Find unknown coefficients from given remainders • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

When $p(x) = x^3 + kx^2 - 4$ is divided by $x - 2$, the remainder is 12. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q1/5

$p(x) = x^3 + ax + b$ leaves remainder 5 on division by $x - 1$ and remainder -1 on division by $x + 1$. Find a and b.

a = _____

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q2/5

True or false?

If p(x) leaves remainder 0 on division by $x - 4$, then $p(-4) = 0$.

A. True

B. False

$p(x) = 2x^2 + kx + 1$ leaves remainder 7 on division by $x + 1$. Then $k =$

A. 4

B. -4

C. -6

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q4/5

For which k is the remainder of $x^4 - kx + 6$ divided by $x + 2$ equal to 0?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSA-APR.B.2

FORWARD 2

Prove the Remainder Theorem • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Which argument proves that the remainder when a polynomial $p(x)$ is divided by $x - a$ is $p(a)$?

A. Write $p(x) = (x - a)q(x) + r$, where r is a constant because it has degree less than 1. Substituting $x = a$ gives $p(a) = 0 \cdot q(a) + r = r$.

B. Dividing x^3 by $x - 2$ leaves remainder 8, and $2^3 = 8$, so the remainder is always $p(a)$.

C. Since $x - a = 0$ exactly when $x = a$, dividing $p(x)$ by $x - a$ is the same as replacing x with a, so the remainder must be $p(a)$.

D. Write $p(x) = (x - a)q(x) + r$. Substituting $x = 0$ gives $p(0) = (0 - a)q(0) + r = -a \cdot q(0) + r$, so the remainder r equals $p(a)$ for every a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q1/5

When a polynomial $p(x)$ is divided by $x - a$, why must the remainder be a constant?

A. Because dividing by $x - a$ always comes out evenly, so the remainder always equals 0.

B. Because a is a constant, everything left over after dividing by $x - a$ must also be a constant.

C. The remainder has a lower degree than the divisor $x - a$ (degree 1), so it is a constant or 0.

D. Because $p(x)$ always has degree 1 when it is divided by $x - a$, so the remainder is a constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q2/5

True or false?

The Remainder Theorem also works for division by $2x - 1$ with remainder $p(1/2)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q3/5

Which step uses the fact that $a - a = 0$?

A. Graphing $p(x)$ and reading its zeros

B. Dividing $p(x)$ by x to find the quotient $q(x)$

C. Factoring $p(x)$ into linear factors

D. Substituting $x = a$ into $(x - a)q(x) + r$

$p(x) = (x - a)q(x) + r$, where r is a constant. Why is $p(a) = 0$ exactly when $x - a$ is a factor of $p(x)$?

A. $p(a) = 0$ for every polynomial written as $(x - a)q(x) + r$, so $x - a$ is always a factor.

B. Substituting $x = a$ gives $p(a) = r$; and $r = 0$ exactly when $p(x) = (x - a)q(x)$, which means $x - a$ is a factor.

C. $x - a$ is a factor exactly when $q(a) = 0$.

D. Substituting $x = a$ gives $p(a) = q(a)$, so $p(a) = 0$ exactly when $q(x)$ is a factor of $p(x)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q5/5

Use p(a) = 0 to find factors

MAIN	Algebra II main question	**HSA-APR.B.2**
BACKWARD 1	Use the Remainder Theorem	**HSA-APR.B.2**
BACKWARD 2	Factor the quadratic quotient	**HSA-SSE.A.2**
BACKWARD 3	Divide a polynomial by a linear factor	**HSA-APR.A.1**
FORWARD 1	Choose parameters so that a given binomial is a factor	**HSA-APR.B.2**
FORWARD 2	Finish factorizations over the complex numbers	**HSN-CN.C.8**

SET 16 • HSA-APR.B.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

$p(x) = x^3 - 3x^2 - 4x + 12$ has $p(2) = 0$, so $x - 2$ is a factor. Factor $p(x)$ completely.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q1/5

$p(x) = x^3 + 2x^2 - 5x - 6$ has $p(-1) = 0$, so $x + 1$ is a factor. Factor $p(x)$ completely.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q2/5

$p(x) = 2x^3 + 5x^2 - 4x - 3$ has $p(-3) = 0$, so $x + 3$ is a factor. Factor $p(x)$ completely.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q3/5

Is $x - 4$ a factor of $p(x) = x^3 - 5x^2 + 2x + 9$? Use the Remainder Theorem: find $p(4)$, then answer Yes or No.

$p(4) =$ _____

Factor? (Yes or No): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q4/5

$p(x) = x^3 - 7x + 6$. Use $p(a) = 0$ to find all the zeros of p. List them separated by commas.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 1

Use the Remainder Theorem • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Find the remainder when $x^3 - 3x^2 - 4x + 12$ is divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q1/5

Find the remainder when $x^3 + 2x^2 - 5x - 6$ is divided by $x - 1$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q2/5

True or false?

Dividing by $x + 3$ means evaluating at $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q3/5

The remainder of $p(x) \div (x + 1)$ is

A. $-p(1)$

B. $p(-1)$

C. $p(0)$

D. $p(1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q4/5

Find $p(-3)$ for $p(x) = 2x^3 + 5x^2 - 4x - 3$.

$p(-3) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 2

Factor the quadratic quotient • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 - x - 6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q1/5

Factor $2x^2 - x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q2/5

$x^2 + x - 6$ equals

A. $(x - 3)(x + 2)$

B. $(x - 6)(x + 1)$

C. $(x + 3)(x - 2)$

D. $(x + 6)(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q3/5

True or false?

$x^2 + 4$ factors over the integers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q4/5

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 3

Divide a polynomial by a linear factor • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Find $q(x)$ if $x^3 - 3x^2 - 4x + 12 = (x - 2)\,q(x)$.

$q(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q1/5

Multiply $(x + 1)(x^2 + x - 6)$ to check a division.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q2/5

True or false?

If $(x - 1)(x^2 + x - 6) = x^3 - 7x + 6$, then $x^2 + x - 6$ is the quotient of $x^3 - 7x + 6 \div (x - 1)$.

A. True

B. False

For $x \neq 2$, $(x^3 - 8) \div (x - 2)$ equals

A. $x^2 + 2x + 4$

B. $x^2 + 4$

C. $x^2 - 4$

D. $x^2 - 2x + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q4/5

Find $q(x)$ if $2x^3 + 5x^2 - 4x - 3 = (x + 3)\, q(x)$.

$q(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q5/5

SET 16 • HSA-APR.B.2

FORWARD 1

Choose parameters so that a given binomial is a factor • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Find k so that $x - 3$ is a factor of $x^3 + kx^2 - 3x + 9$.

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q1/5

Find a and b so that $x - 1$ and $x + 2$ are both factors of $x^3 + ax^2 + bx - 2$.

a = _____

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q2/5

True or false?

If $x - 2$ and $x + 2$ are factors of $p(x)$, then $x^2 + 4$ is a factor of $p(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q3/5

For which k is $x + 1$ a factor of $x^4 + kx + 3$?

A. 2

B. -4

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q4/5

Write a cubic with leading coefficient 1 that has $x - 1$, $x + 1$, and $x - 4$ as factors, in standard form.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSA-APR.B.2

FORWARD 2

Finish factorizations over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

$x - 1$ is a factor of $x^3 - 1$. Factor $x^3 - 1$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q1/5

$x + 2$ is a factor of $x^3 + 2x^2 + 9x + 18$. Factor it completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q2/5

True or false?

Every factor found with the factor theorem gives a zero of the polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q3/5

$x - 2$ is a factor of $x^3 - 2x^2 + x - 2$. The other zeros are

A. i and $-i$

B. -2

C. 1 and -1

D. $2i$ and $-2i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q4/5

$x - 3$ is a factor of $x^3 - 3x^2 + 4x - 12$. List all complex zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q5/5

Identify zeros and multiplicities from a factorization

MAIN	Algebra II main question	**HSA-APR.B.3**
BACKWARD 1	Solve by the zero-product property	**HSA-REI.B.4.b**
BACKWARD 2	Factor to find hidden repeated factors	**HSA-SSE.A.2**
BACKWARD 3	An exponent counts repeated factors	**6.EE.A.1**
FORWARD 1	Build a polynomial from zeros and one more condition	**HSA-APR.B.3**
FORWARD 2	Decide what the zeros determine and what remains free	**HSA-APR.B.3**

SET 17 • HSA-APR.B.3

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

$p(x) = (x - 4)^2(x + 1)(2x - 3)$. List the zero of multiplicity 2, and the zeros of multiplicity 1.

Zero of multiplicity 2: _____

Zeros of multiplicity 1: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q1/5

$p(x) = x^3 - 6x^2 + 9x$ can be written as $x(x - a)^2$ with a positive number a. Find a, then give the zero of multiplicity 1 and the zero of multiplicity 2.

a = _____

Zero of multiplicity 1: _____

Zero of multiplicity 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q2/5

List the zeros of $p(x) = (x^2 - 9)(x^2 + 2x + 1)$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q3/5

List the zeros of $p(x) = -2x(x + 5)^3$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q4/5

Which gives the factorization of $p(x) = x^4 - 8x^2 + 16$ and its zeros with multiplicities?

A. $(x - 2)(x + 2)$; zeros 2 and -2, each of multiplicity 1

B. $(x^2 - 4)^2$; the only zero is 4, of multiplicity 2

C. $(x - 4)^2$; the only zero is 4, of multiplicity 2

D. $(x - 2)^2(x + 2)^2$; zeros 2 and -2, each of multiplicity 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 1

Solve by the zero-product property • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $(x - 4)(x + 1) = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q1/5

Solve $x(2x - 3) = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q2/5

True or false?

If $(x - 2)^2 = 0$, then $x = 2$ or $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q3/5

The solutions of $(x + 5)(x - 5) = 0$ are

A. 25

B. -5 and 5

C. 0 and 5

D. 5 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q4/5

Solve $-2x(x + 5) = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 2

Factor to find hidden repeated factors • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 - 6x + 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q1/5

Factor $x^2 + 2x + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q2/5

$x^4 - 8x^2 + 16$ equals

A. $(x^2 + 4)^2$

B. $(x^2 - 4)^2$

C. $(x^2 - 8)^2$

D. $(x - 4)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q3/5

True or false?

$x^3 - 6x^2 + 9x = x(x + 3)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q4/5

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 3

An exponent counts repeated factors • Grade 6 • Expressions & Equations • 6.EE.A.1

Write $(x - 4)^2$ as a product without an exponent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q1/5

How many factors of $(x + 5)$ are in $(x + 5)^3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q2/5

True or false?

$(x - 2)^2(x - 2) = (x - 2)^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q3/5

In $7(x - 1)^4$, the factor $x - 1$ appears

A. 28 times

B. 4 times

C. 1 time

D. 7 times

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q4/5

Write $(x + 1)(x + 1)(x - 3)$ with an exponent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q5/5

SET 17 • HSA-APR.B.3

FORWARD 1

Build a polynomial from zeros and one more condition • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Write the cubic polynomial with zeros -1, 2, and 4 (each multiplicity 1) whose graph passes through $(0, 16)$.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q1/5

Write a polynomial of degree 3 with a zero of multiplicity 2 at $x = 3$, a zero at $x = -2$, and leading coefficient -1.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q2/5

True or false?

Zeros alone determine a polynomial completely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q3/5

A quartic has zeros 1 and −1, each of multiplicity 2, and p(0) = 3. Then p(x) =

A. $-3(x-1)^2(x+1)^2$

B. $3(x^2-1)$

C. $(x-1)^2(x+1)^2$

D. $3(x-1)^2(x+1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F1-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q4/5

A polynomial has zeros 0 (multiplicity 2) and 5 (multiplicity 1), degree 3, and $p(1) = -8$. Find $p(x)$.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSA-APR.B.3

FORWARD 2

Decide what the zeros determine and what remains free • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Which describes all polynomials of degree 2 whose only zeros are 1 and 3 (each of multiplicity 1)?

A. $a(x + 1)(x + 3)$ for any real number $a \neq 0$

B. $a(x - 1)(x - 3) + c$ for any real numbers a and c

C. $(x - 1)(x - 3)$ only

D. $a(x - 1)(x - 3)$ for any real number $a \neq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q1/5

Which polynomial has 1 and 3 as its only real zeros but is NOT a constant multiple of $(x - 1)(x - 3)$?

A. $2(x - 1)(x - 3)$

B. $(x - 1)(x - 3)(x + 1)$

C. $(x - 1)^2(x - 3)(x^2 + 1)$

D. $(x - 1)(x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q2/5

True or false?

Two cubic polynomials with the same three zeros (each multiplicity 1) must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q3/5

Which extra information fixes a cubic with zeros −2, 0, 2?

A. That it has degree 3

B. That 0 is a zero

C. Its end behavior only

D. Its value at x = 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q4/5

Why may the real zeros of a degree-4 polynomial fail to determine it, even up to a constant multiple?

A. Every degree-4 polynomial with real coefficients has four real zeros counted with multiplicity, so its real zeros always determine it up to a constant multiple.

B. Two polynomials with the same real zeros must be equal.

C. It may have non-real zeros, from a factor such as $x^2 + c$ with $c > 0$, that the real zeros do not show; for example $(x - 1)(x - 2)(x^2 + 1)$ and $(x - 1)(x - 2)(x^2 + 4)$ have the same real zeros.

D. They never fail: the real zeros of a degree-4 polynomial determine it exactly, including its leading coefficient, so $(x - 1)(x - 2)$ gives the whole polynomial.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q5/5

Identify the rough graph of a polynomial from its zeros

MAIN	Algebra II main question	**HSA-APR.B.3**
BACKWARD 1	Read zeros and multiplicities	**HSA-APR.B.3**
BACKWARD 2	End behavior from the leading term	**HSF-IF.B.4**
BACKWARD 3	Find the y-intercept by evaluating at 0	**HSF-IF.A.2**
BACKWARD 4	Sign of a power of a negative or positive number	**7.NS.A.2**
FORWARD 1	Sketch graphs from standard form by factoring first	**HSF-IF.C.7.c**
FORWARD 2	Write a polynomial from its rough graph	**HSA-APR.B.3**

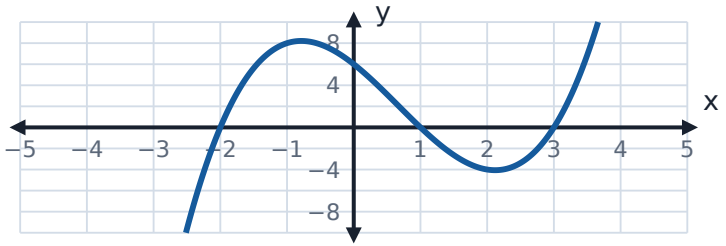
SET 18 • HSA-APR.B.3

MAIN

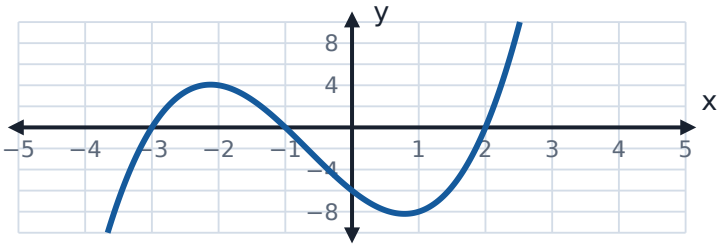
Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Which graph is a rough graph of $y = (x + 2)(x - 1)(x - 3)$?

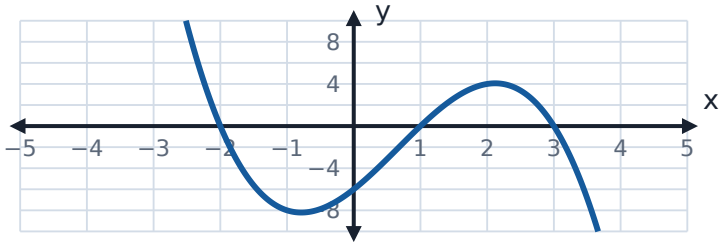
A



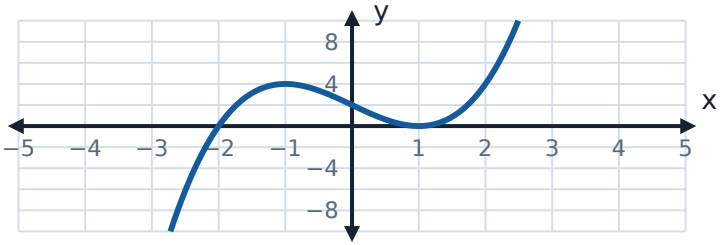
B



C



D



A. Graph A

B. Graph B

C. Graph C

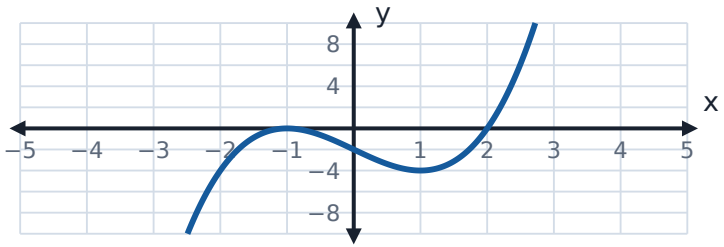
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-M-Q1**

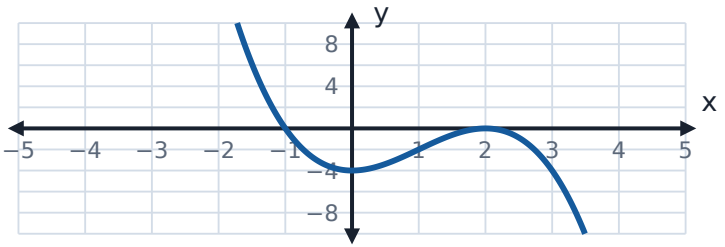
High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q1/5

Which graph is a rough graph of $y = -(x + 1)^2(x - 2)$?

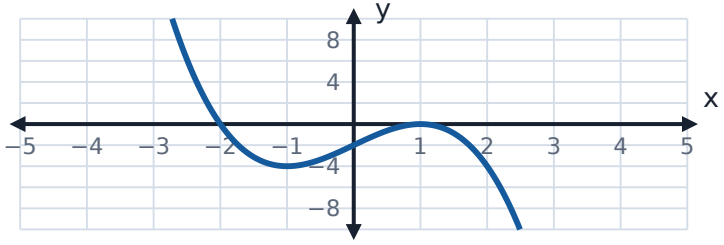
A



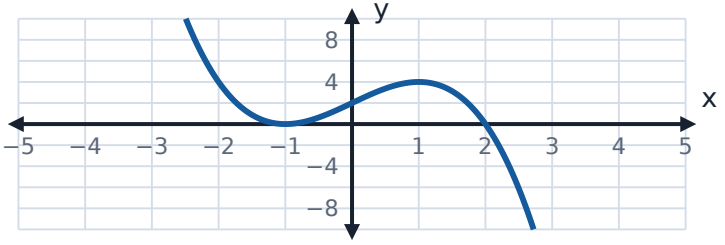
B



C



D



A. Graph A

B. Graph B

C. Graph C

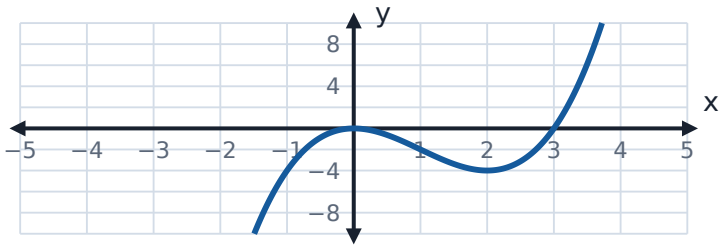
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-M-Q2**

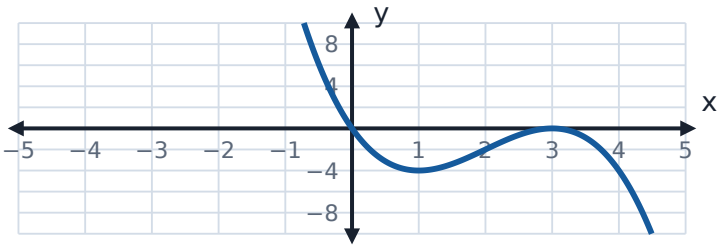
High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q2/5

Which graph is a rough graph of $y = x(x - 3)^2$?

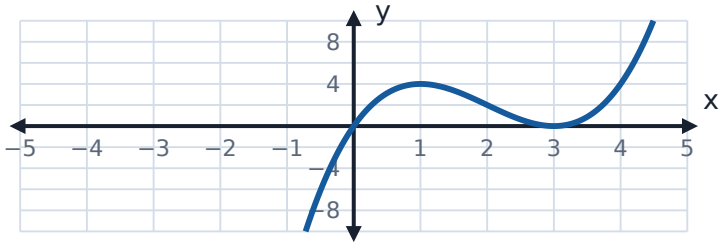
A



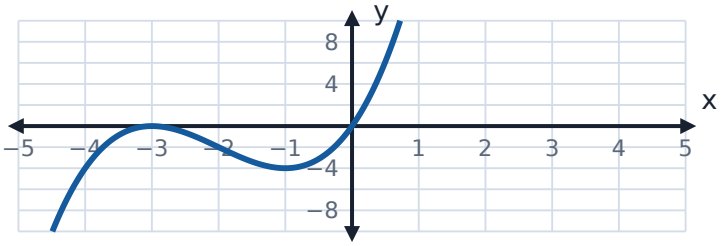
B



C



D



A. Graph A

B. Graph B

C. Graph C

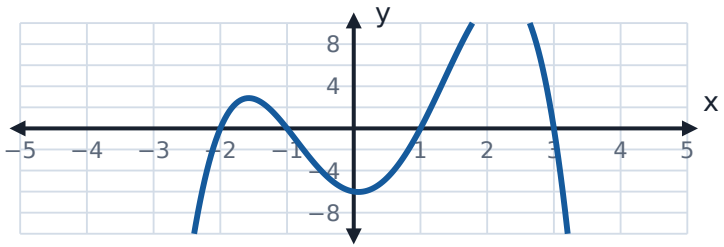
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-M-Q3**

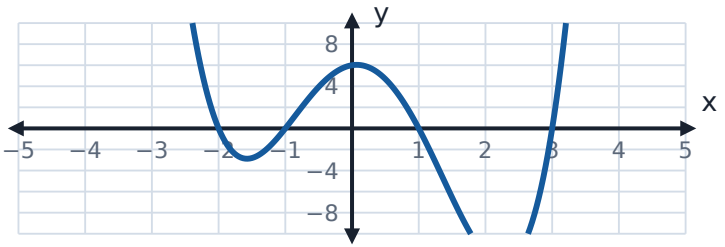
High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q3/5

Which graph is a rough graph of $y = (x + 2)(x + 1)(x - 1)(x - 3)$?

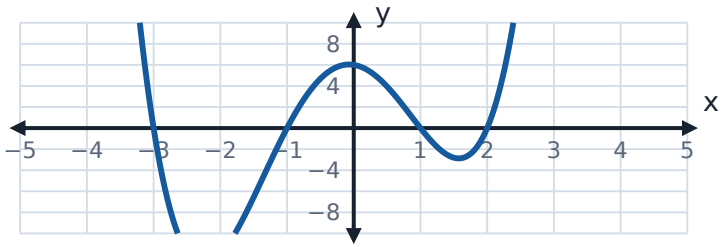
A



B



C



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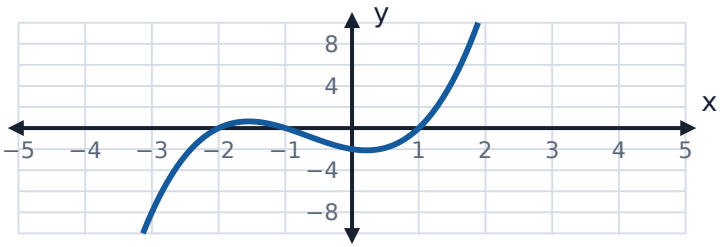


A. Graph A

B. Graph B

C. Graph C

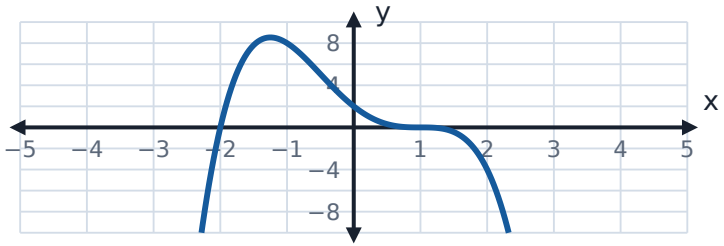
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-M-Q4**

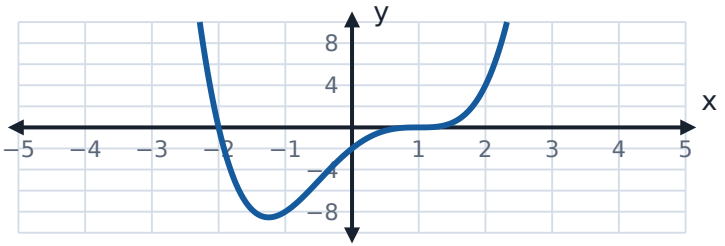
High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q4/5

Which graph is a rough graph of $y = -(x - 1)^3(x + 2)$?

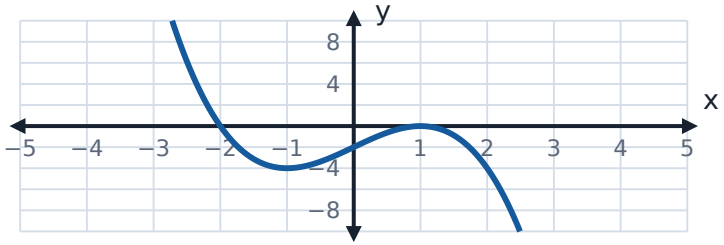
A



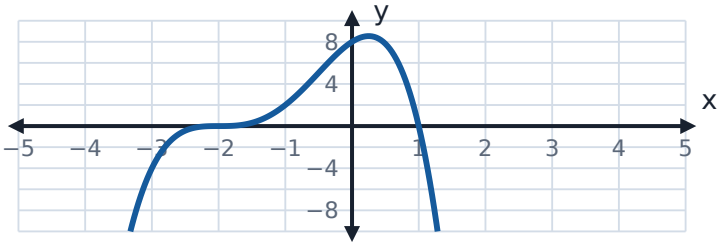
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-M-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 1

Read zeros and multiplicities • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

List the zeros of $(x + 2)(x - 1)(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q1/5

List the zeros of $-(x + 1)^2(x - 2)$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q2/5

True or false?

$x(x - 3)^2$ has a zero of multiplicity 2 at $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q3/5

The zero $x = 1$ of $-(x - 1)^3(x + 2)$ has multiplicity

A. 1

B. 2

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q4/5

List the zeros of $(x + 2)(x + 1)(x - 1)(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 2

End behavior from the leading term • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

As x grows large and positive, what happens to x^3?

A. It grows large and negative.

B. It grows large and positive.

C. It approaches 1.

D. It approaches 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q1/5

As x grows large and negative, what happens to $-x^4$?

A. It becomes large and negative.

B. It approaches -1.

C. It becomes large and positive.

D. It approaches 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q2/5

The leading term of $(x + 2)(x - 1)(x - 3)$ is

A. $-x^3$

B. -6

C. $3x$

D. x^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q3/5

True or false?

A polynomial of even degree with a positive leading coefficient falls at both ends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q4/5

What is the leading term of $-(x - 1)^3(x + 2)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 3

Find the y-intercept by evaluating at 0 • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Find y when x = 0 for $y = (x + 2)(x - 1)(x - 3)$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q1/5

Find y when x = 0 for $y = -(x + 1)^2(x - 2)$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q2/5

True or false?

The y-intercept of $x(x - 3)^2$ is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q3/5

The y-intercept of $-(x-1)^3(x+2)$ is

A. 0

B. -2

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q4/5

Find y when $x = 1$ for $y = x(x - 3)^2$.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 4

Sign of a power of a negative or positive number • Grade 7 • The Number System • 7.NS.A.2

Is $(-0.5)^2$ positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q1/5

Is $(-0.5)^3$ positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q2/5

True or false?

An even power of a negative number is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q3/5

Near $x = 3$, $(x - 3)^2$ is

A. negative on both sides

B. positive on both sides

C. zero everywhere

D. positive on the left and negative on the right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q4/5

Does $(x - 1)^3$ change sign at $x = 1$?

A. No: it is negative on both sides of $x = 1$.

B. No: it is positive on both sides of $x = 1$.

C. Yes: it is positive for $x < 1$ and negative for $x > 1$.

D. Yes: it is negative for $x < 1$ and positive for $x > 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q5/5

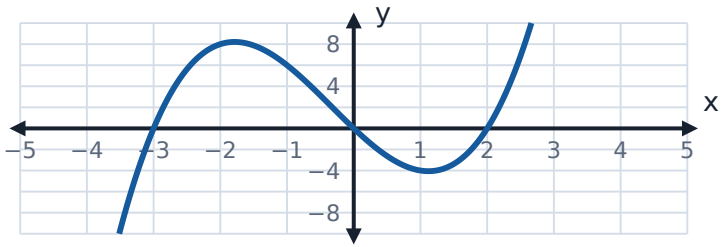
SET 18 • HSA-APR.B.3

FORWARD 1

Sketch graphs from standard form by factoring first • High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c

Factor $y = x^3 - x^2 - 6x$. Which graph is a rough graph of it?

A



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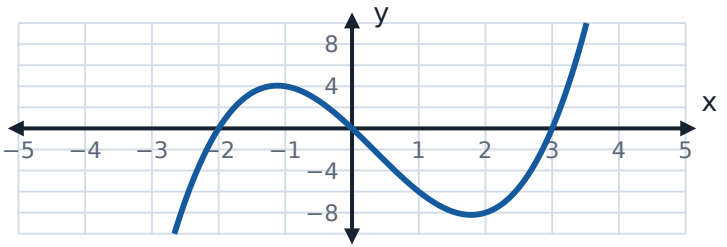

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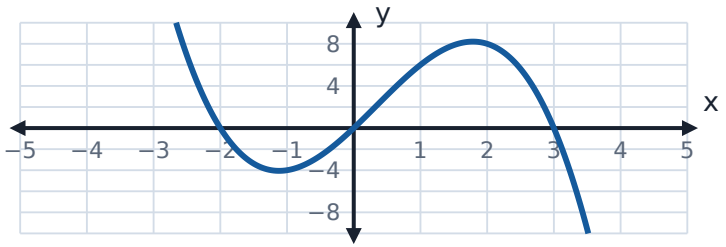
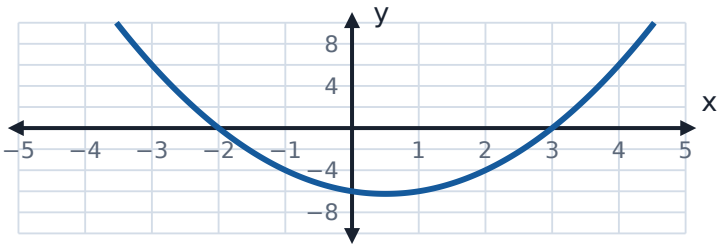


D



A. Graph A

B. Graph B

C. Graph C

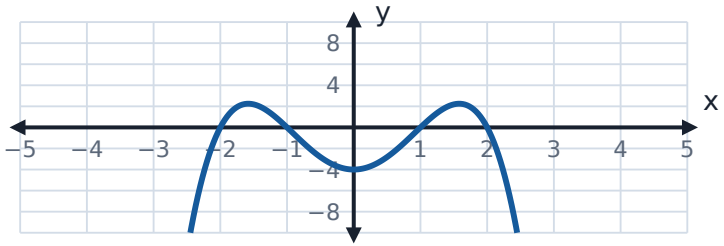
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F1-Q1**

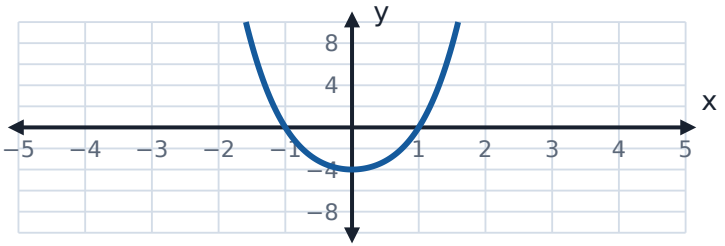
High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q1/5

Factor $y = x^4 - 5x^2 + 4$. Which graph is a rough graph of it?

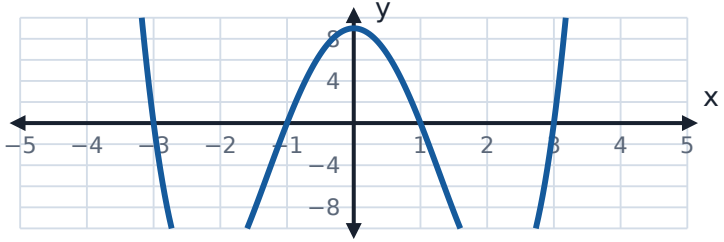
A



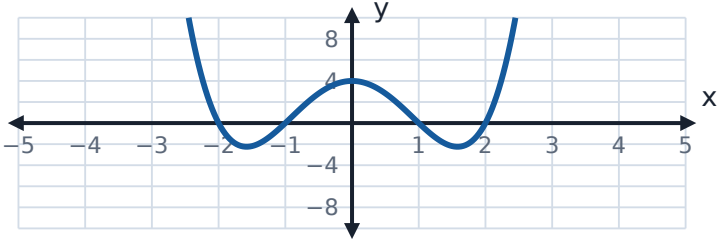
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F1-Q2**

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q2/5

True or false?

$y = x^3 + 4x$ crosses the x-axis at $x = -2$, 0, and 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q3/5

$y = -x^3 + 9x$ crosses the x-axis at

A. 0 and 9

B. 3 only

C. $-3, 0, 3$

D. $-9, 0, 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q4/5

$y = x^3 - 3x^2 + 4 = (x + 1)(x - 2)^2$. What does the graph do at each zero?

A. It crosses the x-axis at $x = 1$ and touches it at $x = -2$.

B. It crosses the x-axis at $x = -1$ and touches it (turns back) at $x = 2$.

C. It crosses the x-axis at both $x = -1$ and $x = 2$.

D. It touches the x-axis at $x = -1$ and crosses it at $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSA-APR.B.3

FORWARD 2

Write a polynomial from its rough graph • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

A cubic's graph crosses the x-axis at −3 and touches it at 2, and passes through (0, 12). Write the polynomial.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q1/5

A quartic rises at both ends, touches the x-axis at -1 and at 3, and passes through (0, 18). Write the polynomial.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q2/5

True or false?

A graph that touches the x-axis at x = a without crossing has a zero of even multiplicity at a.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q3/5

A cubic falls to the right and crosses the x-axis at 0, 1, and 4. Its leading coefficient is

A. zero

B. not determined by the end behavior

C. positive

D. negative

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F2-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q4/5

A polynomial graph crosses at -2, touches at 1, falls to the left and rises to the right, and has y-intercept 4. Give a polynomial of least degree.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q5/5

Prove polynomial identities

MAIN	Algebra II main question	**HSA-APR.C.4**
BACKWARD 1	Expand the square of a binomial	**HSA-APR.A.1**
BACKWARD 2	Recognize a difference of squares	**HSA-SSE.A.2**
BACKWARD 3	What makes an identity	**6.EE.A.4**
FORWARD 1	Prove or disprove proposed identities	**HSA-APR.C.4**
FORWARD 2	Prove identities with the Binomial Theorem	**HSA-APR.C.5**

SET 19 • HSA-APR.C.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Which argument proves the identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$?

A. For $x = 2$ and $y = 1$: $25 = 9 + 16$, so the identity is true.

B. Take the square root of both sides: $x^2 + y^2 = (x^2 - y^2) + 2xy$, and the right side simplifies to $x^2 - y^2 + 2xy$, so the identity is true.

C. The left side is $x^4 + 2x^2y^2 + y^4$. The right side is $x^4 - 2x^2y^2 + y^4 + 4x^2y^2 = x^4 + 2x^2y^2 + y^4$. The sides are equal.

D. The right side is $(x^2 - y^2)^2 + (2xy)^2 = x^4 - y^4 + 4x^2y^2$, and the left side is $x^4 + y^4 + 4x^2y^2$, so the two sides are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q1/5

Which argument proves the identity $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$?

A. The product is $a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3$. The terms $-a^2b$ and a^2b cancel, ab^2 and $-ab^2$ cancel, leaving $a^3 + b^3$.

B. For $a = 1$ and $b = 2$ the left side is $1 + 8 = 9$ and the right side is $3 \cdot (1 - 2 + 4) = 9$, so the identity is true for all a and b.

C. Multiply the first terms and the last terms: $a \cdot a^2 = a^3$ and $b \cdot b^2 = b^3$, so the product is $a^3 + b^3$.

D. The product is $a^3 - 2a^2b + b^3$, which equals $a^3 + b^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q2/5

Which argument proves the identity $(a + b)^2 - (a - b)^2 = 4ab$?

A. For $a = 3$ and $b = 1$: $16 - 4 = 12 = 4 \cdot 3$, so the identity is true.

B. $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = a^2 + 2ab + b^2 - a^2 + 2ab - b^2 = 4ab$.

C. $(a + b)^2 - (a - b)^2 = [(a + b) - (a - b)]^2 = (2b)^2 = 4ab$.

D. $(a + b)^2 - (a - b)^2 = (a^2 + b^2) - (a^2 - b^2) = a^2 + b^2 - a^2 + b^2 = 4ab$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q3/5

Which argument proves the identity $x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2)$?

A. The right side is $((x^2 + 2) + 2x)((x^2 + 2) - 2x) = (x^2 + 2)^2 + (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4$.

B. For $x = 0$: $4 = 2 \cdot 2$, so the identity is true.

C. The right side is $((x^2 + 2) + 2x)((x^2 + 2) - 2x) = (x^2 + 2)^2 - (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4$.

D. Multiply the first terms and the last terms: $x^2 \cdot x^2 = x^4$ and $2 \cdot 2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-M-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q4/5

Which argument proves the identity $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$?

A. For n = 1, 2, and 3 the left side is 7, 19, and 37, and so is the right side, so it is true for every n.

B. $(n + 1)^3 = n^3 + 1$, so $(n + 1)^3 - n^3 = 1$, which is $3n^2 + 3n + 1$.

C. $(n + 1)^3 - n^3 = (n + 1 - n)^3 = 1^3 = 1$.

D. $(n + 1)^3 = n^3 + 3n^2 + 3n + 1$, so $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 1

Expand the square of a binomial • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Expand $(a + b)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q1/5

Expand $(x^2 - y^2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q2/5

True or false?

$(a - b)^2 = a^2 - b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q3/5

$(2xy)^2$ equals

A. $2xy^2$

B. $4xy$

C. $2x^2y^2$

D. $4x^2y^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-B1-Q4**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q4/5

Expand $(n + 1)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 2

Recognize a difference of squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $a^2 - b^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q1/5

Factor $(x^2 + 2)^2 - (2x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q2/5

True or false?

$u^2 - v^2 = (u - v)(u + v)$ for any expressions u and v.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q3/5

Which is a difference of squares?

A. $x^3 - 1$

B. $x^2 - 3x$

C. $9x^2 - 16$

D. $9x^2 + 16$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q4/5

Factor $25 - x^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 3

What makes an identity • Grade 6 • Expressions & Equations • 6.EE.A.4

Test $(a + b)^2 = a^2 + b^2$ with $a = 1$ and $b = 2$. Which conclusion is correct?

A. It is not an identity: the left side is 9 and the right side is 5.

B. The test is inconclusive: one example can never show that a statement is not an identity.

C. It is an identity: both sides equal 9.

D. It is an identity: both sides equal 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q1/5

Which statement describes the difference between an identity and an equation?

A. An identity is true for at least one example; an equation is true for no values.

B. An identity has no variables; an equation has variables.

C. An identity is true for every value of its variables; an equation may be true only for some values.

D. An identity is an equation that has exactly one solution, found by solving it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q2/5

True or false?

Checking an identity with three numbers proves it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q3/5

$2(x + 3) = 2x + 6$ is

A. an identity

B. true only for $x = 3$

C. true only for $x = 0$

D. never true

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q4/5

Which values show that $a^3 + b^3 = (a + b)^3$ is NOT an identity?

A. $a = 0, b = 2$ (left side 8, right side 8)

B. $a = 1, b = -1$ (left side 0, right side 0)

C. $a = 1, b = 1$ (left side 2, right side 8)

D. $a = 2, b = 0$ (left side 8, right side 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q5/5

SET 19 • HSA-APR.C.4

FORWARD 1

Prove or disprove proposed identities • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Is $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$ an identity? Which answer is correct?

A. Yes: $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$, and $3ab(a + b) = 3a^2b + 3ab^2$, so both sides are equal.

B. Yes, because $a = 1$ and $b = 1$ give $8 = 2 + 6$.

C. No: for $a = 1$ and $b = 2$ the left side is $3^3 = 27$ and the right side is $1 + 8 + 3 \cdot 2 = 15$, so the sides differ.

D. No: $(a + b)^3 = a^3 + b^3$, so the extra term makes the two sides different.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q1/5

Is $a^4 - b^4 = (a - b)^4$ an identity? Which answer is correct?

A. Yes: for $a = 1$ and $b = 0$ both sides equal 1.

B. No: for $a = 2$ and $b = 1$, the left side is $16 - 1 = 15$ and the right side is $1^4 = 1$.

C. Yes: a fourth power applies to each term inside the parentheses, so $(a - b)^4 = a^4 - b^4$ for all a and b.

D. No, because $a^4 - b^4 = (a - b)^2(a + b)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q2/5

True or false?

$(x - 1)(x^2 + x + 1) = x^3 + 1$ is an identity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q3/5

Which is an identity?

A. $(x + 2)^2 = x^2 + 4$

B. $(x + 1)^3 = x^3 + 1$

C. $x^2 - 4 = (x - 4)^2$

D. $(x + 2)^2 - (x - 2)^2 = 8x$

Find c so that $(x + c)^2 - (x - c)^2 = 12x$ is an identity.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSA-APR.C.4

FORWARD 2

Prove identities with the Binomial Theorem • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Which argument uses the Binomial Theorem to prove that the entries in row n of Pascal's Triangle add to 2^n?

A. Row n has n + 1 entries, and each entry is the sum of the two entries above it, so every row has twice as many entries as the row before it and the sum of row n is 2^n.

B. Row 3 is 1, 3, 3, 1, which adds to $8 = 2^3$, so every row n adds to 2^n.

C. Set $x = y = 1$ in $(x + y)^n = C(n, 0)x^n + C(n, 1)x^{n-1}y + \ldots + C(n, n)y^n$. The left side is 2^n and the right side is $C(n, 0) + C(n, 1) + \ldots + C(n, n)$, the row sum.

D. Set $x = 1$ and $y = 0$ in $(x + y)^n = C(n, 0)x^n + \ldots + C(n, n)y^n$: the left side is 1 and the right side is $C(n, 0) = 1$, so each row adds to 2^n.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F2-Q1**

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q1/5

Use the Binomial Theorem with $x = 1$ and $y = -1$ to find $C(5,0) - C(5,1) + C(5,2) - C(5,3) + C(5,4) - C(5,5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q2/5

True or false?

$(1 + 1)^4 = 1 + 4 + 6 + 4 + 1.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q3/5

The sum of the coefficients of $(2x + 1)^3$ is

A. 9

B. 8

C. 6

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q4/5

Use $(1 + 0.1)^3 = 1 + 3(0.1) + 3(0.01) + 0.001$ to find 1.1^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q5/5

Use polynomial identities to describe numerical relationships

MAIN	Algebra II main question	**HSA-APR.C.4**
BACKWARD 1	Substitute values into an identity	**6.EE.A.2**
BACKWARD 2	Check a Pythagorean triple	**8.G.B.6**
BACKWARD 3	Choose a and b to fit a product	**HSA-APR.C.4**
FORWARD 1	Generate a triple with a required side	**HSA-APR.C.4**
FORWARD 2	Connect the triple identity to complex numbers	**HSN-CN.A.2**

SET 20 • HSA-APR.C.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

The identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ generates Pythagorean triples. Use $x = 3$ and $y = 2$ to find a triple.

Triple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q1/5

Use the identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ with $x = 5$ and $y = 2$ to find a Pythagorean triple.

Triple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q2/5

Use the identity $(a + b)(a - b) = a^2 - b^2$ to compute 47×53 mentally: write $53 = a + b$ and $47 = a - b$ with whole numbers a and b.

a = _____

b = _____

47×53 = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q3/5

Use the identity $(n + 1)^2 - n^2 = 2n + 1$ to write 41 as $(n + 1)^2 - n^2$, a difference of two consecutive squares. Give $n + 1$ and n.

$n + 1 =$ _____

$n =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q4/5

Use the identity $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$ to compute $13^3 - 12^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 1

Substitute values into an identity • Grade 6 • Expressions & Equations • 6.EE.A.2

Evaluate $x^2 - y^2$ for $x = 3$ and $y = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q1/5

Evaluate 2xy for x = 5 and y = 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q2/5

True or false?

For $n = 12$, $3n^2 + 3n + 1 = 433$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q3/5

For $a = 50$ and $b = 3$, $a^2 - b^2$ is

A. 2491

B. 2500

C. 2494

D. 2509

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q4/5

Evaluate 2n + 1 for n = 20.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 2

Check a Pythagorean triple • Grade 8 • Geometry • 8.G.B.6

Do 5, 12, 13 satisfy $a^2 + b^2 = c^2$?

A. No: 25 + 144 = 196

B. Yes: 25 + 144 = 169

C. No: 25 + 144 ≠ 169

D. Yes: 5 + 12 = 17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q1/5

Do 20, 21, 29 satisfy $a^2 + b^2 = c^2$?

A. Yes: 400 + 441 = 841

B. No: 400 + 441 = 851

C. No: 20 + 21 ≠ 29

D. Yes: 20 + 21 = 41

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q2/5

True or false?

6, 8, 11 is a Pythagorean triple.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q3/5

Which is a Pythagorean triple?

A. 8, 15, 17

B. 6, 7, 9

C. 2, 3, 4

D. 4, 5, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q4/5

Why is a triangle with sides 5, 12, 13 a right triangle?

A. 13 is the longest side, so the angle opposite it must be 90° in any triangle.

B. $5 + 12 > 13$ satisfies the triangle inequality, so it is a right triangle.

C. Its three sides are whole numbers, so it must be a right triangle.

D. $5^2 + 12^2 = 13^2$, so by the converse of the Pythagorean Theorem it is a right triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 3

Choose a and b to fit a product • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Write 47 and 53 as $50 - b$ and $50 + b$. What is b?

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q1/5

Write 98×102 as $(a - b)(a + b)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q2/5

True or false?

$29 \times 31 = 30^2 + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q3/5

64 × 56 equals

A. $(60 - 4)^2$

B. $64^2 - 56^2$

C. $60^2 - 4^2$

D. $60^2 + 4^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-B3-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q4/5

Compute 98×102.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q5/5

SET 20 • HSA-APR.C.4

FORWARD 1

Generate a triple with a required side • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

The identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ gives triples with legs $x^2 - y^2$ and $2xy$ and hypotenuse $x^2 + y^2$. Find whole numbers $x > y$ that give the triple with legs 7 and 24 and hypotenuse 25.

$x =$ _____

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q1/5

The identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ gives triples with legs $x^2 - y^2$ and $2xy$ and hypotenuse $x^2 + y^2$. Find whole numbers $x > y$ that give a leg $2xy = 40$ and hypotenuse 41.

$x =$ _____

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q2/5

True or false?

Every choice of whole numbers $x > y > 0$ in the identity gives a Pythagorean triple.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q3/5

Which choice gives the triple 8, 15, 17?

A. $x = 4, y = 1$

B. $x = 5, y = 3$

C. $x = 4, y = 2$

D. $x = 3, y = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q4/5

The identity gives legs $x^2 - y^2$ and $2xy$. Why does $x = 2y$ (y a positive whole number) give legs in the ratio 3 : 4?

A. With $y = 1$, $x = 2$ gives legs 3 and 4, so every y works the same way without checking.

B. $x^2 - y^2 = 2y^2$ and $2xy = 4y^2$, so the ratio is 1 : 2, which equals 3 : 4.

C. $x^2 - y^2 = 3y$ and $2xy = 4y$, so the legs are $3y$ and $4y$.

D. $x^2 - y^2 = 4y^2 - y^2 = 3y^2$ and $2xy = 4y^2$, so the legs are $3y^2$ and $4y^2$, in the ratio 3 : 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSA-APR.C.4

FORWARD 2

Connect the triple identity to complex numbers • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Compute $(3 + 2i)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q1/5

Compute $(x + yi)^2$ in the form $a + bi$, where x and y are real. Give the real part and the imaginary part.

Real part: _____

Imaginary part: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q2/5

True or false?

For positive integers $x > y$, the real and imaginary parts of $(x + yi)^2$ are $x^2 - y^2$ and $2xy$, the legs of a Pythagorean triple whose hypotenuse is $x^2 + y^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q3/5

$(5 + 2i)^2$ equals

A. $25 + 4i$

B. $21 + 20i$

C. $21 + 10i$

D. $29 + 20i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F2-Q4**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q4/5

Use $(4 + i)^2$ to find a Pythagorean triple.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q5/5

Expand with the Binomial Theorem

MAIN	Algebra II main question	**HSA-APR.C.5**
BACKWARD 1	Multiply binomials repeatedly	**HSA-APR.A.1**
BACKWARD 2	Build Pascal's Triangle by a rule	**4.OA.C.5**
BACKWARD 3	Powers of products and of negatives	**8.EE.A.1**
FORWARD 1	Find a specific term or coefficient with a condition	**HSA-APR.C.5**
FORWARD 2	Use binomial coefficients to compute probabilities	**HSS-CP.B.9**

SET 21 • HSA-APR.C.5

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Use the Binomial Theorem to expand $(x + 2)^5$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q1/5

Use the Binomial Theorem to expand $(2a - b)^4$. Write every term multiplied out.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q2/5

Use the Binomial Theorem to find the coefficient of x^3 in $(x + 3)^7$.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q3/5

Use the Binomial Theorem to find the term containing x^3y^3 in $(x - 2y)^6$.

Term: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q4/5

Use the first three terms of the binomial expansion of $(1 + 0.02)^5$ to estimate 1.02^5 to four decimal places.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q5/5

SET 21 • HSA-APR.C.5

BACKWARD 1

Multiply binomials repeatedly • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Expand $(x + 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q1/5

Expand $(x + 2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q2/5

True or false?

$(x + y)^3 = x^3 + y^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q3/5

The number of terms in the expansion of $(x + y)^5$ is

A. 6

B. 5

C. 10

D. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q4/5

Expand $(a - b)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSA-APR.C.5

BACKWARD 2

Build Pascal's Triangle by a rule • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

Row 4 of Pascal's Triangle is 1, 4, 6, 4, 1. Write row 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q1/5

Write row 7 of Pascal's Triangle (row 6 is 1, 6, 15, 20, 15, 6, 1).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q2/5

True or false?

Each inner entry of Pascal's Triangle is the sum of the two entries above it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q3/5

The fourth entry of row 7 (counting from the first 1) is

A. 20

B. 7

C. 35

D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q4/5

What is the sum of the entries in row 5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q5/5

BACKWARD 3

Powers of products and of negatives • Grade 8 • Expressions & Equations • 8.EE.A.1

Simplify $(2a)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q1/5

Simplify $(-2y)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q2/5

True or false?

$(-b)^4 = -b^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q3/5

3^4 equals

A. 27

B. 12

C. 64

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q4/5

Simplify $(0.02)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q5/5

SET 21 • HSA-APR.C.5

FORWARD 1

Find a specific term or coefficient with a condition • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Find the constant term of $(x + 1/x)^6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q1/5

The coefficient of x^2 in $(1 + kx)^5$ is 90. Find k ($k > 0$).

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q2/5

True or false?

The middle term of $(x + y)^8$ is $56x^4y^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q3/5

The term with x^4 in $(x^2 - 1)^5$ is

A. $-5x^4$

B. $-10x^4$

C. $10x^4$

D. $5x^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F1-Q4**

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q4/5

Find the coefficient of x in $(2x - 1)^6$.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSA-APR.C.5

FORWARD 2

Use binomial coefficients to compute probabilities • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

A fair coin is tossed 5 times. Use C(5, k) to find P(exactly 3 heads). Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.B.9 • Set 21 • FORWARD 2 • Q1/5

A spinner lands on red with probability 0.3. Find P(exactly 2 reds in 4 spins).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.B.9 • Set 21 • FORWARD 2 • Q2/5

True or false?

The probabilities of 0, 1, ..., n successes add to 1 because $(p + (1 - p))^n = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.B.9 • Set 21 • FORWARD 2 • Q3/5

For 6 fair coin tosses, P(exactly 3 heads) is

A. 6/64

B. 3/6

C. 15/64

D. 20/64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q4**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q4/5

Why does C(n, k) count the outcomes with exactly k successes in n trials?

A. C(n, k) = n · k, the number of trials times the number of successes, counts every outcome with k successes.

B. Each such outcome is determined by choosing which k of the n trials are successes, and there are C(n, k) such choices.

C. The successes must come first, so there is exactly C(n, k) = 1 way.

D. Each of the trials has 2 outcomes, success or failure, so there are C(n, k) = 2k outcomes with k successes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q5**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q5/5

Rewrite rational expressions by polynomial division

MAIN	Algebra II main question	**HSA-APR.D.6**
BACKWARD 1	Write a division with a remainder as a mixed number	**4.NF.B.3**
BACKWARD 2	Multiply to check a division	**HSA-APR.A.1**
BACKWARD 3	Use the Remainder Theorem to check the remainder	**HSA-APR.B.2**
FORWARD 1	Divide with unknown coefficients and higher-degree divisors	**HSA-APR.D.6**
FORWARD 2	Use division to find slant asymptotes	**HSF-IF.C.7.d**

SET 22 • HSA-APR.D.6

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Which shows $(x^3 + 2x^2 - 5x + 3)/(x - 1)$ written in the form $q(x) + r(x)/b(x)$?

A. $x^2 + x - 6 + 9/(x - 1)$

B. $x^2 + 3x - 2 + 1/(x + 1)$

C. $x^2 + 3x - 2 - 1/(x - 1)$

D. $x^2 + 3x - 2 + 1/(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q1/5

Which shows $(2x^3 - 3x^2 + 4x - 7)/(x^2 + 1)$ written in the form $q(x) + r(x)/b(x)$?

A. $2x - 3 + (2x + 4)/(x^2 + 1)$

B. $2x - 3 + (2x - 10)/(x^2 + 1)$

C. $2x - 3 + (2x - 4)/(x^2 + 1)$

D. $2x - 3 + (6x - 10)/(x^2 + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q2/5

Which shows $(x^4 - 16)/(x - 2)$ written in the form $q(x) + r(x)/b(x)$?

A. $x^3 + 2x^2 + 4x + 8$ (remainder 0)

B. $x^3 + 2x^2 + 4x + 8 - 32/(x - 2)$

C. $x^3 - 2x^2 + 4x - 8$ (remainder 0)

D. $x^3 + 8$ (remainder 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q3/5

Which shows $(6x^2 + x - 4)/(2x + 1)$ written in the form $q(x) + r(x)/b(x)$?

A. $6x - 2 - 3/(2x + 1)$

B. $3x - 1 - 4/(2x + 1)$

C. $3x - 1 + 3/(2x + 1)$

D. $3x - 1 - 3/(2x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q4/5

Which shows $(x^3 - 4x + 5)/(x + 3)$ written in the form $q(x) + r(x)/b(x)$?

A. $x - 7 + 26/(x + 3)$

B. $x^2 - 3x + 5 - 10/(x + 3)$

C. $x^2 - 3x + 5 + 10/(x + 3)$

D. $x^2 + 3x + 5 + 20/(x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q5/5

SET 22 • HSA-APR.D.6

BACKWARD 1

Write a division with a remainder as a mixed number • Grade 4 • Number & Operations–Fractions • 4.NF.B.3

Write 23/4 as a mixed number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q1/5

Write 47/5 in the form q + r/5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q2/5

True or false?

17/3 = 5 + 1/3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q3/5

31/7 equals

A. 4 + 1/7

B. 4 + 3/7

C. 3 + 4/7

D. 4 + 3/31

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q4/5

Write 100/9 in the form q + r/9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSA-APR.D.6

BACKWARD 2

Multiply to check a division • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Compute $(x - 1)(x^2 + 3x - 2) + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q1/5

Compute $(x^2 + 1)(2x - 3) + (2x - 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q2/5

True or false?

$(2x + 1)(3x - 1) - 3 = 6x^2 + x - 4.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q3/5

$(x + 3)(x^2 - 3x + 5)$ equals

A. $x^3 - 4x + 15$

B. $x^3 + 15$

C. $x^3 - 4x + 5$

D. $x^3 + 6x^2 - 4x + 15$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q4/5

Compute $(x - 2)(x^3 + 2x^2 + 4x + 8)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSA-APR.D.6

BACKWARD 3

Use the Remainder Theorem to check the remainder • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Find the remainder of $x^3 + 2x^2 - 5x + 3$ divided by $x - 1$ by evaluating.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q1/5

Find the remainder of $x^3 - 4x + 5$ divided by $x + 3$ by evaluating.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q2/5

True or false?

The remainder of $x^4 - 16$ divided by $x - 2$ is 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q3/5

The remainder of $6x^2 + x - 4$ divided by $2x + 1$ is

A. 3

B. -4

C. 1

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q4/5

Find the remainder of $2x^3 + x - 1$ divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q5/5

SET 22 • HSA-APR.D.6

FORWARD 1

Divide with unknown coefficients and higher-degree divisors • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

When $x^3 + ax^2 + 2x + 4$ is divided by $x^2 - x + 1$, the remainder is a constant. Find a and the remainder.

a = _____

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q1/5

Divide $x^4 + x^2 + 1$ by $x^2 + x + 1$. Give the quotient $q(x)$ and the remainder.

$q(x) =$ _____

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q2/5

True or false?

Dividing a degree-5 polynomial by a degree-2 polynomial gives a quotient of degree 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q3/5

The remainder of a division by $x^2 + 4$ can have degree at most

A. 0

B. 1

C. 2

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F1-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q4/5

Find the polynomial $a(x)$ for which $a(x)/(x - 2) = x^2 + 1 + 3/(x - 2)$.

$a(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSA-APR.D.6

FORWARD 2

Use division to find slant asymptotes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Use division to find the slant asymptote of $f(x) = (x^2 + 3x + 1)/(x + 1)$.

Asymptote: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q1/5

Find the slant asymptote of $f(x) = (2x^2 - x - 4)/(x - 1)$.

Asymptote: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q2/5

True or false?

A rational function with numerator degree exactly one more than the denominator degree has a slant asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q3/5

The slant asymptote of $y = (x^2 + 1)/x$ is

A. $y = x$

B. $y = 1$

C. $x = 0$

D. $y = x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q4/5

Why does the graph of $y = x + 2 - 1/(x + 1)$ get close to the line $y = x + 2$ for large $|x|$?

A. $1/(x + 1)$ approaches 0 as $|x|$ grows, so the vertical distance $y - (x + 2) = -1/(x + 1)$ approaches 0.

B. $1/(x + 1)$ grows as $|x|$ grows, so the graph gets pushed toward the line $y = x + 2$ for large $|x|$.

C. The graph and the line $y = x + 2$ meet at $x = -1$, so they stay close for large $|x|$.

D. At $x = 0$, $y = x + 2$ exactly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q5/5

Rewrite simple rational expressions by inspection

MAIN	Algebra II main question	**HSA-APR.D.6**
BACKWARD 1	Split a fraction with a sum in the numerator	**7.EE.A.1**
BACKWARD 2	Rewrite a numerator as a multiple of the denominator plus a cons…	**6.EE.A.3**
BACKWARD 3	Divide by long division when inspection fails	**HSA-APR.D.6**
FORWARD 1	Use the rewritten form to reason about values	**HSA-APR.D.6**
FORWARD 2	Graph rational functions as transformations of y = 1/x	**HSF-BF.B.3**

SET 23 • HSA-APR.D.6

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Rewrite $(x + 5)/(x + 2)$ by inspection in the form $q + r/(x + 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q1/5

Rewrite $(2x - 1)/(x - 3)$ by inspection in the form $q + r/(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q2/5

Rewrite $(x^2 + 1)/x$ by inspection in the form $q(x) + r/x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q3/5

Rewrite $(3x + 7)/(x + 1)$ by inspection in the form $q + r/(x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q4/5

Rewrite $x^2/(x - 1)$ by inspection in the form $q(x) + r/(x - 1)$. (Hint: $x^2 = (x^2 - 1) + 1$.)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q5/5

SET 23 • HSA-APR.D.6

BACKWARD 1

Split a fraction with a sum in the numerator • Grade 7 • Expressions & Equations • 7.EE.A.1

Which writes (x + 3)/x as a sum of a whole number and a fraction?

A. 1/x + 3/x

B. x + 3/x

C. 1 + 3/x

D. 1 + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q1/5

Simplify $(2(x - 3) + 5)/(x - 3)$ as a sum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q2/5

True or false?

$(a + b)/c = a/c + b/c$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q3/5

$(x^2 + 1)/x$ equals

A. $x + 1/x$

B. $1 + 1/x$

C. $x + 1$

D. $x^2 + 1/x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q4/5

True or false?

$c/(a + b) = c/a + c/b$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • HSA-APR.D.6

BACKWARD 2

Rewrite a numerator as a multiple of the denominator plus a constant • Grade 6 • Expressions & Equations • 6.EE.A.3

Find c so that $x + 5 = (x + 2) + c$.

$c =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q1/5

Find c so that $2x - 1 = 2(x - 3) + c$.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q2/5

True or false?

$3x + 7 = 3(x + 1) + 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q3/5

$x^2 - 1$ equals

A. $x(x - 1)$

B. $(x + 1)^2$

C. $(x - 1)^2$

D. $(x - 1)(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q4/5

Find c so that $4x + 1 = 4(x - 2) + c$.

$c =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • HSA-APR.D.6

BACKWARD 3

Divide by long division when inspection fails • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Divide x^2 by $x - 1$ with long division. Give the quotient and the remainder.

Quotient: _____

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q1/5

Divide $3x + 7$ by $x + 1$. Give the quotient and the remainder.

Quotient: _____

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q2/5

True or false?

Long division and inspection give the same result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q3/5

The remainder of $(2x - 1) \div (x - 3)$ is

A. -5

B. 2

C. -1

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q4/5

Divide x + 5 by x + 2. Give the quotient and the remainder.

Quotient: _____

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q5/5

SET 23 • HSA-APR.D.6

FORWARD 1

Use the rewritten form to reason about values • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

For which integers x is $(x + 5)/(x + 2)$ an integer? (Use $(x + 5)/(x + 2) = 1 + 3/(x + 2)$.)

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q1/5

Why can $(2x - 1)/(x - 3)$ never equal 2?

A. It is undefined for every value of x.

B. Its numerator $2x - 1$ is never 2.

C. At $x = 0$ it equals $1/3$, which is not 2.

D. It equals $2 + 5/(x - 3)$, and $5/(x - 3)$ is never 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q2/5

True or false?

$(3x + 7)/(x + 1) > 3$ for every $x > -1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q3/5

As x grows large, (x + 5)/(x + 2) approaches

A. 3

B. 0

C. 5/2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q4/5

For $x > 0$, what is the least value of $x + 1/x$, and at which x does it occur? (Use $x + 1/x - 2 = (x - 1)^2/x$.)

Least value: _____

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSA-APR.D.6

FORWARD 2

Graph rational functions as transformations of y = 1/x • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

How is the graph of $y = 1 + 3/(x + 2)$ obtained from the graph of $y = 3/x$?

A. Shift it left 2 and down 1.

B. Shift it left 2 and up 1.

C. Shift it right 2 and up 1.

D. Shift it up 2 and left 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q1/5

Give the vertical and horizontal asymptotes of $y = 2 + 5/(x - 3)$ as equations.

Vertical asymptote: _____

Horizontal asymptote: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q2/5

True or false?

$y = (3x + 7)/(x + 1)$ has horizontal asymptote $y = 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q3/5

$y = (x + 5)/(x + 2)$ is a translation of $y = 3/x$ by

A. left 2, up 1

B. right 2, up 1

C. left 2, up 5

D. left 5, up 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q4/5

Write $(4x - 3)/(x - 1)$ in the form $q + r/(x - 1)$ with constants q and r. Then give its vertical and horizontal asymptotes as equations.

q = _____

r = _____

Vertical asymptote: _____

Horizontal asymptote: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q5/5

Multiply and divide rational expressions

MAIN	Algebra II main question	**HSA-APR.D.7**
BACKWARD 1	Multiply and divide fractions	**6.NS.A.1**
BACKWARD 2	Factor numerators and denominators	**HSA-SSE.A.2**
BACKWARD 3	Division by zero is undefined	**7.NS.A.2.b**
FORWARD 1	Simplify complex fractions	**HSA-APR.D.7**
FORWARD 2	Explain closure of rational expressions	**HSA-APR.D.7**

SET 24 • HSA-APR.D.7

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Simplify $(x^2 - 9)/(x^2 + x - 6) \cdot (x - 2)/(x - 3)$. Give the result and all excluded values.

Result: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q1/5

Simplify $(x + 2)/(3x^2) \div (x^2 - 4)/x$. Write the result as one fraction with its denominator multiplied out, and give all excluded values.

Result: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q2/5

Simplify $(2x - 6)/(x + 1) \cdot (x + 1)/(x^2 - 9)$. Give the result and all excluded values.

Result: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q3/5

Simplify $(3x + 3)/(x - 4) \div (x + 1)/(x - 4)$. Give the result and all excluded values.

Result: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q4/5

Simplify $(5x^2 + 5x)/(x^2 - 4) \div (x + 1)/(x - 2)$. Give the result and all excluded values.

Result: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q5/5

SET 24 • HSA-APR.D.7

BACKWARD 1

Multiply and divide fractions • Grade 6 • The Number System • 6.NS.A.1

Compute 2/3 · 9/4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q1/5

Compute 5/6 ÷ 10/9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q2/5

True or false?

$a/b \div c/d = a/b \cdot c/d$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q3/5

3/8 ÷ 3/4 equals

A. 9/32

B. 2

C. 3/4

D. 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q4/5

Compute 14/15 · 5/7.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSA-APR.D.7

BACKWARD 2

Factor numerators and denominators • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 + x - 6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q1/5

Factor $5x^2 + 5x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q2/5

$x^2 - 4$ equals

A. $(x - 4)(x + 1)$

B. $(x - 2)^2$

C. $(x - 2)(x + 2)$

D. $x(x - 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q3/5

True or false?

$2x - 6 = 2(x - 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q4/5

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSA-APR.D.7

BACKWARD 3

Division by zero is undefined • Grade 7 • The Number System • 7.NS.A.2.b

For which x is 1/(x − 4) undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q1/5

For which x is $(x + 2)/(x^2 - 9)$ undefined?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q2/5

True or false?

After canceling $(x - 3)$, $x = 3$ is allowed in the original expression.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q3/5

Dividing by $(x + 1)/(x - 2)$ is impossible when

A. $x = -1$ only

B. $x = 0$

C. $x = -1$ or $x = 2$

D. $x = 2$ only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q4/5

Why is $x = 0$ excluded from $(x + 2)/(3x^2)$?

A. Every expression excludes $x = 0$.

B. The value of the expression is 0 when $x = 0$.

C. The denominator $3x^2$ equals 0 when $x = 0$.

D. The numerator $x + 2$ equals 0 when $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q5/5

SET 24 • HSA-APR.D.7

FORWARD 1

Simplify complex fractions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Which is $(1/x + 1)/(1/x - 1)$ simplified, with all excluded values?

A. $(1 + x)/(1 - x)$; $x \neq 0, 1$

B. -1; $x \neq 0$

C. $(x + 1)/(x - 1)$; $x \neq 0, 1$

D. $(1 + x)/(1 - x)$; $x \neq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q1/5

Which is $(1 - 4/x^2)/(1 + 2/x)$ simplified, with all excluded values?

A. $(x - 2)/x;\ x \neq 0, -2$

B. $x - 2;\ x \neq -2$

C. $(x + 2)/x;\ x \neq 0, -2$

D. $(x - 2)/x;\ x \neq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q2/5

True or false?

$(a/b)/(c/d) = ad/(bc)$ when $b, c, d \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q3/5

$(x/2)/(x/4)$ for $x \neq 0$ equals

A. 2

B. 1/2

C. $x^2/8$

D. $8/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q4/5

Which is $(x - 1/x)/(x + 1)$ simplified, with all excluded values?

A. $(x + 1)/x;\ x \neq 0, -1$

B. $(x - 1)/x;\ x \neq 0, -1$

C. $x - 1;\ x \neq -1$

D. $(x - 1)/x;\ x \neq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q5**

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSA-APR.D.7

FORWARD 2

Explain closure of rational expressions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Why is the product of two rational expressions a rational expression?

A. Rational expressions are always polynomials, and a product of polynomials is a polynomial.

B. $(a/b)(c/d) = (a + c)/(b + d)$, and $a + c$ and $b + d$ are polynomials, so the product is a ratio of polynomials.

C. The product of any two expressions is a polynomial.

D. $(a/b)(c/d) = ac/(bd)$, and ac and bd are polynomials (bd is not the zero polynomial), so the product is a ratio of polynomials.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q1**

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q1/5

Why is the quotient of a rational expression by a nonzero rational expression a rational expression?

A. Any quotient is undefined, so the question does not apply.

B. (a/b) ÷ (c/d) = (a ÷ c)/(b ÷ d), and quotients of polynomials are polynomials.

C. (a/b) ÷ (c/d) = ad/(bc), a ratio of polynomials, and bc is not the zero polynomial because c is not.

D. Dividing never changes an expression.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q2**

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q2/5

True or false?

Rational expressions are closed under division by the zero expression.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q3/5

Which system is closed under addition, subtraction, multiplication, and division by nonzero elements?

A. Whole numbers

B. Rational expressions

C. Integers

D. Polynomials

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q4/5

Which example shows that polynomials are NOT closed under division?

A. $6x^2 \div 3x = 2x$ for $x \neq 0$, which is a polynomial

B. $(x^3 + x) \div x = x^2 + 1$ for $x \neq 0$, a polynomial

C. $x \div (x + 1) = x/(x + 1)$, which is not a polynomial

D. $(x^2 - 1) \div (x - 1) = x + 1$ for $x \neq 1$, a polynomial

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q5**

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q5/5

Add and subtract rational expressions

MAIN	Algebra II main question	**HSA-APR.D.7**
BACKWARD 1	Add fractions with unlike denominators	**5.NF.A.1**
BACKWARD 2	Find a least common denominator by factoring	**HSA-SSE.A.2**
BACKWARD 3	Subtract a whole numerator	**6.EE.A.3**
FORWARD 1	Decompose into partial fractions	**HSA-APR.D.7**
FORWARD 2	Solve rational equations	**HSA-REI.A.2**

SET 25 • HSA-APR.D.7

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Simplify $(x + 1)/(x^2 + x) + 1/x$. Give the result and all excluded values.

Result: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q1/5

Write $\frac{2}{(x + 1)} + \frac{3}{(x - 2)}$ as one fraction with denominator $(x + 1)(x - 2)$. Give the numerator and all excluded values.

Numerator: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q2/5

Simplify $x/(x-3) - 3/(x-3)$. Give the result and all excluded values.

Result: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q3/5

Write $1/(x^2 - 1) - 1/(x - 1)$ as one fraction with denominator $(x - 1)(x + 1)$. Give the numerator and all excluded values.

Numerator: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q4/5

Write $4/(x^2 - 4x) - 1/x$ as one fraction with denominator $x(x - 4)$. Give the numerator and all excluded values.

Numerator: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q5/5

SET 25 • HSA-APR.D.7

BACKWARD 1

Add fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Compute 2/3 + 3/4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q1/5

Compute 5/6 − 1/4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q2/5

True or false?

1/2 + 1/3 = 2/5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q3/5

1/4 + 1/6 equals

A. 5/12

B. 2/10

C. 2/12

D. 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q4/5

Compute 3/10 + 2/15.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSA-APR.D.7

BACKWARD 2

Find a least common denominator by factoring • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 + x$ and $x^2 - 4x$.

$x^2 + x =$ _____

$x^2 - 4x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q1/5

What is the least common denominator of $1/(x^2 - 1)$ and $1/(x - 1)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q2/5

The LCD of $1/x$ and $1/(x(x-4))$ is

A. $x - 4$

B. $x^2(x - 4)$

C. $x(x - 4)$

D. x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q3/5

True or false?

The LCD of $2/(x + 1)$ and $3/(x - 2)$ is $x - 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q4/5

What is the LCD of $1/(x^2 + x)$ and $1/x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSA-APR.D.7

BACKWARD 3

Subtract a whole numerator • Grade 6 • Expressions & Equations • 6.EE.A.3

Simplify $4 - (x - 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q1/5

Simplify $1 - (x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q2/5

True or false?

$3 - (x - 3) = -x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q3/5

$2(x - 2) - 3(x + 1)$ equals

A. $5x - 7$

B. $-x + 1$

C. $-x - 7$

D. $-x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q4/5

Simplify $x - (3 - x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q5/5

SET 25 • HSA-APR.D.7

FORWARD 1

Decompose into partial fractions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Find A and B so that

$$\frac{5x + 1}{(x - 1)(x + 2)} = \frac{A}{x - 1} + \frac{B}{x + 2}.$$

A = _____

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q1/5

Find A and B so that $1/(x(x + 1)) = A/x + B/(x + 1)$.

A = _____

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q2/5

True or false?

$1/(x(x + 1)) = 1/x + 1/(x + 1)$ for $x \neq 0, -1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q3/5

$4/(x^2 - 4)$ equals

A. $2/(x - 2) - 2/(x + 2)$

B. $1/(x - 2) + 1/(x + 2)$

C. $4/(x - 2) - 4/(x + 2)$

D. $1/(x - 2) - 1/(x + 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q4**

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q4/5

Use $1/(n(n+1)) = 1/n - 1/(n+1)$ to find $1/(1\cdot2) + 1/(2\cdot3) + 1/(3\cdot4) + 1/(4\cdot5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSA-APR.D.7

FORWARD 2

Solve rational equations • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $2/(x + 1) + 3/(x - 2) = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q1/5

Which is the solution set of $x/(x - 3) - 3/(x - 3) = 1$?

A. $x = 3$ only

B. Every real number

C. No solution

D. Every real number except 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q2**

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q2/5

True or false?

A solution of a rational equation that makes a denominator 0 must be rejected.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q3/5

The solution of $1/x + 1/(2x) = 3$ is

A. $x = 1/2$

B. $x = 2$

C. $x = 3/2$

D. no solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q4/5

Solve $4/(x^2 - 4x) - 1/x = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q5/5

Create and solve rational, radical, and exponential equations from contexts

MAIN	Algebra II main question	**HSA-CED.A.1**
BACKWARD 1	Express rates of work and unit rates	**6.RP.A.3.b**
BACKWARD 2	Write exponential growth models	**HSF-LE.A.2**
BACKWARD 3	Solve linear equations after clearing a denominator	**HSA-REI.B.3**
FORWARD 1	Create equations whose algebraic solutions must be checked aga.	**HSA-CED.A.1**
FORWARD 2	Solve exponential context equations with logarithms	**HSF-LE.A.4**

SET 26 • HSA-CED.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

Ana can paint a room in 4 hours. Working together, Ana and Ben paint it in 2.4 hours. Write an equation for Ben's time b (in hours) working alone, and use it to find b.

b (hours) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q1/5

A car's speed s (in mph) before braking is estimated from its skid length d (in feet) by $s = \sqrt{(22.5d)}$. Write an equation for the skid length of a car going 45 mph, and use it to find d.

d (feet) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q2/5

A town of 1,200 people doubles in population every 5 years. Write an equation for the time t (in years) when the population is 9,600, and use it to find t.

t (years) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q3/5

A printer charges $500 plus $4 per shirt. Write an equation for the number of shirts x that makes the average cost per shirt $9, and use it to find x.

x (shirts) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q4/5

A tank holds 10 L of a 20% salt solution. Write an equation for the liters x of pure water to add to make an 8% solution, and use it to find x.

x (liters) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q5/5

SET 26 • HSA-CED.A.1

BACKWARD 1

Express rates of work and unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

Ana paints a room in 4 hours. What fraction of the room does she paint per hour?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q1/5

A pump fills a pool in 6 hours. What fraction does it fill in 2 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q2/5

True or false?

If two people work together, their times to finish the job add.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q3/5

A machine makes 300 parts in 5 hours. Its rate is

A. 1500 parts per hour

B. 300 parts per hour

C. 60 parts per hour

D. 5/300 parts per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q4/5

Working together, two people finish a job in 2.4 hours. What fraction of the job do they finish per hour? Give a fraction.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSA-CED.A.1

BACKWARD 2

Write exponential growth models • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

A population of 1,200 doubles every 5 years. Write the population after t years.

P(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q1/5

A $500 deposit grows 3% per year. Write its value after t years.

V(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q2/5

True or false?

A quantity that doubles every 5 years is multiplied by 2^5 after 20 years.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q3/5

A quantity triples every year starting at 7. After t years it is

A. $7 + 3t$

B. $21t$

C. $3 \cdot 7^t$

D. $7 \cdot 3^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q4/5

How many doublings happen in 15 years if the doubling time is 5 years?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSA-CED.A.1

BACKWARD 3

Solve linear equations after clearing a denominator • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve 500 + 4x = 9x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q1/5

Solve $2 = 0.08(10 + x)$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q2/5

True or false?

To solve $(500 + 4x)/x = 9$ we may multiply both sides by x when $x \neq 0$.

A. True

B. False

2025 = 22.5d gives d =

A. 2002.5

B. 45

C. 9

D. 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q4/5

Solve $5x/12 = 1/2.4$ for x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q5/5

SET 26 • HSA-CED.A.1

FORWARD 1

Create equations whose algebraic solutions must be checked against the context • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

A boat travels 24 km downstream and 24 km back upstream in 5 hours. The current is 2 km/h. The still-water speed b satisfies $24/(b + 2) + 24/(b - 2) = 5$. Solve the equation and give the speed that fits the context, in km/h.

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q1/5

The length of a rectangle is 3 more than its width w, and its diagonal is 15. Then $w^2 + (w + 3)^2 = 225$. Find the width that fits the context.

w = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q2/5

True or false?

A negative solution of an equation for a time must be rejected.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q3/5

A model gives $t = -3$ or $t = 4$ for the time (in seconds) of an event. The answer is

A. neither

B. both

C. 4 seconds

D. -3 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q4/5

A pipe fills a tank in x hours and a second pipe takes 2 hours longer. Together they fill it in 2.4 hours. Write a rational equation and solve it for x (reject any value that makes no sense in the context).

x (hours) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSA-CED.A.1

FORWARD 2

Solve exponential context equations with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

A town of 1,200 people doubles every 5 years, so its population is $1200 \cdot 2^{t/5}$ after t years. Solve $1200 \cdot 2^{t/5} = 5000$ for t, to the nearest tenth of a year. A calculator may be used.

t (years) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q1/5

Solve $500(1.04)^t = 800$ for t, to the nearest hundredth. A calculator may be used.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q2/5

True or false?

$2^{t/5} = 8$ can be solved without logarithms, but

$2^{t/5} = 5$ cannot be solved exactly without them.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q3/5

The exact solution of $3^x = 20$ is

A. 20/3

B. log(20/3)

C. log 20/log 3

D. 3 log 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q4/5

An amount grows 4% per year, so after t years it is multiplied by $(1.04)^t$. Why is t = log 1.6/log 1.04 the time for it to grow by 60%?

A. Growing by 60% means $(1.04)^t = 60$; taking logarithms gives t log 1.04 = log 60, so t = log 60/log 1.04.

B. Growing by 60% means $(1.04)^t = 1.6$; taking logarithms gives t log 1.04 = log 1.6, so t = log 1.6/log 1.04.

C. Growing by 60% means 1.04t = 1.6, because the amount is multiplied by 1.04 for each year, so t = 1.6/1.04.

D. log 1.6/log 1.04 = log(1.6 − 1.04), the difference of the rates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q5/5

SET 27 • HSA-CED.A.1 • CREATING EQUATIONS AND REASONING WITH EQUATIONS

Create and solve nonlinear inequalities from contexts

MAIN	Algebra II main question	**HSA-CED.A.1**
BACKWARD 1	Solve quadratic equations by factoring	**HSA-REI.B.4.b**
BACKWARD 2	Sign of a product on intervals	**7.NS.A.2**
BACKWARD 3	Reverse the inequality when multiplying by a negative	**HSA-REI.B.3**
FORWARD 1	Solve rational inequalities with sign charts and excluded values	**HSA-CED.A.1**
FORWARD 2	Write systems of constraints and judge viability	**HSA-CED.A.3**

SET 27 • HSA-CED.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

A ball's height is $h = -16t^2 + 64t$ feet after t seconds ($0 \le t \le 4$). Use an inequality to find the times when the ball is at least 48 feet high. Write them as a compound inequality.

Times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q1/5

A company sells x items at a price of $100 - 2x$ dollars each ($0 \le x \le 50$). Use an inequality to find the values of x that give revenue of at least $1,200. Write them as a compound inequality.

Values of x: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q2/5

The average cost of making x posters ($x > 0$) is $(300 + 2x)/x$ dollars. Use an inequality to find the numbers of posters with an average cost of at most $5.

Values of x: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q3/5

A drug concentration is $C(t) = 20t/(t^2 + 4)$ mg/L after t hours ($t \geq 0$). Use an inequality to find the times when C is at least 4 mg/L. Write them as a compound inequality.

Times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q4/5

$500 is invested at 4% interest, credited once at the end of each year. Use an inequality to find the least whole number of years t after which the value is at least $800. A calculator may be used.

t (whole years) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q5/5

SET 27 • HSA-CED.A.1

BACKWARD 1

Solve quadratic equations by factoring • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $t^2 - 4t + 3 = 0$.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q1/5

Solve $x^2 - 50x + 600 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q2/5

True or false?

The solutions of $t^2 - 5t + 4 = 0$ are -1 and -4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q3/5

The solutions of $x^2 - x - 12 = 0$ are

A. 4 and −3

B. 6 and −2

C. 12 and −1

D. −4 and 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q4/5

Solve $2x^2 - 100x + 1200 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSA-CED.A.1

BACKWARD 2

Sign of a product on intervals • Grade 7 • The Number System • 7.NS.A.2

Is $(t - 1)(t - 3)$ positive or negative when $t = 2$? Answer Positive or Negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q1/5

Is $(t - 1)(t - 3)$ positive or negative when $t = 5$? Answer Positive or Negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q2/5

True or false?

A product of two negative numbers is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q3/5

$(x - 20)(x - 30) \leq 0$ when

A. $x \leq 20$ or $x \geq 30$

B. $x \leq 20$

C. $20 \leq x \leq 30$

D. $x \geq 30$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q4/5

For which t is $(t - 1)(t - 4) < 0$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSA-CED.A.1

BACKWARD 3

Reverse the inequality when multiplying by a negative • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Both sides of $-16t^2 + 64t - 48 \geq 0$ are divided by -16. Which inequality results?

A. $t^2 - 4t + 3 \leq 0$

B. $t^2 - 4t - 3 \geq 0$

C. $t^2 + 4t + 3 \leq 0$

D. $t^2 - 4t + 3 \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q1/5

Solve $-3x \geq 12$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q2/5

True or false?

Multiplying both sides of an inequality by x is always allowed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q3/5

If $x > 0$, then $(300 + 2x)/x \leq 5$ is equivalent to

A. $300 + 2x \leq 5x$

B. $300 + 2x \geq 5x$

C. $2x \leq 5$

D. $300 \leq 2x$

Solve $2 - x < 7$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q5/5

SET 27 • HSA-CED.A.1

FORWARD 1

Solve rational inequalities with sign charts and excluded values • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

Solve $(x - 3)/(x + 1) \leq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q1/5

Which is the solution set of $(x + 2)/(x - 4) > 0$?

A. $-2 < x < 4$

B. $x < -2$ or $x > 4$

C. $x > 4$ only

D. $x < -2$ or $x \geq 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q2/5

True or false?

Multiplying $(x - 3)/(x + 1) \leq 0$ by $x + 1$ gives $x - 3 \leq 0$ for all x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q3/5

The solution set of $1/x > 2$ is

A. $x > 1/2$

B. $x < 1/2$

C. $x > 0$

D. $0 < x < 1/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q4/5

A model's average cost is (400 + 3x)/x dollars. For how many items x > 0 is the average cost under $7?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSA-CED.A.1

FORWARD 2

Write systems of constraints and judge viability • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3

A ball's height is $h(t) = -16t^2 + 64t$ feet after t seconds, for $0 \le t \le 4$ (while it is in the air). Its height must be at least 48 ft and the time must be at most 2.5 s. Give the viable times as a compound inequality.

Viable times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q1/5

A company sells x items (x a whole number) at a price of 100 − 2x dollars each. Revenue x(100 − 2x) must be at least $1,200 and at most 24 items can be made. Give the viable values of x as a compound inequality.

Viable x: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q2/5

True or false?

A value that solves the inequality may still not be viable in the context.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q3/5

Revenue $x(100 - 2x) \geq 1200$ with x a whole number is satisfied by

A. $x = 19$

B. $x = 25$

C. $x = 31$

D. $x = 50$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q4/5

$500 is invested at 4% interest credited once at the end of each year. The equation $500(1.04)^t = 800$ gives $t \approx 11.98$. After how many years is the account first worth at least $800, and why?

A. 12 years: because 11.98 rounds to 12 when rounded to the nearest whole number of years.

B. 11.98 years: the balance reaches exactly $800 at that time, as the equation shows.

C. 11 years: round 11.98 down, because after 11 years the account is already close to $800.

D. 12 years: interest is credited only at year ends, so the balance is below $800 until the 12th.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q5/5

Create and graph nonlinear two-variable models

MAIN	Algebra II main question	**HSA-CED.A.2**
BACKWARD 1	Make a table of values from an equation	**6.EE.C.9**
BACKWARD 2	Choose scales and label axes	**HSN-Q.A.1**
BACKWARD 3	Write a growth or decay model from a percent	**HSF-LE.A.2**
FORWARD 1	Create equations in more than two variables and graph a member	**HSA-CED.A.2**
FORWARD 2	Interpret key features of the created models	**HSF-IF.B.4**

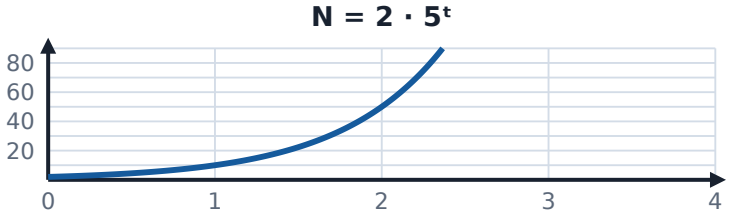
SET 28 • HSA-CED.A.2

MAIN

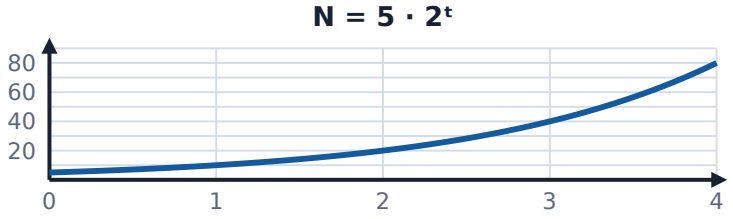
Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2

A rabbit population starts at 5 and doubles every year. Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le t \le 4$ (t in years, N rabbits)?

A



B



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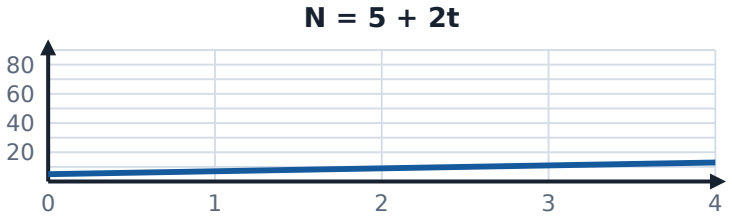
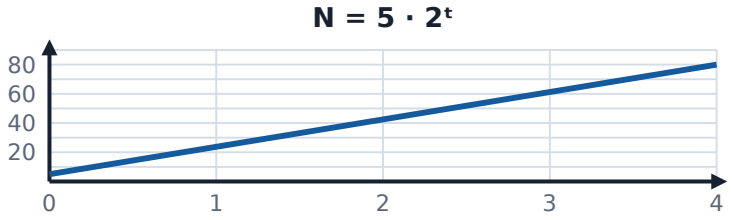


D



A. Graph A

B. Graph B

C. Graph C

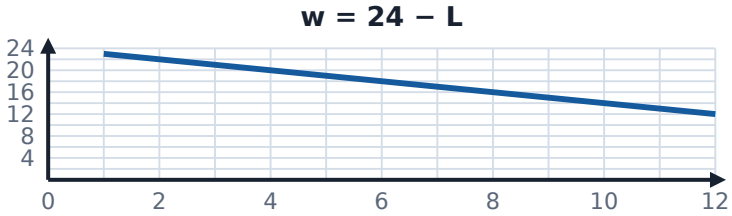
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q1**

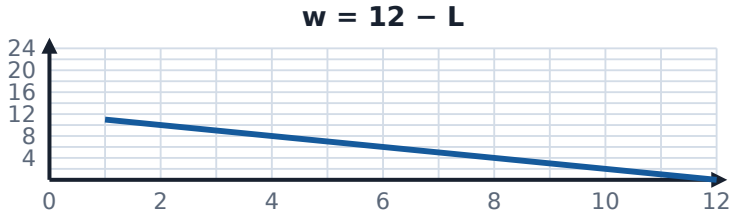
High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q1/5

A rectangle has area 24 square units. Which graph shows a correct equation for the situation with a correct graph of that equation for $1 \le L \le 12$ (length L, width w)?

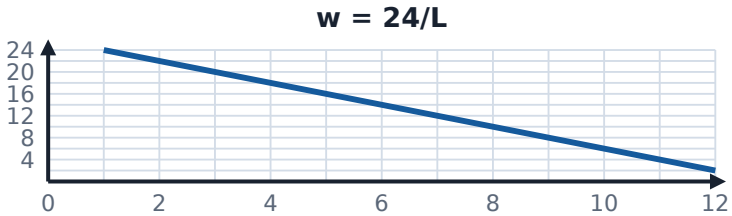
A



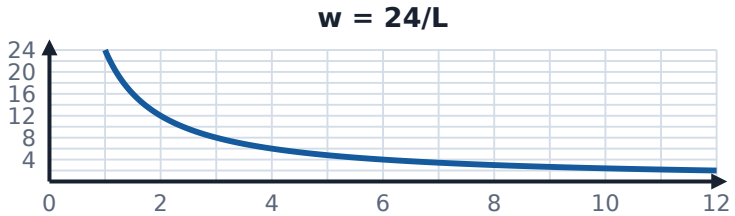
B



C



D



A. Graph A

B. Graph B

C. Graph C

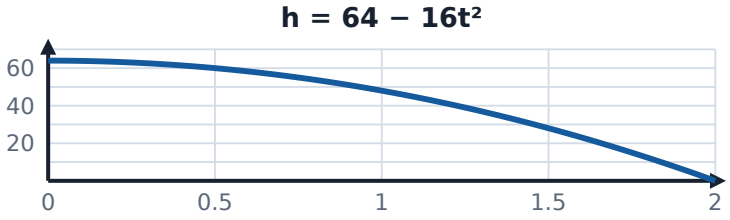
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q2**

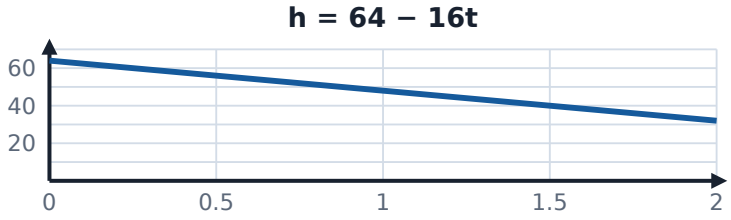
High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q2/5

A stone is dropped from 64 feet. Its height after t seconds is 64 minus 16 times the square of t. Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le t \le 2$ (h in feet)?

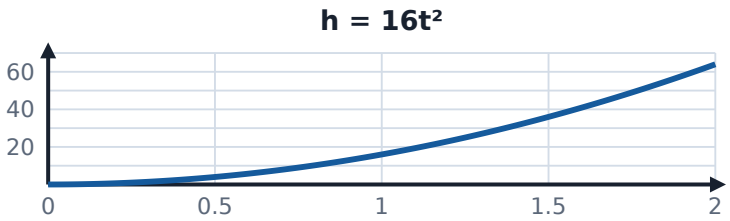
A



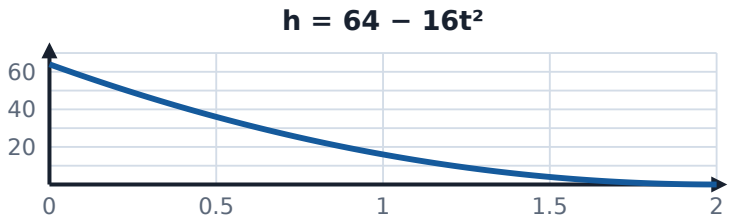
B



C



D



A. Graph A

B. Graph B

C. Graph C

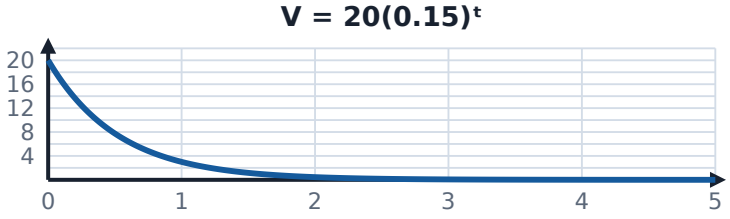
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q3**

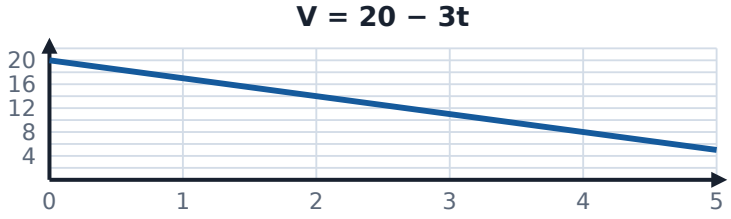
High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q3/5

A $20,000 car loses 15% of its value each year. Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le t \le 5$ (V in thousands of dollars, t in years)?

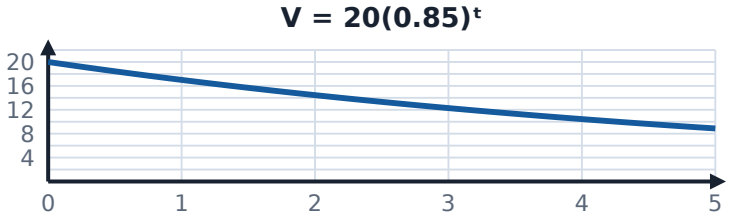
A



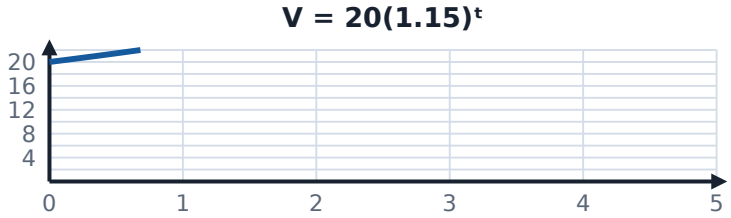
B



C



D



A. Graph A

B. Graph B

C. Graph C

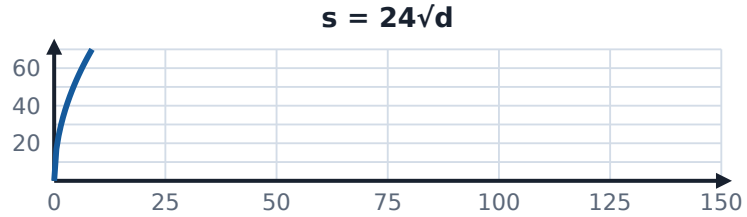
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q4**

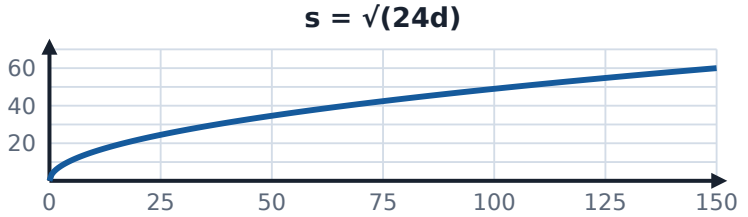
High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q4/5

A car's speed s (mph) is the square root of 24 times its skid length d (ft). Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le d \le 150$?

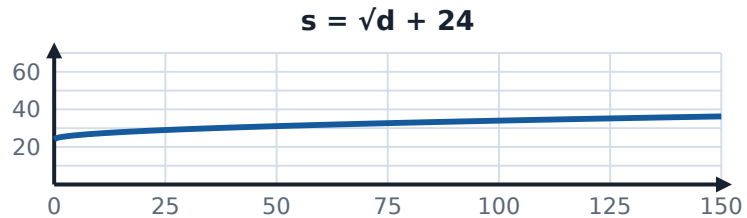
A



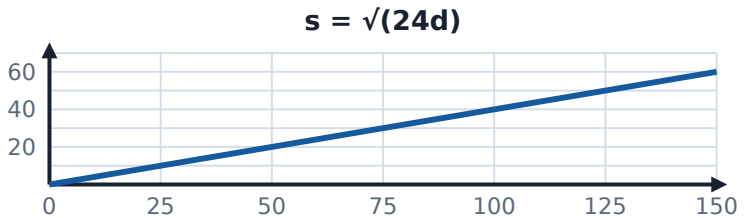
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q5/5

SET 28 • HSA-CED.A.2

BACKWARD 1

Make a table of values from an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

For $N = 200 \cdot 2^t$, find N when t = 0, 1, 2, 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q1/5

For w = 24/L, find w when L = 2, 3, 4, 6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q2/5

True or false?

For $h = 64 - 16t^2$, $h = 48$ when $t = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q3/5

For $s = \sqrt{(24d)}$, s when $d = 96$ is

A. 96

B. 48

C. 2304

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q4/5

For $V = 20000(0.85)^t$, find V when t = 1 and when t = 2.

V at t = 1: _____

V at t = 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSA-CED.A.2

BACKWARD 2

Choose scales and label axes • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1

Values of N go from 200 to 3,200. Which step for the vertical axis, starting at 0, shows all of them using about 7 grid lines?

A. 500

B. 50

C. 1,000

D. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q1/5

V depends on t, the time in years. Which quantity goes on the horizontal axis?

A. V, the value

B. Either one; it makes no difference

C. t, the time in years

D. Neither; time is not graphed

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q2/5

True or false?

Axis labels should include units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q3/5

The best vertical scale for values from 0 to 64 is

A. steps of 0.5

B. steps of 1 from 0 to 10

C. steps of 100 from 0 to 1000

D. steps of 10 from 0 to 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q4/5

A horizontal axis runs from d = 0 to d = 150 with tick marks every 10. How many intervals of 10 are there?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSA-CED.A.2

BACKWARD 3

Write a growth or decay model from a percent • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the value of $20,000 after losing 15% per year for t years.

$V =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q1/5

What is the growth factor for 4% growth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q2/5

True or false?

Losing 15% each year means multiplying by 0.15 each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q3/5

A quantity doubles every hour starting at 200. After t hours it is

A. $2 \cdot 200^{t}$

B. $200 \cdot 2^{t}$

C. $200 + 2t$

D. $400t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q4/5

What is the decay factor for 30% decay?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q5/5

SET 28 • HSA-CED.A.2

FORWARD 1

Create equations in more than two variables and graph a member of the family • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2

The current I (amps) through a resistor is the voltage V divided by the resistance R. Write I in terms of V and R, then write I as a function of R when V = 12.

I = _____

I (when V = 12) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q1/5

The volume of a cylinder is $V = \pi r^2 h$. Write h in terms of r when $V = 100\pi$ $(r > 0)$.

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q2/5

True or false?

Fixing one variable of a three-variable equation gives a two-variable equation that can be graphed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q3/5

For $A = P(1.05)^t$ with $P = 1000$, the graph of A against t is

A. a line

B. an increasing exponential curve

C. a parabola

D. a decreasing curve

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q4/5

For the area formula $A = \pi r^2$ with $r \geq 0$, which describes the graph of A against r?

A. A full parabola opening down from its vertex

B. An increasing exponential curve through (0, 1)

C. Half of a parabola, starting at the origin and increasing

D. A straight line through the origin with slope π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSA-CED.A.2

FORWARD 2

Interpret key features of the created models • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

A bacteria population is $N = 200 \cdot 2^t$ after t hours. What does the N-intercept, (0, 200), mean?

A. The population never goes below 200 in the past.

B. There are 200 bacteria at the start (t = 0).

C. The population doubles every 200 hours.

D. There are 200 bacteria after 1 hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q1/5

A stone's height is $h = 64 - 16t^2$ feet after t seconds. At what time (the t-intercept, $t > 0$) does it hit the ground?

t (seconds) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q2/5

True or false?

The graph of $w = 24/L$ never reaches the L-axis because a rectangle of area 24 cannot have width 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q3/5

For $V = 20000(0.85)^t$, the graph

A. decreases and approaches 0

B. increases

C. is a line

D. decreases to 0 at t = 6.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q4/5

A car's speed is $s = \sqrt{(24d)}$ mph for a skid length of d ft. How does the rate of change of s change as d grows?

A. It increases: the speed grows faster and faster, so the graph gets steeper.

B. It becomes negative after a while: the speed starts to decrease for long skids, so the graph turns down.

C. It stays positive but decreases: the speed increases more and more slowly.

D. It is constant: s increases by the same amount for each foot of skid length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q5/5

Represent nonlinear constraints and judge viable options

MAIN	Algebra II main question	**HSA-CED.A.3**
BACKWARD 1	Represent constraints with a linear system	**HSA-CED.A.3**
BACKWARD 2	Test whether a point satisfies an inequality	**6.EE.B.5**
BACKWARD 3	Solve quadratic inequalities	**HSA-CED.A.1**
FORWARD 1	Optimize within nonlinear constraints	**HSA-CED.A.3**
FORWARD 2	Locate constraint boundaries as graph intersections	**HSA-REI.D.11**

SET 29 • HSA-CED.A.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3

A garden has length L and width W (feet). The fencing gives L + W = 30, and the area must be at least 200 ft². Give the viable widths as a compound inequality, and decide whether W = 8 is viable.

Viable widths: _____

Is W = 8 viable? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q1/5

A box has a square base of side x in and a volume of 32 in^3; both x and the height h must be at most 4 in. Give the viable x as a compound inequality with exact values, and decide whether x = 3 is viable.

Viable x: _____

Is x = 3 viable? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q2/5

A plan invests $1,000 at 5% compounded yearly and needs at least $1,500 within 10 years. Find the value after 10 years to the nearest cent, and decide whether the plan is viable. A calculator may be used.

Value after 10 years ($): _____

Viable? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q3/5

Attendance at a show is $500 - 10p$ people when tickets cost p dollars ($0 \le p \le 50$). Revenue must be at least $6,000. Give the viable prices as a compound inequality, and decide whether $35 is viable.

Viable prices: _____

Is $p = 35$ viable? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q4/5

A rectangular pen with sides x and y (m) needs area at least 50 m² and perimeter at most 40 m. Decide whether (5, 10) and (4, 12) are viable.

Is (5, 10) viable? (Yes or No) _____

Is (4, 12) viable? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q5/5

SET 29 • HSA-CED.A.3

BACKWARD 1

Represent constraints with a linear system • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3

Write a constraint for "the total of L and W is 30".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q1/5

A rectangle has sides x and y (m), and at most 40 m of fence can be used for its perimeter. Which constraint is correct?

A. $2x + 2y \leq 40$

B. $x + y \leq 40$

C. $2x + 2y \geq 40$

D. $xy \leq 40$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q2/5

True or false?

"At least 200" is written ≤ 200.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q3/5

"The height h must be at most 4" is

A. $h \leq 4$

B. $h \geq 4$

C. $h < 4$

D. $h = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q4/5

Is (12, 20) a solution of $x + y \leq 30$? Find $x + y$, then answer Yes or No.

$x + y =$ _____

Solution? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSA-CED.A.3

BACKWARD 2

Test whether a point satisfies an inequality • Grade 6 • Expressions & Equations • 6.EE.B.5

Does $W = 8$ satisfy $W(30 - W) \geq 200$? Find $W(30 - W)$, then answer Yes or No.

$W(30 - W) =$ _____

Satisfies? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q1/5

Does $p = 25$ satisfy $p(500 - 10p) \geq 6000$? Find $p(500 - 10p)$, then answer Yes or No.

$p(500 - 10p) =$ _____

Satisfies? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q2/5

True or false?

(4, 12) satisfies $xy \geq 50$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q3/5

Which x satisfies $x^2 \geq 8$?

A. 3

B. 2

C. 1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q4/5

Does $t = 10$ satisfy $1000(1.05)^t \geq 1500$? Find $1000(1.05)^{10}$ to the nearest cent, then answer Yes or No. A calculator may be used.

$1000(1.05)^{10} =$ _____

Satisfies? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSA-CED.A.3

BACKWARD 3

Solve quadratic inequalities • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

Solve $W^2 - 30W + 200 \leq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q1/5

Solve $p^2 - 50p + 600 \leq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q2/5

True or false?

$x^2 \geq 8$ with $x > 0$ means $x \geq 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q3/5

The solution of $(x - 2)(x - 6) \leq 0$ is

A. $x = 2$ or $x = 6$

B. $2 \leq x \leq 6$

C. $x \leq 2$ or $x \geq 6$

D. $x \geq 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q4/5

Solve $32/x^2 \leq 4$ for $x > 0$.

A. $0 < x \leq 2\sqrt{2}$

B. $x \geq 8$

C. $x \geq \sqrt{2}$

D. $x \geq 2\sqrt{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q5/5

SET 29 • HSA-CED.A.3

FORWARD 1

Optimize within nonlinear constraints • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3

A rectangular garden has length L and width W (feet) with L + W = 30. Which width gives the greatest area, and what is that area?

W (ft) = _____

Greatest area (ft²) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q1/5

At a ticket price of p dollars, revenue is $p(500 - 10p)$ dollars ($0 \le p \le 50$). Which price gives the greatest revenue, and what is that revenue?

p ($) = _____

Greatest revenue ($) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q2/5

True or false?

A maximum found from the model must still satisfy every other constraint to be a viable choice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q3/5

For the pen with $xy \geq 50$ and $2x + 2y \leq 40$, which is viable?

A. (10, 10)

B. (15, 10)

C. (2, 20)

D. (4, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q4/5

A rectangular garden has length L and width W (feet) with L + W = 30, where L and W are positive whole numbers. The area must be at least 216 ft². Give all viable widths as a compound inequality.

Viable W: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q5/5

SET 29 • HSA-CED.A.3

FORWARD 2

Locate constraint boundaries as graph intersections • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

A garden has width W and area $y = W(30 - W)$ ft². How do the graphs of $y = W(30 - W)$ and $y = 200$ show that the widths with area at least 200 ft² are $10 \le W \le 20$?

A. The line $y = 200$ meets the parabola only at its vertex, $W = 15$, so 15 ft is the only width.

B. The parabola crosses the W-axis at $W = 10$ and $W = 20$, so those are the boundary widths.

C. The parabola is below the line $y = 200$ between $W = 10$ and $W = 20$, and above it outside.

D. The graphs intersect at $W = 10$ and $W = 20$, and the parabola is on or above the line $y = 200$ between them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q1/5

Where do $y = p(500 - 10p)$ and $y = 6000$ intersect? Give both values of p.

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q2/5

True or false?

The x-coordinates where $y = 32/x^2$ meets $y = 4$ solve $32/x^2 = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q3/5

The graphs of $y = 1000(1.05)^t$ and $y = 1500$ meet at $t \approx$

A. 15

B. 8.3

C. 10

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q4/5

A garden of width W ft has area $A = W(30 - W)$ ft² $(0 < W < 30)$. On axes with W horizontal and area vertical, the parabola $y = W(30 - W)$ and the line $y = 200$ are drawn. Why do the widths where the parabola is on or above the line give the viable widths for an area of at least 200 ft²?

A. At each such W, the height of the parabola is the area $W(30 - W)$, and it is at least 200.

B. At each such W, the parabola is below 200, so the area is small enough.

C. At each such W, the width W is at least 200.

D. At each such W, the area equals exactly 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q5/5

Rearrange rational, radical, and exponential formulas

MAIN	Algebra II main question	**HSA-CED.A.4**
BACKWARD 1	Rearrange linear formulas	**HSA-CED.A.4**
BACKWARD 2	Undo squares and square roots	**8.EE.A.2**
BACKWARD 3	Clear fractions in an equation	**HSA-REI.B.3**
FORWARD 1	Rearrange formulas where conditions decide the result	**HSA-CED.A.4**
FORWARD 2	Rearrange exponential formulas with logarithms	**HSF-LE.A.4**

SET 30 • HSA-CED.A.4

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4

The lens formula is $1/f = 1/a + 1/b$, where a, b, and f are nonzero and $a \neq f$. Solve for b.

$b =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q1/5

A pendulum's period is $T = 2\pi\sqrt{(L/g)}$, where $g > 0$ and $L \geq 0$ (so $T \geq 0$). Solve for L.

$L =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q2/5

An investment grows by $A = P(1 + r)^t$, where $P > 0$, $A > 0$, the growth base $1 + r$ is positive, and $t \neq 0$. Solve for r.

A. $r = (A - P)^{1/t}$

B. $r = (A/P)^t - 1$

C. $r = (A/P)^{1/t} - 1$

D. $r = (A/P - 1)^{1/t}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q3/5

The speed of a falling object is $v = \sqrt{(2gh)}$, where $g > 0$ and $h \geq 0$ (so $v \geq 0$). Solve for h.

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q4/5

In a circuit, $I = V/(R + r)$, where $R + r \neq 0$ and $I \neq 0$. Solve for R.

$R =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q5/5

SET 30 • HSA-CED.A.4

BACKWARD 1

Rearrange linear formulas • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

Solve P = 2L + 2W for W.

W = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q1/5

Solve d = rt for t.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q2/5

True or false?

Solving V = IR for R gives R = VI.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q3/5

Solving $y = mx + b$ for x gives

A. $(y - b)/m$

B. $y/m - b$

C. $(y + b)/m$

D. $m(y - b)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q4/5

Solve A = (1/2)bh for h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSA-CED.A.4

BACKWARD 2

Undo squares and square roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve $x^2 = 49$ for $x > 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q1/5

Solve $\sqrt{y} = 6$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q2/5

True or false?

If $\sqrt{(L/g)} = k$, then $L/g = \sqrt{k}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q3/5

$(\sqrt{(2gh)})^2$ equals

A. $2g\sqrt{h}$

B. $2gh$

C. $\sqrt{(2gh)}$

D. $4g^2h^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q4/5

Solve $T/(2\pi) = \sqrt{m}$ for m.

m = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSA-CED.A.4

BACKWARD 3

Clear fractions in an equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve 1/2 + 1/x = 1 for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q1/5

Write 1/a − 1/f as one fraction with denominator af (a and f nonzero). Give the numerator.

Numerator: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q2/5

True or false?

If $1/b = (a - f)/(af)$, then $b = af/(a - f)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q3/5

Solving 3 = 12/(R + 2) for R gives

A. 36

B. 4

C. 6

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q4/5

Solve $I = V/(R + r)$ for $R + r$.

$R + r =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q5/5

SET 30 • HSA-CED.A.4

FORWARD 1

Rearrange formulas where conditions decide the result • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4

Solve $A = \pi r^2$ for r, where r is a length ($r \geq 0$) and $A \geq 0$.

A. $r = \sqrt{A/\pi}$

B. $r = \sqrt{(A/\pi)}$

C. $r = \pm\sqrt{(A/\pi)}$

D. $r = A/(2\pi)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q1/5

Solve $y = (x + 1)/(x - 2)$ for x, where $x \neq 2$ and $y \neq 1$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q2/5

True or false?

Solving $s = \sqrt{(24d)}$ for d gives $d = s^2/24$ for every real s.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q3/5

Solving $E = (1/2)mv^2$ for v ($v \geq 0$) gives

A. $\sqrt{(2E/m)}$

B. $(2E/m)^2$

C. $\sqrt{(E/2m)}$

D. $2E/m$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q4/5

Solve $P = k/V^2$ for V, where $k > 0$, $P > 0$, and $V > 0$.

A. $V = k/(2P)$

B. $V = \pm\sqrt{(k/P)}$

C. $V = \sqrt{(k/P)}$

D. $V = \sqrt{(P/k)}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q5/5

SET 30 • HSA-CED.A.4

FORWARD 2

Rearrange exponential formulas with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$A = Pe^{rt}$, where $P \neq 0$, $A/P > 0$, and $r \neq 0$. Which formula gives t?

A. $t = \ln(A/P)/r$

B. $t = \ln(A - P)/r$

C. $t = r \ln(A/P)$

D. $t = (A/P)/(r\ e)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q1/5

$A = P(1 + r)^t$, where $P > 0$, $A > 0$, and the growth base $1 + r$ is positive and not 1. Which formula gives t?

A. $t = (A/P)/(1 + r)$

B. $t = \log(A/P)/\log(1 + r)$

C. $t = \log(A/P) - \log(1 + r)$

D. $t = \log(A - P)/\log(1 + r)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q2/5

True or false?

Solving $N = N_0 \cdot 2^{t/d}$ for t gives $t = d \cdot \log_2(N/N_0)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q3/5

Solving $y = 5 \cdot 10^x$ for x gives

A. y/50

B. log(y/5)

C. $10^{y/5}$

D. log y − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q4/5

Use $t = \log(A/P)/\log(1 + r)$ to find t for $A = 2000$, $P = 1000$, $r = 0.05$. Round to the nearest hundredth. A calculator may be used.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q5/5

Solve rational equations and identify extraneous solutions

MAIN	Algebra II main question	**HSA-REI.A.2**
BACKWARD 1	Solve proportions	**7.RP.A.2**
BACKWARD 2	Find a common denominator of rational expressions	**HSA-APR.D.7**
BACKWARD 3	Values that make a denominator zero	**7.NS.A.2.b**
FORWARD 1	Explain and construct extraneous solutions	**HSA-REI.A.2**
FORWARD 2	See rational equations as graph intersections	**HSA-REI.D.11**

SET 31 • HSA-REI.A.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $3/(x - 1) = 6/(x + 2)$. Give every value found after clearing the denominators, and the actual solutions (enter none if there are none).

Values found after clearing: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q1/5

Solve $x/(x-2)+1=2/(x-2)$. Give every value found after clearing the denominators, and the actual solutions (enter none if there are none).

Values found after clearing: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q2/5

Solve $1/x + 1/(x + 2) = 4/3$. Give every value found after clearing the denominators, and the actual solutions (enter none if there are none).

Values found after clearing: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q3/5

Solve $x^2/(x - 3) = 9/(x - 3)$. Give every value found after clearing the denominators, and the actual solutions (enter none if there are none).

Values found after clearing: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q4/5

Solve $2/(x + 1) - 1/(x - 1) = 2/(x^2 - 1)$. Give every value found after clearing the denominators, and the actual solutions (enter none if there are none).

Values found after clearing: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q5/5

SET 31 • HSA-REI.A.2

BACKWARD 1

Solve proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

Solve $3/x = 6/10$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q1/5

Solve 4/7 = x/21.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q2/5

True or false?

$a/b = c/d$ implies $ac = bd$ when $b, d \neq 0$.

A. True

B. False

2/5 = 6/x gives x =

A. 12

B. 2.4

C. 3

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q4/5

Solve $3/(x - 1) = 1$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSA-REI.A.2

BACKWARD 2

Find a common denominator of rational expressions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

What is the LCD of 1/x and 1/(x + 2)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q1/5

What is the LCD of $\frac{2}{x+1}$, $\frac{1}{x-1}$, and $\frac{2}{x^2-1}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q2/5

True or false?

The LCD of $1/(x-2)$ and $2/(x-2)$ is $(x-2)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q3/5

Multiplying $1/x + 1/(x + 2) = 4/3$ by $3x(x + 2)$ gives

A. $3 + 3 = 4$

B. $3(x + 2) + 3x = 4$

C. $3(x + 2) + 3x = 4x(x + 2)$

D. $x + 2 + x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q4/5

Write 1/x + 1/(x + 2) as one fraction.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSA-REI.A.2

BACKWARD 3

Values that make a denominator zero • Grade 7 • The Number System • 7.NS.A.2.b

Which x-values make $x/(x - 2)$ undefined?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q1/5

Which x-values make $2/(x^2 - 1)$ undefined?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q2/5

True or false?

A rational equation cannot have a solution that makes any denominator zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q3/5

Which value is excluded from $1/x + 1/(x + 2) = 4/3$?

A. 2

B. 1

C. 4/3

D. -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q4/5

List the excluded values of $x^2/(x - 3) = 9/(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q5/5

SET 31 • HSA-REI.A.2

FORWARD 1

Explain and construct extraneous solutions • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solving $x/(x - 2) + 1 = 2/(x - 2)$, Leo multiplies both sides by $x - 2$ and gets $x = 2$. Why is $x = 2$ extraneous?

A. $x = 2$ is extraneous because it is positive, and solutions of rational equations must be negative.

B. $x = 2$ makes both sides equal after multiplying by $x - 2$, so it is a solution of the original equation and is not extraneous.

C. Leo made an arithmetic error when he multiplied; the correct solution of the equation is $x = -2$.

D. Multiplying by $x - 2$ is not reversible when $x - 2 = 0$, so the new equation can be true at $x = 2$ even though the original equation is undefined there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F1-Q1**

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q1/5

Which rational equation has $x = 4$ as an extraneous solution and no actual solutions?

A. $1/(x - 4) = 1$

B. $x/(x - 4) = 2$

C. $x/(x - 4) = 4/(x - 4)$

D. $(x - 4)/x = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q2/5

True or false?

Every rational equation has at least one extraneous solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q3/5

For which a does $a/(x - 1) = 2/(x - 1)$ have infinitely many solutions?

A. a = 2

B. no value

C. a = 0

D. a = 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q4/5

Which is the solution set of
$(x + 1)/(x - 1) = 2/(x - 1) + 1$?

A. $x = 1$ only

B. Every real number except 1

C. Every real number

D. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSA-REI.A.2

FORWARD 2

See rational equations as graph intersections • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

Why does $x/(x - 2) + 1 = 2/(x - 2)$ have no solution, in terms of the graphs of $y = x/(x - 2) + 1$ and $y = 2/(x - 2)$?

A. For $x \neq 2$ the left side minus the right side is $(2x - 4)/(x - 2) = 2$, so the graphs never meet.

B. The graphs meet at $x = 2$, where both expressions have the same value, so $x = 2$ is a solution.

C. The two graphs are the same curve, so every x except 2 is a solution of the equation.

D. The graphs cross at $x = 0$ and nowhere else, so $x = 0$ is the only solution of the equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q1/5

Where do $y = 3/(x - 1)$ and $y = 6/(x + 2)$ intersect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q2/5

True or false?

An extraneous solution corresponds to an intersection point of the two graphs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q3/5

$y = x^2/(x - 3)$ and $y = 9/(x - 3)$ intersect at

A. $(3, 3/2)$

B. $(-3, 3/2)$

C. $(-3, -3/2)$

D. $(3, 0)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q4/5

Use a graph or table to approximate the solution of $1/x = x - 1$ with $x > 0$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q5/5

Solve radical equations and identify extraneous solutions

MAIN	Algebra II main question	**HSA-REI.A.2**
BACKWARD 1	Square and cube both sides; principal roots	**8.EE.A.2**
BACKWARD 2	Solve the resulting quadratic	**HSA-REI.B.4.b**
BACKWARD 3	Check a candidate in the original equation	**6.EE.B.5**
FORWARD 1	Solve equations with two radicals, rational exponents, and $\sqrt{(x^2)}$	**HSA-REI.A.2**
FORWARD 2	Explain extraneous solutions with graphs of radical functions	**HSF-IF.C.7.b**

SET 32 • HSA-REI.A.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $\sqrt{(x + 7)} = x - 5$. Give every value found after squaring, and the actual solutions (enter none if there are none).

Values found after squaring: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q1/5

Solve $\sqrt{(2x - 1)} = 3$. Give every value found after squaring, and the actual solutions (enter none if there are none).

Values found after squaring: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q2/5

Solve $\sqrt{(x + 3)} + 3 = x$. Give every value found after squaring, and the actual solutions (enter none if there are none).

Values found after squaring: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q3/5

Solve $\sqrt[3]{(x-2)} = -3$. Give every value found after cubing, and the actual solutions (enter none if there are none).

Values found after cubing: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q4/5

Solve $\sqrt{(x - 1)} + 5 = 2$. Give every value found after squaring, and the actual solutions (enter none if there are none).

Values found after squaring: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q5/5

SET 32 • HSA-REI.A.2

BACKWARD 1

Square and cube both sides; principal roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve $\sqrt{y} = 3$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q1/5

Solve $\sqrt[3]{y} = -3$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q2/5

True or false?

$\sqrt{9} = 3$ or -3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q3/5

$\sqrt{y} = -2$ has

A. $y = 2$

B. no solution

C. $y = 4$

D. $y = -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q4/5

Square both sides of $\sqrt{(x + 7)} = x - 5$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSA-REI.A.2

BACKWARD 2

Solve the resulting quadratic • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $x^2 - 11x + 18 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q1/5

Solve $x^2 - 7x + 6 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q2/5

True or false?

$(x - 5)^2 = x^2 + 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q3/5

$x^2 - 6x + 9 = x + 3$ gives

A. $x^2 - 7x + 12 = 0$

B. $x^2 - 5x + 12 = 0$

C. $x^2 - 6x + 6 = 0$

D. $x^2 - 7x + 6 = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q4/5

Solve $x^2 - 4x = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSA-REI.A.2

BACKWARD 3

Check a candidate in the original equation • Grade 6 • Expressions & Equations • 6.EE.B.5

Check $x = 2$ in $\sqrt{(x + 7)} = x - 5$: evaluate each side, then say whether $x = 2$ is a solution.

Left side: _____

Right side: _____

Solution? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q1/5

Check $x = 6$ in $\sqrt{(x + 3)} + 3 = x$: evaluate each side, then say whether $x = 6$ is a solution.

Left side: _____

Right side: _____

Solution? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q2/5

True or false?

$x = 4$ satisfies $\sqrt{(2x - 1)} = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q3/5

Which value satisfies $\sqrt{(x + 3)} + 3 = x$?

A. 1

B. 0

C. 6

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q4/5

Check $x = 10$ in $\sqrt{(x - 1)} + 5 = 2$: evaluate each side, then say whether $x = 10$ is a solution.

Left side: _____

Right side: _____

Solution? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q5/5

SET 32 • HSA-REI.A.2

FORWARD 1

Solve equations with two radicals, rational exponents, and $\sqrt{(x^2)}$ • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $\sqrt{(x + 5)} - \sqrt{x} = 1$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q1/5

Solve $x^{3/2} = 8$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q2/5

Solve $\sqrt{(x^2)} = 5$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q3/5

True or false?

$\sqrt{(x^2)} = x$ for every real x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q4/5

The solution set of $\sqrt{(x^2)} = x$ is

A. $x \geq 0$

B. $x \leq 0$

C. $x = 0$ only

D. all real x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSA-REI.A.2

FORWARD 2

Explain extraneous solutions with graphs of radical functions • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

The graphs of $y = \sqrt{(x + 7)}$ and $y = x - 5$ meet at one point. Give it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q1/5

Squaring $\sqrt{(x + 7)} = x - 5$ gives $x + 7 = (x - 5)^2$, with solutions $x = 9$ and $x = 2$. Squaring cannot tell $\sqrt{(x + 7)}$ from $-\sqrt{(x + 7)}$. At $x = 2$, the line $y = x - 5$ meets which graph?

A. $y = (x - 5)^2$

B. $y = x + 7$

C. $y = \sqrt{(x + 7)}$

D. $y = -\sqrt{(x + 7)}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q2/5

True or false?

The graph of $y = \sqrt{(x + 7)}$ starts at $(7, 0)$ and increases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q3/5

The graph of $y = \sqrt{(x - 1)} + 5$ lies

A. below $y = 2$

B. below the x-axis

C. at or below $y = 5$

D. at or above $y = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q4/5

Why does $\sqrt{(x - 1)} + 5 = 2$ have no solution, in terms of the graphs of $y = \sqrt{(x - 1)} + 5$ and $y = 2$?

A. $y = \sqrt{(x - 1)} + 5$ is always below the line $y = 2$, so the graphs never meet.

B. The graph of $y = \sqrt{(x - 1)} + 5$ is not defined for any x, so there is nothing to meet $y = 2$.

C. The graphs meet at $x = 10$, because squaring $\sqrt{(x - 1)} = -3$ gives $x = 10$.

D. $y = \sqrt{(x - 1)} + 5$ is always at least 5 (a square root is never negative), so it never meets the line $y = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q5/5

Explain why intersections of nonlinear graphs solve f(x) = g(x)

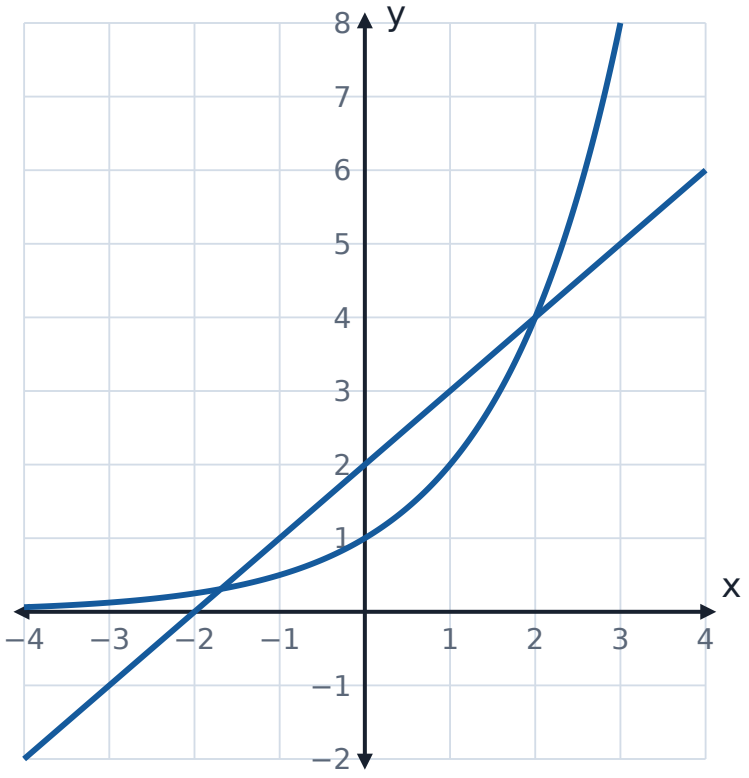
MAIN	Algebra II main question	**HSA-REI.D.11**
BACKWARD 1	A graph is the set of its solutions	**HSA-REI.D.10**
BACKWARD 2	Solve a linear system by graphing	**8.EE.C.8.a**
BACKWARD 3	Evaluate nonlinear functions at an input	**HSF-IF.A.2**
FORWARD 1	Relate intersections to zeros of f - g	**HSA-REI.D.11**
FORWARD 2	Find all solutions of a trigonometric equation on an interval from a…	**HSF-TF.B.7**

SET 33 • HSA-REI.D.11

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

The graphs of $y = 2^x$ and $y = x + 2$ are shown. They meet at $x = 2$ and at a negative x. Why does $x = 2$ solve $2^x = x + 2$?



A. Because 2^x and $x + 2$ are equal for every $x \geq 0$, including $x = 2$.

B. At an intersection both functions have the same output for the same input; at $x = 2$, $2^2 = 4 = 2 + 2$.

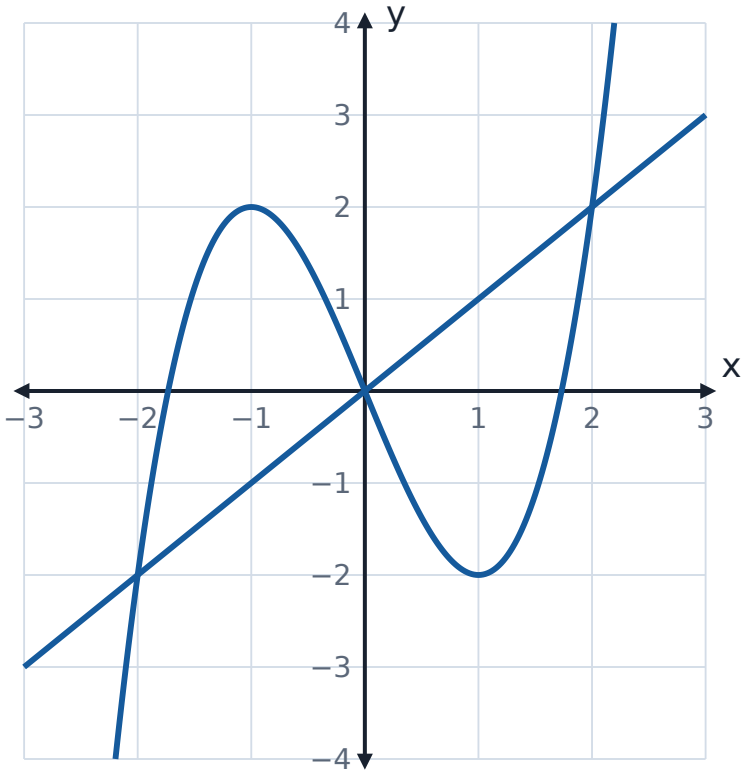
C. Because $x = 2$ is where the graph of $y = 2^x$ crosses the x-axis.

D. Because the y-coordinate of the intersection point is 2, not its x-coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q1/5

The graphs of $y = x^3 - 3x$ and $y = x$ are shown. Which gives every solution of $x^3 - 3x = x$ with a correct reason?



A. $x = 0$ only: that is where both graphs pass through the origin.

B. $x = -\sqrt{3}, 0, \sqrt{3}$: these are where $y = x^3 - 3x$ crosses the x-axis.

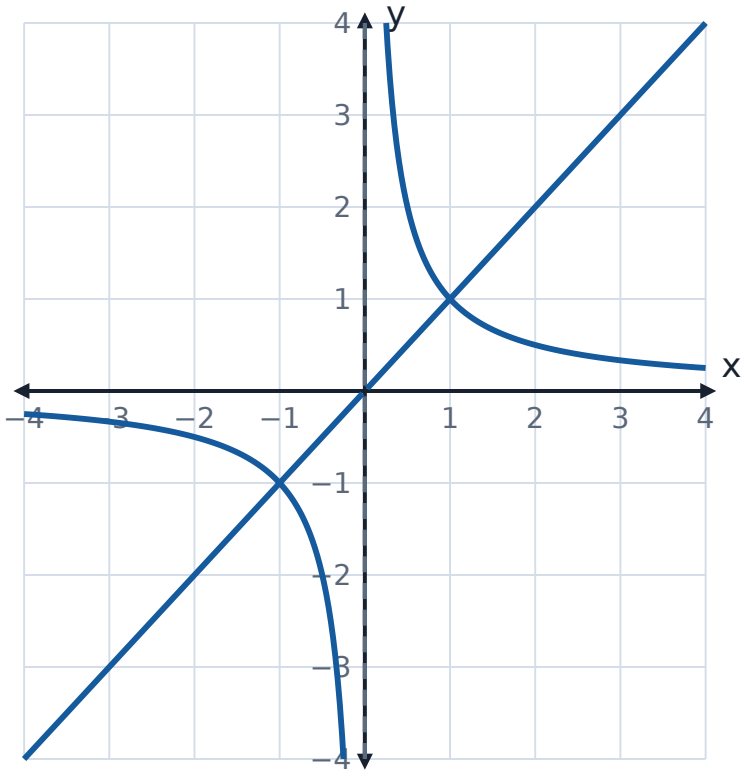
C. $x = -2, 0, 2$: the graphs meet at these x-values, where the two outputs are equal.

D. $y = -2, 0, 2$: these are the heights where the graphs meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q2/5

The graphs of $y = 1/x$ and $y = x$ are shown. Which gives every solution of $1/x = x$ with a correct reason?



A. $x = 0$: the graphs are closest near the origin, where $1/x$ and x both approach 0.

B. $x = 1$ only: the graphs meet only in the first quadrant, at $(1, 1)$.

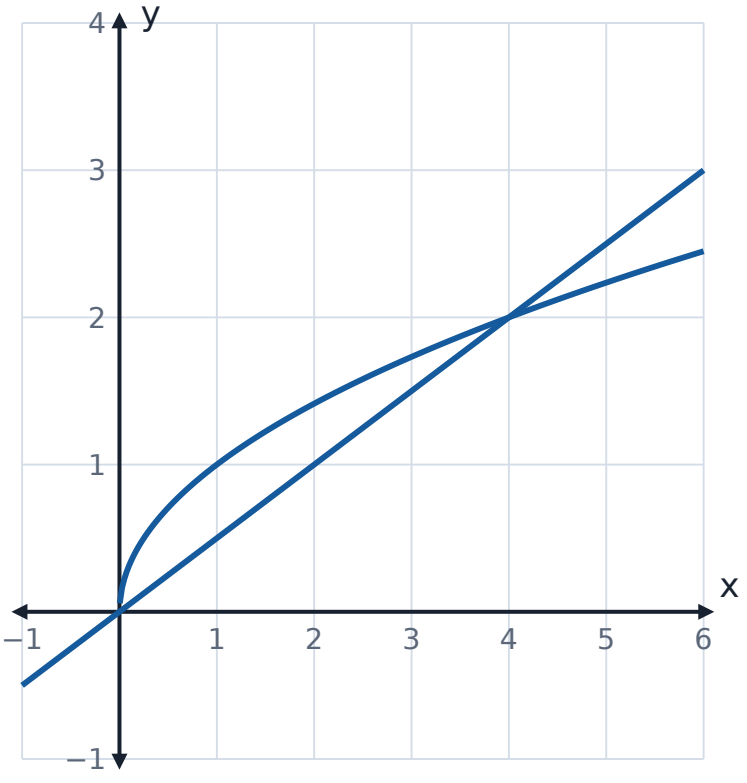
C. $x = -1$ and $x = 1$: the graphs meet at $(-1, -1)$ and $(1, 1)$, where $1/x$ and x are equal.

D. There is no solution, because $y = 1/x$ has a vertical asymptote at $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q3/5

The graphs of $y = \sqrt{x}$ and $y = x/2$ are shown. Which gives every solution of $\sqrt{x} = x/2$ with a correct check?



A. $x = 0$ and $x = 4$: $\sqrt{0} = 0 = 0/2$ and $\sqrt{4} = 2 = 4/2$.

B. $x = 2$ and $x = 4$: these are the heights and the widths of the intersection.

C. $x = 4$ only: $\sqrt{4} = 2 = 4/2$; the origin does not count.

D. $x = 0$ and $x = 2$: $\sqrt{2} = 1 = 2/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q4/5

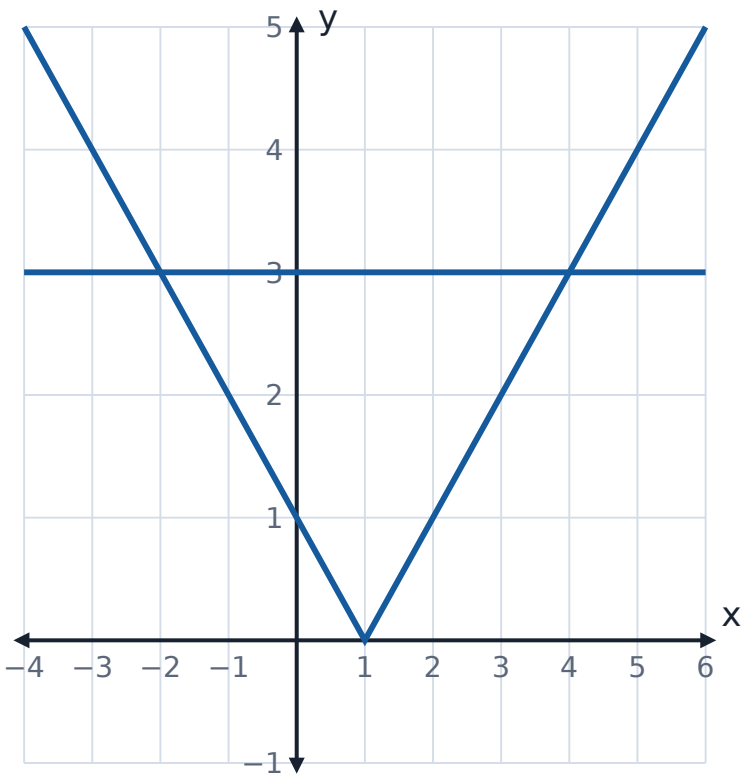
The graphs of $y = |x - 1|$ and $y = 3$ are shown. Which gives every solution of $|x - 1| = 3$ with a correct reason?

A. $x = -4$ and $x = 2$: these x-values are 3 units from 1 on the number line.

B. $x = -2$ and $x = 4$: the V-shaped graph is at height 3 at these x-values.

C. $x = 3$: the horizontal line is at height 3, so $x = 3$ is the solution.

D. $x = 4$ only: $|x - 1|$ is positive only for $x > 1$, so only one solution.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q5/5

SET 33 • HSA-REI.D.11

BACKWARD 1

A graph is the set of its solutions • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

Is the point (2, 4) on the graph of $y = 2^x$? (Yes or No)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q1/5

Is the point (3, 6) on the graph of $y = x + 2$? (Yes or No)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q2/5

True or false?

Every point on the graph of $y = f(x)$ has the form $(x, f(x))$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q3/5

Which point is on $y = x^3 - 3x$?

A. (0, 3)

B. (2, 6)

C. (2, 2)

D. (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q4/5

Is the point $(-1, -1)$ on the graph of $y = 1/x$? (Yes or No)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSA-REI.D.11

BACKWARD 2

Solve a linear system by graphing • Grade 8 • Expressions & Equations • 8.EE.C.8.a

The lines $y = x + 1$ and $y = -x + 5$ meet at $(2, 3)$. Why does $x = 2$ solve $x + 1 = -x + 5$?

A. Because the two lines have opposite slopes, 1 and -1.

B. Because the y-coordinate of the intersection point $(2, 3)$ is 2.

C. At $x = 2$ both expressions equal 3, so both sides are equal.

D. Because $x = 2$ is where both lines cross the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q1/5

Where do y = 2x and y = 6 meet?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q2/5

True or false?

Two lines that do not intersect give an equation with no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q3/5

$y = 3$ and $y = x - 1$ meet at $x =$

A. 4

B. 3

C. -1

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q4/5

Solve $x + 1 = -x + 5$ algebraically.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSA-REI.D.11

BACKWARD 3

Evaluate nonlinear functions at an input • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Find 2^x and $x + 2$ when $x = 2$.

$2^x =$ _____

$x + 2 =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q1/5

Find $x^3 - 3x$ when $x = -2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q2/5

True or false?

$\sqrt{9} = 9/2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q3/5

$|x - 1|$ when $x = -2$ is

A. 3

B. 1

C. -1

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q4/5

Find $1/x$ when $x = -1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q5/5

SET 33 • HSA-REI.D.11

FORWARD 1

Relate intersections to zeros of f − g • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

Why are the solutions of $x^3 - 3x = x$ the zeros of $h(x) = x^3 - 4x$?

A. Because $h(x) = f(x) \cdot g(x)$ and a product is 0 when a factor is 0.

B. Because adding x to both sides gives $x^3 - 4x = 0$.

C. Because $x^3 - 4x$ has three terms.

D. $f(x) = g(x)$ exactly when $f(x) - g(x) = 0$, and $(x^3 - 3x) - x = x^3 - 4x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q1/5

Find the zeros of $h(x) = x^3 - 4x$ algebraically.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q2/5

True or false?

If $y = f(x)$ and $y = g(x)$ meet at $x = a$, then the graph of $y = f(x) - g(x)$ crosses or touches the x-axis at a.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q3/5

The solutions of $\sqrt{x} = x/2$ are the zeros of

A. $\sqrt{x} + x/2$

B. $\sqrt{x} - x/2$

C. $\sqrt{x} \cdot x/2$

D. $x/2$

$h(x) = 2^x - x - 2$, with $h(-2) = 0.25$, $h(0) = -1$, $h(3) = 3$. h decreases and then increases (it has one turning point). Why does $2^x = x + 2$ have exactly two solutions?

A. h is continuous and changes sign twice (on -2 to 0 and 0 to 3), and with one turning point it has at most two zeros.

B. Every equation that sets an exponential function equal to a line has exactly two solutions, so $2^x = x + 2$ has two.

C. $h(-2)$ and $h(3)$ are both positive, so h cannot change sign anywhere between them, and therefore it has no zeros at all.

D. h has degree 2 because of the 2^x term, so like every quadratic it has exactly two zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSA-REI.D.11

FORWARD 2

Find all solutions of a trigonometric equation on an interval from a graph • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

On $0 \le x < 2\pi$, where does $y = \sin x$ meet $y = 1/2$? Give exact values.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q1/5

On $0 \le x < 2\pi$, where does $y = \cos x$ meet $y = 0$?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q2/5

True or false?

sin x = 1/2 has exactly two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q3/5

How many solutions does $\sin x = 0.3$ have on $0 \leq x < 2\pi$?

A. 1

B. 0

C. 2

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q4/5

Why does $\sin x = 2$ have no solution?

A. The graph of $y = \sin x$ meets $y = 2$ only at $x = \pi/2$.

B. $\sin x = 2$ has solutions only when x is in degrees.

C. The graph of $y = \sin x$ is not periodic, so it never returns to the height 2.

D. The graph of $y = \sin x$ stays between -1 and 1, so it never meets $y = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q5/5

Approximate solutions of f(x) = g(x)

MAIN	Algebra II main question	**HSA-REI.D.11**
BACKWARD 1	Read solutions of linear-quadratic systems from graphs	**HSA-REI.D.11**
BACKWARD 2	Evaluate exponential and logarithmic expressions with a calculat…	**HSF-IF.A.2**
BACKWARD 3	Round to a place value	**5.NBT.A.4**
FORWARD 1	Guarantee the accuracy of an approximation	**HSA-REI.D.11**
FORWARD 2	Compare an approximation with an exact logarithmic solution	**HSF-LE.A.4**

SET 34 • HSA-REI.D.11

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

Use a table or successive approximation to solve $2^x = 5 - x$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q1/5

Solve $x^3 = 7 - x$ to the nearest hundredth with a table or successive approximation. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q2/5

Solve $\log x = 2 - x$ (log base 10) to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q3/5

The table shows $f(x) = 3(0.8)^x$ and $g(x) = x$. Between which two consecutive table inputs is the solution of $f(x) = g(x)$? Then estimate the solution to the nearest tenth.

x	0	0.5	1	1.5	2	2.5
f(x)	3	2.683	2.4	2.147	1.92	1.717
g(x)	0	0.5	1	1.5	2	2.5

Between x = _____

and x = _____

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q4/5

Use a graph or table to find the negative solution of $2^x = x + 2$ to the nearest tenth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q5/5

SET 34 • HSA-REI.D.11

BACKWARD 1

Read solutions of linear-quadratic systems from graphs • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

Where do $y = x^2$ and $y = 4$ meet? Give both x-values.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q1/5

Where do $y = x^2$ and $y = x + 2$ meet? Give both x-values.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q2/5

True or false?

If $f - g$ is continuous on an interval and changes sign between two of its table values, then $f(x) = g(x)$ has a solution between those two inputs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q3/5

f and g are continuous on [1, 2], with $f(1) - g(1) = -2$ and $f(2) - g(2) = 1$. A solution of $f(x) = g(x)$ lies

A. below 1

B. at 1.5 exactly

C. above 2

D. between 1 and 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q4/5

$f(x) = x^2$ and $g(x) = 3$. Between which two consecutive integers is the positive solution of $f(x) = g(x)$?

Smaller integer: _____

Larger integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • HSA-REI.D.11

BACKWARD 2

Evaluate exponential and logarithmic expressions with a calculator • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Find $2^{1.7}$ to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q1/5

Find log 1.75 (base 10) to the nearest thousandth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q2/5

True or false?

$1.7^3 = 5.1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q3/5

$3(0.8)^2$ equals

A. 0.192

B. 4.8

C. 1.92

D. 2.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q4/5

Find $2^{-1.7}$ to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSA-REI.D.11

BACKWARD 3

Round to a place value • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 1.7156 to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q1/5

Round 1.944 to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q2/5

True or false?

1.755 rounded to the nearest hundredth is 1.76.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q3/5

-1.690 to the nearest tenth is

A. -1.6

B. -1.69

C. -2

D. -1.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q4/5

Round 1.7392 to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q5/5

SET 34 • HSA-REI.D.11

FORWARD 1

Guarantee the accuracy of an approximation • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

$h(x) = 2^x + x - 5$. Find h(1.71) and h(1.72), each to 3 decimal places. A calculator may be used.

h(1.71) = _____

h(1.72) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q1/5

$h(x) = x^3 + x - 7$ is continuous. One step of bisection on [1, 2] evaluates $h(1.5)$. Which half contains the solution of $h(x) = 0$, and why?

A. [1, 1.5], because $h(1.5)$ is closer to 0 than $h(1)$.

B. [1.5, 2], because $h(1.5) = -2.125 < 0$ and $h(2) = 3 > 0$.

C. [1.5, 2], because 1.5 is the midpoint.

D. [1, 1.5], because $h(1) = -5 < 0$ and $h(1.5) = -2.125 < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q2/5

True or false?

After 4 bisection steps starting from an interval of length 1, the interval has length 1/8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q3/5

How many bisection steps from [1, 2] guarantee an interval shorter than 0.01?

A. 5

B. 100

C. 10

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q4/5

$h(x) = 2^x + x - 5$ is continuous and increasing, so $h(x) = 0$ has exactly one solution. $h(1.715) \approx -0.002$ and $h(1.725) \approx 0.031$. What can be concluded?

A. The solution is strictly between 1.715 and 1.725, so it rounds to 1.72 to two decimal places.

B. Nothing can be concluded about the solution, because neither $h(1.715)$ nor $h(1.725)$ is exactly 0.

C. The solution is exactly 1.72, because 1.72 is halfway between 1.715 and 1.725.

D. The solution could round to 1.73, because $h(1.725)$ is positive and 1.725 rounds up to 1.73.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q5/5

SET 34 • HSA-REI.D.11

FORWARD 2

Compare an approximation with an exact logarithmic solution • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $2^x = 5$. Give x to the nearest hundredth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q1/5

Solve $3(0.8)^x = 1$. Give x to the nearest hundredth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q2/5

True or false?

$2^x = 5 - x$ can be solved exactly with one logarithm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q3/5

Which equation can be solved exactly with logarithms?

A. $\log x = 2 - x$

B. $2^x = x + 2$

C. $5(1.2)^x = 30$

D. $3(0.8)^x = x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q4/5

$2^x = x + 2$ has the exact solution $x = 2$ ($2^2 = 4 = 2 + 2$) and a second, negative solution. Why is the negative solution found by approximation, while $2^x = 4$ is solved exactly?

A. Negative numbers cannot be exact solutions of an exponential equation, so any negative solution of $2^x = x + 2$ must be approximated, while $x = 2$ is exact.

B. The negative solution is approximated because 2^x is negative for $x < 0$, so logarithms cannot be taken of it; $2^x = 4$ has a positive power, so $x = \log_2 4 = 2$.

C. $2^x = 4$ isolates the power, so $x = \log_2 4 = 2$; in $2^x = x + 2$ the variable is both in an exponent and outside it, and no algebra step isolates it, so the negative solution (about -1.69) is approximated.

D. Approximation is needed only because $x = 2$ is already a solution: once one exact solution is known, the other solution of the equation cannot also be found exactly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q5**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q5/5

SET 35 • HSF-TF.A.1 • TRIGONOMETRIC FUNCTIONS

Use radian measure as arc length on the unit circle

MAIN	Algebra II main question	**HSF-TF.A.1**
BACKWARD 1	Circumference of a circle	**7.G.B.4**
BACKWARD 2	Arc length is proportional to the angle	**HSG-C.B.5**
BACKWARD 3	Degrees as fractions of a turn	**4.MD.C.5**
FORWARD 1	Measure negative angles and angles beyond one turn	**HSF-TF.A.1**
FORWARD 2	Read sine and cosine of radian angles from the unit circle	**HSF-TF.A.2**

SET 35 • HSF-TF.A.1

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

On the unit circle, a central angle subtends an arc of length 2.5. What is the radian measure of the angle?

Radians: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q1/5

Convert 150° to radians. Give an exact answer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q2/5

Convert 7π/4 radians to degrees.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q3/5

A central angle of $2\pi/3$ radians is drawn in a circle of radius 6 cm. Find the exact length of the arc it subtends, in cm, in terms of π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q4/5

Why does one full turn measure 2π radians?

A. A full turn subtends the whole unit circle, whose circumference is $2\pi \cdot 1 = 2\pi$, and radian measure equals arc length on the unit circle.

B. Because π radians is a quarter turn of the unit circle, so a full turn of four quarters measures 2π radians.

C. Because the area of the unit circle is $2\pi \cdot 1 = 2\pi$, and radian measure equals the area swept out by the angle.

D. Because a full turn is $360°$, and $360 = 2\pi$ when degrees are written as numbers of radians.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q5/5

SET 35 • HSF-TF.A.1

BACKWARD 1

Circumference of a circle • Grade 7 • Geometry • 7.G.B.4

Find the circumference of a circle of radius 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q1/5

Find the circumference of a circle of radius 6 cm.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q2/5

True or false?

The circumference of a circle is πr^2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q3/5

Half of the unit circle has length

A. 2π

B. 1

C. π

D. $\pi/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q4/5

What length is one quarter of the unit circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSF-TF.A.1

BACKWARD 2

Arc length is proportional to the angle • High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5

A 60° angle subtends what fraction of a full circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q1/5

Find the arc length of a 90° angle in a circle of radius 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q2/5

True or false?

Doubling the radius doubles the arc length for the same angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q3/5

The arc length for angle θ (radians) in a circle of radius r is

A. θ/r

B. $r^2\theta$

C. $2\pi\theta$

D. $r\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q4/5

What fraction of the circle is an angle of π/3 radians?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q5/5

SET 35 • HSF-TF.A.1

BACKWARD 3

Degrees as fractions of a turn • Grade 4 • Measurement & Data • 4.MD.C.5

How many degrees are in a full turn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q1/5

What fraction of a turn is 150°?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q2/5

True or false?

A 315° angle is 3/4 of a turn.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q3/5

A quarter turn is

A. 90°

B. 180°

C. 45°

D. 360°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q4/5

How many degrees are in 2/3 of a turn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q5/5

SET 35 • HSF-TF.A.1

FORWARD 1

Measure negative angles and angles beyond one turn • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

Give a positive angle less than 2π that is coterminal with $-\pi/3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q1/5

Give the angle between 0 and 2π that is coterminal with 13π/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q2/5

True or false?

An arc of length 10 on the unit circle goes more than once around the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q3/5

An angle of $-\pi/2$ radians is traced

A. a quarter turn clockwise

B. one full turn

C. half a turn

D. a quarter turn counterclockwise

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q4/5

A point travels 10 units counterclockwise around the unit circle from (1, 0). How far past one full turn is it (along the circle)? Round to the nearest hundredth. A calculator may be used.

Distance past one turn: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q5/5

SET 35 • HSF-TF.A.1

FORWARD 2

Read sine and cosine of radian angles from the unit circle • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

The point of the unit circle at angle π is (cos π, sin π). Give that point.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q1/5

Find $\cos(3\pi/2)$ and $\sin(3\pi/2)$.

$\cos(3\pi/2) =$ _____

$\sin(3\pi/2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q2/5

True or false?

$\cos(2\pi) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q3/5

$\sin(\pi/2)$ equals

A. -1

B. 0

C. $\pi/2$

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q4/5

Why do cos θ and sin θ make sense for θ = 13π/6?

A. 13π/6 ends at the same point as 13π/6 − π = 7π/6, so the values are those at 7π/6.

B. 13π/6 = 2π + π/6 ends at the same point as π/6, so cos and sin take the values at that point.

C. 13π/6 is greater than 2π, so cos and sin are both equal to 0 at that angle.

D. cos and sin are defined only for 0 ≤ θ ≤ 2π, so 13π/6 is first divided by 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q5/5

SET 36 • HSF-TF.A.2 • TRIGONOMETRIC FUNCTIONS

Find exact trigonometric values beyond the first quadrant

MAIN	Algebra II main question	**HSF-TF.A.2**
BACKWARD 1	Special right triangle ratios	**HSG-SRT.C.6**
BACKWARD 2	Reference angles for radian measures	**HSF-TF.A.1**
BACKWARD 3	Signs of coordinates by quadrant	**6.NS.C.6.b**
FORWARD 1	Values for angles beyond 2π and solving for angles	**HSF-TF.A.2**
FORWARD 2	Verify the Pythagorean identity at exact values	**HSF-TF.C.8**

SET 36 • HSF-TF.A.2

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Use the unit circle to find $\sin(5\pi/6)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q1/5

Use the unit circle to find cos(4π/3) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q2/5

Use the unit circle to find tan(7π/4) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q3/5

Use the unit circle to find $\sin(-\pi/2)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q4/5

Use the unit circle to find cos(11π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q5/5

SET 36 • HSF-TF.A.2

BACKWARD 1

Special right triangle ratios • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

In a 30°−60°−90° triangle with hypotenuse 1, give the side lengths.

Side opposite 30°: _____

Side opposite 60°: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q1/5

Find sin 45° exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q2/5

True or false?

$\cos 60° = \sqrt{3}/2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q3/5

sin 60° equals

A. √2/2

B. √3/2

C. √3

D. 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q4/5

Find tan 30° exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q5/5

SET 36 • HSF-TF.A.2

BACKWARD 2

Reference angles for radian measures • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

Find the reference angle of 5π/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q1/5

Find the reference angle of $4\pi/3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q2/5

True or false?

The reference angle of $7\pi/4$ is $3\pi/4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q3/5

Which quadrant contains 11π/6?

A. I

B. IV

C. III

D. II

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q4/5

Convert 4π/3 to degrees.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSF-TF.A.2

BACKWARD 3

Signs of coordinates by quadrant • Grade 6 • The Number System • 6.NS.C.6.b

In which quadrant are both coordinates negative? Answer I, II, III, or IV.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q1/5

What are the signs of the coordinates of a point in quadrant II?

A. both positive

B. both negative

C. x positive, y negative

D. x negative, y positive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q2/5

True or false?

A point in quadrant IV has a positive y-coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q3/5

The point $(-\sqrt{3}/2, 1/2)$ is in quadrant

A. I

B. II

C. III

D. IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q4/5

What are the signs of the coordinates of a point in quadrant IV?

A. both negative

B. x negative, y positive

C. x positive, y negative

D. both positive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q5/5

SET 36 • HSF-TF.A.2

FORWARD 1

Values for angles beyond 2π and solving for angles • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Find sin(13π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q1/5

Find $\cos(-3\pi/4)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q2/5

Find every θ with $0 \le \theta < 2\pi$ and $\cos\theta = -1/2$.

$\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q3/5

True or false?

$\tan(\pi/2)$ is undefined.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q4/5

Which angle has $\sin\theta = -\sqrt{3}/2$ and $\cos\theta = 1/2$?

A. $5\pi/3$

B. $4\pi/3$

C. $2\pi/3$

D. $7\pi/6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSF-TF.A.2

FORWARD 2

Verify the Pythagorean identity at exact values • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

For $\theta = 5\pi/6$, find $\sin^2\theta$ and $\cos^2\theta$, then their sum.

$\sin^2\theta =$ _____

$\cos^2\theta =$ _____

Sum = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q1/5

For $\theta = 7\pi/4$, find $\sin^2\theta$ and $\cos^2\theta$, then their sum.

$\sin^2\theta =$ _____

$\cos^2\theta =$ _____

Sum = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q2/5

True or false?

$\sin^2\theta + \cos^2\theta = 1$ is true only for first-quadrant angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q3/5

If $\cos\theta = -1/2$ and θ is in quadrant III, then $\sin\theta =$

A. $\sqrt{3}/2$

B. $-\sqrt{3}/2$

C. $1/2$

D. $-1/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q4/5

Why do cos θ and sin θ, read from the unit circle, always satisfy $\sin^2\theta + \cos^2\theta = 1$?

A. Because sin θ + cos θ = 1 for every θ.

B. Because sin θ and cos θ are always equal.

C. Because the values were rounded so that they work.

D. The point (cos θ, sin θ) lies on the unit circle $x^2 + y^2 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q5/5

SET 37 • HSF-TF.A.2 • TRIGONOMETRIC FUNCTIONS

Explain the extension of sine and cosine to all real numbers

MAIN	Algebra II main question	**HSF-TF.A.2**
BACKWARD 1	Right-triangle definitions of sine and cosine	**HSG-SRT.C.6**
BACKWARD 2	Points on the unit circle	**HSG-GPE.A.1**
BACKWARD 3	Radian measure as directed arc length	**HSF-TF.A.1**
FORWARD 1	Explain where tangent is defined and how values repeat	**HSF-TF.A.2**
FORWARD 2	Explain symmetry and periodicity	**HSF-TF.A.4**

SET 37 • HSF-TF.A.2

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Which statement correctly explains how the unit circle defines cos t and sin t for every real number t?

A. Start at (1, 0) and move t units straight up; cos t and sin t are the coordinates of the point reached.

B. Start at (1, 0) and travel |t| units along the unit circle, counterclockwise if t > 0 and clockwise if t < 0; cos t and sin t are the x- and y-coordinates of the endpoint.

C. Travel t units counterclockwise from (1, 0); for t < 0 the values are undefined.

D. Draw a right triangle with an acute angle of t radians; cos t and sin t are the ratios adjacent/hypotenuse and opposite/hypotenuse, so the definition works only for t between 0 and π/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q1/5

Which explanation shows why $\sin(t + 2\pi) = \sin t$ for every real number t?

A. For $t = \pi/2$ both sides equal 1, so they are always equal.

B. Sine distributes over addition, so $\sin(t + 2\pi) = \sin t + \sin 2\pi$, and $\sin 2\pi = 0$ because the point at 2π is $(1, 0)$; therefore $\sin(t + 2\pi) = \sin t$.

C. Adding 2π to t reflects the endpoint on the unit circle over the x-axis, and a reflection over the x-axis keeps the y-coordinate, so $\sin(t + 2\pi) = \sin t$.

D. The unit circle has circumference 2π, so adding 2π adds one full trip around it: t and $t + 2\pi$ end at the same point, which has the same y-coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q2/5

Why is $\cos t < 0$ for $\pi/2 < t < \pi$?

A. For these t the endpoint on the unit circle is in quadrant II, where x-coordinates are negative, and cos t is the x-coordinate.

B. For these t the endpoint is in quadrant II, where y-coordinates are negative, and cos t is the y-coordinate.

C. $\cos(2\pi/3) = -1/2$, so cos t is negative on this interval.

D. On the unit circle cos t decreases steadily as t increases from 0, so once t passes $\pi/2$, where $\cos t = 0$, the value of cos t is negative for every larger t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q3/5

Why is $-1 \le \sin t \le 1$ for every real number t?

A. $\sin(\pi/2) = 1$ and $\sin(3\pi/2) = -1$, so every value lies between them.

B. In a right triangle the hypotenuse is longer than either leg, so opposite/hypotenuse is less than 1, which means $\sin t < 1$ for every real number t.

C. $\sin t$ is the y-coordinate of a point on the unit circle $x^2 + y^2 = 1$, so $y^2 \le 1$, which means $-1 \le y \le 1$.

D. $\sin t$ is an arc length on the unit circle, and those arcs are at most 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q4/5

How does the unit-circle definition give $\sin(-\pi/6)$?

A. Travel $\pi/6$ clockwise from $(1, 0)$ to $(1/2, -\sqrt{3}/2)$, so $\sin(-\pi/6) = -\sqrt{3}/2$.

B. Negative inputs are not on the unit circle, so $\sin(-\pi/6)$ is undefined.

C. Travel $\pi/6$ clockwise from $(1, 0)$ to the point $(\sqrt{3}/2, -1/2)$, so $\sin(-\pi/6) = -1/2$.

D. Travel $\pi/6$ counterclockwise from $(1, 0)$ to the point $(\sqrt{3}/2, 1/2)$, so $\sin(-\pi/6) = 1/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q5/5

SET 37 • HSF-TF.A.2

BACKWARD 1

Right-triangle definitions of sine and cosine • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

In a right triangle, sin A (A an acute angle) is defined as

A. opposite/adjacent

B. adjacent/hypotenuse

C. opposite/hypotenuse

D. hypotenuse/opposite

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q1/5

In a right triangle with hypotenuse 1, what are the lengths of the leg adjacent to an acute angle θ and the leg opposite θ?

A. adjacent cos θ, opposite sin θ

B. adjacent 1, opposite tan θ

C. adjacent sin θ, opposite cos θ

D. adjacent tan θ, opposite 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q2/5

True or false?

The right-triangle definition works for $\theta = 120°$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q3/5

In a right triangle with hypotenuse 1, the leg opposite θ is

A. $\sin \theta$

B. $\tan \theta$

C. 1

D. $\cos \theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q4/5

Why does the right-triangle definition of sine fail for negative angles?

A. Because sine is always negative for negative angles, and side ratios in a triangle are positive.

B. Because the hypotenuse of a triangle with a negative angle would have negative length.

C. The acute angles of a right triangle are between 0° and 90°, so there is no right triangle with a negative angle.

D. It does not fail: a negative angle just flips the right triangle over, and the same side ratio is used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B1-Q5**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSF-TF.A.2

BACKWARD 2

Points on the unit circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Write the equation of the circle with center (0, 0) and radius 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q1/5

Is $(-1/2, \sqrt{3}/2)$ on the unit circle $x^2 + y^2 = 1$? Find $x^2 + y^2$, then answer Yes or No.

$x^2 + y^2 =$ _____

On the circle? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q2/5

True or false?

Every point on the unit circle has $-1 \leq x \leq 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q3/5

Which point is on the unit circle?

A. (0.9, 0.9)

B. (0.5, 0.5)

C. (1, 1)

D. (0.6, −0.8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B2-Q4**

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q4/5

A point on the unit circle has x = 0. Give both possible points.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSF-TF.A.2

BACKWARD 3

Radian measure as directed arc length • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

What arc length on the unit circle does t = π/6 cut off?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q1/5

On the unit circle, which direction does $t = -\pi/6$ travel from (1, 0)? Answer Clockwise or Counterclockwise.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q2/5

True or false?

$t = \pi$ returns to the starting point (1, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q3/5

An arc of length π from (1, 0) ends at

A. (0, 1)

B. (0, −1)

C. (1, 0)

D. (−1, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q4/5

Find the value of t with $-2\pi < t < 0$ for which travel of t units from (1, 0) on the unit circle ends at (0, −1). Give it exactly, in terms of π.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q5/5

SET 37 • HSF-TF.A.2

FORWARD 1

Explain where tangent is defined and how values repeat • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Why is tan t undefined at $t = \pi/2 + k\pi$ (k an integer)?

A. At these t the endpoint on the unit circle is (0, 1) or (0, −1), so cos t = 0 and tan t = sin t/cos t would divide by 0.

B. tan t is undefined whenever sin t = 1.

C. At these t the endpoint on the unit circle is (1, 0) or (−1, 0), so sin t = 0 and tan t = sin t/cos t would have to divide by 0.

D. tan t = cos t/sin t, and sin t = 0 at these t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q1/5

What are the domain and range of $y = \sin t$?

A. Domain: $0 \leq t \leq 2\pi$; range: $-1 \leq y \leq 1$

B. Domain: all real numbers; range: $-1 \leq y \leq 1$

C. Domain: all real numbers; range: all real numbers

D. Domain: $-1 \leq t \leq 1$; range: all real numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q2/5

True or false?

cos t and sin t can both equal 0 for the same t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q3/5

The unit-circle definition agrees with the triangle definition for

A. $t < 0$ only

B. $\pi/2 < t < \pi$ only

C. all t

D. $0 < t < \pi/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q4/5

Why is $\cos(-t) = \cos t$, using the unit circle?

A. Because $-t$ and t are the same angle, so they end at the same point on the unit circle.

B. Because cos is always positive, so $\cos(-t)$ and $\cos t$ cannot have different signs.

C. The endpoints for t and $-t$ are reflections of each other over the y-axis, which keeps the x-coordinate (the cosine).

D. The endpoints for t and $-t$ are reflections of each other over the x-axis, which keeps the x-coordinate (the cosine).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q5/5

SET 37 • HSF-TF.A.2

FORWARD 2

Explain symmetry and periodicity • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

Which unit-circle argument shows that $\sin(-t) = -\sin t$?

A. $\sin(-\pi/2) = -1 = -\sin(\pi/2)$, so the rule holds for every t.

B. The endpoints for t and $-t$ are reflections of each other over the x-axis, so their y-coordinates (the sines) are opposites.

C. The endpoints for t and $-t$ are the same point, so their sines are opposites.

D. The endpoints for t and $-t$ are always reflections of each other over the y-axis, so their y-coordinates (the sines) are opposites.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q1/5

Is sine an even function, an odd function, or neither?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q2/5

True or false?

$\tan(t + \pi) = \tan t$ for every t where it is defined.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q3/5

The period of $y = \tan t$ is

A. π

B. 4π

C. $\pi/2$

D. 2π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q4/5

Why is cos an even function?

A. The points for t and −t are reflections over the x-axis, which keeps the x-coordinate, so cos(−t) = cos t.

B. The points for t and −t are reflections over the y-axis, which keeps the x-coordinate, so cos(−t) = cos t.

C. cos t is positive for every t, so cos(−t) and cos t are always equal.

D. The points for t and −t are reflections over the x-axis, which negates the x-coordinate, so cos(−t) = −cos t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q5/5

SET 38 • HSF-TF.B.5 • TRIGONOMETRIC FUNCTIONS

Write sinusoidal models with given amplitude, period, and midline

MAIN	Algebra II main question	**HSF-TF.B.5**
BACKWARD 1	Amplitude and midline from the maximum and minimum	**7.NS.A.3**
BACKWARD 2	Period and frequency	**6.RP.A.2**
BACKWARD 3	Vertical stretches, shifts, and reflections	**HSF-BF.B.3**
FORWARD 1	Write models from data or with a phase shift	**HSF-TF.B.5**
FORWARD 2	Solve for times in sinusoidal models	**HSF-TF.B.7**

SET 38 • HSF-TF.B.5

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

Write a cosine function with amplitude 3, period 2π, midline $y = 1$, and a maximum at $x = 0$, in the form $y = A \cdot \cos(Bx) + D$ with $A > 0$ and $B > 0$.

$A =$ _____

$B =$ _____

$D =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q1/5

A Ferris wheel has diameter 40 m, its center is 25 m above the ground, and it turns once every 4 minutes. A rider starts at the bottom. Write the height h after t minutes in the form $h = D - A\cdot\cos(Bt)$ with $A > 0$ and $B > 0$. Give B exactly, in terms of π.

A = _____

B = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q2/5

Water depth at a harbor varies between 1 ft (low tide) and 9 ft (high tide) with a period of 12 hours. High tide is at t = 0. Write the depth d after t hours in the form d = A·cos(Bt) + D with A > 0 and B > 0. Give B exactly, in terms of π.

A = _____

B = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q3/5

A town's average temperature is lowest, 10°C, in month t = 0 and highest, 30°C, at t = 6, with a 12-month period. Write T(t) in the form $T = D - A\cdot\cos(Bt)$ with $A > 0$ and $B > 0$. Give B exactly, in terms of π.

A = _____

B = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q4/5

Write a sine function with amplitude 2, period π, and midline $y = -1$ that is at the midline and increasing at $x = 0$, in the form $y = A \cdot \sin(Bx) + D$ with $A > 0$ and $B > 0$.

A = _____

B = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q5/5

SET 38 • HSF-TF.B.5

BACKWARD 1

Amplitude and midline from the maximum and minimum • Grade 7 • The Number System • 7.NS.A.3

Find $(9 + 1)/2$ and $(9 - 1)/2$.

$(9 + 1)/2 =$ _____

$(9 - 1)/2 =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q1/5

A quantity varies between 10 and 30. Find the midline value and the amplitude.

Midline value: _____

Amplitude: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q2/5

True or false?

The amplitude is the maximum minus the minimum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q3/5

For max 45 and min 5, the midline is

A. y = 20

B. y = 50

C. y = 25

D. y = 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q4/5

A Ferris wheel has diameter 40 m with its center 25 m above the ground. Give its maximum and minimum heights in meters.

Maximum (m): _____

Minimum (m): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSF-TF.B.5

BACKWARD 2

Period and frequency • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

A wheel turns once every 4 minutes. How many turns per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q1/5

A spring oscillates 3 times per second. What is the period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q2/5

True or false?

Frequency is equal to the period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q3/5

A period of 12 hours means a frequency of

A. 12 cycles per hour

B. 2 cycles per hour

C. 1/2 cycle per hour

D. 1/12 cycle per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q4/5

$y = \sin(Bx)$ has period $2\pi/B$. Find B for period 4.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSF-TF.B.5

BACKWARD 3

Vertical stretches, shifts, and reflections • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How is $y = 3\cos x + 1$ obtained from $y = \cos x$?

A. Stretch vertically by 3, then shift up 1

B. Stretch vertically by 3, then shift right 1

C. Stretch horizontally by 3, then shift up 1

D. Shift up 3, then stretch vertically by 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q1/5

How does the graph of $y = -\cos x$ compare with the graph of $y = \cos x$?

A. It is the reflection over the y-axis

B. It is shifted down 1

C. It is the reflection over the x-axis

D. It is the same graph

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q2/5

True or false?

$y = \cos x + 5$ has a maximum of 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q3/5

$y = -20\cos x$ starts (at $x = 0$) at

A. its minimum

B. its maximum

C. the midline going up

D. the midline going down

First stretch the graph of $y = \sin x$ vertically by 2, then shift it down 1. Which equation results?

A. $y = \sin(2x) - 1$

B. $y = 2\sin(x - 1)$

C. $y = 2(\sin x - 1)$

D. $y = 2\sin x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q5/5

SET 38 • HSF-TF.B.5

FORWARD 1

Write models from data or with a phase shift • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

A rider's height is 3 m at t = 0 (the bottom), 27 m at the top after 15 s, and the motion repeats every 30 s. Write h(t) in the form h = D − A·cos(Bt) with A > 0 and B > 0. Give B exactly, in terms of π.

A = _____

B = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q1/5

Sunrise times vary between 5:30 and 7:30 (written as 5.5 and 7.5 hours), with period 365 days and the latest sunrise at day t = 0. Write S(t) in the form S = D + A·cos(Bt) with A > 0 and B > 0. Give B exactly, in terms of π.

A = _____

B = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q2/5

True or false?

$y = 4\sin(x - \pi/2)$ is the same as $y = 4\cos x$.

A. True

B. False

A sine model with period 10, amplitude 3, midline 7, and a maximum at t = 2.5 is

A. $7 + 3\sin(10t)$

B. $7 + 3\sin(\pi t/5)$

C. $3 + 7\sin(\pi t/5)$

D. $7 + 3\cos(\pi t/5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q4/5

A sinusoidal model has a maximum of 18 at t = 3 and the next minimum of 6 at t = 7. Write it in the form $y = D + A\cdot\cos(B(t - 3))$ with $A > 0$ and $B > 0$. Give B exactly, in terms of π.

A = _____

B = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSF-TF.B.5

FORWARD 2

Solve for times in sinusoidal models • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

For $h = 25 - 20\cos(\pi t/2)$, find the first time $t > 0$ when the rider is at 35 m. Give an exact fraction. A calculator may be used.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q1/5

For $d = 4\cos(\pi t/6) + 5$, find all times in $0 \leq t < 12$ when the depth is 5 ft.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q2/5

True or false?

A sinusoidal model usually reaches a given value between its min and max twice per period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q3/5

For $T = 20 - 10\cos(\pi t/6)$, $T = 25$ first when

A. $t = 2$

B. $t = 4$

C. $t = 3$

D. $t = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q4/5

Why does a sinusoidal model take each value strictly between its minimum and maximum exactly twice in each period?

A. Because every function takes each value exactly twice in each interval of the same length as the period.

B. In each period the graph rises from the minimum to the maximum and falls back, passing each height once each way.

C. Because the period is 2π, and 2π contains each value exactly twice, once in each half of the period.

D. Because the graph crosses its midline twice in each period, and each crossing gives one of the two times the graph takes the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q5/5

SET 39 • HSF-TF.B.5 • TRIGONOMETRIC FUNCTIONS

Interpret the parameters of sinusoidal models

MAIN	Algebra II main question	**HSF-TF.B.5**
BACKWARD 1	Build the model from amplitude, period, and midline	**HSF-TF.B.5**
BACKWARD 2	Evaluate a sinusoidal model	**HSF-TF.A.2**
BACKWARD 3	Convert rates between units	**6.RP.A.3.d**
FORWARD 1	Predict how changing a parameter changes the situation	**HSF-TF.B.5**
FORWARD 2	Interpret periodic features from graphs and tables	**HSF-IF.B.4**

SET 39 • HSF-TF.B.5

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

A Ferris wheel rider's height is $h(t) = 12 + 10\sin(\pi t/15)$ meters after t seconds. Give the maximum height, the minimum height, and the time for one revolution.

Maximum (m): _____

Minimum (m): _____

One revolution (s): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q1/5

Hours of daylight are $D(t) = 12 + 3\sin(2\pi(t - 80)/365)$, t in days. What do 12, 3, and 365 represent?

A. 12: average daylight (midline, hours); 3: greatest change from 12 (amplitude, hours); 365: day of greatest daylight.

B. 12: average daylight (midline, hours); 3: days in one cycle (period); 365: greatest daylight (amplitude, hours).

C. 12: greatest daylight (maximum, hours); 3: average daylight (midline, hours); 365: days in one cycle (period).

D. 12: average daylight (midline, hours); 3: greatest change from 12 (amplitude, hours); 365: days in one cycle (period).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q2/5

A model of blood pressure is $P(t) = 100 + 20\sin(2.5\pi t)$, t in seconds. Find the period and the number of heartbeats per minute.

Period (s): _____

Beats per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q3/5

A spring's position is $y = -4\cos(3\pi t)$ cm, t in seconds. Give the amplitude, the period, and the frequency.

Amplitude (cm): _____

Period (s): _____

Frequency (cycles per second): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q4/5

Two sounds are $y = \sin(880\pi t)$ and $y = \sin(440\pi t)$, t in seconds. Give the frequency of each in cycles per second.

Frequency of $\sin(880\pi t)$: _____

Frequency of $\sin(440\pi t)$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q5/5

SET 39 • HSF-TF.B.5

BACKWARD 1

Build the model from amplitude, period, and midline • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

What is the amplitude of $y = 12 + 10\sin(\pi t/15)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q1/5

What is the midline of $y = 12 + 10\sin(\pi t/15)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q2/5

True or false?

The period of $y = \sin(Bt)$ is $B/(2\pi)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q3/5

The period of $y = \sin(\pi t/15)$ is

A. 30

B. 15

C. $\pi/15$

D. 2π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q4/5

Write a sine model with amplitude 4, period 2/3, and midline 0, in the form $y = A \cdot \sin(Bt)$ with $A > 0$ and $B > 0$. Give B exactly, in terms of π.

A = _____

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • HSF-TF.B.5

BACKWARD 2

Evaluate a sinusoidal model • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Evaluate $12 + 10\sin(\pi/2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q1/5

Evaluate $12 + 10\sin(3\pi/2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q2/5

True or false?

$-4\cos(0) = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q3/5

100 + 20sin(π/2) equals

A. 100

B. 20

C. 120

D. 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q4/5

Evaluate 12 + 3sin(0).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSF-TF.B.5

BACKWARD 3

Convert rates between units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

A cycle lasts 0.8 s. How many cycles are in 60 s?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q1/5

1.5 cycles per second is how many cycles per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q2/5

True or false?

A period of 30 s means 30 revolutions per minute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q3/5

440 cycles per second is how many cycles per millisecond?

A. 4.4

B. 44

C. 440

D. 0.44

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q4/5

A tide has period 12 hours. How many cycles in 3 days?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q5/5

SET 39 • HSF-TF.B.5

FORWARD 1

Predict how changing a parameter changes the situation • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

A Ferris wheel rider's height is $h(t) = 12 + 10\sin(\pi t/15)$ m after t seconds. The wheel is replaced by one of the same size that turns twice as fast. In the new model $h(t) = 12 + 10\sin(Bt)$, what is B? Give B exactly, in terms of π.

$B =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q1/5

Daylight in one city is $D(t) = 12 + 3\sin(2\pi(t - 80)/365)$ hours. A city farther from the equator has daylight between 8 and 16 hours with the same period and timing. Which parameter changes, and to what?

A. The amplitude changes to 4

B. The amplitude changes to 8

C. The midline changes to 16

D. The period changes to 8 days

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q2/5

True or false?

Changing the midline changes the period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q3/5

Multiplying B by 3 makes the period

A. 3 units longer

B. unchanged

C. three times as long

D. one third as long

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q4/5

A heart rate rises from 75 to 100 beats per minute. What is the new period, in seconds?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q5/5

SET 39 • HSF-TF.B.5

FORWARD 2

Interpret periodic features from graphs and tables • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

A sinusoidal model of a tide's depth has consecutive maxima at t = 2 and t = 14 hours. What is its period, in hours?

Period (hours): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q1/5

A temperature model has a high of 30 and a low of 10. What are its midline value and amplitude?

Midline value: _____

Amplitude: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q2/5

True or false?

For a nonconstant sinusoidal function, consecutive minima are one period apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q3/5

On which interval does $h(t) = 12 + 10\sin(\pi t/15)$ increase?

A. $0 < t < 15$

B. $0 < t < 7.5$

C. $15 < t < 30$

D. $7.5 < t < 22.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q4/5

How can the frequency be read from the graph of a periodic function?

A. Read the maximum height of the graph, which is the number of cycles per unit of the input.

B. Find the length of one cycle (the period) and take its reciprocal, the number of cycles per unit of the input.

C. Find half the difference between the maximum and the minimum, which is the number of cycles per unit.

D. Count the number of times the graph crosses the x-axis in the window shown; that count is the frequency.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q5/5

SET 40 • HSF-TF.C.8 • TRIGONOMETRIC FUNCTIONS

Prove the Pythagorean identity

MAIN	Algebra II main question	**HSF-TF.C.8**
BACKWARD 1	The Pythagorean Theorem	**8.G.B.7**
BACKWARD 2	Coordinates of the point at angle θ	**HSF-TF.A.2**
BACKWARD 3	Expand and simplify algebraic squares	**HSA-SSE.A.2**
FORWARD 1	Prove identities derived from the Pythagorean identity	**HSF-TF.C.8**
FORWARD 2	Use the angle-addition formulas	**HSF-TF.C.9**

SET 40 • HSF-TF.C.8

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

Which argument proves $\sin^2\theta + \cos^2\theta = 1$ for every angle θ, using the unit circle?

A. On the unit circle $\sin\theta + \cos\theta = 1$ for every angle θ, because the coordinates of a point add to the radius; squaring both sides gives $\sin^2\theta + \cos^2\theta = 1$.

B. The point at angle θ is $(\sin\theta, \cos\theta)$, so $\sin\theta + \cos\theta = 1$.

C. For $\theta = 30°$: $(1/2)^2 + (\sqrt{3}/2)^2 = 1$, so the identity holds.

D. The point at angle θ is $(\cos\theta, \sin\theta)$, and every point on the unit circle satisfies $x^2 + y^2 = 1$; substituting gives $\cos^2\theta + \sin^2\theta = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q1/5

A right triangle has legs a and b, hypotenuse c, and acute angle θ opposite a. Which argument proves $\sin^2\theta + \cos^2\theta = 1$?

A. $\sin\theta = a/c$ and $\cos\theta = b/c$, so $\sin^2\theta + \cos^2\theta = (a + b)^2/c^2$, and $(a + b)^2 = c^2$ by the Pythagorean Theorem, so the sum is 1.

B. $\sin\theta = b/c$ and $\cos\theta = a/c$, so $\sin\theta + \cos\theta = (a + b)/c = 1$.

C. $\sin\theta = a/c$ and $\cos\theta = b/c$, so $\sin^2\theta + \cos^2\theta = (a^2 + b^2)/c^2 = c^2/c^2 = 1$ by the Pythagorean Theorem.

D. In a 3-4-5 triangle, $(3/5)^2 + (4/5)^2 = 1$, so the identity holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q2/5

Which argument proves $1 + \tan^2\theta = 1/\cos^2\theta$ wherever $\cos\theta \neq 0$?

A. Divide $\sin^2\theta + \cos^2\theta = 1$ by $\sin^2\theta$ to get $1 + \cos^2\theta/\sin^2\theta = 1/\sin^2\theta$; since $\cos^2\theta/\sin^2\theta = \tan^2\theta$, this gives $1 + \tan^2\theta = 1/\cos^2\theta$.

B. For $\theta = 0$: $1 + 0 = 1/1$, so the identity holds.

C. Square $\tan\theta = \sin\theta/\cos\theta$ to get $\tan^2\theta = \sin^2\theta/\cos^2\theta = 1/\cos^2\theta$.

D. Divide $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$, which is allowed because $\cos\theta \neq 0$: $\sin^2\theta/\cos^2\theta + 1 = 1/\cos^2\theta$, and $\sin^2\theta/\cos^2\theta = \tan^2\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q3/5

θ is in quadrant II, and its point on the unit circle is (x, y) with $x < 0 < y$. Which argument proves $\sin^2\theta + \cos^2\theta = 1$ with a reference triangle?

A. The reference triangle has legs x and y, so $x + y = 1$, which means $\cos\theta + \sin\theta = 1$.

B. The reference triangle has legs $|x|$ and $|y|$ and hypotenuse 1, so $|x|^2 + |y|^2 = 1$. Squaring removes the signs, so $x^2 + y^2 = \cos^2\theta + \sin^2\theta = 1$.

C. The legs of the reference triangle are x and y, and x is negative in quadrant II, so $x^2 + y^2 = -1$ there, which means $\cos^2\theta + \sin^2\theta = -1$ instead of 1.

D. The identity was proved for acute angles, so it is true in quadrant II with no further work.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-M-Q4**

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q4/5

Which argument proves $(\sin\theta + \cos\theta)^2 = 1 + 2\sin\theta\cos\theta$?

A. $(\sin\theta + \cos\theta)^2 = \sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta$, and $\sin^2\theta + \cos^2\theta = 1$.

B. $(\sin\theta + \cos\theta)^2 = \sin^2\theta + \cos^2\theta = 1$, so the term $2\sin\theta\cos\theta$ is extra.

C. For $\theta = 0$: $(0 + 1)^2 = 1 + 0$, so the identity holds.

D. $(\sin\theta + \cos\theta)^2 = 1^2 = 1$ because $\sin\theta + \cos\theta = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q5/5

SET 40 • HSF-TF.C.8

BACKWARD 1

The Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

A right triangle has legs 3 and 4. Find the hypotenuse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q1/5

A right triangle has hypotenuse 1 and a leg 0.6. Find the other leg.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q2/5

True or false?

In a right triangle, $a^2 + b^2 = c^2$ where c is the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q3/5

Legs 5 and 12 give hypotenuse

A. 60

B. 7

C. 17

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q4/5

Legs a and b and hypotenuse c: write c^2 in terms of a and b.

$c^2 =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSF-TF.C.8

BACKWARD 2

Coordinates of the point at angle θ • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Which gives the coordinates of the point on the unit circle at angle θ?

A. (sin θ, cos θ)

B. (θ, sin θ)

C. (cos θ, sin θ)

D. (cos θ, tan θ)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q1/5

Give cos(2π/3) and sin(2π/3) exactly.

cos(2π/3) = _____

sin(2π/3) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q2/5

True or false?

sin θ is the x-coordinate of the point at angle θ.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q3/5

The point at angle π/4 is

A. (1/2, √3/2)

B. (1, 1)

C. (√3/2, 1/2)

D. (√2/2, √2/2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q4/5

What is the radius of the unit circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • HSF-TF.C.8

BACKWARD 3

Expand and simplify algebraic squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Expand $(a + b)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q1/5

Simplify $(a/c)^2 + (b/c)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q2/5

True or false?

$\sin^2\theta$ means $\sin(\theta^2)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q3/5

Dividing $x^2 + y^2 = 1$ by x^2 gives

A. $1 + y^2/x^2 = 1/x^2$

B. $1 + y^2 = 1/x$

C. $y^2 = 1$

D. $x^2 + y^2 = 1/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q4/5

Expand $(s + c)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q5/5

SET 40 • HSF-TF.C.8

FORWARD 1

Prove identities derived from the Pythagorean identity • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

Which argument proves $\sin^4\theta - \cos^4\theta = \sin^2\theta - \cos^2\theta$?

A. Divide both sides by $\sin^2\theta - \cos^2\theta$ to get $\sin^2\theta + \cos^2\theta = 1$, which is the Pythagorean identity, so the identity is true.

B. $\sin^4\theta - \cos^4\theta = (\sin\theta - \cos\theta)^4$, which equals $\sin^2\theta - \cos^2\theta$.

C. For $\theta = \pi/4$ both sides are 0, so the identity holds.

D. $\sin^4\theta - \cos^4\theta = (\sin^2\theta - \cos^2\theta)(\sin^2\theta + \cos^2\theta) = (\sin^2\theta - \cos^2\theta)(1) = \sin^2\theta - \cos^2\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q1/5

Which argument proves $(1 - \cos\theta)(1 + \cos\theta) = \sin^2\theta$?

A. For $\theta = \pi/2$: $(1 - 0)(1 + 0) = 1 = \sin^2(\pi/2)$, so the identity holds.

B. $(1 - \cos\theta)(1 + \cos\theta) = 1 - 2\cos\theta + \cos^2\theta$, and by the Pythagorean identity this equals $\sin^2\theta$.

C. $(1 - \cos\theta)(1 + \cos\theta) = 1 - \cos^2\theta$, and $1 - \cos^2\theta = \sin^2\theta$ by the Pythagorean identity.

D. $(1 - \cos\theta)(1 + \cos\theta) = 1 + \cos^2\theta$, which equals $\sin^2\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q2/5

True or false?

$\sin\theta + \cos\theta = 1$ for every θ.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q3/5

Which equals $1 - \sin^2\theta$?

A. $\tan^2\theta$

B. $1 - \cos^2\theta$

C. $\cos\theta$

D. $\cos^2\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q4/5

Which argument proves $\cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1$?

A. Replace $\sin^2\theta$ with $1 - \cos^2\theta$:
$\cos^2\theta - (1 - \cos^2\theta) = \cos^2\theta - 1 + \cos^2\theta = 2\cos^2\theta - 1$.

B. Add $\sin^2\theta$ to both sides to get $\cos^2\theta = 2\cos^2\theta - 1 + \sin^2\theta$, which is always true.

C. For $\theta = 0$: $1 - 0 = 2 - 1$, so the identity holds.

D. Replace $\sin^2\theta$ with $1 - \cos^2\theta$: $\cos^2\theta - 1 - \cos^2\theta = -1$, so the identity is false for every θ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSF-TF.C.8

FORWARD 2

Use the angle-addition formulas • High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9

Use sin(a + b) = sin a cos b + cos a sin b to find sin(75°) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q1/5

Use $\cos(a + b) = \cos a \cos b - \sin a \sin b$ with $a = b = \theta$. Which expression equals $\cos 2\theta$?

A. $\cos^2\theta + \sin^2\theta$

B. $2\cos\theta$

C. $2\sin\theta \cos\theta$

D. $\cos^2\theta - \sin^2\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q2/5

True or false?

$\cos 2\theta = 1 - 2\sin^2\theta$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q3/5

$\sin 2\theta$ equals

A. $\sin^2\theta - \cos^2\theta$

B. $2\sin\theta$

C. $2\sin\theta\cos\theta$

D. $\sin\theta + \cos\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q4**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q4/5

$\sin 2\theta = 2\sin\theta\cos\theta$. Use $(\sin\theta + \cos\theta)^2 = 1 + 2\sin\theta\cos\theta$ to write $(\sin\theta + \cos\theta)^2$ in terms of $\sin 2\theta$.

A. $2 + \sin 2\theta$

B. $\sin 2\theta$

C. $1 + \sin 2\theta$

D. $1 + 2\sin 2\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q5**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q5/5

SET 41 • HSF-TF.C.8 • TRIGONOMETRIC FUNCTIONS

Use the Pythagorean identity to find trigonometric values

MAIN	Algebra II main question	**HSF-TF.C.8**
BACKWARD 1	Signs of sine, cosine, and tangent by quadrant	**HSF-TF.A.2**
BACKWARD 2	Square roots of fractions	**8.EE.A.2**
BACKWARD 3	Solve $x^2 = c$	**HSA-REI.B.4.b**
FORWARD 1	Find values when the quadrant is not given, and combine values	**HSF-TF.C.8**
FORWARD 2	Find the angles with inverse trigonometric functions	**HSF-TF.B.7**

SET 41 • HSF-TF.C.8

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

$\sin \theta = 3/5$ and θ is in quadrant II. Find $\cos \theta$ and $\tan \theta$ exactly.

$\cos \theta =$ _____

$\tan \theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q1/5

$\cos\theta = -5/13$ and θ is in quadrant III. Find $\sin\theta$ and $\tan\theta$ exactly.

$\sin\theta =$ _____

$\tan\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q2/5

$\tan\theta = -2$ and θ is in quadrant IV. Find $\sin\theta$ and $\cos\theta$ exactly.

$\sin\theta =$ _____

$\cos\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q3/5

$\sin\theta = -1/3$ and θ is in quadrant IV. Find $\cos\theta$ and $\tan\theta$ exactly.

$\cos\theta =$ _____

$\tan\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q4/5

$\cos\theta = 0.6$ and $\sin\theta < 0$. Find $\sin\theta$ and $\tan\theta$.

$\sin\theta =$ _____

$\tan\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q5/5

SET 41 • HSF-TF.C.8

BACKWARD 1

Signs of sine, cosine, and tangent by quadrant • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

In quadrant II, which of sin, cos, and tan are positive?

A. sin only

B. sin and tan

C. cos only

D. all three

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q1/5

In quadrant III, what is the sign of tan θ? Answer Positive or Negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q2/5

True or false?

In quadrant IV, $\cos\theta < 0$ and $\sin\theta > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q3/5

If $\sin\theta < 0$ and $\cos\theta > 0$, θ is in quadrant

A. I

B. II

C. IV

D. III

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q4/5

If $\tan\theta < 0$ and $\sin\theta > 0$, which quadrant is θ in? Answer I, II, III, or IV.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSF-TF.C.8

BACKWARD 2

Square roots of fractions • Grade 8 • Expressions & Equations • 8.EE.A.2

Find √(16/25).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q1/5

Find √(8/9) in simplest radical form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q2/5

True or false?

$\sqrt{(144/169)} = 12/169$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q3/5

√(1/5) equals

A. 5

B. √5

C. 1/25

D. √5/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q4/5

Find $\sqrt{0.64}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • HSF-TF.C.8

BACKWARD 3

Solve $x^2 = c$ • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $x^2 = 16/25$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q1/5

Solve $x^2 = 1 - 9/25$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q2/5

True or false?

$x^2 = 0.36$ has only the solution 0.6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q3/5

$x^2 + (3/5)^2 = 1$ gives $x =$

A. $\pm 2/5$

B. $2/5$

C. $4/5$

D. $\pm 4/5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q4/5

Solve $5x^2 = 1$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q5/5

SET 41 • HSF-TF.C.8

FORWARD 1

Find values when the quadrant is not given, and combine values • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

$\sin\theta = 3/5$ and $0 \leq \theta < 2\pi$. Give $\cos\theta$ when θ is in quadrant I and when θ is in quadrant II.

$\cos\theta$ in quadrant I: _____

$\cos\theta$ in quadrant II: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q1/5

$\cos\theta = -5/13$ and θ is in quadrant III. Find $\sin\theta \cdot \cos\theta$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q2/5

True or false?

Knowing $\tan \theta = 1$ determines $\sin \theta$ uniquely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q3/5

If $\sin\theta = 0.8$ and $\tan\theta < 0$, then $\cos\theta =$

A. -0.8

B. 0.2

C. 0.6

D. -0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q4/5

Use $1 + \tan^2\theta = 1/\cos^2\theta$ to find $\cos\theta$ if $\tan\theta = 3$ and θ is in quadrant III.

$\cos\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q5/5

SET 41 • HSF-TF.C.8

FORWARD 2

Find the angles with inverse trigonometric functions • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

$\sin\theta = 3/5$ with $0 \le \theta < 2\pi$. Find both values of θ, in radians, to the nearest thousandth. A calculator may be used.

$\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q1/5

$\cos\theta = -5/13$ and θ is in quadrant III ($0 \leq \theta < 2\pi$). Find θ, in radians, to the nearest thousandth. A calculator may be used.

$\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q2/5

True or false?

arcsin gives only angles between $-\pi/2$ and $\pi/2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q3/5

The quadrant II angle with $\sin\theta = 3/5$ is

A. $\pi + \arcsin(3/5)$

B. $\pi - \arcsin(3/5)$

C. $\arcsin(3/5)$

D. $-\arcsin(3/5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q4/5

$\sin\theta = 3/5$ and θ is in quadrant II. Why does the Pythagorean identity give $\cos\theta$, while an inverse function is needed to find θ itself?

A. $\cos\theta$ cannot be found from $\sin\theta$ alone; θ must be found first with an inverse function, and only then can $\cos\theta$ be computed.

B. The identity relates the values $\sin\theta$ and $\cos\theta$ to each other; finding the angle from a value means undoing sine, which needs arcsin (and the quadrant).

C. The identity gives θ directly: $\theta = 1 - \sin^2\theta = 16/25$, so no inverse function is needed for θ either.

D. The inverse function is needed only for angles measured in degrees; in radians the identity gives θ directly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q5/5

SET 42 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

Interpret key features of polynomial and rational models

MAIN	Algebra II main question	**HSF-IF.B.4**
BACKWARD 1	Interpret key features of linear and quadratic models	**HSF-IF.B.4**
BACKWARD 2	Describe increasing and decreasing behavior	**8.F.B.5**
BACKWARD 3	Zeros and sign from factors	**HSA-APR.B.3**
FORWARD 1	Sketch a model from a verbal description of key features	**HSF-IF.B.4**
FORWARD 2	Compare key features across representations	**HSF-IF.C.9**

SET 42 • HSF-IF.B.4

MAIN

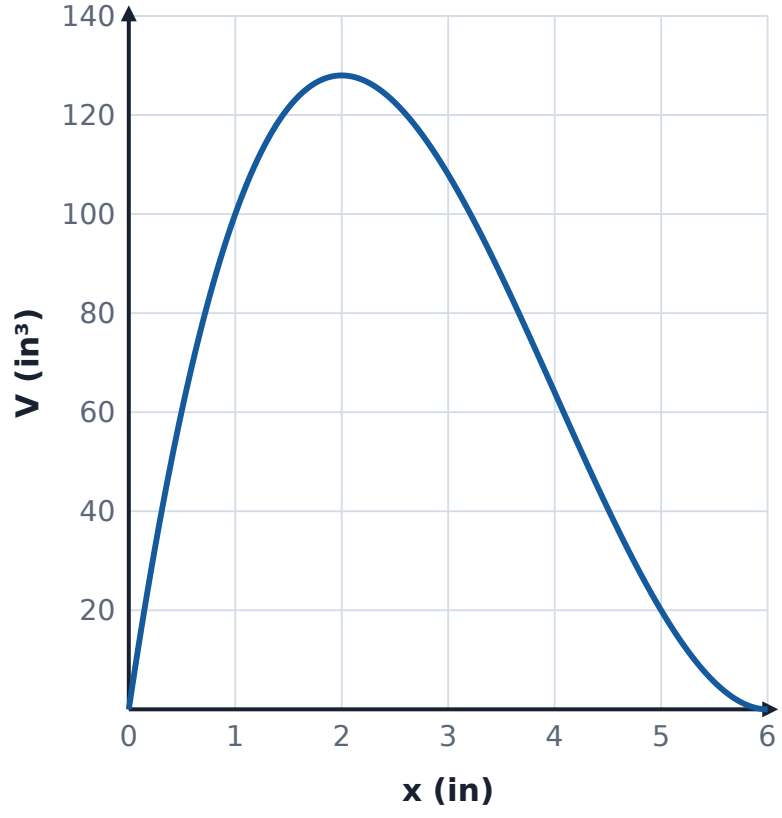
Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

An open box is made from a 12-in square sheet by cutting squares of side x from the corners. Its volume is $V(x) = x(12 - 2x)^2$. Use the graph to give the x that maximizes V, the maximum volume, and the interval where V increases.

x (in) = _____

Maximum volume (in^3): _____

V increases for: _____

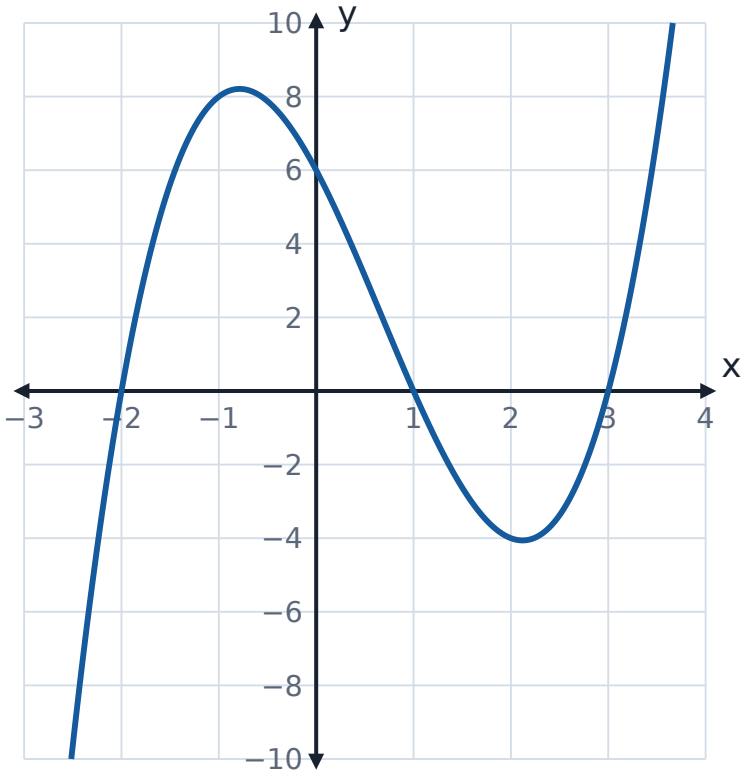


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q1/5

The graph of $f(x) = (x + 2)(x - 1)(x - 3)$ is shown. Give the two intervals where $f(x) > 0$ and the two intervals where $f(x) < 0$, each pair from left to right.



$f > 0$, first interval: _____

$f > 0$, second interval: _____

$f < 0$, first interval: _____

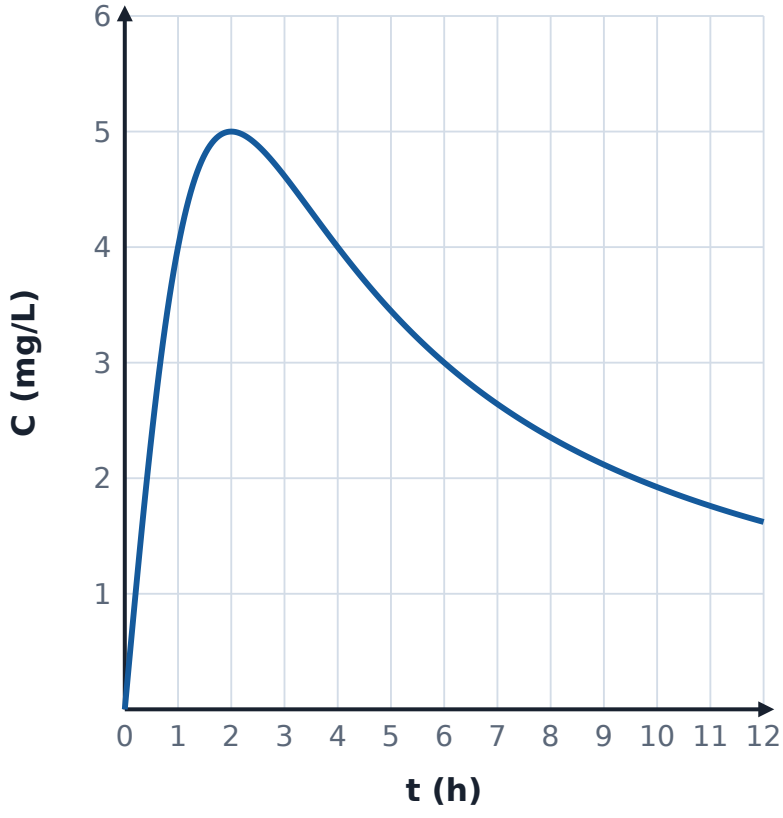
$f < 0$, second interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q2/5

A drug concentration is $C(t) = 20t/(t^2 + 4)$ mg/L after t hours. Use the graph to give the maximum concentration, when it occurs, and the value C approaches as t grows.



Maximum (mg/L): _____

t (hours) = _____

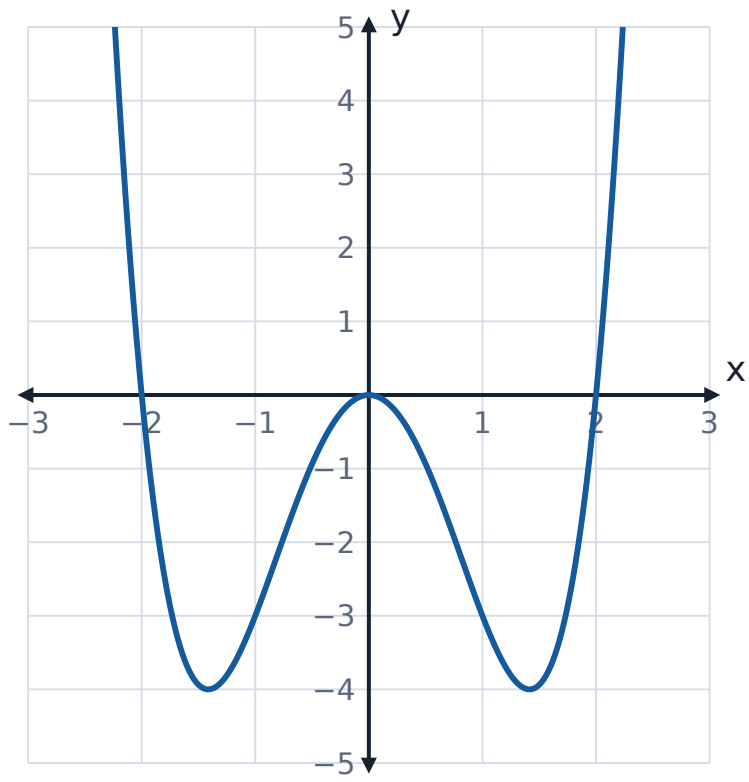
C approaches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q3/5

The graph of $f(x) = x^4 - 4x^2$ is shown. Give its x-intercepts, the y-value of its relative maximum, and the y-value of its relative minima.



x-intercepts: _____

Relative maximum value: _____

Relative minimum value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q4/5

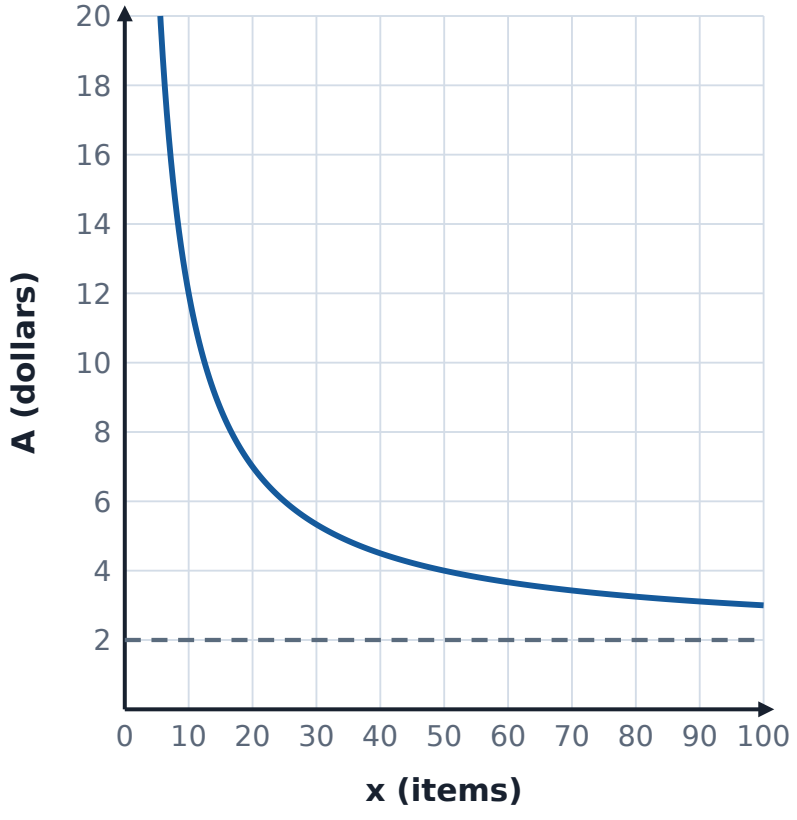
The average cost of making x items is A(x) = (100 + 2x)/x dollars. The graph is shown. Which describes the end behavior and its meaning?

A. As x grows, A(x) increases without bound: each extra item adds $2, so the average cost keeps rising.

B. As x grows, A(x) approaches $0 (the asymptote y = 0): with enough items the fixed cost and the item cost both vanish.

C. As x grows, A(x) approaches $100 (the asymptote y = 100): the $100 fixed cost dominates the average for large x.

D. As x grows, A(x) approaches $2 (the asymptote y = 2): the $100 fixed cost is spread over more items, so the average approaches the $2 cost of each item.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q5/5

SET 42 • HSF-IF.B.4

BACKWARD 1

Interpret key features of linear and quadratic models • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

A ball's height is $h(t) = -16t^2 + 32t + 4$ feet after t seconds. When is the height greatest, and what is the greatest height?

t (seconds) = _____

Greatest height (ft): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q1/5

$C = 25 + 3x$ is the cost in dollars of x rides. What does the C-intercept (0, 25) mean?

A. A fixed cost of $25 before any rides

B. After 25 rides the cost is $3

C. Each ride costs $25

D. The cost of 3 rides is $25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q2/5

True or false?

A parabola that opens down has a minimum.

A. True

B. False

$h(t) = -16t^2 + 32t + 4$ increases for

A. all t

B. $0 < t < 1$

C. $t > 1$

D. $1 < t < 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q4/5

Where is $y = (x - 2)(x - 6)$ negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSF-IF.B.4

BACKWARD 2

Describe increasing and decreasing behavior • Grade 8 • Functions • 8.F.B.5

A graph rises from $x = 0$ to $x = 2$ and then falls. Which description is correct?

A. Increasing for every $x > 0$

B. Constant for $0 < x < 2$

C. Increasing for $0 < x < 2$ and decreasing for $x > 2$

D. Decreasing for $0 < x < 2$ and increasing for $x > 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q1/5

A drug's concentration rises after a dose, reaches a peak, and then levels off toward 0. Which describes its graph against time?

A. It stays constant at its peak value.

B. It increases to a peak, then decreases and approaches 0.

C. It decreases to a minimum, then increases without bound.

D. It increases without bound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q2/5

True or false?

A highest point between rising and falling parts is a relative maximum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q3/5

A function that falls on $1 < x < 3$ is

A. constant there

B. decreasing there

C. undefined there

D. increasing there

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q4/5

What is a relative minimum of a graph?

A. The lowest point of the whole graph, always

B. A point lower than all nearby points of the graph

C. A point higher than all nearby points

D. A point where the graph crosses the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSF-IF.B.4

BACKWARD 3

Zeros and sign from factors • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

List the zeros of $(x + 2)(x - 1)(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q1/5

List the zeros of $x^4 - 4x^2 = x^2(x - 2)(x + 2)$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q2/5

True or false?

$V(x) = x(12 - 2x)^2$ has zeros 0 and 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q3/5

$20t/(t^2 + 4) = 0$ when

A. never

B. $t = -2$

C. $t = 2$

D. $t = 0$

Is $(x + 2)(x - 1)(x - 3)$ positive or negative at $x = 2$?
Answer Positive or Negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q5/5

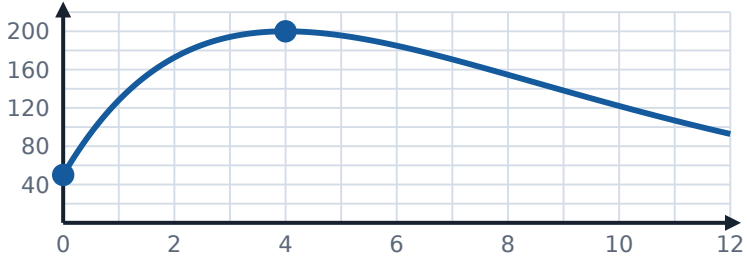
SET 42 • HSF-IF.B.4

FORWARD 1

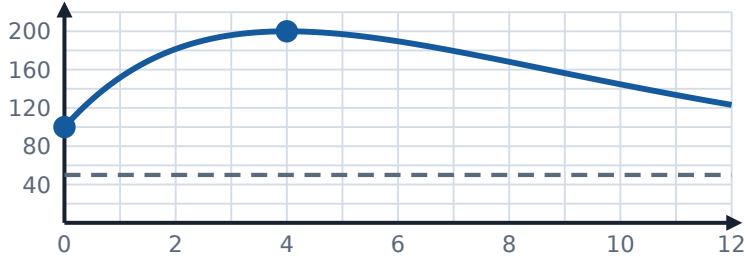
Sketch a model from a verbal description of key features • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

Which graph could show a population that is 50 at t = 0, rises to a maximum of 200 at t = 4, and then decreases toward a level of 100 as t grows?

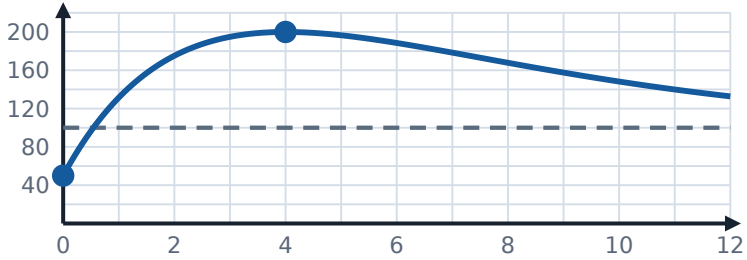
A



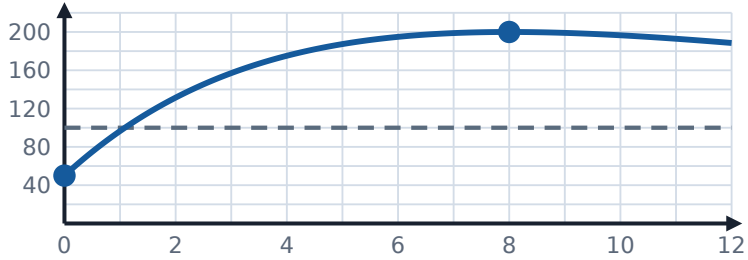
B



C



D



A. Graph A

B. Graph B

C. Graph C

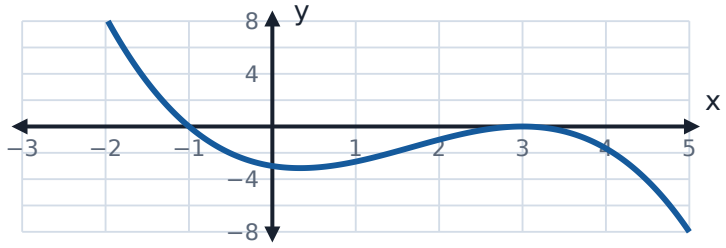
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q1**

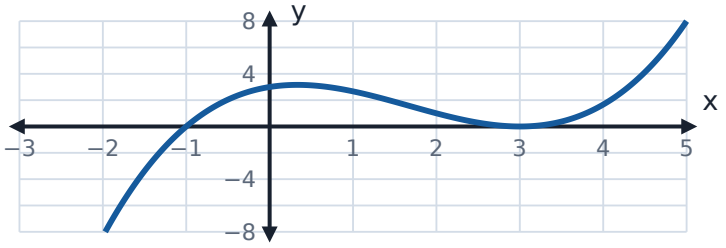
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q1/5

Which graph could show a cubic with zeros at −1 and 3 only, a negative y-intercept, falling to the right, and touching the x-axis at 3?

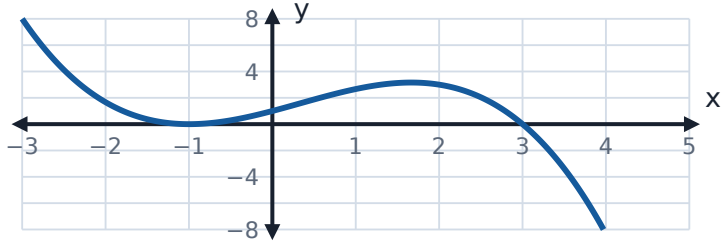
A



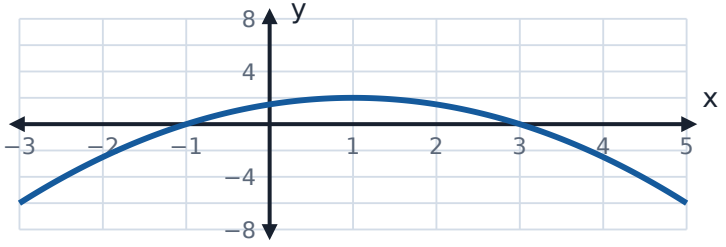
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q2/5

True or false?

A graph can have a relative maximum that is not the absolute maximum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q3/5

A model rises to a peak and approaches 0 as t grows. Which formula fits?

A. $20t/(t^2 + 4)$

B. $20t$

C. t^2

D. $20 - t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q4/5

Which quadratic function has zeros 0 and 6 and is positive between them (so it has a relative maximum between the zeros)?

A. $f(x) = (x - 6)(x + 6)$

B. $f(x) = 6 - x^2$

C. $f(x) = x(x - 6)$

D. $f(x) = x(6 - x)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSF-IF.B.4

FORWARD 2

Compare key features across representations • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

$f(x) = x(12 - 2x)^2$, and g is a quadratic that opens downward with vertex (3, 120). On $0 < x < 6$, f has its maximum at $x = 2$. Give the maximum value of each function on $0 < x < 6$.

Maximum of f: _____

Maximum of g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q1/5

$C(t) = 20t/(t^2 + 4)$ and a table of $D(t)$ gives $D(1) = 3$, $D(2) = 6$, $D(3) = 4$. Among these values ($t = 1, 2, 3$), which function has the greater peak, and when?

A. C, with 6 at $t = 2$

B. D, with 4 at $t = 3$

C. D, with 6 at $t = 2$

D. C, with 5 at $t = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q2/5

True or false?

Two functions with the same zeros have the same relative extrema.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q3/5

Which has a horizontal asymptote?

A. $(x + 2)(x - 1)(x - 3)$

B. $x^4 - 4x^2$

C. $x(12 - 2x)^2$

D. $(100 + 2x)/x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q4/5

A polynomial is given by a formula and a rational function is given by a graph. How can their end behavior be compared?

A. Use the polynomial's leading term for large |x|, and read the rational function's asymptote or trend from the graph.

B. Compare their y-intercepts, read from the formula and from the graph, since the larger intercept wins for large |x| in both directions.

C. Compare their values at x = 1, using the formula and the graph; the larger value at 1 stays larger for large |x|.

D. Compare the number of zeros of each: the function with more zeros has the stronger end behavior for large |x|.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q5/5

SET 43 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

Interpret periodic models from graphs and tables

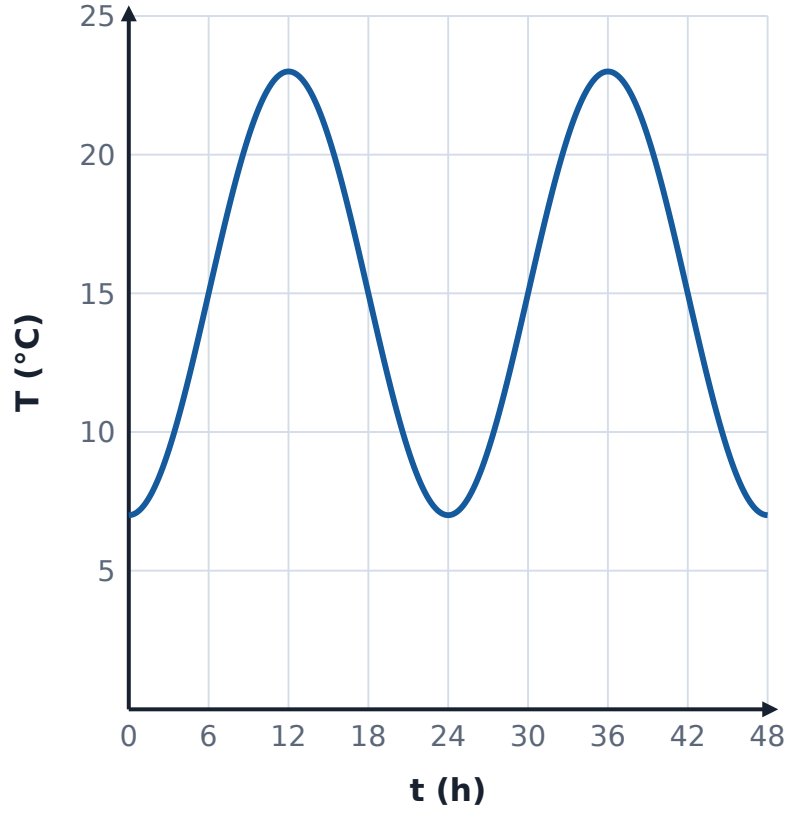
MAIN	Algebra II main question	**HSF-IF.B.4**
BACKWARD 1	Interpret maximums and minimums on a graph	**HSF-IF.B.4**
BACKWARD 2	Read values from a table	**6.EE.C.9**
BACKWARD 3	Amplitude, period, and midline of a sinusoidal model	**HSF-TF.B.5**
FORWARD 1	Predict with periodicity	**HSF-IF.B.4**
FORWARD 2	Write a model from periodic features	**HSF-TF.B.5**

SET 43 • HSF-IF.B.4

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

The graph shows a city's temperature T (°C) t hours after midnight on two days. Give the period in hours, the maximum temperature in °C, and the first time t (in hours) when it occurs.



Period: _____

Maximum: _____

First time t: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q1/5

The table shows the water depth d (ft) at a harbor every 2 hours. Give the period, the depth at low tide, and the first time of low tide.

t (h)	0	2	4	6	8	10	12	14	16	18
d (ft)	9	7	3	1	3	7	9	7	3	1

Period (h): _____

Depth at low tide (ft): _____

First low tide at t (h) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q2/5

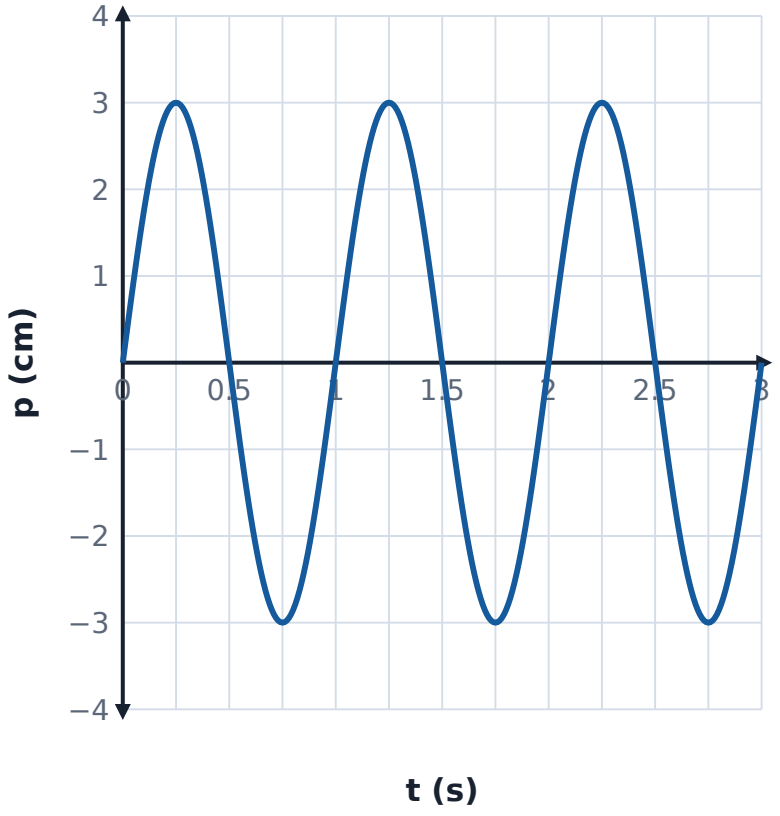
A piston's position p (cm) is graphed for $0 \le t \le 3$ seconds. Give the amplitude, the period, and the two intervals of the first cycle ($0 \le t \le 1$) where p increases, from left to right.

Amplitude (cm): _____

Period (s): _____

First interval: _____

Second interval: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q3/5

The table shows hours of daylight D in month m (m = 0 is January). Give the maximum, the minimum, the period, and the interval of m (from the table) in which daylight increases.

m	0	1	2	3	4	5	6	7	8	9	10	11	12
D (h)	9	9.4	10.5	12	13.5	14.6	15	14.6	13.5	12	10.5	9.4	9

Maximum (h): _____

Minimum (h): _____

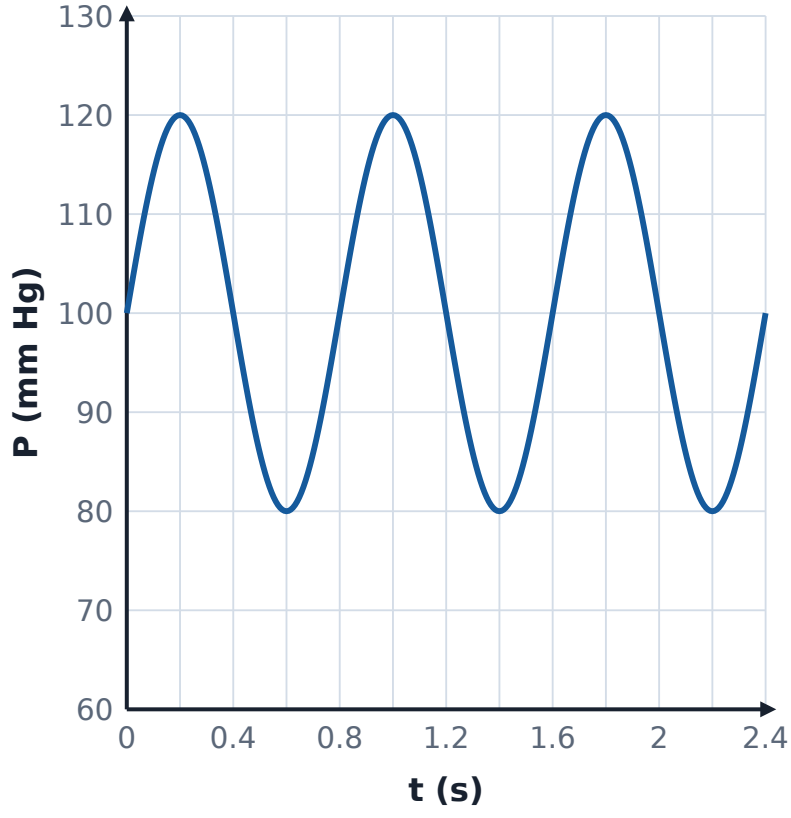
Period (months): _____

Increasing for: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q4/5

The graph shows blood pressure P (mm Hg) over 2.4 seconds. Give the maximum and minimum pressures and the number of heartbeats per minute.



Maximum: _____

Minimum: _____

Beats per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q5/5

SET 43 • HSF-IF.B.4

BACKWARD 1

Interpret maximums and minimums on a graph • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

A graph of a ball's height (m) against time peaks at 45. What does the peak mean?

A. The greatest height reached is 45 m

B. The ball rises 45 m per second

C. The ball lands at 45 m

D. The ball is in the air for 45 s

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q1/5

A graph of harbor depth d (ft) against time t (h) is lowest at the point (6, 1). Give the least depth and the time it occurs.

Least depth (ft): _____

Time (h): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q2/5

True or false?

The minimum value of a graph is its lowest y-value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q3/5

A relative maximum on a graph is

A. the last point

B. any x-intercept

C. a point higher than nearby points

D. the y-intercept

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q4/5

A temperature graph rises on $0 < t < 2$. What does this mean?

A. The temperature stays constant

B. The temperature is above 2 degrees

C. The temperature is increasing during that time

D. The temperature is decreasing during that time

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSF-IF.B.4

BACKWARD 2

Read values from a table • Grade 6 • Expressions & Equations • 6.EE.C.9

Harbor depths d (ft) at t = 0, 2, 4, 6, 8, 10, 12 hours are 9, 7, 3, 1, 3, 7, 9. What is d when t = 4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q1/5

Harbor depths d (ft) at t = 0, 2, 4, 6, 8, 10, 12 hours are 9, 7, 3, 1, 3, 7, 9. At which listed times is d = 9?

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q2/5

True or false?

A repeated pattern in a table suggests a periodic relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q3/5

Hours of daylight in months m = 0, 3, 6, 12 are 9, 12, 15, 9. Daylight of 15 h occurs at

A. m = 12

B. m = 6

C. m = 0

D. m = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q4/5

Harbor depth is 7 ft at t = 2 h and 3 ft at t = 4 h. By how much does the depth d change from t = 2 to t = 4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSF-IF.B.4

BACKWARD 3

Amplitude, period, and midline of a sinusoidal model • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

Give the amplitude and the midline of $T = 15 - 8\cos(\pi t/12)$.

Amplitude: _____

Midline: $y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q1/5

Give the period of $p = 3\sin(2\pi t)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q2/5

True or false?

A wave between 80 and 120 has amplitude 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q3/5

A wave with maximum 9 and minimum 1 has midline

A. y = 8

B. y = 5

C. y = 10

D. y = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q4/5

A period of 0.8 s means how many cycles per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q5/5

SET 43 • HSF-IF.B.4

FORWARD 1

Predict with periodicity • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

Use the temperature model $T(t) = 15 - 8\cos(\pi t/12)$ to find T at t = 100 hours.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q1/5

Harbor depths repeat every 12 hours, and the depth is 1 ft at t = 6 h. Predict the depth at t = 30 hours.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q2/5

True or false?

If f has period 12, then f(50) = f(10).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q3/5

A piston's position is $p = 3\sin(2\pi t)$ cm, with period 1 s. At $t = 7.25$ s, p is

A. 0

B. 3

C. -3

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q4/5

Daylight follows a 12-month cycle with a maximum of 15 h at m = 6 (m = 0 is January). In which month after m = 12 does daylight next reach 15 h?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSF-IF.B.4

FORWARD 2

Write a model from periodic features • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

Harbor depth varies between 9 ft (at t = 0) and 1 ft (at t = 6 h) and repeats every 12 hours. Write the depth in the form d = D + A·cos(Bt) with A > 0 and B > 0. Give B exactly, in terms of π.

A = _____

B = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q1/5

Daylight varies between 9 h (month m = 0) and 15 h (m = 6) and repeats every 12 months. Write daylight in the form k − A·cos(Bm) with A > 0 and B > 0. Give B exactly, in terms of π.

A = _____

B = _____

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q2/5

True or false?

A temperature that is 7°C at t = 0, 23°C at t = 12 h, and repeats every 24 hours is modeled by $15 - 8\cos(\pi t/12)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q3/5

A blood pressure that starts at 100 mm Hg and rises, varies between 80 and 120, and repeats every 0.8 s is modeled by

A. $100 + 20\sin(0.8t)$

B. $20 + 100\sin(2.5\pi t)$

C. $120\sin(2.5\pi t)$

D. $100 + 20\sin(2.5\pi t)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q4/5

A piston's position starts at 0 cm and increases, varies between −3 and 3 cm, and repeats every 1 s. Write it in the form $p = A \cdot \sin(Bt)$ with $A > 0$ and $B > 0$. Give B exactly, in terms of π.

A = _____

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q5/5

SET 44 • HSF-IF.B.5 • INTERPRETING FUNCTIONS

State domains of radical, rational, logarithmic, and polynomial models

MAIN	Algebra II main question	**HSF-IF.B.5**
BACKWARD 1	Domains of linear, quadratic, and exponential models	**HSF-IF.B.5**
BACKWARD 2	Division by zero	**7.NS.A.2.b**
BACKWARD 3	Solve linear inequalities	**HSA-REI.B.3**
FORWARD 1	Domains defined by nonlinear conditions	**HSF-IF.B.5**
FORWARD 2	Domains and ranges of inverse functions	**HSF-BF.B.4.a**

SET 44 • HSF-IF.B.5

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5

A car's speed is $s(d) = \sqrt{(22.5d)}$ mph for a skid of d feet. Which gives the domain of the function and a domain that makes sense in context?

A. Function: $d > 22.5$; context: $d > 22.5$, up to the longest skid that can occur

B. Function: $d \geq 0$; context: $d \geq 0$, up to the longest skid that can actually occur

C. Function: $d \geq 0$; context: all real numbers, since any skid length is possible

D. Function: all real numbers; context: $d \geq 0$, up to the longest skid that can occur

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q1/5

The average cost of x shirts is $A(x) = (500 + 4x)/x$ dollars. Which gives the domain of the function and the domain in context?

A. Function: $x > 0$; context: all real numbers

B. Function: $x \neq 500$; context: $x \geq 500$

C. Function: all real numbers; context: whole numbers $x \geq 0$

D. Function: all real $x \neq 0$; context: whole numbers $x \geq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q2/5

$pH = -\log[H^+]$, where $[H^+]$ is the hydrogen ion concentration. What is the domain?

A. $[H^+] > 0$

B. $[H^+] \geq 0$

C. All real numbers

D. $0 < [H^+] < 14$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q3/5

A box made from a 12-in square sheet has volume $V(x) = x(12 - 2x)^2$. Which gives the domain of the polynomial and the domain in context?

A. Polynomial: all real numbers; context: $0 < x < 12$

B. Polynomial: all real numbers; context: $0 < x < 6$

C. Polynomial: $x \geq 0$; context: $x \geq 6$

D. Polynomial: $0 < x < 6$; context: all real numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q4/5

Which is the domain of $f(x) = \sqrt{(x - 2)/(x - 5)}$?

A. $x > 5$

B. $x \geq 2$

C. $x \geq 2$ and $x \neq 5$

D. All real numbers except 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q5/5

SET 44 • HSF-IF.B.5

BACKWARD 1

Domains of linear, quadratic, and exponential models • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

$h(t) = -16t^2 + 64t$ models height until the ball lands. Give the domain in context.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q1/5

$C(n) = 2n + 5$ is the cost in dollars of n tickets. Which domain makes sense in context?

A. Negative numbers

B. All real numbers

C. All real numbers $n \geq 5$

D. Whole numbers $n \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q2/5

True or false?

The domain of $f(x) = 2^x$ is $x > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q3/5

The input n of a model counts people. Which domain fits?

A. Fractions between 0 and 1

B. All real numbers

C. Nonnegative whole numbers

D. Negative numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q4/5

Why might the domain that makes sense for a model be smaller than the domain of its formula?

A. The context restricts the inputs, for example to nonnegative numbers or whole numbers.

B. The formula is always wrong outside small inputs, so only small inputs make sense.

C. The domain of a model must always be all real numbers, as for its formula.

D. Formulas never have restricted domains, so only the context can restrict them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSF-IF.B.5

BACKWARD 2

Division by zero • Grade 7 • The Number System • 7.NS.A.2.b

For which x is 1/(x − 5) undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q1/5

For which x is (500 + 4x)/x undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q2/5

True or false?

$3/0 = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q3/5

$(x + 1)/(x^2 - 9)$ is undefined at

A. $x = 0$

B. $x = 9$

C. $x = -1$

D. $x = \pm 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q4/5

Give the value of 0/5.

0/5 = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSF-IF.B.5

BACKWARD 3

Solve linear inequalities • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $x - 2 \geq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q1/5

Solve $12 - 2x > 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q2/5

True or false?

$22.5d \geq 0$ means $d \geq 0$.

A. True

B. False

$3 - x \geq 0$ means

A. $x \geq 3$

B. $x \leq 3$

C. $x \geq -3$

D. $x \leq -3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q4/5

Solve $2x + 8 > 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q5/5

SET 44 • HSF-IF.B.5

FORWARD 1

Domains defined by nonlinear conditions • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5

Give the domain of $f(x) = \sqrt{(9 - x^2)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q1/5

Which is the domain of $g(x) = \log(x^2 - 4)$?

A. $x \geq 2$

B. $-2 < x < 2$

C. $x > 2$ only

D. $x < -2$ or $x > 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q2/5

$h(x) = 1/(x^2 - x - 6)$. List the real numbers excluded from the domain.

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q3/5

True or false?

The domain of $\sqrt{(x^2)}$ is $x \geq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q4/5

The domain of $\log(x - 1) + \sqrt{(5 - x)}$ is

A. $1 < x \leq 5$

B. $x \leq 5$

C. $1 \leq x \leq 5$

D. $x > 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSF-IF.B.5

FORWARD 2

Domains and ranges of inverse functions • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

$f(x) = \sqrt{(x - 2)}$ has domain $x \geq 2$ and range $y \geq 0$. Write $f^{-1}(x)$ and give its domain.

$f^{-1}(x) =$ _____

Domain of f^{-1}: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q1/5

$f(x) = 2^x$ has range $y > 0$. What is the domain of $f^{-1}(x) = \log_2 x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q2/5

True or false?

The domain of an inverse function is the range of the original function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q3/5

$f(x) = 1/x + 3$ $(x \neq 0)$ has an inverse with domain

A. $x \neq 0$

B. $x > 3$

C. all real x

D. $x \neq 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q4/5

$s(d) = \sqrt{(22.5d)}$ gives a car's speed s from its skid length d. Why does its inverse $d = s^2/22.5$ need the restriction $s \geq 0$?

A. Because $s^2/22.5$ is undefined for $s < 0$, so the formula cannot be used for negative speeds.

B. Because d must be a whole number of feet, and $s \geq 0$ makes $s^2/22.5$ a whole number.

C. The range of $s(d)$ is $s \geq 0$, so $d = s^2/22.5$ undoes s only for $s \geq 0$; without the restriction it is not one-to-one.

D. Because 22.5 is positive, so s must also be positive for the quotient $s^2/22.5$ to make sense.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q5/5

SET 45 • HSF-IF.B.6 • INTERPRETING FUNCTIONS

Calculate and interpret average rates of change of nonlinear models

MAIN	Algebra II main question	**HSF-IF.B.6**
BACKWARD 1	Average rate of change of simpler functions	**HSF-IF.B.6**
BACKWARD 2	Slope between two points	**8.F.B.4**
BACKWARD 3	Evaluate nonlinear functions	**HSF-IF.A.2**
FORWARD 1	Compare average rates on different intervals	**HSF-IF.B.6**
FORWARD 2	Compare rates of change across representations	**HSF-IF.C.9**

SET 45 • HSF-IF.B.6

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6

A culture has $N(t) = 200 \cdot 2^t$ bacteria after t hours. Find the average rate of change from $t = 1$ to $t = 3$.

Average rate (bacteria per hour): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q1/5

A stone's height is $h(t) = 64 - 16t^2$ ft after t seconds. Find the average rate of change from t = 0.5 to t = 1.5 s.

Average rate (ft per second): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q2/5

The table shows a car's value. Find the average rate of change from t = 2 to t = 4 years, to the nearest cent.

t (years)	0	1	2	3	4	5
V ($)	20000	17000	14450	12282.50	10440.06	8874.11

Average rate (dollars per year): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q3/5

Speed from skid length is $s(d) = \sqrt{(24d)}$ mph. Find the average rate of change from $d = 24$ to $d = 96$ ft.

Average rate (mph per foot): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q4/5

Temperature is $T(t) = 15 - 8\cos(\pi t/12)$ °C after t hours. Find the average rate of change from $t = 0$ to $t = 6$.

Average rate (°C per hour): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q5/5

SET 45 • HSF-IF.B.6

BACKWARD 1

Average rate of change of simpler functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = x^2$. Find the average rate of change from $x = 1$ to $x = 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q1/5

A table gives f(2) = 10 and f(5) = 1. Find the average rate of change.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q2/5

True or false?

The average rate of change from a to b is $(f(b) - f(a))/(b - a)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q3/5

The average rate of change of f(x) = 3x + 1 on any interval is

A. it varies

B. 1

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q4/5

f is in dollars and x is in years. What are the units of the average rate of change of f?

A. Years per dollar

B. Dollars per year

C. Dollar-years

D. Dollars

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSF-IF.B.6

BACKWARD 2

Slope between two points • Grade 8 • Functions • 8.F.B.4

Find the slope between (1, 400) and (3, 1600).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q1/5

Find the slope between (0.5, 60) and (1.5, 28).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q2/5

True or false?

A negative slope means the output increases as the input increases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q3/5

The slope between (24, 24) and (96, 48) is

A. 72/24

B. 1/3

C. 2

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q4/5

Find the slope between (0, 7) and (6, 15).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSF-IF.B.6

BACKWARD 3

Evaluate nonlinear functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Find $N(3)$ for $N(t) = 200 \cdot 2^t$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q1/5

Find $h(1.5)$ for $h(t) = 64 - 16t^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q2/5

True or false?

$\sqrt{(24 \cdot 96)} = 2304$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q3/5

$15 - 8\cos(\pi/2)$ equals

A. 15

B. 7

C. 23

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q4/5

Find $T(0)$ for $T(t) = 15 - 8\cos(\pi t/12)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q5/5

SET 45 • HSF-IF.B.6

FORWARD 1

Compare average rates on different intervals • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6

For $N(t) = 200 \cdot 2^t$ bacteria after t hours, find the average rates of change on [0, 1] and on [3, 4].

Rate on [0, 1] (bacteria per hour): _____

Rate on [3, 4] (bacteria per hour): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q1/5

For $s(d) = \sqrt{(24d)}$, find the average rates of change on $[0, 24]$ and on $[96, 150]$.

Rate on $[0, 24]$: _____

Rate on $[96, 150]$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q2/5

True or false?

For a function with period 24, the average rate of change over [0, 24] is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q3/5

For $f(x) = x^3$, the average rate on $[-1, 1]$ is

A. 3

B. 0

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q4/5

$T(t) = 15 - 8\cos(\pi t/12)$. On which interval of length 6 is the average rate of change $-4/3$?

A. [6, 12]

B. [3, 9]

C. [12, 18]

D. [0, 6]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSF-IF.B.6

FORWARD 2

Compare rates of change across representations • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

$f(t) = 200 \cdot 2^t$, and g is given by $g(1) = 500$ and $g(3) = 1500$. Find the average rate of change of each on $[1, 3]$.

Rate of f: _____

Rate of g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q1/5

A graph of a cooling liquid falls from 90°C at t = 0 to 50°C at t = 20 min. Find the average rate of change of the graph and of $T(t) = 90 - 1.5t$ on [0, 20].

Graph (°C per min): _____

Formula (°C per min): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q2/5

True or false?

A function with a greater average rate on one interval must be greater everywhere.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q3/5

A table shows $h(0) = 0$, $h(2) = 40$; the formula $k(t) = 5t^2$ gives

A. no rate for k

B. a greater rate for k

C. a greater rate for h

D. equal average rates on [0, 2]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q4/5

One function is given by a graph and another by a formula. How can their average rates of change on the same interval be compared?

A. Compare the two functions at the right endpoint only, since the larger final value means the larger average rate of change.

B. Read the two endpoint values from the graph, evaluate the formula at the same inputs, and compute both quotients (change in output)/(change in input).

C. Compare the steepest part of the graph with the leading coefficient of the formula, since both measure the rate of change on the interval.

D. Compare the y-intercepts of the graph and the formula, since the larger starting value has the larger average rate of change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q5/5

SET 46 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph square root and cube root functions with transformations

MAIN	Algebra II main question	**HSF-IF.C.7.b**
BACKWARD 1	Graph $y = \sqrt{x}$ and $y = \sqrt[3]{x}$ from a table	**HSF-IF.C.7.b**
BACKWARD 2	Translations and reflections of graphs	**HSF-BF.B.3**
BACKWARD 3	Square and cube roots of numbers	**8.EE.A.2**
FORWARD 1	Graph radical functions with horizontal stretches and reflections	**HSF-IF.C.7.b**
FORWARD 2	Graph a radical function and its inverse	**HSF-BF.B.4.a**

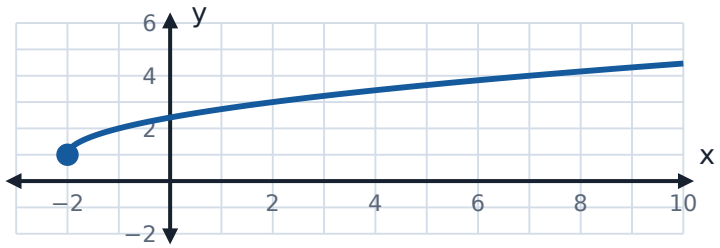
SET 46 • HSF-IF.C.7.b

MAIN

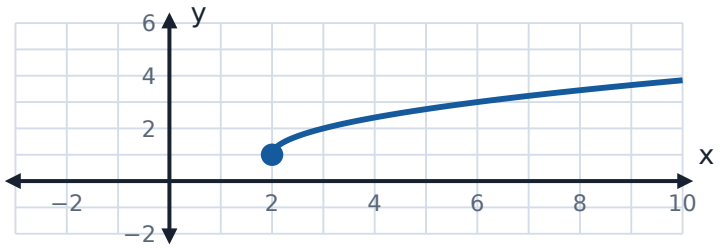
Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

Which graph shows $y = \sqrt{(x - 2)} + 1$?

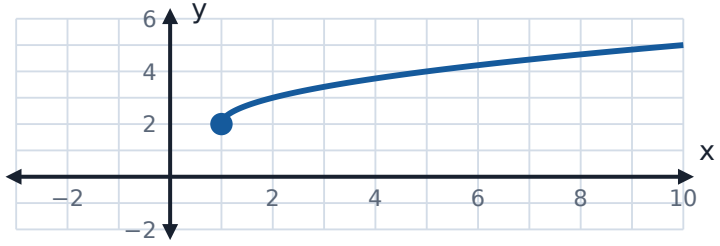
A



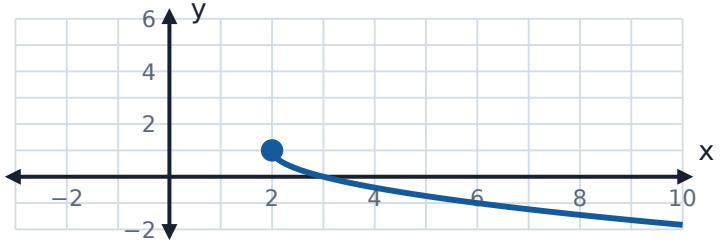
B



C



D



A. Graph A

B. Graph B

C. Graph C

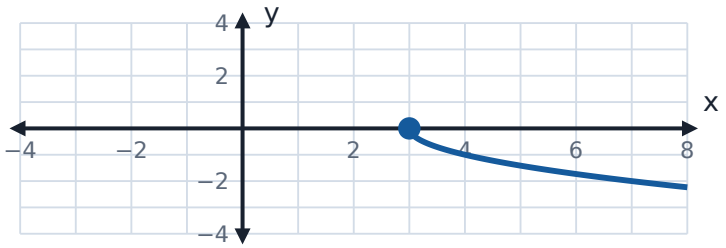
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-M-Q1**

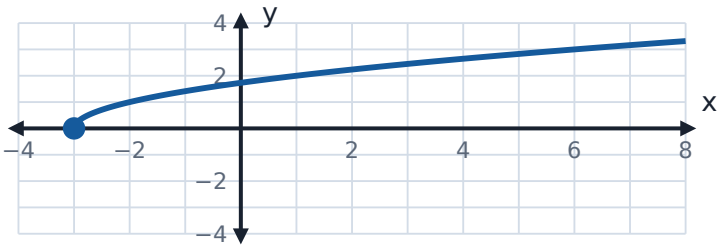
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q1/5

Which graph shows $y = -\sqrt{(x + 3)}$?

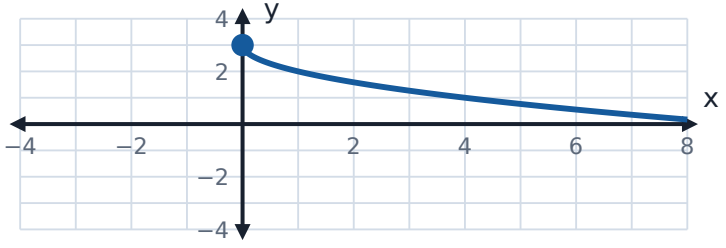
A



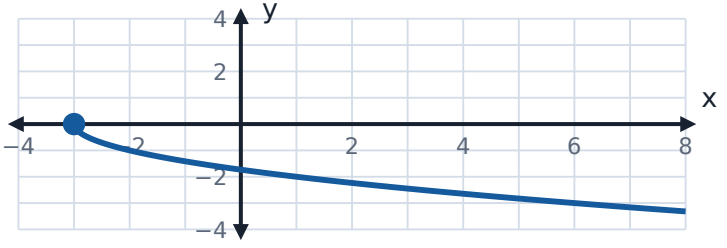
B



C



D



A. Graph A

B. Graph B

C. Graph C

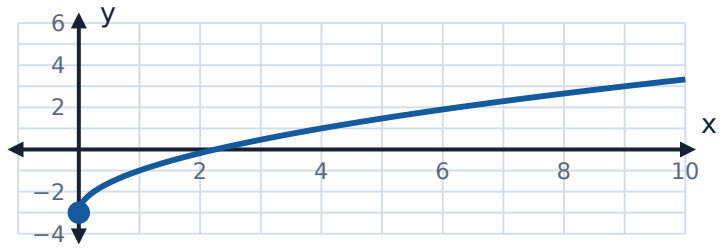
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-M-Q2**

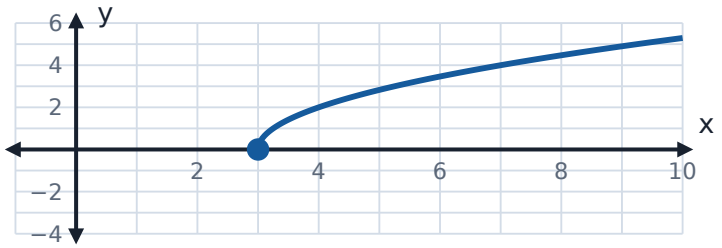
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q2/5

Which graph shows $y = 2\sqrt{x} - 3$?

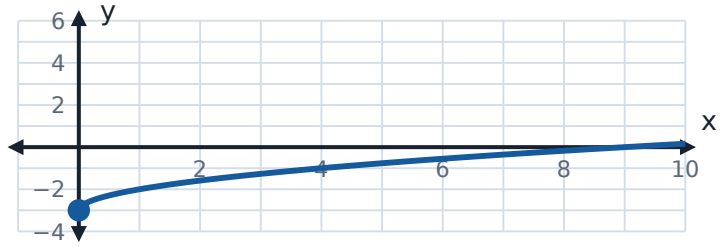
A



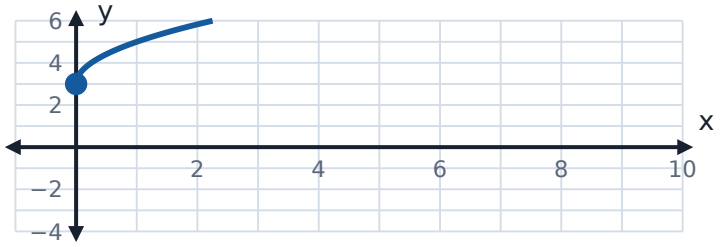
B



C



D



A. Graph A

B. Graph B

C. Graph C

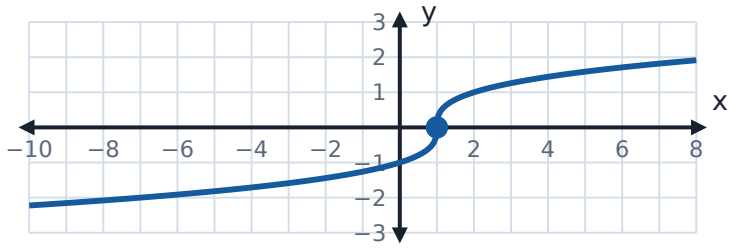
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-M-Q3**

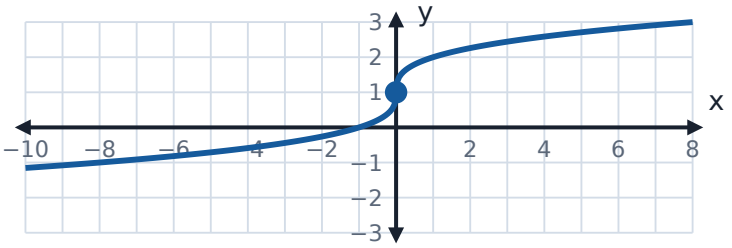
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q3/5

Which graph shows $y = \sqrt[3]{(x + 1)}$?

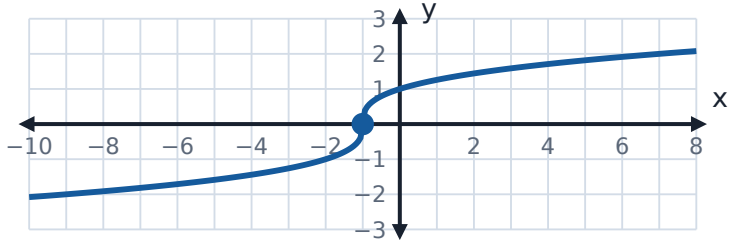
A



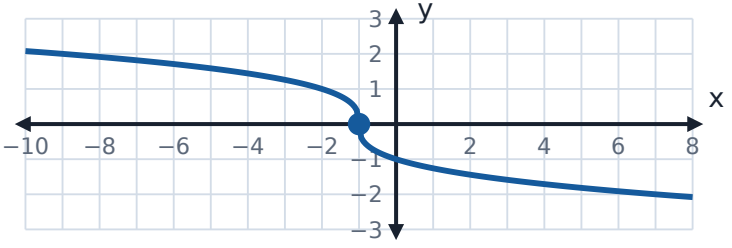
B



C



D



A. Graph A

B. Graph B

C. Graph C

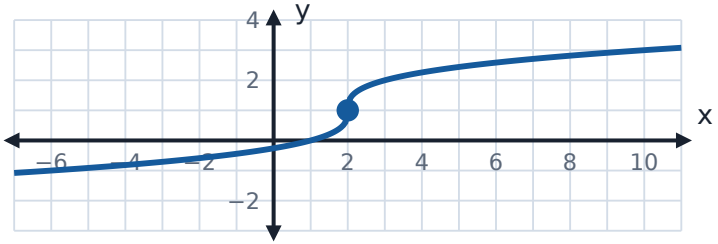
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-M-Q4**

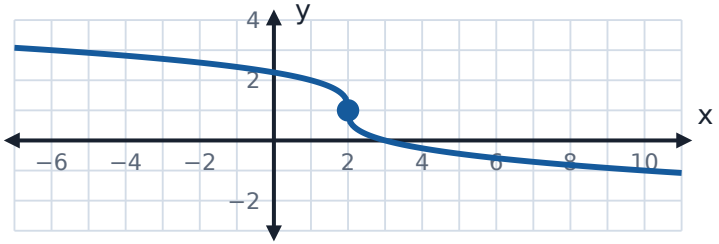
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q4/5

Which graph shows $y = -\sqrt[3]{(x - 2)} + 1$?

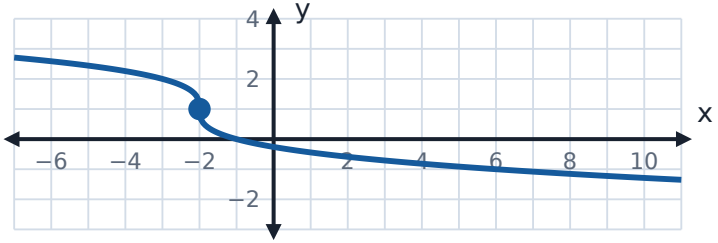
A



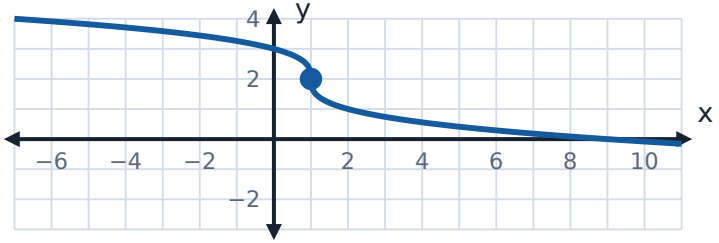
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q5/5

SET 46 • HSF-IF.C.7.b

BACKWARD 1

Graph y = √x and y = ∛x from a table • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

Find $y = \sqrt{x}$ at $x = 0$, 1, 4, and 9, in that order.

y-values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q1/5

Find $y = \sqrt[3]{x}$ at $x = -8, -1, 0, 1,$ and 8, in that order.

y-values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q2/5

True or false?

The graph of $y = \sqrt{x}$ starts at the origin and decreases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q3/5

The domain of $y = \sqrt[3]{x}$ is

A. $x > 0$

B. all real numbers

C. $x \leq 0$

D. $x \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q4/5

Give the range of $y = \sqrt{x}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSF-IF.C.7.b

BACKWARD 2

Translations and reflections of graphs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How is the graph of $y = \sqrt{(x - 2)} + 1$ obtained from the graph of $y = \sqrt{x}$?

A. Shift right 2 and up 1

B. Shift up 2 and right 1

C. Shift left 2 and up 1

D. Shift right 2 and down 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q1/5

How is the graph of $y = -\sqrt{(x + 3)}$ obtained from the graph of $y = \sqrt{x}$?

A. Shift right 3, then reflect over the x-axis

B. Shift down 3, then reflect over the x-axis

C. Shift left 3, then reflect over the y-axis

D. Shift left 3, then reflect over the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q2/5

True or false?

$y = 2\sqrt{x}$ is a vertical stretch of $y = \sqrt{x}$ by 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q3/5

The point (0, 0) of $y = \sqrt{x}$ moves to which point for $y = \sqrt{(x - 2)} + 1$?

A. (2, 1)

B. (1, 2)

C. (−2, 1)

D. (2, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q4/5

Where does (1, 1) of $y = \sqrt[3]{x}$ go under $y = \sqrt[3]{(x + 1)}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSF-IF.C.7.b

BACKWARD 3

Square and cube roots of numbers • Grade 8 • Expressions & Equations • 8.EE.A.2

Find $\sqrt{9}$ and $\sqrt[3]{(-8)}$.

$\sqrt{9} =$ _____

$\sqrt[3]{(-8)} =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q1/5

Find $\sqrt[3]{27}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q2/5

True or false?

$\sqrt{(-4)}$ is a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q3/5

$\sqrt[3]{(-1)}$ equals

A. i

B. undefined

C. -1

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q4/5

Find $\sqrt{16} - 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q5/5

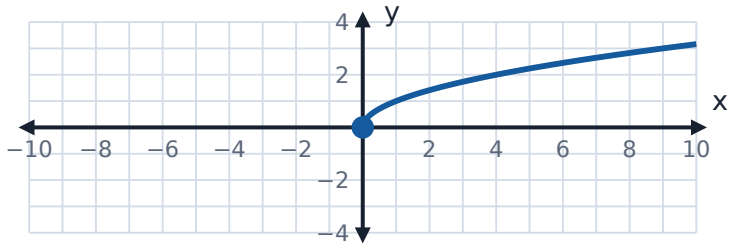
SET 46 • HSF-IF.C.7.b

FORWARD 1

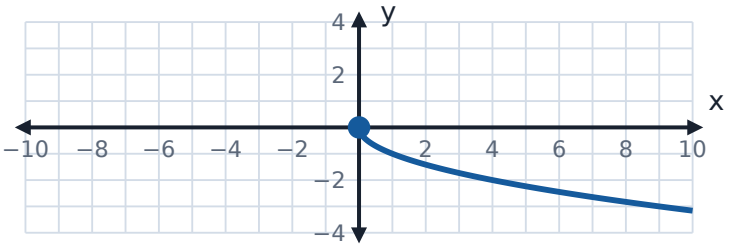
Graph radical functions with horizontal stretches and reflections • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

Which graph shows $y = \sqrt{(-x)}$?

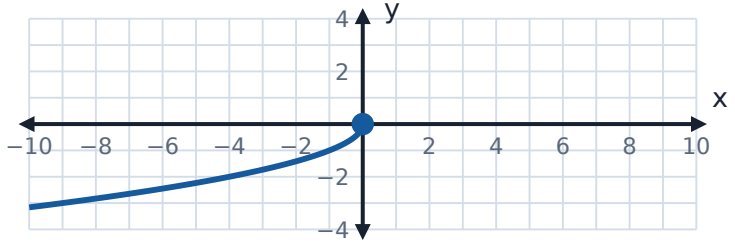
A



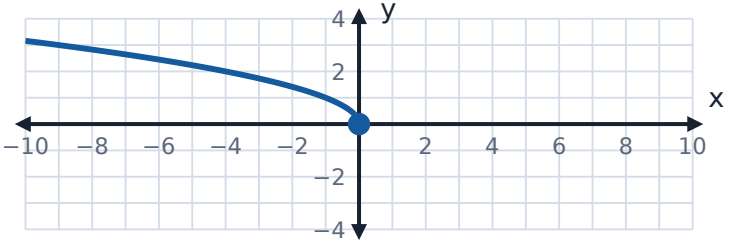
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q1/5

For $x \geq 0$, $\sqrt{(4x)} = 2\sqrt{x}$. What does this say about the graphs of $y = \sqrt{(4x)}$ and $y = 2\sqrt{x}$?

A. They are the same graph: a horizontal compression of $y = \sqrt{x}$ by a factor of 1/4 gives the same result as a vertical stretch by 2.

B. They are reflections of each other over the y-axis, because one changes the input and the other the output.

C. They are different graphs that meet only at the origin, because a horizontal change never equals a vertical one.

D. They are different graphs: $y = \sqrt{(4x)}$ is $y = \sqrt{x}$ shifted right 4, and $y = 2\sqrt{x}$ is $y = \sqrt{x}$ stretched up by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q2/5

True or false?

$y = \sqrt[3]{(-x)}$ is the same as $y = \sqrt[3]{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q3/5

The domain of $y = \sqrt{(3 - x)}$ is

A. all reals

B. $x \geq 3$

C. $x \leq 3$

D. $x \geq -3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q4/5

Which square root function has a graph that starts at (4, −2) and goes up to the left?

A. $y = \sqrt{(4 - x)} + 2$

B. $y = -\sqrt{(4 - x)} - 2$

C. $y = \sqrt{(x - 4)} - 2$

D. $y = \sqrt{(4 - x)} - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSF-IF.C.7.b

FORWARD 2

Graph a radical function and its inverse • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

$f(x) = \sqrt{(x - 2)} + 1$. Write $f^{-1}(x)$ and give its domain.

$f^{-1}(x) =$ _____

Domain of f^{-1}: _____

Find the inverse of $g(x) = \sqrt[3]{(x + 1)}$.

$g^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q2/5

True or false?

The graph of an inverse is the reflection of the graph over y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q3/5

The point (6, 3) on $f(x) = \sqrt{(x - 2)} + 1$ gives which point on f^{-1}?

A. (6, 3)

B. (3, 6)

C. (−6, −3)

D. (3, −6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q4/5

Why does the inverse of $f(x) = \sqrt{(x - 2)} + 1$ need the domain $x \geq 1$?

A. Because $f(x) = \sqrt{(x - 2)} + 1$ is undefined for $x < 1$, so its inverse is undefined there too.

B. Because every inverse of a square root function has domain $x \geq 1$, whatever the shifts are.

C. The range of f is $y \geq 1$, so f^{-1} has domain $x \geq 1$; otherwise $(x - 1)^2 + 2$ is not one-to-one.

D. Because the inverse of f is $\sqrt{(x - 1)} + 2$, and $\sqrt{(x - 1)}$ is undefined for $x < 1$, so the inverse needs $x \geq 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q5/5

SET 47 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph piecewise-defined and absolute value functions

MAIN	Algebra II main question	**HSF-IF.C.7.b**
BACKWARD 1	Evaluate and graph simple piecewise functions	**HSF-IF.C.7.b**
BACKWARD 2	Absolute value as distance	**6.NS.C.7**
BACKWARD 3	Open and closed endpoints	**6.EE.B.8**
FORWARD 1	Write piecewise functions from graphs and contexts	**HSF-IF.C.7.b**
FORWARD 2	Describe absolute value graphs as transformations, and identify…	**HSF-BF.B.3**

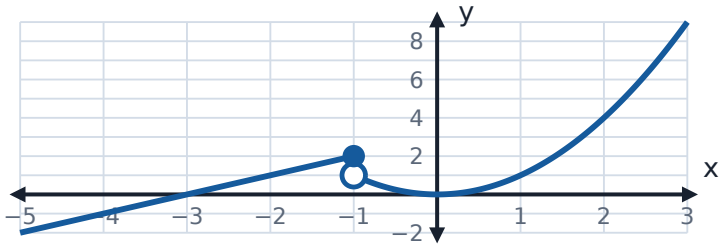
SET 47 • HSF-IF.C.7.b

MAIN

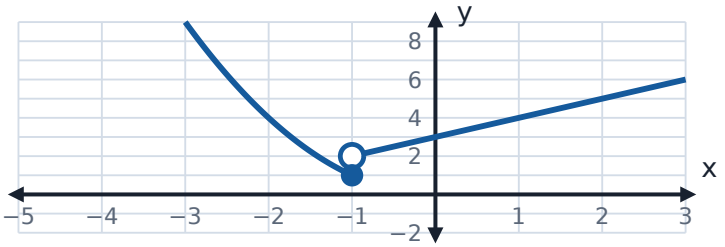
Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

Which graph shows f(x) = x + 3 for x < −1, and f(x) = x² for x ≥ −1?

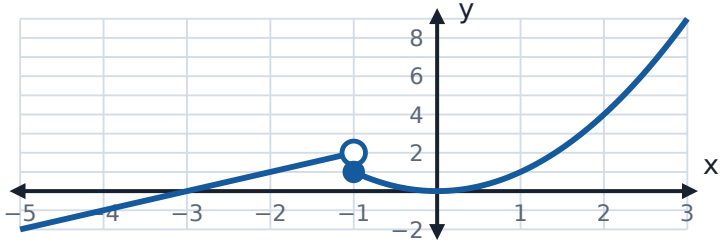
A



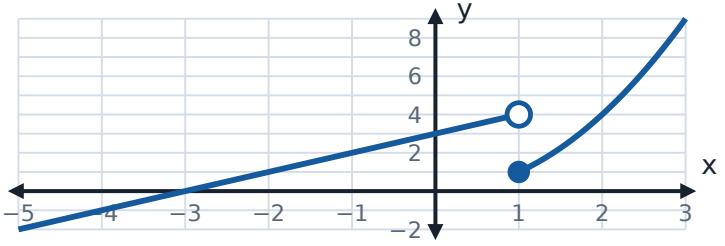
B



C



D



A. Graph A

B. Graph B

C. Graph C

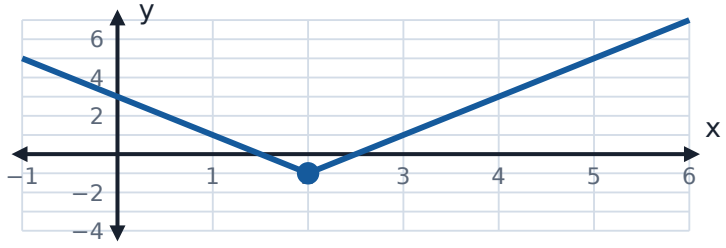
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-M-Q1**

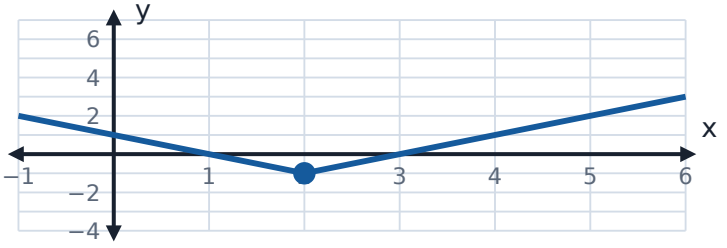
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q1/5

Which graph shows $y = |2x - 4| - 1$?

A



B



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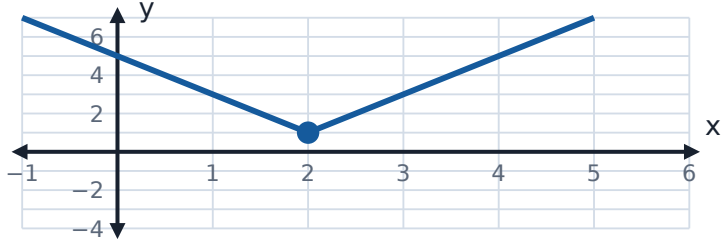
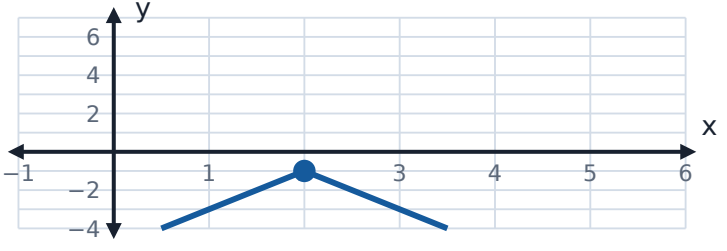


D



A. Graph A

B. Graph B

C. Graph C

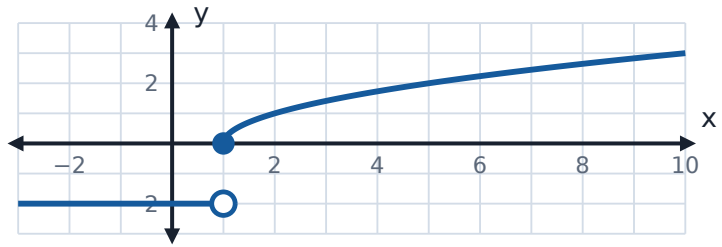
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-M-Q2**

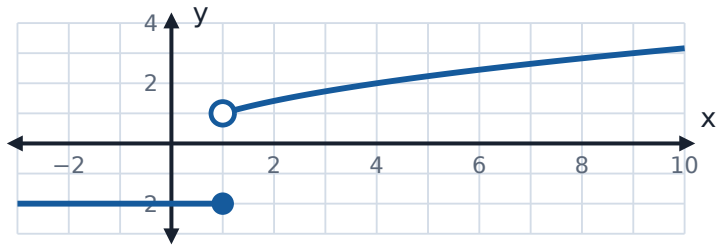
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q2/5

Which graph shows f(x) = −2 for x ≤ 1, and f(x) = √(x − 1) for x > 1?

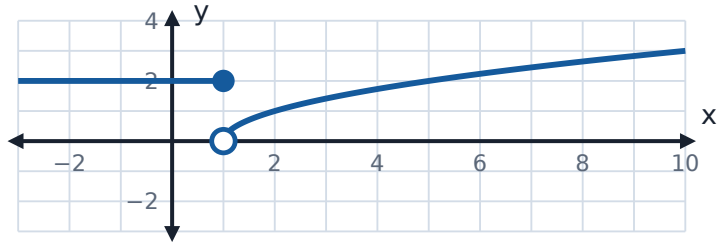
A



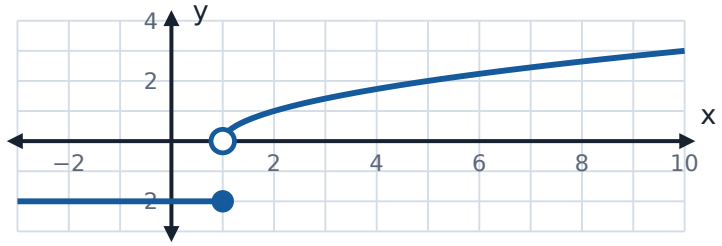
B



C



D



A. Graph A

B. Graph B

C. Graph C

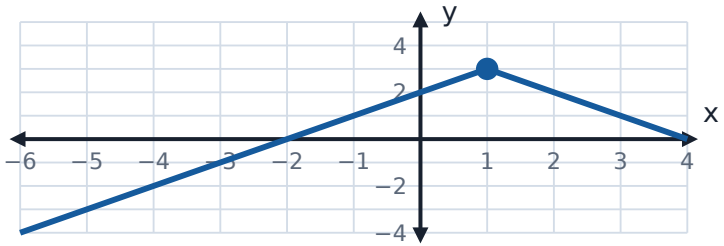
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-M-Q3**

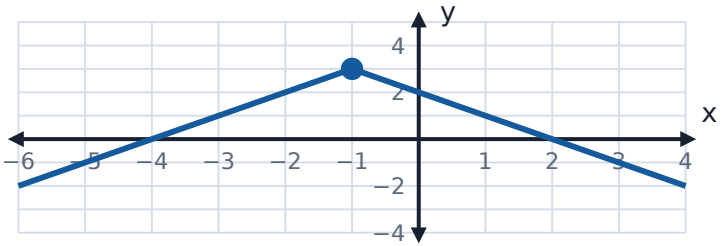
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q3/5

Which graph shows $y = -|x + 1| + 3$?

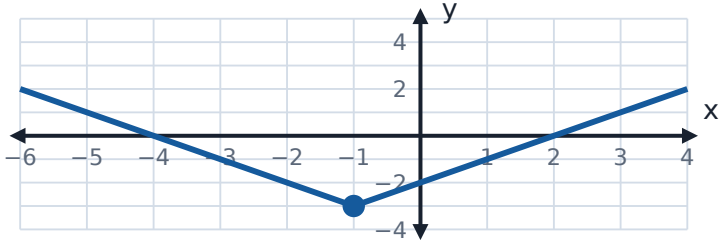
A



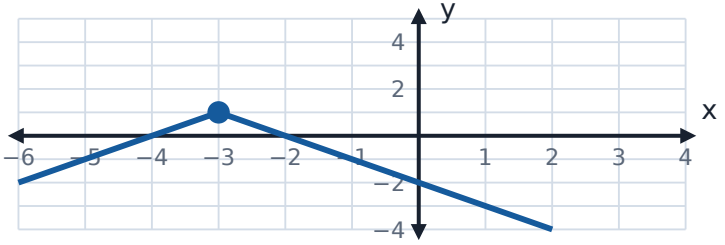
B



C



D



A. Graph A

B. Graph B

C. Graph C

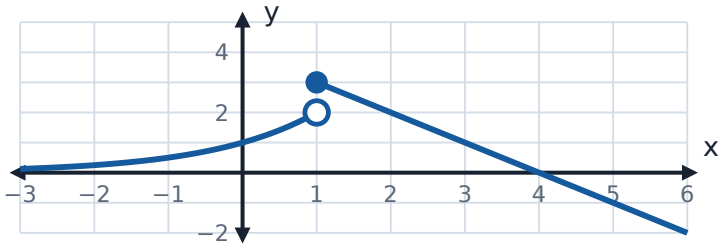
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-M-Q4**

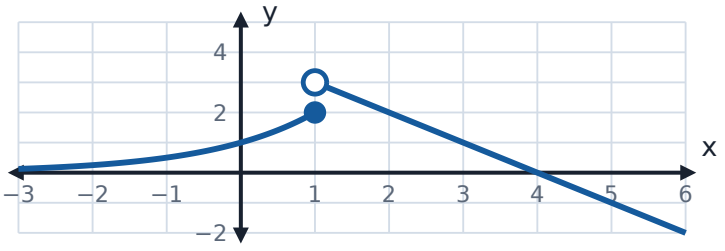
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q4/5

Which graph shows $f(x) = 2^x$ for $x < 1$, and $f(x) = 4 - x$ for $x \geq 1$?

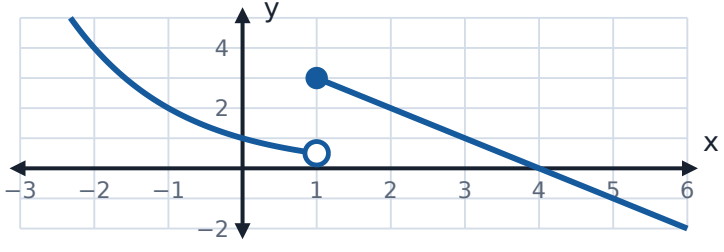
A



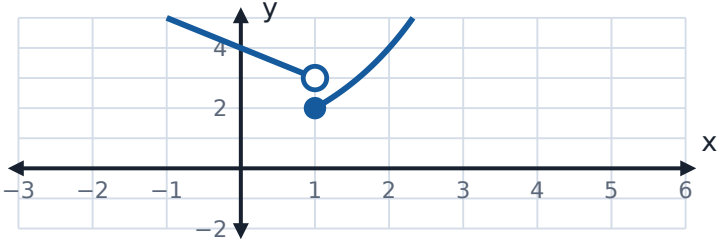
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q5/5

SET 47 • HSF-IF.C.7.b

BACKWARD 1

Evaluate and graph simple piecewise functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

$f(x) = 2$ for $x < 0$ and $f(x) = x$ for $x \geq 0$. Find $f(-3)$ and $f(5)$.

$f(-3) =$ _____

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q1/5

$f(x) = 2$ for $x < 0$ and $f(x) = x$ for $x \geq 0$. Find $f(0)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q2/5

True or false?

A piecewise function uses different rules on different parts of its domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q3/5

For $f(x) = x + 3$ $(x < -1)$, x^2 $(x \geq -1)$, $f(-1)$ is

A. 3

B. 2

C. -1

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q4/5

$f(x) = x + 3$ for $x < -1$, and $f(x) = x^2$ for $x \geq -1$. Find $f(-2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSF-IF.C.7.b

BACKWARD 2

Absolute value as distance • Grade 6 • The Number System • 6.NS.C.7

Find $|-5|$ and $|3|$.

$|-5| =$ _____

$|3| =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q1/5

Find $|2(1) - 4| - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q2/5

True or false?

$|x|$ can be negative for negative x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q3/5

$|x + 1| = 0$ when

A. $x = 0$

B. $x = -1$

C. $x = 1$

D. never

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q4/5

Find $-|0 + 1| + 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSF-IF.C.7.b

BACKWARD 3

Open and closed endpoints • Grade 6 • Expressions & Equations • 6.EE.B.8

On a number line, is $x < -1$ drawn with an open or a closed circle at -1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q1/5

Is $x \geq 1$ drawn with an open or a closed circle at 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q2/5

True or false?

An open circle means the endpoint is included.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q3/5

On a number line, $x \leq 1$ is drawn with

A. an open circle and shading right

B. a closed circle at 1 and shading right

C. an open circle at 1 and shading left

D. a closed circle at 1 and shading left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q4/5

Write the inequality for "all numbers greater than 1".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q5/5

SET 47 • HSF-IF.C.7.b

FORWARD 1

Write piecewise functions from graphs and contexts • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

A shipping company charges $5 for packages up to 2 lb and $5 plus $1.50 per pound over 2 lb. Which gives the cost C(w) for a package of w pounds?

A. $C(w) = 5$ for $0 < w \leq 2$, and $C(w) = 1.5(w - 2)$ for $w > 2$

B. $C(w) = 5$ for $0 < w \leq 2$, and $C(w) = 5 + 1.5(w - 2)$ for $w > 2$

C. $C(w) = 5$ for $0 < w < 2$, and $C(w) = 5 + 1.5w$ for $w \geq 2$

D. $C(w) = 5 + 1.5w$ for every $w > 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q1/5

Which writes $|x - 3|$ as a piecewise function without absolute value bars?

A. $3 - x$ for $x < 3$, and $x - 3$ for $x \geq 3$

B. $x - 3$ for every x

C. $3 - x$ for $x < 0$, and $x - 3$ for $x \geq 0$

D. $x - 3$ for $x < 3$, and $3 - x$ for $x \geq 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q2/5

True or false?

A piecewise function can have a jump and still be a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q3/5

Which piecewise function is continuous at $x = 1$?

A. 2^x for $x < 1$, $4 - x$ for $x \geq 1$

B. x^2 for $x < 1$, $2x - 1$ for $x \geq 1$

C. 2 for $x < 1$, 3 for $x \geq 1$

D. x for $x < 1$, $x + 1$ for $x \geq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q4/5

Find k so that $f(x) = kx$ for $x < 2$ and $f(x) = x^2$ for $x \geq 2$ has no jump at 2.

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSF-IF.C.7.b

FORWARD 2

Describe absolute value graphs as transformations, and identify symmetry • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Which describes $y = |2x - 4| - 1$ as a transformation of $y = |x|$?

A. $y = 2|x - 4| - 1$: shift right 4, stretch vertically by 2, shift down 1

B. $y = 2|x - 2| - 1$: shift right 2, stretch vertically by 2, shift down 1

C. $y = 2|x + 2| - 1$: shift left 2, stretch vertically by 2, shift down 1

D. $y = |x - 2| - 1$: shift right 2, shift down 1, no vertical stretch

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q1/5

An absolute value function $y = a|x + 1| + 3$ has vertex $(-1, 3)$, opens down, and passes through $(2, 0)$. Find a.

$a =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q2/5

True or false?

$y = |x|$ is an odd function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q3/5

$y = |x - 2|$ is symmetric about

A. the origin

B. $y = 2$

C. $x = 0$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q4/5

$f(x) = -|x + 1| + 3$. Find $f(1)$ and $f(-1)$, and decide whether f is even.

$f(1) =$ _____

$f(-1) =$ _____

Is f even? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q5/5

SET 85 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph step functions

MAIN	Algebra II main question	**HSF-IF.C.7.b**
BACKWARD 1	Evaluate step functions	**HSF-IF.C.7.b**
BACKWARD 2	Included and excluded endpoints of intervals	**6.EE.B.8**
BACKWARD 3	Integers just below a number	**6.NS.C.7**
FORWARD 1	Write and interpret step-function models	**HSF-IF.C.7.b**
FORWARD 2	Transform step functions	**HSF-BF.B.3**

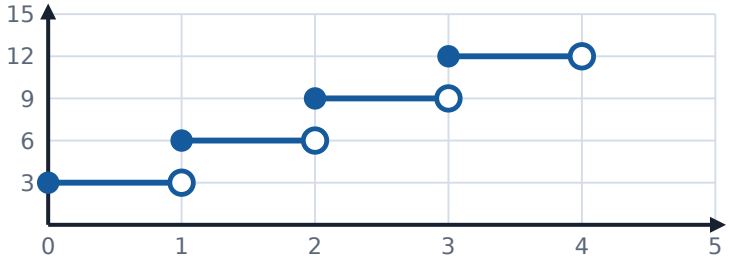
SET 85 • HSF-IF.C.7.b

MAIN

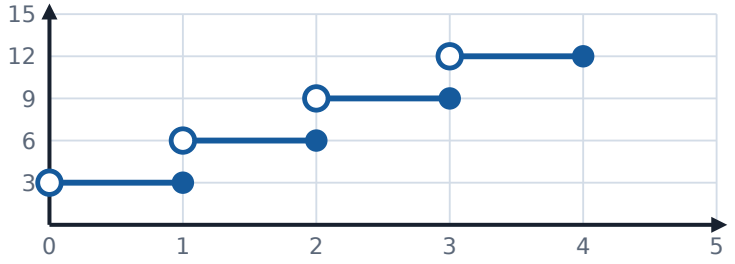
Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

A parking garage charges $3 for each hour or part of an hour, so C(t) = 3[t] dollars for 0 < t ≤ 4 hours, where [t] is the least integer greater than or equal to t. Which graph shows C?

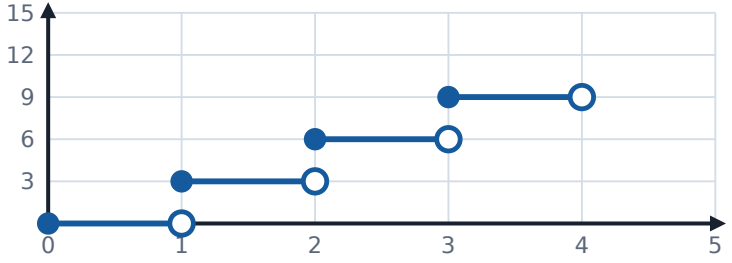
A



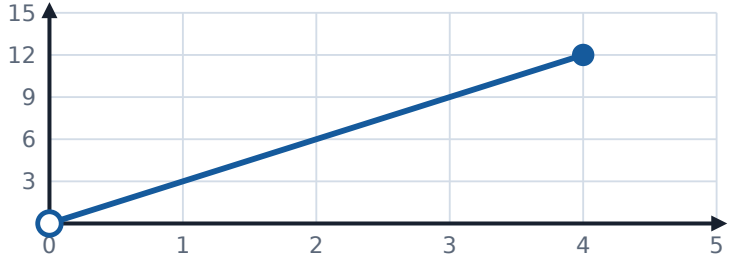
B



C



D



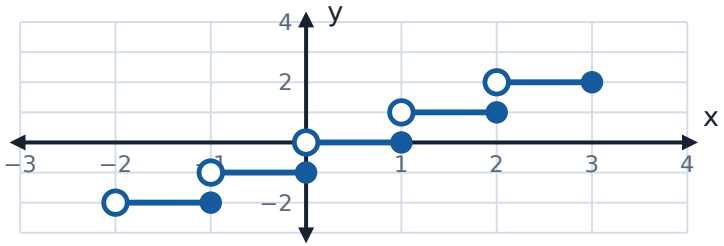
A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-M-Q1**

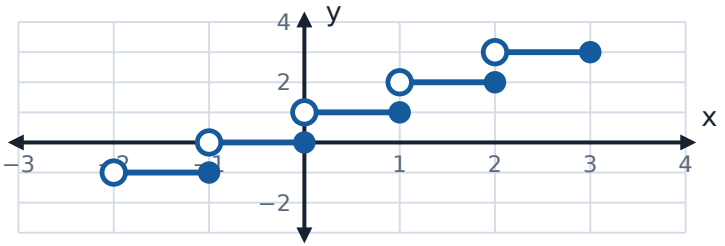
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q1/5

$f(x) = \lfloor x \rfloor$, the greatest integer less than or equal to x, for $-2 \le x < 3$. Which graph shows f?

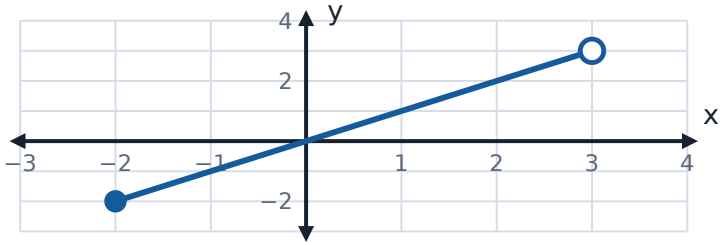
A



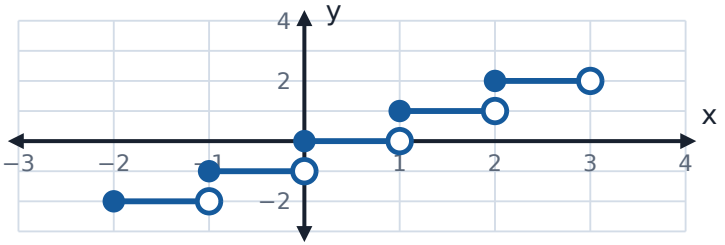
B



C



D



A. Graph A

B. Graph B

C. Graph C

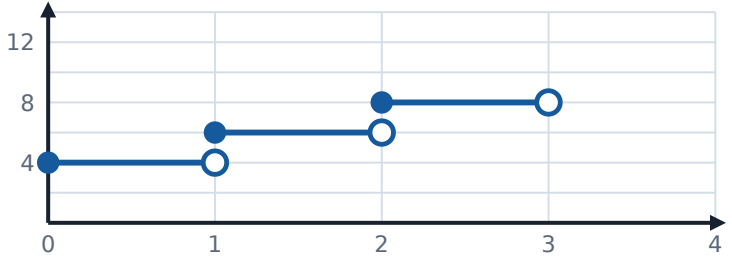
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q2/5

A courier charges $4 for a package up to 1 lb, $6 for more than 1 lb up to 2 lb, and $8 for more than 2 lb up to 3 lb. Which graph shows the cost for weights $0 < w \leq 3$?

A



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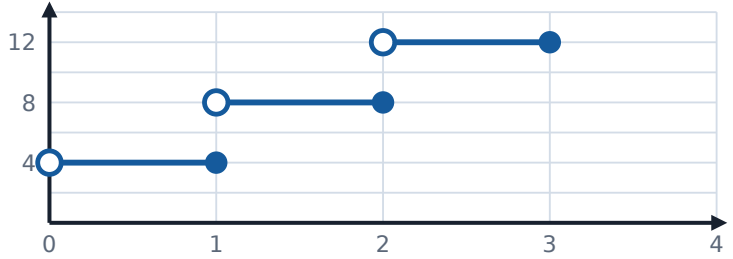
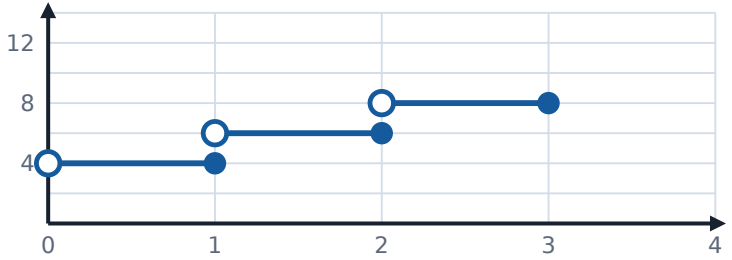
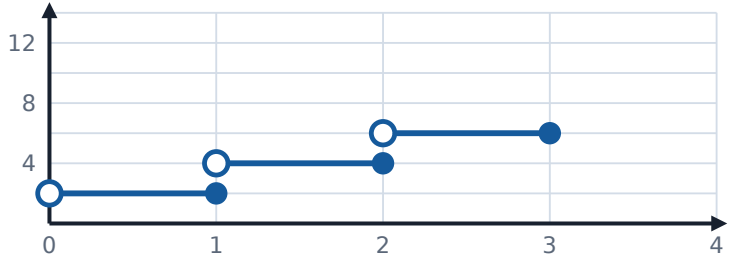


C



D



A. Graph A

B. Graph B

C. Graph C

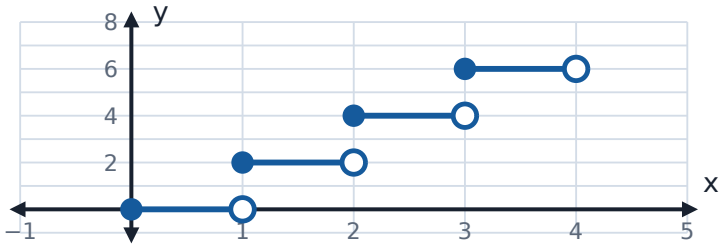
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-M-Q3**

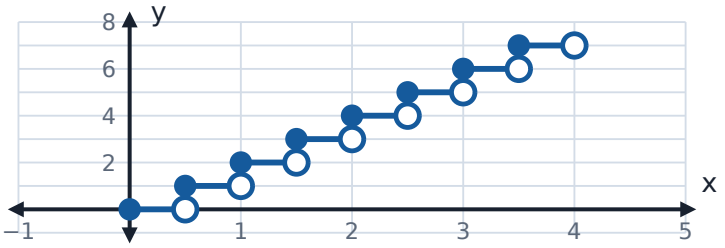
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q3/5

$g(x) = 2\lfloor x \rfloor$ for $0 \le x < 4$, where $\lfloor x \rfloor$ is the greatest integer less than or equal to x. Which graph shows g?

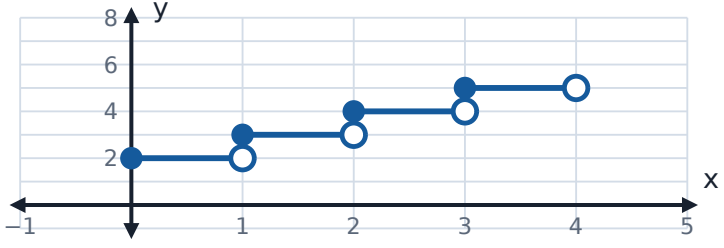
A



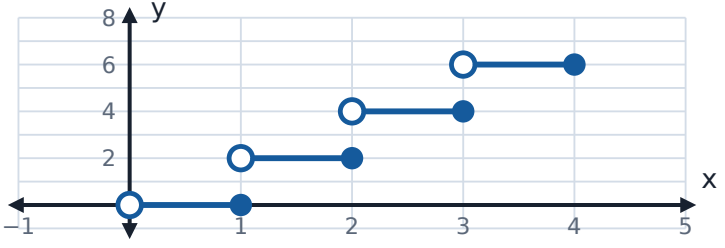
B



C



D



A. Graph A

B. Graph B

C. Graph C

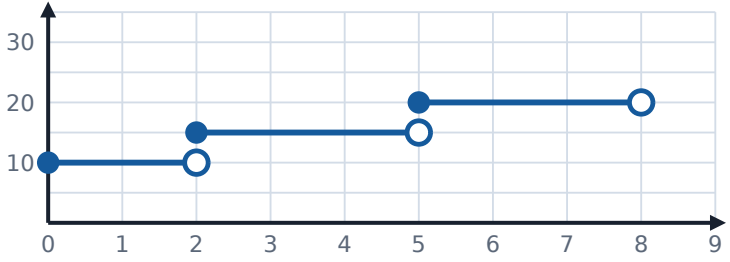
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-M-Q4**

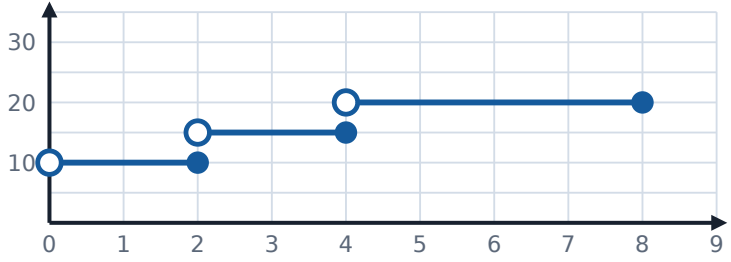
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q4/5

A phone plan costs $10 for up to 2 GB of data, $15 for more than 2 GB up to 5 GB, and $20 for more than 5 GB up to 8 GB. Which graph shows the monthly cost for data use $0 < d \leq 8$?

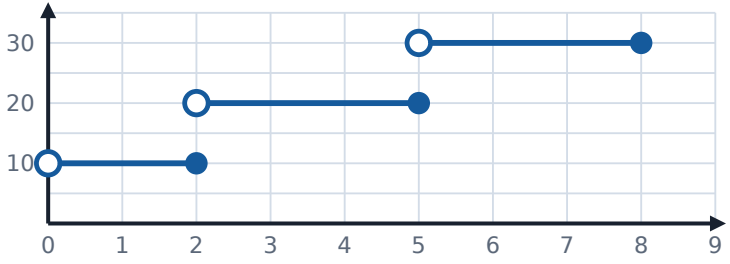
A



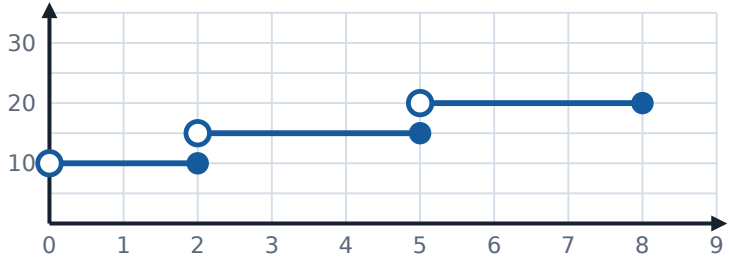
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q5/5

SET 85 • HSF-IF.C.7.b

BACKWARD 1

Evaluate step functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

$\lfloor x \rfloor$ is the greatest integer less than or equal to x. Find $\lfloor 2.7 \rfloor$ and $\lfloor -1.3 \rfloor$.

$\lfloor 2.7 \rfloor =$ _____

$\lfloor -1.3 \rfloor =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q1/5

A parking garage charges $3 for each hour or part of an hour. How many dollars does it charge for 2.5 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q2/5

True or false?

$\lfloor x \rfloor$, the greatest integer less than or equal to x, equals x exactly when x is an integer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q3/5

$\lfloor x \rfloor$ is the greatest integer less than or equal to x. What is $\lfloor -0.5 \rfloor$?

A. 1

B. 0

C. -1

D. -0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q4/5

A parking garage charges $3 for each hour or part of an hour. How many dollars does it charge for exactly 4 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q5/5

SET 85 • HSF-IF.C.7.b

BACKWARD 2

Included and excluded endpoints of intervals • Grade 6 • Expressions & Equations • 6.EE.B.8

Is 2 included in the interval $1 < t \leq 2$? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q1/5

Is 1 included in the interval $1 < t \leq 2$? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q2/5

True or false?

On a graph, a closed dot at the end of a segment means that endpoint is included.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q3/5

Which inequality describes all t greater than 2 and at most 3?

A. $2 < t < 3$

B. $2 \leq t < 3$

C. $2 \leq t \leq 3$

D. $2 < t \leq 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q4/5

A horizontal segment from (1, 6) to (2, 6) has an open circle at (1, 6) and a closed dot at (2, 6). Which inputs does it represent?

A. $1 < x \leq 2$

B. $1 \leq x < 2$

C. $1 < x < 2$

D. $1 \leq x \leq 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q5/5

SET 85 • HSF-IF.C.7.b

BACKWARD 3

Integers just below a number • Grade 6 • The Number System • 6.NS.C.7

What is the greatest integer that is less than or equal to 3.2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q1/5

What is the greatest integer that is less than or equal to -3.2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q2/5

True or false?

On a number line, -4 is to the left of -3.2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q3/5

Which integer is the greatest integer less than or equal to -0.1?

A. -0.1

B. 1

C. -1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q4/5

What is the least integer that is greater than or equal to 2.1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q5/5

SET 85 • HSF-IF.C.7.b

FORWARD 1

Write and interpret step-function models • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

A parking garage charges $3 for each hour or part of an hour, up to a maximum of $15 per day. $\lceil t \rceil$ is the least integer greater than or equal to t. Which rule gives the cost C(t) for $0 < t \leq 24$ hours?

A. $C(t) = 3\lceil t \rceil$ for $0 < t \leq 5$, and $C(t) = 15$ for $5 < t \leq 24$

B. $C(t) = 3\lceil t \rceil$ for every t with $0 < t \leq 24$

C. $C(t) = 3t$ for $0 < t \leq 5$, and $C(t) = 15$ for $5 < t \leq 24$

D. $C(t) = 3\lfloor t \rfloor$ for $0 < t \leq 5$, and $C(t) = 15$ for $5 < t \leq 24$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q1/5

Which function has a jump of 2 at every integer and steps of width 1?

A. $y = \lfloor 2x \rfloor$

B. $y = \lfloor x \rfloor + 2$

C. $y = \lfloor x/2 \rfloor$

D. $y = 2\lfloor x \rfloor$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q2/5

True or false?

For $f(x) = \lfloor x \rfloor$, the average rate of change over any interval inside one step, such as $1.2 \leq x \leq 1.8$, is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q3/5

For $f(x) = \lfloor x \rfloor$, how many solutions does $f(x) = 3$ have?

A. none

B. infinitely many: every x with $3 \leq x < 4$

C. exactly one: $x = 3$

D. exactly two: $x = 3$ and $x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q4/5

Postage is $0.70 for a letter up to 1 oz plus $0.25 for each additional ounce or part of an ounce. What is the postage, in dollars, for a 2.3-oz letter? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q5/5

SET 85 • HSF-IF.C.7.b

FORWARD 2

Transform step functions • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

How is the graph of $y = 3\lfloor x \rfloor$ related to the graph of $y = \lfloor x \rfloor$?

A. shifted right 3: each step moves 3 units to the right

B. compressed horizontally by 3: each step is 1/3 as wide

C. stretched vertically by 3: each step is 3 times as high

D. shifted up 3: each step is 3 units higher than before

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q1/5

How is the graph of $y = \lfloor x/2 \rfloor$ related to the graph of $y = \lfloor x \rfloor$?

A. stretched horizontally by 2: each step is 2 units wide

B. shifted right 2: each step moves 2 units to the right

C. stretched vertically by 2: each step is 2 times as high

D. compressed horizontally by 2: each step is 1/2 unit wide

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q2/5

True or false?

The graphs of $y = \lfloor x \rfloor + 1$ and $y = \lfloor x + 1 \rfloor$ are the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q3/5

Which function has steps of width 1 at heights ..., −2, 0, 2, 4, ..., with a closed dot at the left end of each step?

A. $y = 2\lceil x \rceil$

B. $y = \lfloor 2x \rfloor$

C. $y = \lfloor x \rfloor + 2$

D. $y = 2\lfloor x \rfloor$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q4/5

$g(x) = 2\lfloor x - 1 \rfloor + 3$, where $\lfloor x \rfloor$ is the greatest integer less than or equal to x. Find g(2.5).

g(2.5) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q5/5

SET 48 • HSF-IF.C.7.c • INTERPRETING FUNCTIONS

Graph polynomial functions showing zeros and end behavior

MAIN	Algebra II main question	**HSF-IF.C.7.c**
BACKWARD 1	Factor out common factors and special products	**HSA-SSE.A.2**
BACKWARD 2	Crossing and touching from multiplicity	**HSA-APR.B.3**
BACKWARD 3	Evaluate polynomials at test points	**HSF-IF.A.2**
FORWARD 1	Locate relative extrema of polynomial graphs	**HSF-IF.C.7.c**
FORWARD 2	Interpret polynomial graphs in context	**HSF-IF.B.4**

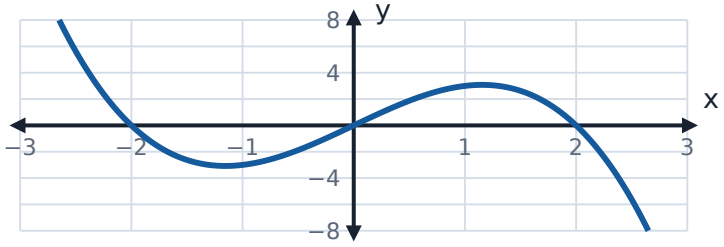
SET 48 • HSF-IF.C.7.c

MAIN

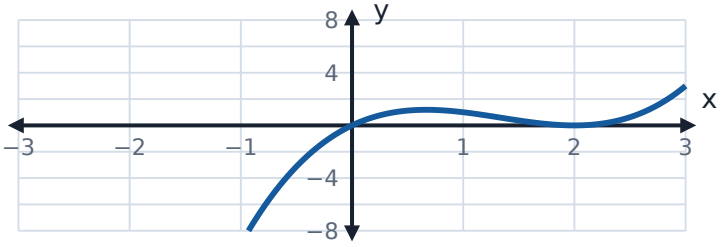
Algebra II main question • High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c

Which graph shows $y = x^3 - 4x$?

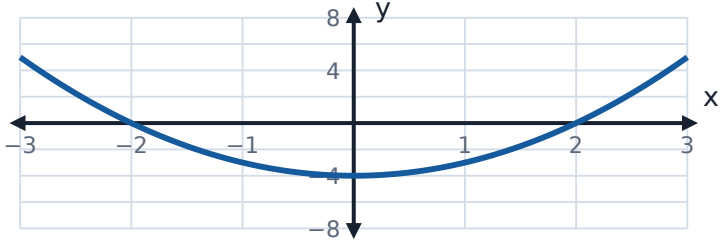
A



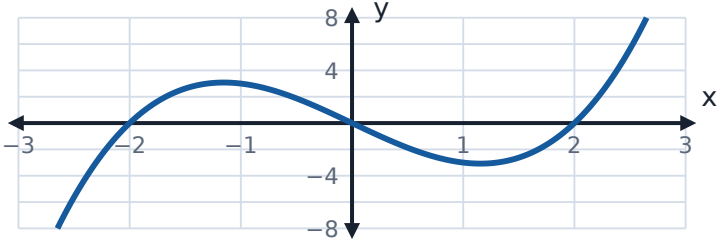
B



C



D



A. Graph A

B. Graph B

C. Graph C

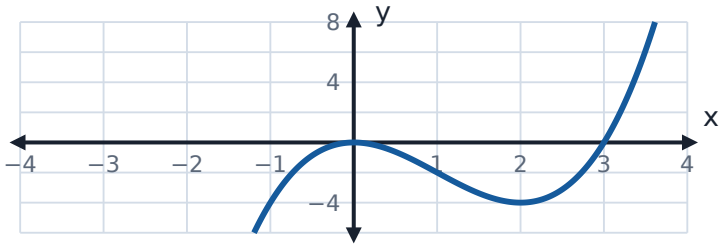
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-M-Q1**

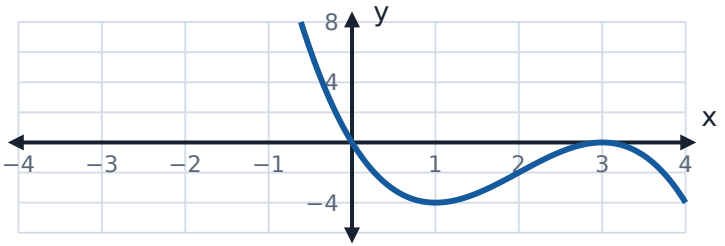
High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q1/5

Which graph shows $y = -x^3 + 3x^2$?

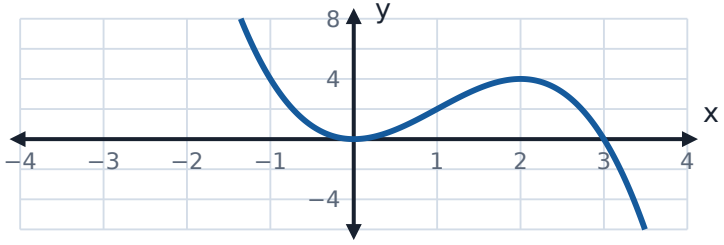
A



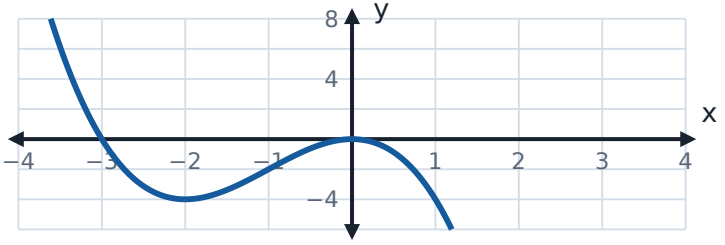
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

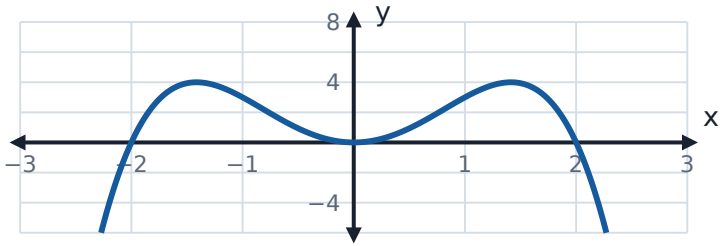
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

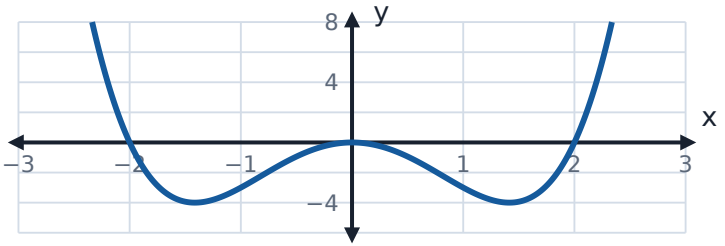
High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q2/5

Which graph shows $y = x^4 - 4x^2$?

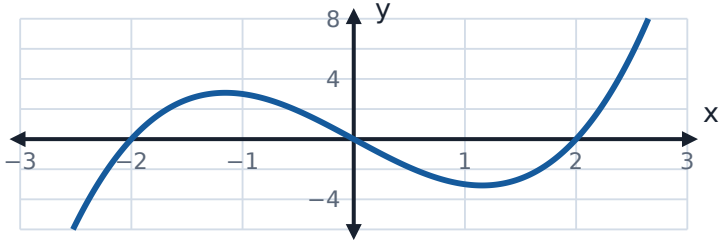
A



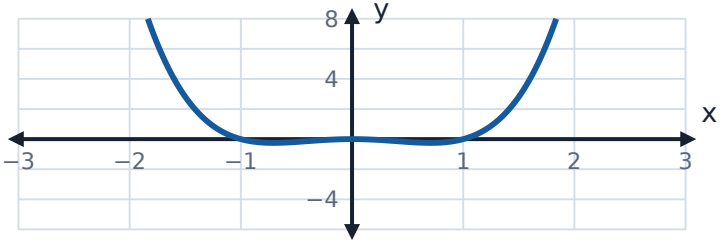
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

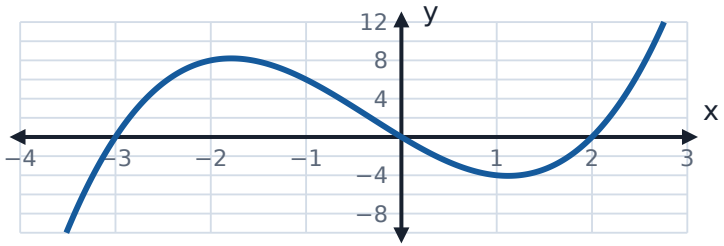
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

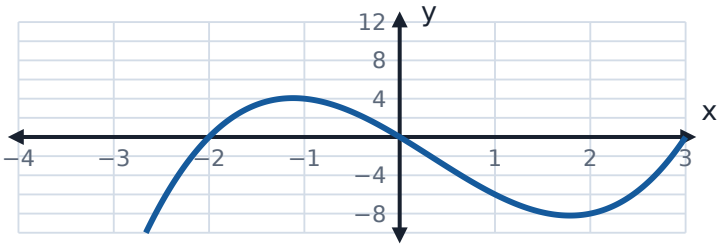
High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q3/5

Which graph shows $y = x^3 + x^2 - 6x$?

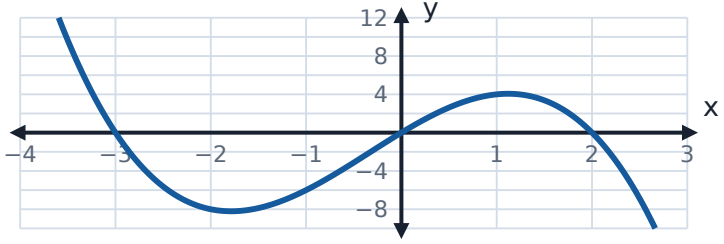
A



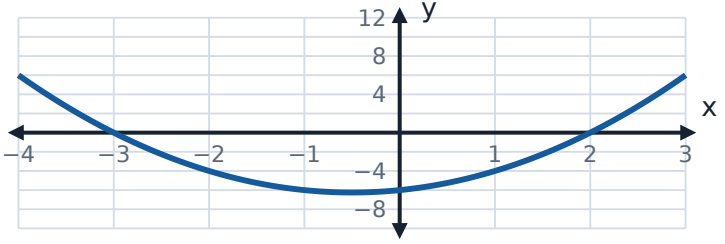
B



C



D



A. Graph A

B. Graph B

C. Graph C

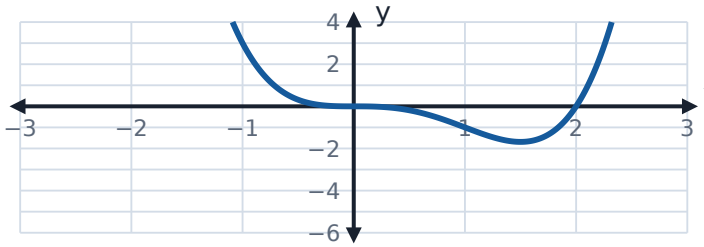
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-M-Q4**

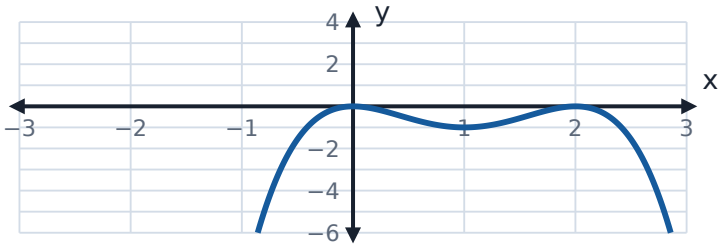
High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q4/5

Which graph shows $y = -x^4 + 2x^3$?

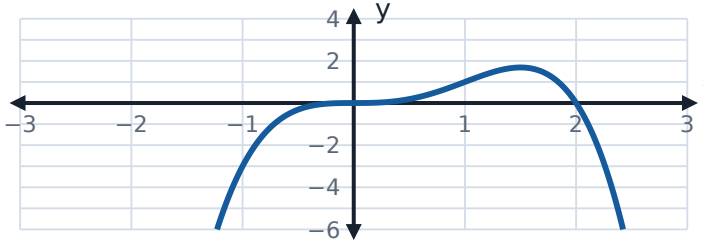
A



B



C



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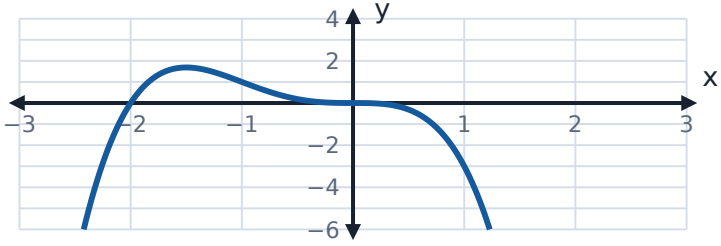


A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q5/5

SET 48 • HSF-IF.C.7.c

BACKWARD 1

Factor out common factors and special products • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^3 - 4x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q1/5

Factor $-x^3 + 3x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q2/5

True or false?

$$x^4 - 4x^2 = x^2(x^2 - 4) = x^2(x - 2)(x + 2).$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q3/5

$x^3 + x^2 - 6x$ equals

A. $(x + 3)(x - 2)$

B. $x(x + 3)(x - 2)$

C. $x(x - 3)(x + 2)$

D. $x(x + 6)(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q4/5

Factor $-x^4 + 2x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSF-IF.C.7.c

BACKWARD 2

Crossing and touching from multiplicity • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Does $y = x^2(x - 3)$ cross or touch the x-axis at 0?
Answer Crosses or Touches.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q1/5

Does the graph of $y = x^3(x - 2)$ cross the x-axis at $x = 0$?

A. Yes, as a straight line (multiplicity 1).

B. Yes, and it flattens there (odd multiplicity 3).

C. No, $x = 0$ is not a zero.

D. No, it only touches there (even multiplicity).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q2/5

True or false?

A zero of odd multiplicity is a crossing point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q3/5

The graph of $y = (x + 2)^2(x - 1)$ at $x = -2$

A. crosses

B. touches

C. has an asymptote

D. is undefined

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q4/5

Which describes the end behavior of $y = -x^4 + 2x^3$?

A. Rises at both ends

B. Rises to the left and falls to the right

C. Falls to the left and rises to the right

D. Falls at both ends

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-B2-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSF-IF.C.7.c

BACKWARD 3

Evaluate polynomials at test points • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Evaluate $x^3 - 4x$ at $x = -1$ and at $x = 1$.

At $x = -1$: _____

At $x = 1$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q1/5

Evaluate $-x^3 + 3x^2$ at $x = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q2/5

True or false?

$x^4 - 4x^2$ at $x = 1$ equals 3.

A. True

B. False

$x^3 + x^2 - 6x$ at $x = -2$ equals

A. 8

B. 0

C. -8

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q4/5

Evaluate $-x^4 + 2x^3$ at $x = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q5/5

SET 48 • HSF-IF.C.7.c

FORWARD 1

Locate relative extrema of polynomial graphs • High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c

Use a table with steps of 0.1 (or technology) to estimate the relative maximum of $y = x^3 - 4x$. Give the x-coordinate to the nearest tenth and the y-value to the nearest hundredth. A calculator may be used.

$x \approx$ _____

$y \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q1/5

Use technology or a table to estimate the relative minimum of $y = x^3 + x^2 - 6x$ on $0 < x < 2$. Give the x-coordinate to the nearest tenth and the y-value to the nearest hundredth. A calculator may be used.

$x \approx$ _____

$y \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q2/5

True or false?

A polynomial of degree 4 always has exactly 3 turning points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q3/5

How many turning points can $y = x^4 - 4x^2$ have?

A. 1

B. 4

C. 3

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q4/5

Why does the graph of $y = x^3 - 4x$ have a turning point between $x = 0$ and $x = 2$?

A. Because y is positive between 0 and 2 (for example $y(1) = 3$), so the graph must come back down to 0 at $x = 2$.

B. Because the graph crosses the x-axis at $x = 1$, halfway between 0 and 2, and a graph turns at every crossing.

C. y is 0 at $x = 0$ and at $x = 2$ and negative between them (for example $y(1) = -3$), so the graph must turn back up somewhere between them.

D. Because every cubic with a positive leading coefficient turns at $x = 1$, which lies between 0 and 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSF-IF.C.7.c

FORWARD 2

Interpret polynomial graphs in context • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

A box's volume is $V(x) = x(12 - 2x)^2$ in³, where x is the side (in) of the squares cut from a 12-in square sheet. Which interval of x is meaningful, and what is the maximum volume there (it occurs at x = 2)?

Meaningful interval: _____

Maximum volume (in³): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q1/5

A company's profit (thousands of dollars) is $P(x) = -x^3 + 9x^2 - 15x$ for x hundred items, $0 \le x \le 8$. For which x is the profit positive? Give the endpoints to the nearest hundredth.

Profit positive for: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q2/5

True or false?

A zero of a profit model is a break-even point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q3/5

A population model $P(t) = -t^3 + 6t^2 + 15t + 100$ rises then falls. The relative maximum means

A. the largest population before decline

B. the starting population

C. the end behavior

D. the time the population is 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q4/5

Why does the end behavior of $V(x) = x(12 - 2x)^2$ not matter for the box made from a 12-in sheet?

A. The context uses only $0 < x < 6$, so the behavior for large $|x|$ is never used.

B. Because $V(x) = x(12 - 2x)^2$ has no end behavior at all.

C. Because the volume of the box is always 128 cubic inches.

D. Because x must be negative for the box, so large positive x is never used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q5/5

SET 49 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph exponential functions with intercepts, asymptote, and end behavior

MAIN	Algebra II main question	**HSF-IF.C.7.e**
BACKWARD 1	Graph basic exponential functions	**HSF-IF.C.7.e**
BACKWARD 2	Zero and negative exponents	**8.EE.A.1**
BACKWARD 3	Vertical and horizontal shifts and reflections	**HSF-BF.B.3**
FORWARD 1	Compare and describe exponential graphs	**HSF-IF.C.7.e**
FORWARD 2	Graph the inverse of an exponential function	**HSF-BF.B.4.a**

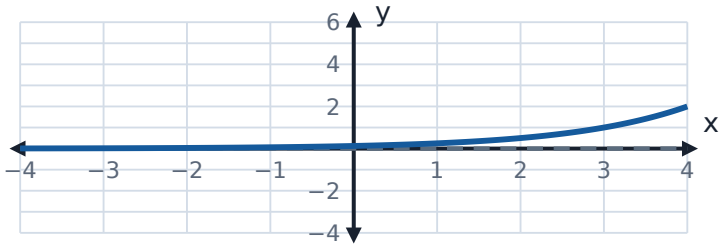
SET 49 • HSF-IF.C.7.e

MAIN

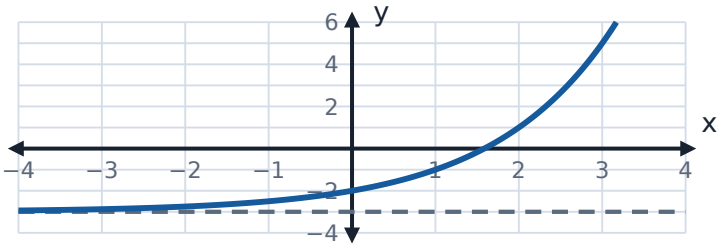
Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Which graph shows $y = 2^x - 3$?

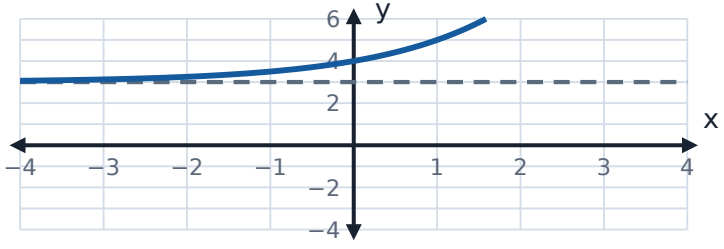
A



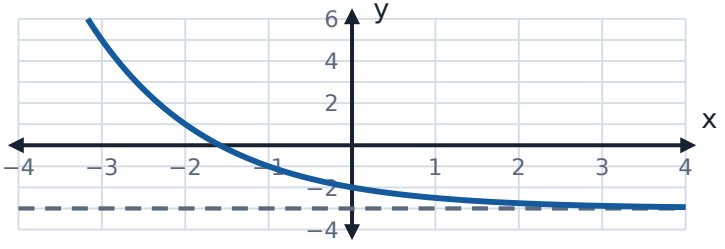
B



C



D



A. Graph A

B. Graph B

C. Graph C

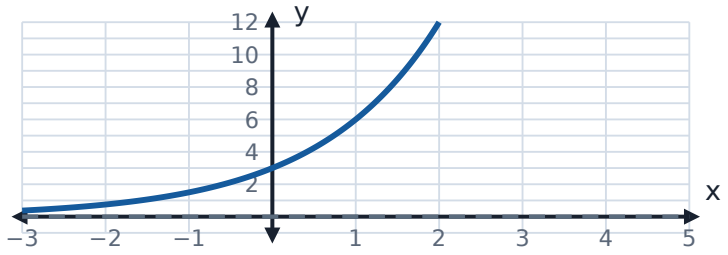
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-M-Q1**

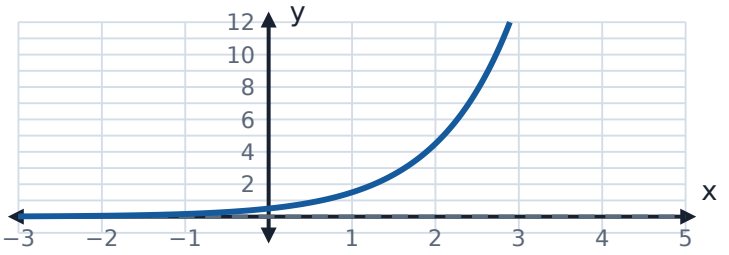
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q1/5

Which graph shows $y = 3(0.5)^x$?

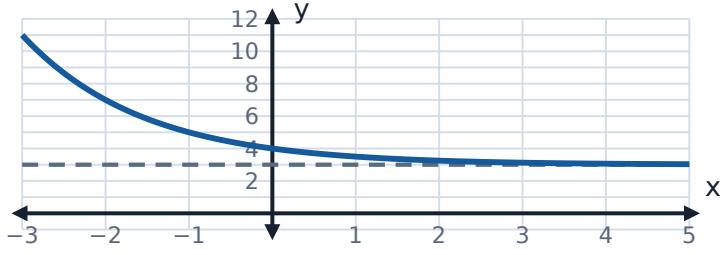
A



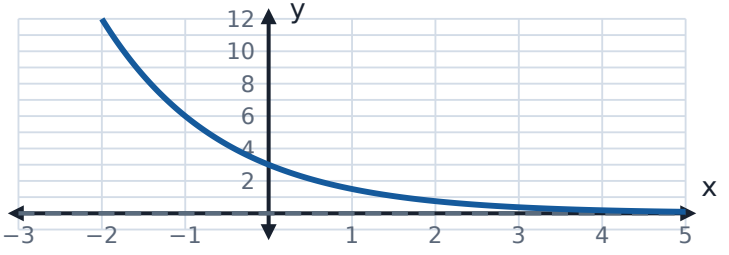
B



C



D



A. Graph A

B. Graph B

C. Graph C

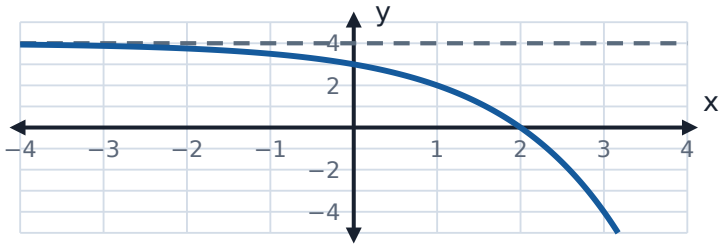
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-M-Q2**

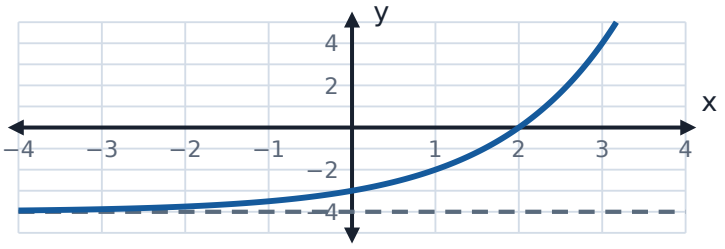
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q2/5

Which graph shows $y = -2^x + 4$?

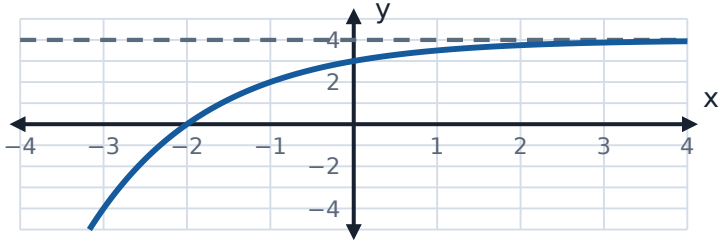
A



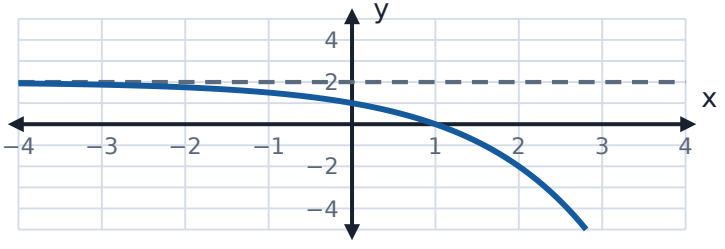
B



C



D



A. Graph A

B. Graph B

C. Graph C

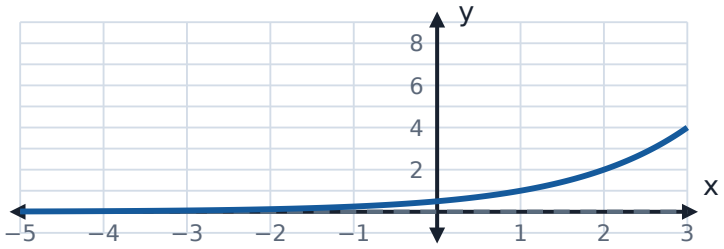
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-M-Q3**

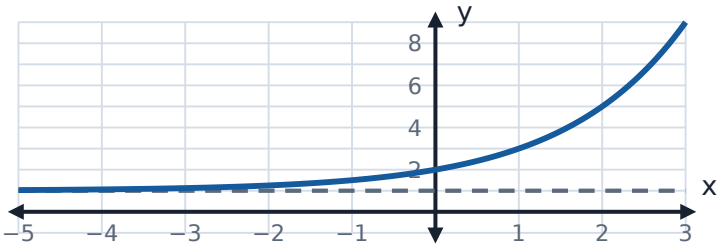
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q3/5

Which graph shows $y = 2^{x+1}$?

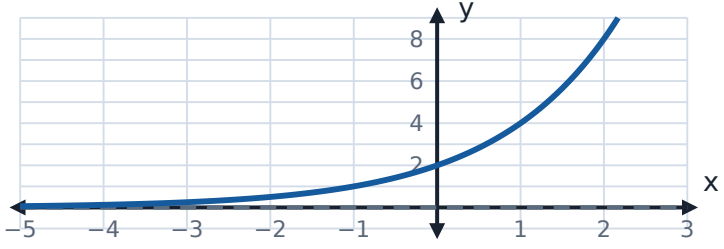
A



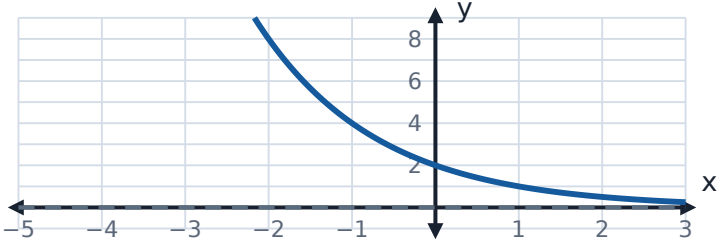
B



C



D



A. Graph A

B. Graph B

C. Graph C

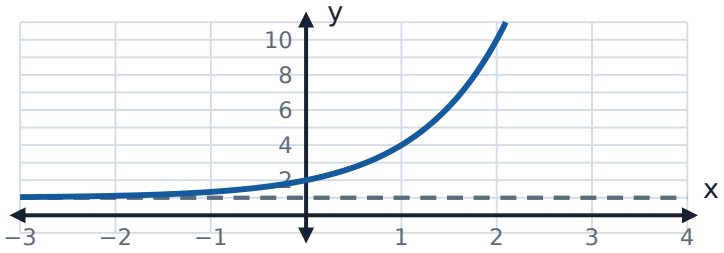
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-M-Q4**

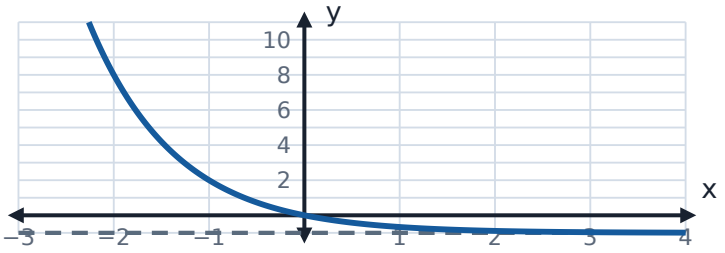
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q4/5

Which graph shows $y = (1/3)^x + 1$?

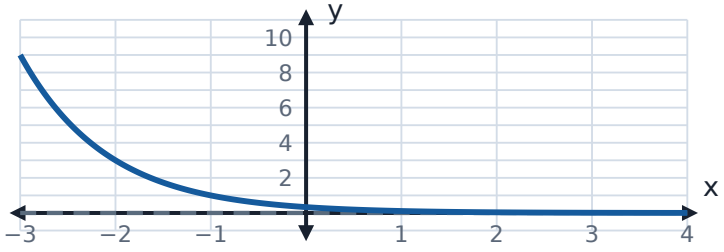
A



B



C



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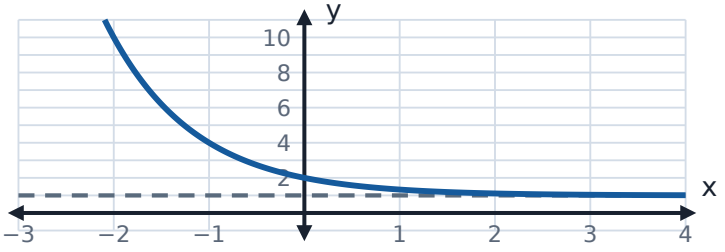


A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q5/5

SET 49 • HSF-IF.C.7.e

BACKWARD 1

Graph basic exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

Find $y = 2^x$ at $x = -2, -1, 0, 1$, and 2, in that order.

y-values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q1/5

What is the y-intercept of $y = 3(0.5)^x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q2/5

True or false?

$y = 2^x$ has the horizontal asymptote $y = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q3/5

$y = (0.5)^x$ is

A. linear

B. increasing

C. constant

D. decreasing

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q4/5

Does the graph of $y = 2^x$ have an x-intercept?

A. Yes: at $x = 0$

B. No: $2^x > 0$ for every x, so y is never 0

C. Yes: at $x = -1$

D. No: the graph does not cross the y-axis either

SET 49 • HSF-IF.C.7.e

BACKWARD 2

Zero and negative exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Evaluate 2^{-2}.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q1/5

Evaluate $(1/3)^{-2}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q2/5

True or false?

$b^0 = 0$ for $b \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q3/5

$3(0.5)^{-1}$ equals

A. 6

B. 3

C. -1.5

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q4/5

Evaluate $2^{-1} - 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSF-IF.C.7.e

BACKWARD 3

Vertical and horizontal shifts and reflections • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How is the graph of $y = 2^x - 3$ related to the graph of $y = 2^x$?

A. Shifted left 3

B. Shifted up 3

C. Shifted right 3

D. Shifted down 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q1/5

How is the graph of $y = 2^{x+1}$ related to the graph of $y = 2^{x}$?

A. Shifted left 1

B. Shifted up 1

C. Shifted down 1

D. Shifted right 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q2/5

True or false?

A vertical shift moves the horizontal asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q3/5

$y = -2^x + 4$ has asymptote

A. $y = 4$

B. $y = 0$

C. $y = -4$

D. $x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q4/5

Give the asymptote of $y = (1/3)^x + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q5/5

SET 49 • HSF-IF.C.7.e

FORWARD 1

Compare and describe exponential graphs • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Why are the graphs of $y = 2^x$ and $y = (1/2)^x$ reflections of each other over the y-axis?

A. Because 1/2 is less than 1, so $y = (1/2)^x$ decreases while $y = 2^x$ increases.

B. $(1/2)^x = 2^{-x}$, so $y = (1/2)^x$ is $y = 2^x$ with x replaced by $-x$.

C. $(1/2)^x = -2^x$, so $y = (1/2)^x$ is $y = 2^x$ with every output made negative.

D. Because both graphs pass through (0, 1), which lies on the y-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q1/5

$2^{x+1} = 2 \cdot 2^x$. What does this say about two transformations of $y = 2^x$?

A. A shift right by 1 gives the same graph as a vertical stretch by 2.

B. A shift left by 1 gives the same graph as a shift up by 2.

C. A shift left by 1 gives the same graph as a vertical stretch by 2.

D. They are never the same graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q2/5

True or false?

$y = 5 \cdot 3^x$ and $y = 3^x$ have different horizontal asymptotes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q3/5

Which exponential graph passes through (0, 4) and (1, 2)?

A. $y = 2^x + 3$

B. $y = 4 \cdot 2^x$

C. $y = 0.5 \cdot 4^x$

D. $y = 4(0.5)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q4/5

Which exponential function is increasing, has asymptote $y = -1$, has y-intercept 2, and passes through (1, 5)?

A. $y = 3(1/2)^x - 1$

B. $y = 3 \cdot 3^x - 1$

C. $y = 3 \cdot 2^x - 1$

D. $y = 2 \cdot 2^x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSF-IF.C.7.e

FORWARD 2

Graph the inverse of an exponential function • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

Which is the inverse of $f(x) = 2^x$?

A. $f^{-1}(x) = \log_2 x$

B. $f^{-1}(x) = x^2$

C. $f^{-1}(x) = (1/2)^x$

D. $f^{-1}(x) = x/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q1/5

The point (3, 8) is on $y = 2^x$. Which point is on its inverse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q2/5

True or false?

The inverse of $y = 2^x$ has the vertical asymptote $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q3/5

The inverse of $y = 2^x - 3$ is

A. $y = \log_2(x - 3)$

B. $y = \log_2 x - 3$

C. $y = 2^{x+3}$

D. $y = \log_2(x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q4/5

Which gives the domain and range of $y = \log_2 x$?

A. Domain: all real numbers; range: $y > 0$

B. Domain: $x \geq 0$; range: $y \geq 0$

C. Domain: $x > 1$; range: $y > 0$

D. Domain: $x > 0$; range: all real numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q5/5

SET 50 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph logarithmic functions with intercept, asymptote, and end behavior

MAIN	Algebra II main question	**HSF-IF.C.7.e**
BACKWARD 1	Graph exponential functions	**HSF-IF.C.7.e**
BACKWARD 2	Find inverses by swapping inputs and outputs	**HSF-BF.B.4.a**
BACKWARD 3	Shift and reflect graphs	**HSF-BF.B.3**
FORWARD 1	Graph logarithms with reflections that change the domain	**HSF-IF.C.7.e**
FORWARD 2	Solve equations with logarithms	**HSF-LE.A.4**

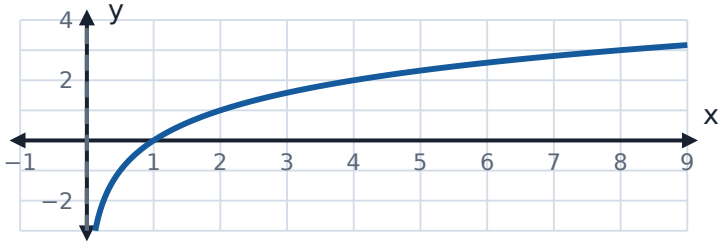
SET 50 • HSF-IF.C.7.e

MAIN

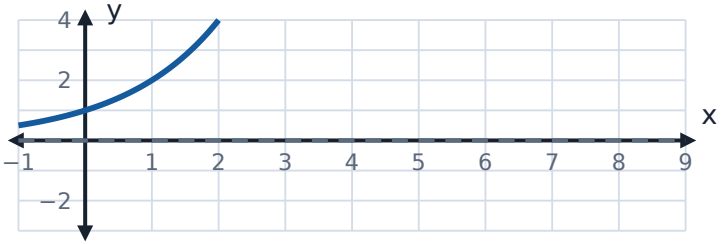
Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Which graph shows $y = \log_2 x$?

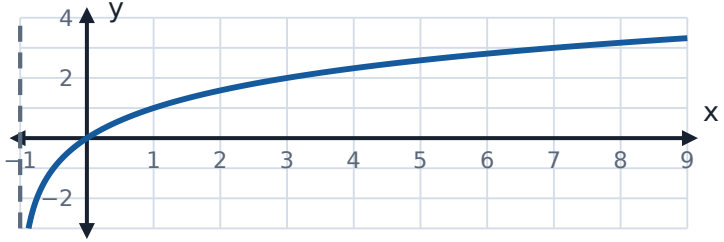
A



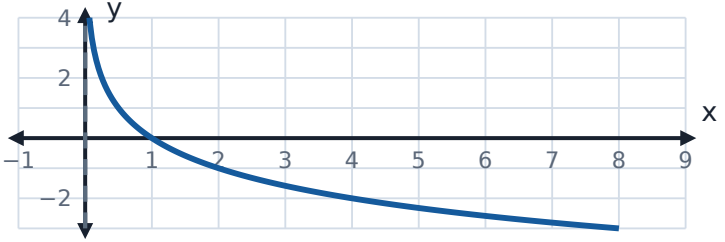
B



C



D



A. Graph A

B. Graph B

C. Graph C

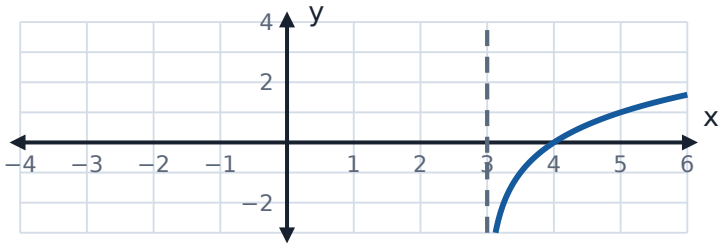
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-M-Q1**

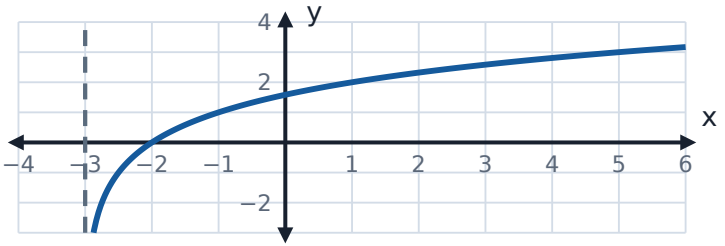
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q1/5

Which graph shows $y = \log_2(x + 3)$?

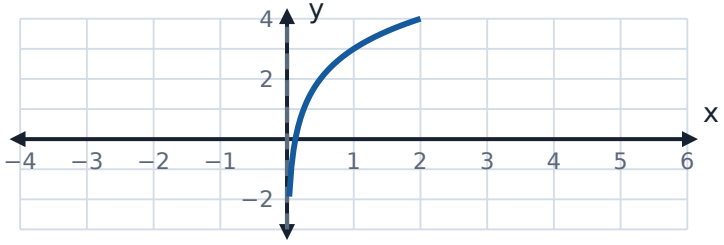
A



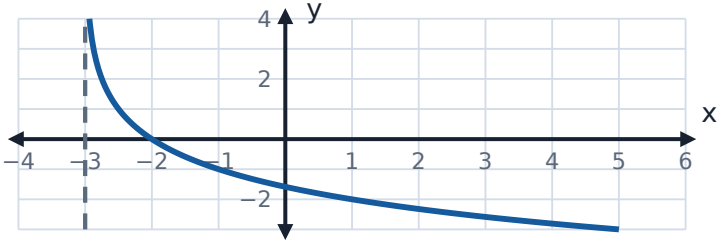
B



C



D



A. Graph A

B. Graph B

C. Graph C

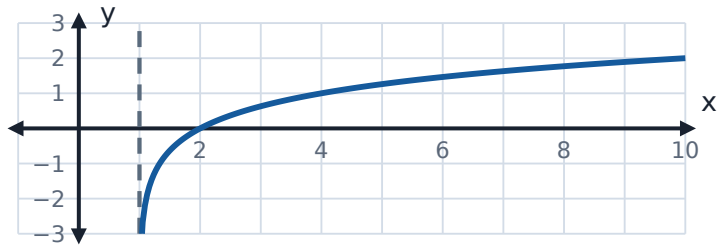
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-M-Q2**

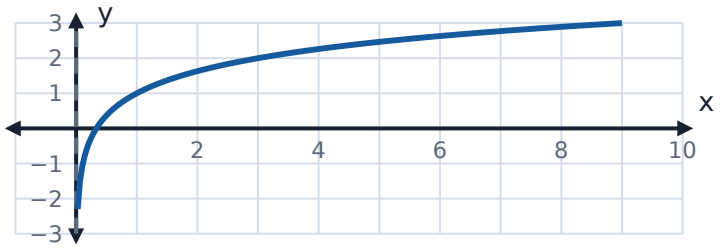
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q2/5

Which graph shows $y = \log_3 x - 1$?

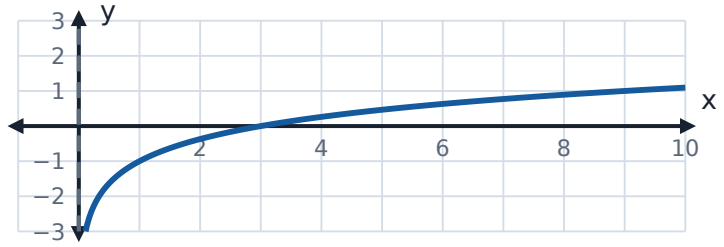
A



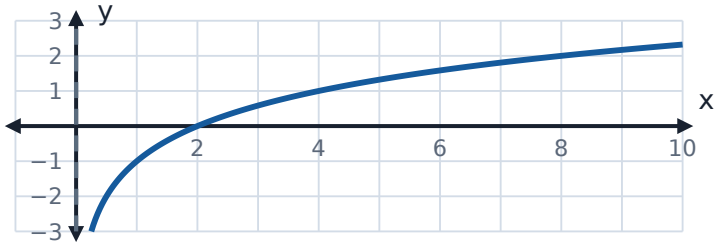
B



C



D



A. Graph A

B. Graph B

C. Graph C

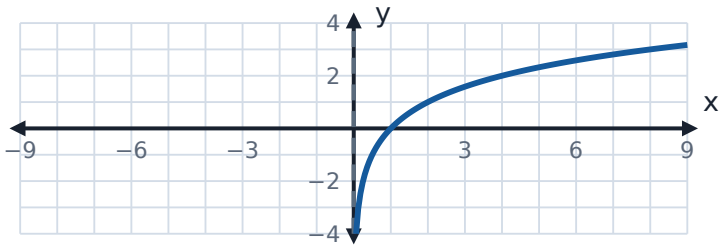
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-M-Q3**

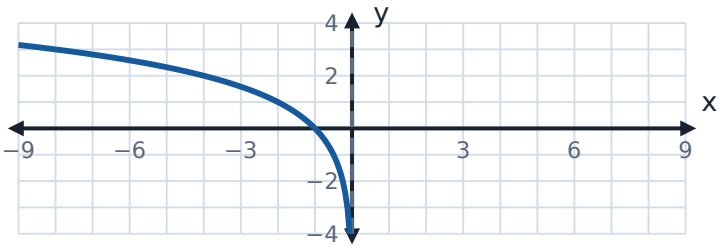
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q3/5

Which graph shows $y = -\log_2 x$?

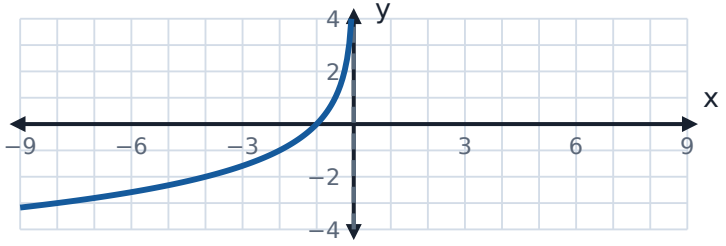
A



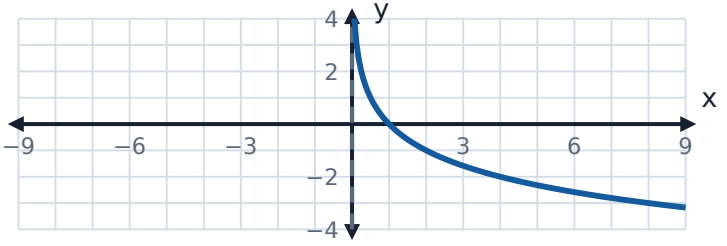
B



C



D



A. Graph A

B. Graph B

C. Graph C

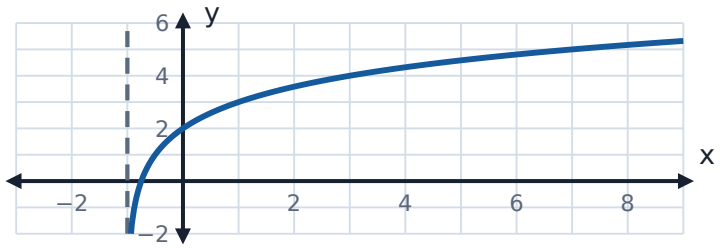
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-M-Q4**

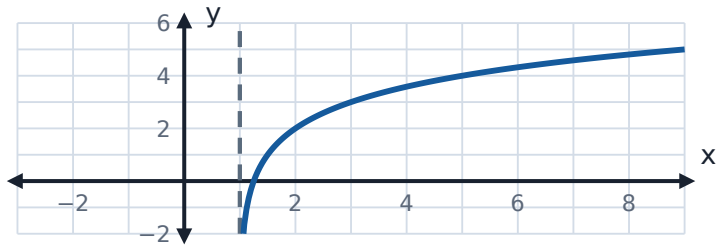
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q4/5

Which graph shows $y = \log_2(x - 1) + 2$?

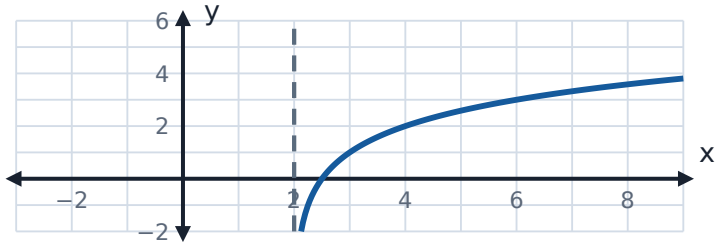
A



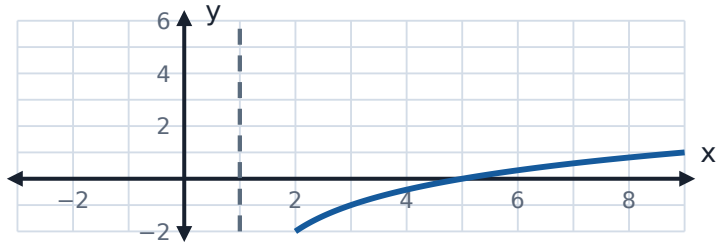
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q5/5

SET 50 • HSF-IF.C.7.e

BACKWARD 1

Graph exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

Find the y-values of $y = 2^x$ at $x = 0$, 1, and 2, in that order, to get three points on the graph.

y-values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q1/5

Give the asymptote of $y = 2^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q2/5

True or false?

$y = 3^x$ passes through (3, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q3/5

$y = 2^x$ passes through

A. (2, 1)

B. (−1, −2)

C. (1/2, −1)

D. (−1, 1/2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q4/5

Give the range of $y = 3^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSF-IF.C.7.e

BACKWARD 2

Find inverses by swapping inputs and outputs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

(2, 4) is on $y = 2^x$. Which point is on its inverse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q1/5

Which is the inverse of $y = 3^x$?

A. $y = x^3$

B. $y = \log_3 x$

C. $y = x/3$

D. $y = 3^{-x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q2/5

True or false?

Swapping x and y in $y = 2^x$ gives $x = 2^y$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q3/5

Which point is on $y = \log_2 x$?

A. (3, 8)

B. (2, 4)

C. (8, 3)

D. (0, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q4/5

What is $\log_2 16$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q5/5

SET 50 • HSF-IF.C.7.e

BACKWARD 3

Shift and reflect graphs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How is the graph of $y = \log_2(x + 3)$ related to the graph of $y = \log_2 x$?

A. Shifted right 3

B. Shifted down 3

C. Shifted left 3

D. Shifted up 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q1/5

How is the graph of $y = -\log_2 x$ related to the graph of $y = \log_2 x$?

A. Reflected over the line $y = x$

B. Reflected over the y-axis

C. Reflected over the x-axis

D. Shifted down 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q2/5

True or false?

A horizontal shift moves the vertical asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q3/5

$y = \log_2(x - 1) + 2$ has asymptote

A. $x = -1$

B. $y = 2$

C. $x = 1$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q4/5

Where does (1, 0) of $y = \log_3 x$ move under $y = \log_3 x - 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q5/5

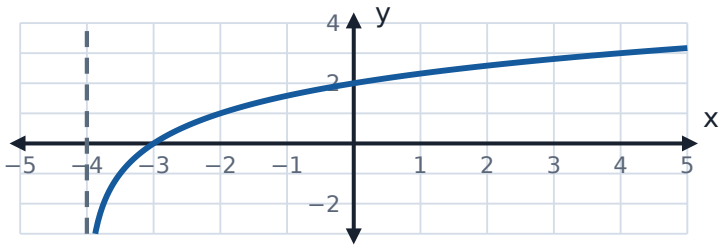
SET 50 • HSF-IF.C.7.e

FORWARD 1

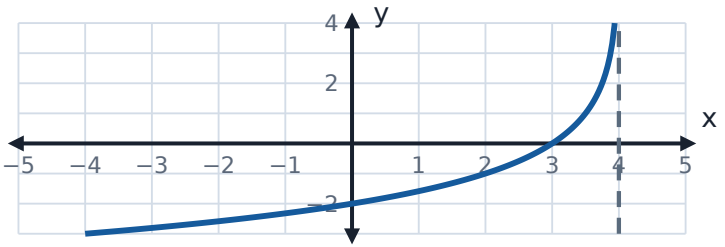
Graph logarithms with reflections that change the domain • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Which graph shows $y = \log_2(4 - x)$?

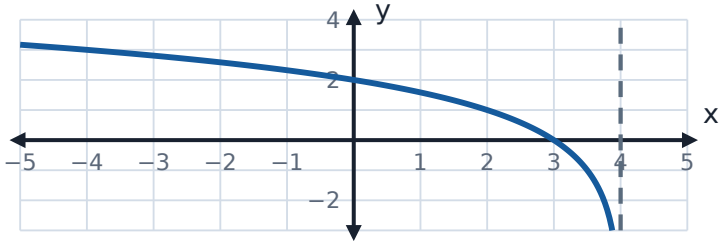
A



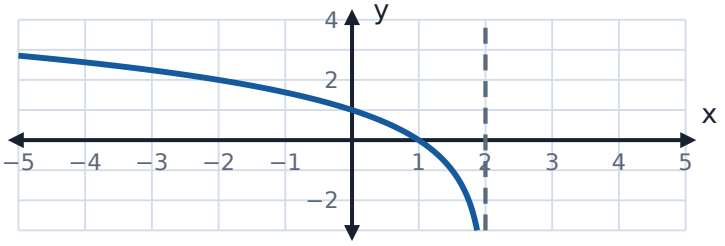
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q1/5

Why does the graph of $y = \log(x^2 - 1)$ have two separate branches?

A. Every logarithm graph has two branches, one on each side of its asymptote.

B. Its domain is $x < -1$ or $x > 1$: $x^2 - 1 > 0$ on two separate intervals.

C. $\log(x^2 - 1)$ is undefined at $x = 0$ only, which splits the graph in two.

D. Its domain is $-1 < x < 1$, an interval split in two at $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q2/5

True or false?

$y = \log_2(2x)$ is $y = \log_2 x$ shifted up 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q3/5

The x-intercept of $y = \log_2(x + 3)$ is

A. -2

B. 1

C. -3

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q4/5

Find the x-intercept of $y = \log_2(x - 1) + 2$ exactly.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSF-IF.C.7.e

FORWARD 2

Solve equations with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $2^x = 12$. Give x to the nearest hundredth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q1/5

Solve $5 \cdot 10^{x} = 40$. Give x to the nearest thousandth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q2/5

True or false?

$\log_2 12$ is between 4 and 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q3/5

The solution of $3^x = 7$ is

A. log 7/log 3

B. log 3/log 7

C. 7/3

D. log 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q4/5

Use a graphing calculator (or the logarithm key) to estimate $\log_2 6$ to the nearest tenth. A calculator may be used.

$\log_2 6 \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q5/5

SET 51 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph trigonometric functions showing period, midline, and amplitude

MAIN	Algebra II main question	**HSF-IF.C.7.e**
BACKWARD 1	Unit-circle values at quarter turns	**HSF-TF.A.2**
BACKWARD 2	Read amplitude, period, and midline from a formula	**HSF-TF.B.5**
BACKWARD 3	Vertical stretches, shifts, and reflections	**HSF-BF.B.3**
FORWARD 1	Graph shifted sinusoids and the tangent function	**HSF-IF.C.7.e**
FORWARD 2	Horizontal stretches f(kx) of trigonometric graphs	**HSF-BF.B.3**

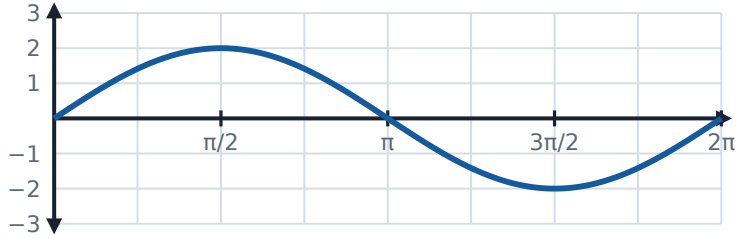
SET 51 • HSF-IF.C.7.e

MAIN

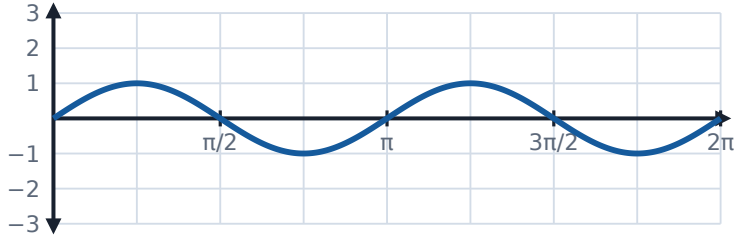
Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Which graph shows $y = 2\sin x$ on $0 \le x \le 2\pi$?

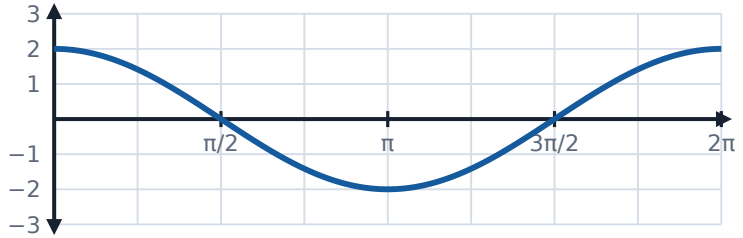
A



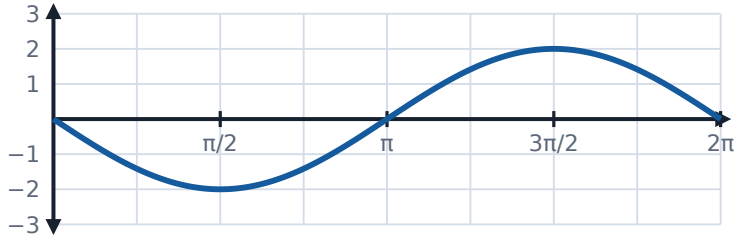
B



C



D



A. Graph A

B. Graph B

C. Graph C

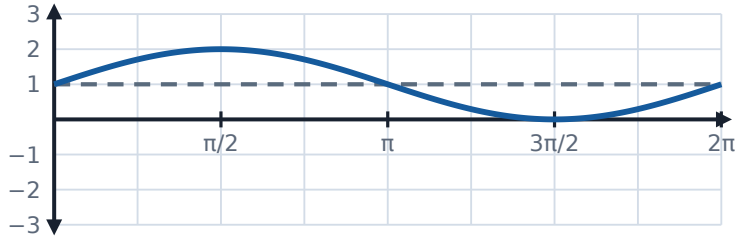
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-M-Q1**

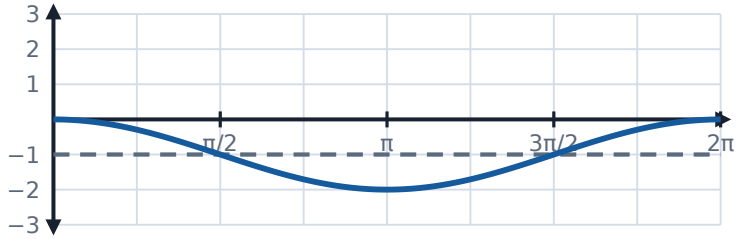
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q1/5

Which graph shows $y = \cos x + 1$ on $0 \le x \le 2\pi$?

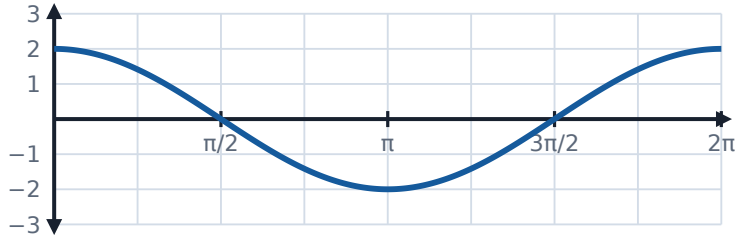
A



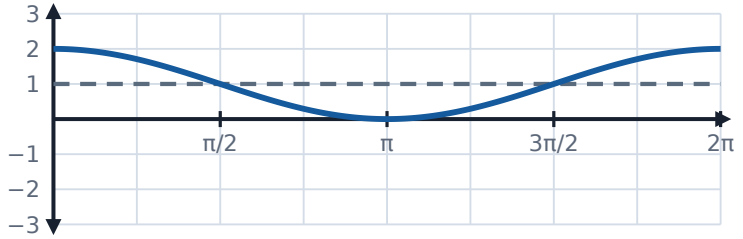
B



C



D



A. Graph A

B. Graph B

C. Graph C

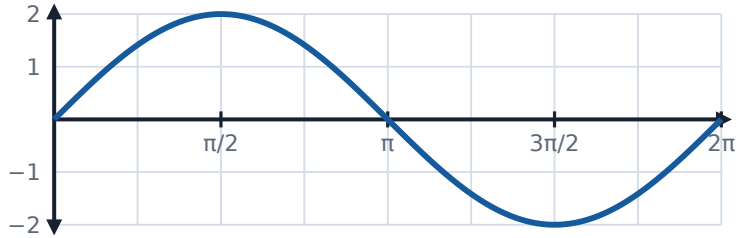
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-M-Q2**

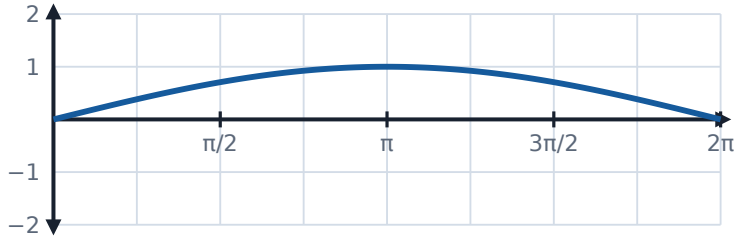
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q2/5

Which graph shows $y = \sin(2x)$ on $0 \le x \le 2\pi$?

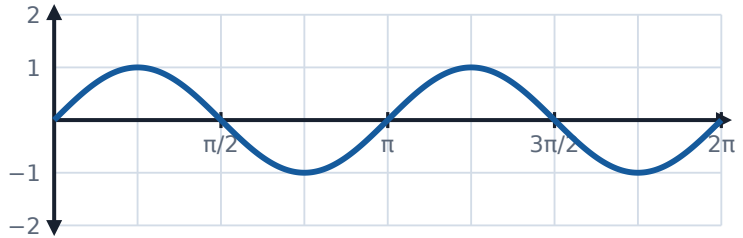
A



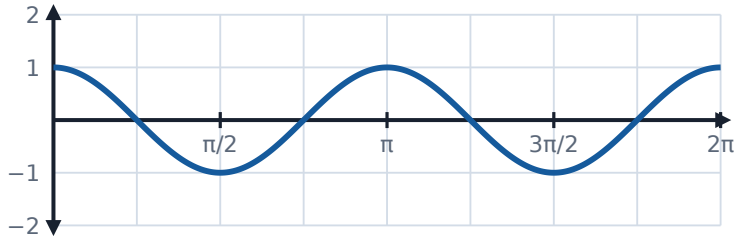
B



C



D



A. Graph A

B. Graph B

C. Graph C

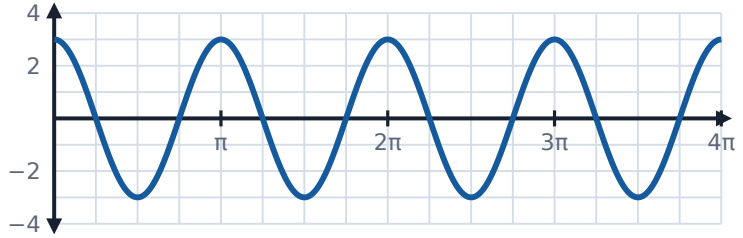
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-M-Q3**

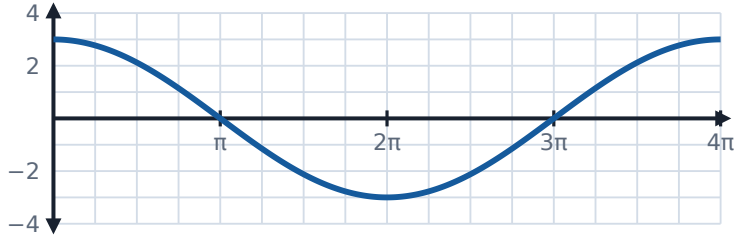
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q3/5

Which graph shows $y = 3\cos(x/2)$ on $0 \le x \le 4\pi$?

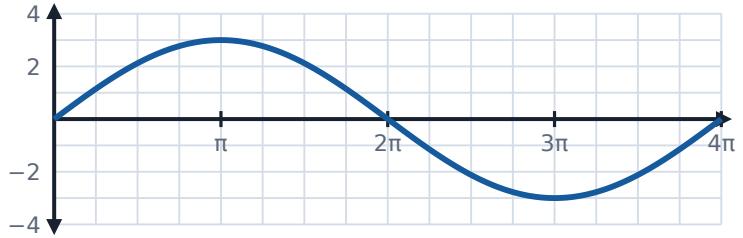
A



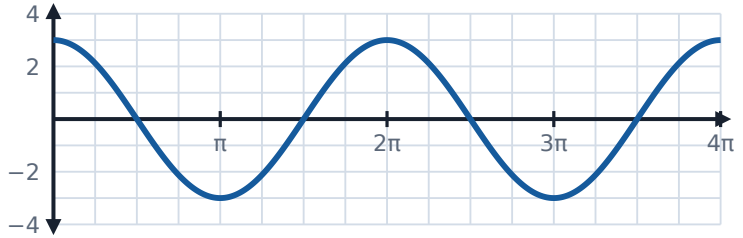
B



C



D



A. Graph A

B. Graph B

C. Graph C

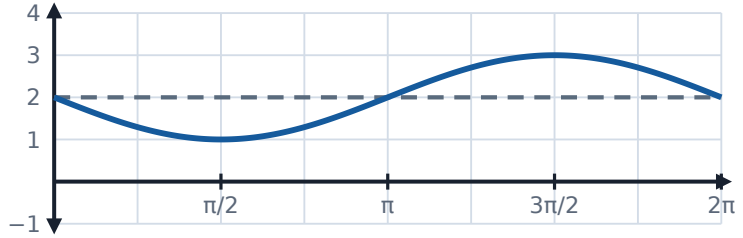
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-M-Q4**

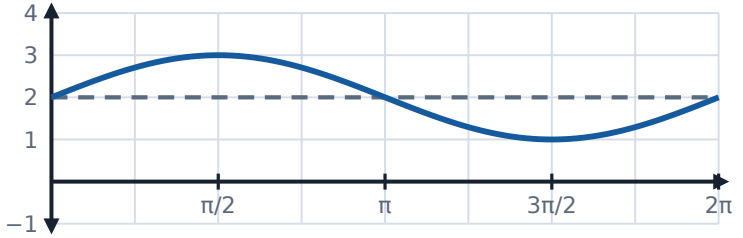
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q4/5

Which graph shows $y = -\sin x + 2$ on $0 \le x \le 2\pi$?

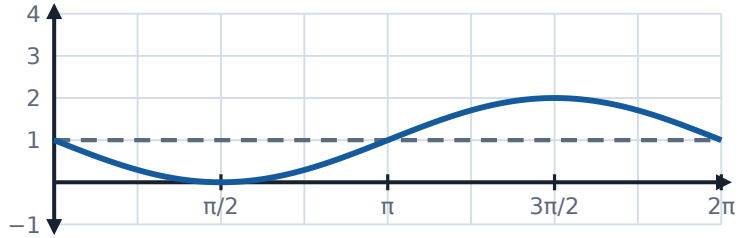
A



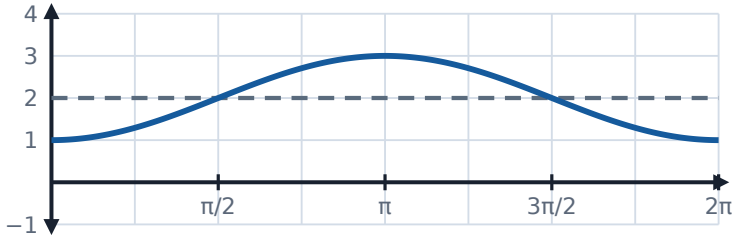
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q5/5

SET 51 • HSF-IF.C.7.e

BACKWARD 1

Unit-circle values at quarter turns • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Give sin x at x = 0, π/2, π, 3π/2, 2π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q1/5

Give cos x at x = 0, π/2, π, 3π/2, 2π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q2/5

True or false?

$\sin(3\pi/2) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q3/5

cos(π) equals

A. π

B. 1

C. −1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q4/5

Give $\sin(\pi/4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSF-IF.C.7.e

BACKWARD 2

Read amplitude, period, and midline from a formula • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

Give the amplitude and the period of $y = 3\cos(x/2)$.

Amplitude: _____

Period: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q1/5

Give the midline of $y = -\sin x + 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q2/5

True or false?

$y = \sin(2x)$ has period 4π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q3/5

$y = 2\sin x$ has amplitude

A. 1

B. 2

C. π

D. 2π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q4/5

Give the period of $y = \cos(3x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q5/5

SET 51 • HSF-IF.C.7.e

BACKWARD 3

Vertical stretches, shifts, and reflections • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How does the graph of $y = \cos x + 1$ compare with the graph of $y = \cos x$?

A. Shifted left 1

B. Stretched vertically by 1

C. Shifted down 1

D. Shifted up 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q1/5

How does the graph of $y = -\sin x$ compare with the graph of $y = \sin x$?

A. Reflected over the x-axis

B. Stretched vertically by 2

C. Shifted down 1

D. Unchanged

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q2/5

True or false?

$y = 2\sin x$ has the same zeros as $y = \sin x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q3/5

The maximum of $y = -\sin x + 2$ is

A. 1

B. 2

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q4/5

What is the minimum of $y = \cos x + 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q5/5

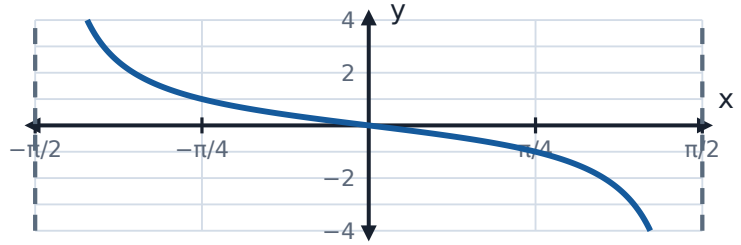
SET 51 • HSF-IF.C.7.e

FORWARD 1

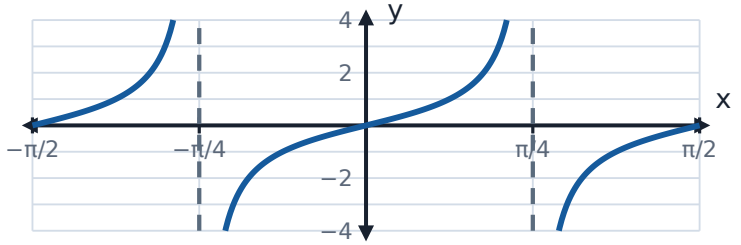
Graph shifted sinusoids and the tangent function • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Which graph shows $y = \tan x$ for $-\pi/2 < x < \pi/2$?

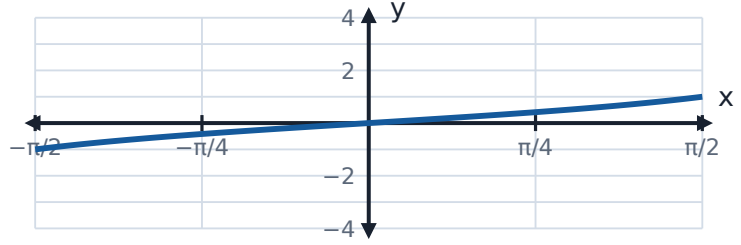
A



B



C



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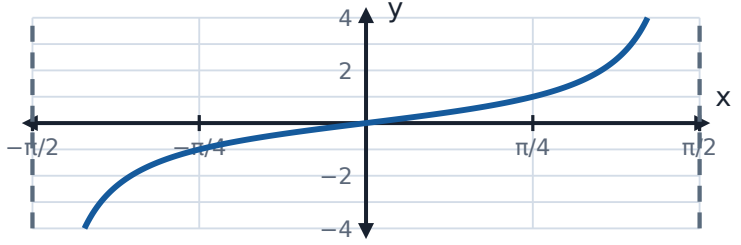


A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q1/5

How does $y = \sin(x - \pi/2)$ compare with $y = \sin x$, and which simpler formula describes it?

A. It is shifted left $\pi/2$, and it equals $\cos x$.

B. It is shifted right $\pi/2$, and it equals $-\cos x$.

C. It is shifted down $\pi/2$, and it equals $\sin x - \pi/2$.

D. It is shifted right $\pi/2$, and it equals $\cos x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q2/5

True or false?

The period of $y = \tan x$ is 2π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q3/5

The first maximum of $y = \cos(x - \pi/3)$ for $x \geq 0$ is at

A. $x = 2\pi/3$

B. $x = \pi/3$

C. $x = 0$

D. $x = -\pi/3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q4/5

Give the x-values of the maximum points of $y = \sin(x + \pi/4)$ on $0 \leq x < 2\pi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSF-IF.C.7.e

FORWARD 2

Horizontal stretches f(kx) of trigonometric graphs • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

How is the graph of $y = \sin(2x)$ obtained from the graph of $y = \sin x$?

A. Horizontal compression by a factor of 1/2

B. Horizontal stretch by a factor of 2

C. Shift left 2

D. Vertical stretch by a factor of 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q1/5

How is the graph of $y = \cos(x/2)$ obtained from the graph of $y = \cos x$?

A. Shift right 2

B. Vertical compression by a factor of 1/2

C. Horizontal stretch by a factor of 2

D. Horizontal compression by a factor of 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q2/5

True or false?

Replacing x by kx (k > 1) shortens the period by the factor k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q3/5

$y = \sin(-x)$ is the same as

A. $\cos x$

B. $\sin x$

C. $-\cos x$

D. $-\sin x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q4/5

For which $k > 0$ does $y = \cos(kx)$ have period 6π?

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q5/5

SET 52 • HSF-IF.C.8.b • INTERPRETING FUNCTIONS

Rewrite exponential functions to reveal rates over other time units

MAIN	Algebra II main question	**HSF-IF.C.8.b**
BACKWARD 1	Read growth rates from exponential functions	**HSF-IF.C.8.b**
BACKWARD 2	Powers of powers and rational exponents	**HSN-RN.A.1**
BACKWARD 3	Percent change and growth factors	**7.RP.A.3**
FORWARD 1	Compare growth offers by rewriting to the same time unit	**HSF-IF.C.8.b**
FORWARD 2	Find doubling times and half-lives with logarithms	**HSF-LE.A.4**

SET 52 • HSF-IF.C.8.b

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

$y = 100(1.08)^t$ with t in years. Rewrite it as $y = 100(b)^{12t}$ with a monthly growth factor b. Give b to 6 decimal places and the monthly percent rate to the nearest hundredth of a percent. A calculator may be used.

b = _____

Monthly rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q1/5

$y = 500(0.97)^t$ with t in years. Rewrite it as $y = 500(b)^{t/10}$ with a growth factor b per decade. Give b to 4 decimal places and the percent decrease per decade to the nearest hundredth of a percent. A calculator may be used.

b = _____

Percent decrease per decade: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q2/5

$Q = 50(2)^{t/3}$ with t in hours. Rewrite it as $Q = 50(b)^{t}$ with an hourly growth factor b. Give b to 4 decimal places and the hourly percent rate to the nearest hundredth of a percent. A calculator may be used.

b = _____

Hourly rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q3/5

$A = 1000(1.005)^{12t}$ with t in years. Rewrite it as $A = 1000(b)^{t}$ with an annual growth factor b. Give b to 6 decimal places and the annual percent rate to the nearest hundredth of a percent. A calculator may be used.

b = _____

Annual rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q4/5

$N = 800(0.5)^{t/10}$ with t in years. Rewrite it as $N = 800(b)^t$ with a yearly factor b. Give b to 6 decimal places and the yearly percent decrease to the nearest hundredth of a percent. A calculator may be used.

b = _____

Yearly decrease: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q5/5

SET 52 • HSF-IF.C.8.b

BACKWARD 1

Read growth rates from exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b

Is $y = 100(1.08)^t$ growth or decay, and by what percent per year? Answer Growth or Decay, then the percent.

Growth or Decay: _____

Percent per year: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q1/5

Is $y = 500(0.97)^t$ growth or decay, and by what percent per year? Answer Growth or Decay, then the percent.

Growth or Decay: _____

Percent per year: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q2/5

True or false?

$y = 2(1.25)^t$ grows 125% per unit of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q3/5

$y = 40(0.9)^t$ decreases by

A. 9% per unit

B. 10% per unit

C. 90% per unit

D. 0.9% per unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q4/5

What is the growth factor for 6.17% growth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q5/5

SET 52 • HSF-IF.C.8.b

BACKWARD 2

Powers of powers and rational exponents • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

Rewrite 1.08^{t} as a power of $1.08^{1/12}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q1/5

Rewrite $2^{t/3}$ as $(2^{1/3})^{t}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q2/5

True or false?

$(b^m)^n = b^{mn}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q3/5

1.005^{12t} equals

A. $(1.005^{12})^{t}$

B. $(1.005^{t})/12$

C. $12 \cdot 1.005^{t}$

D. $1.005^{12} + t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q4/5

Write $8^{1/3}$ as a whole number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSF-IF.C.8.b

BACKWARD 3

Percent change and growth factors • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

Write the growth factor for a 25.99% increase.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q1/5

A factor 0.7374 means what percent decrease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q2/5

True or false?

A factor 1.006434 means about 0.64% growth.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q3/5

A 3% decrease corresponds to the factor

A. 0.03

B. 0.97

C. 0.7

D. 1.03

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q4/5

An amount is multiplied by 0.933. By what percent does it decrease?

Percent decrease: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q5/5

SET 52 • HSF-IF.C.8.b

FORWARD 1

Compare growth offers by rewriting to the same time unit • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

Account A grows 0.5% per month. Account B grows 6.1% per year. Find A's annual growth factor (to 4 decimal places), and say which account grows faster (A or B). A calculator may be used.

A's annual factor: _____

Faster account (A or B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q1/5

Culture 1 doubles every 3 hours; culture 2 triples every 5 hours. Find each hourly growth factor to 3 decimal places, and say which grows faster (1 or 2). A calculator may be used.

Hourly factor of culture 1: _____

Hourly factor of culture 2: _____

Faster culture (1 or 2): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q2/5

True or false?

Growing 1% per month is the same as growing 12% per year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q3/5

Which is the greatest annual growth?

A. 4% per half year

B. 0.6% per month

C. 8% per year

D. 2% per quarter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q4/5

A substance with a half-life of 10 years and a substance that decays 7% per year are compared. Which decays faster? A calculator may be used.

A. 7% per year: its yearly factor 0.93 is less than $0.5^{1/10} \approx 0.933$.

B. 7% per year: 7 is less than 10.

C. The half-life substance: its yearly factor 0.5 is less than 0.93.

D. They decay at the same rate, because $0.5^{1/10} \approx 0.93$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q5/5

SET 52 • HSF-IF.C.8.b

FORWARD 2

Find doubling times and half-lives with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Find the doubling time of $y = 100(1.08)^t$, in years to the nearest hundredth. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q1/5

Find the half-life of $y = 500(0.97)^t$, in years to the nearest hundredth. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q2/5

True or false?

The doubling time of $y = 50(2)^{t/3}$ is 6 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q3/5

The doubling time of a quantity growing 6.17% per year is about

A. 16.2 years

B. 11.6 years

C. 6.17 years

D. 3.1 years

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q4/5

$y = 100(1.08)^t$ is rewritten as $y = 100 \cdot 2^{t/9.01}$. Why does the rewritten form show the doubling time?

A. Because 9.01 is the growth rate in percent per year, and doubling depends only on the rate.

B. Because 2 is the starting value, and the exponent t/9.01 counts how many times it is used.

C. Because $y = 200$ when $t = 1$, so y doubles from 100 to 200 in the first year and in each 9.01-year period.

D. The exponent t/9.01 increases by 1 every 9.01 years, and each such increase multiplies y by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q5/5

SET 53 • HSF-IF.C.9 • INTERPRETING FUNCTIONS

Compare properties of Algebra II functions in different representations

MAIN	Algebra II main question	**HSF-IF.C.9**
BACKWARD 1	Compare linear, quadratic, and exponential functions	**HSF-IF.C.9**
BACKWARD 2	Recognize a pattern in a table	**8.F.A.3**
BACKWARD 3	Domains, asymptotes, and periods of Algebra II families	**HSF-IF.B.5**
FORWARD 1	Compare rates of change and long-run behavior across represent…	**HSF-IF.C.9**
FORWARD 2	Build a difference function to compare	**HSF-BF.A.1.b**

SET 53 • HSF-IF.C.9

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

$f(x) = 3(1.1)^x$. The table shows the linear function g. Give the greater y-intercept, f(20) to the nearest hundredth, and g(20). A calculator may be used.

x	0	1	2	3	4
g(x)	10	15	20	25	30

Greater y-intercept: _____

f(20) = _____

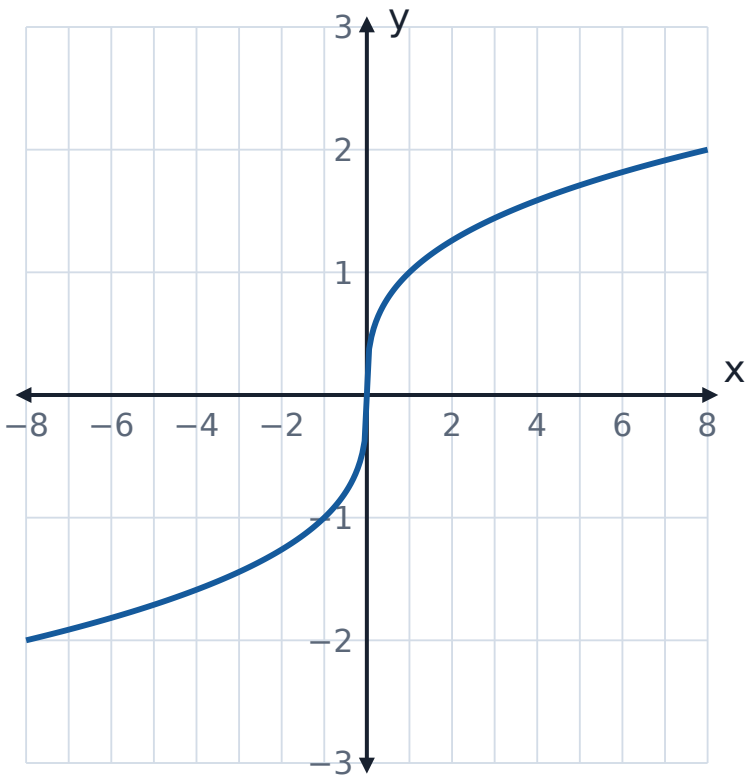
g(20) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q1/5

$f(x) = \sqrt{(x - 1)}$. The graph of g is shown. Compare their domains.



A. f: $x > 1$; g: all real numbers

B. f: $x \geq 1$; g: $x \geq 0$

C. f: $x \geq 1$; g: all real numbers

D. f: all real numbers; g: $x \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q2/5

$h(t) = 12 + 10\sin(\pi t/15)$ gives one Ferris wheel's height (m, t in s). A second wheel has radius 8 m, its center 10 m high, and turns once every 40 s. Give the maximum height and the period of each wheel.

First wheel maximum (m): _____

First wheel period (s): _____

Second wheel maximum (m): _____

Second wheel period (s): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q3/5

$f(x) = \log_2 x$. The table shows g. Find f(9) to the nearest hundredth and g(9). A calculator may be used.

x	1	3	9	27
g(x)	0	1	2	3

f(9) = _____

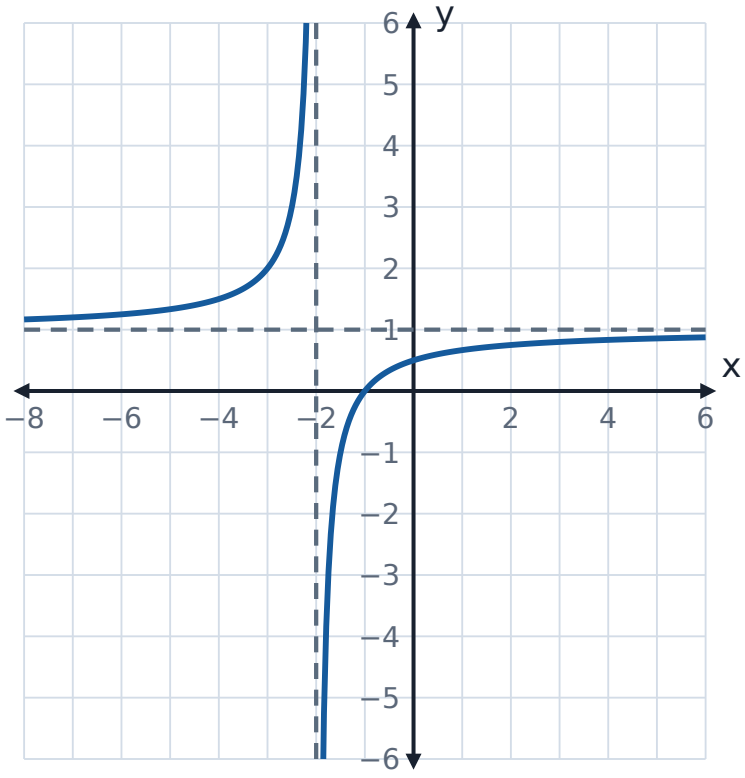
g(9) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q4/5

$f(x) = (2x + 1)/(x - 3)$. The graph of g shows its asymptotes. Give the horizontal and vertical asymptotes of f and of g as equations.



f, horizontal: _____

f, vertical: _____

g, horizontal: _____

g, vertical: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q5/5

SET 53 • HSF-IF.C.9

BACKWARD 1

Compare linear, quadratic, and exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9

$f(x) = 2x + 1$, and g has the table $g(0) = 3$, $g(1) = 4$. Which function has the greater y-intercept?

A. f, with y-intercept 2

B. f, with y-intercept 1

C. g, with y-intercept 3

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q1/5

$f(x) = x^2$ and $g(x) = 2^x$. Find f(10) and g(10).

f(10) = _____

g(10) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q2/5

True or false?

An increasing exponential eventually exceeds any linear function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q3/5

$f(x) = -(x - 1)^2 + 5$ and g with maximum 4 in a graph. The greater maximum is

A. f: 5

B. cannot tell

C. equal

D. g: 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q4/5

A table's outputs increase by 5 at each equal step of the input. What kind of function fits it?

A. Exponential

B. Linear

C. Quadratic

D. Logarithmic

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSF-IF.C.9

BACKWARD 2

Recognize a pattern in a table • Grade 8 • Functions • 8.F.A.3

Is the table x: 0, 1, 2 and y: 10, 15, 20 linear?

A. No: it starts at 10, not 0, and a linear table must start at 0

B. Yes: every table of numbers can be joined by straight segments, so every table is linear

C. No: the outputs 10, 15, 20 are not all multiples of 10, so they are not linear

D. Yes: the outputs have a constant difference of 5 for equal steps in x

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q1/5

Is the table x: 1, 3, 9, 27 and y: 0, 1, 2, 3 linear?

A. No: each output step of 1 needs the input multiplied by 3, so it is logarithmic ($y = \log_3 x$)

B. Yes: the inputs increase each time, and any table with increasing inputs is linear

C. No: it is exponential, because each input is 3 times the one before it ($y = 3^x$)

D. Yes: the outputs increase by 1 each time, so the table has a constant rate of change

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q2/5

True or false?

A table with constant ratios of outputs is exponential.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q3/5

Which table is exponential?

A. 1, 4, 9, 16

B. 1, 2, 4, 8

C. 2, 2, 2, 2

D. 1, 3, 5, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q4/5

What is the slope of a linear table with outputs 10, 15, 20 at x = 0, 1, 2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • HSF-IF.C.9

BACKWARD 3

Domains, asymptotes, and periods of Algebra II families • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

Give the domain of $\sqrt{(x - 1)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q1/5

Give the vertical and horizontal asymptotes of $y = (2x + 1)/(x - 3)$ as equations.

Vertical asymptote: _____

Horizontal asymptote: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q2/5

True or false?

The period of $12 + 10\sin(\pi t/15)$ is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q3/5

The domain of $\log_2 x$ is

A. all reals

B. $x \geq 0$

C. $x > 0$

D. $x \neq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q4/5

A wheel of radius 8 m with center 10 m high has what maximum height?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q5/5

SET 53 • HSF-IF.C.9

FORWARD 1

Compare rates of change and long-run behavior across representations • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

$f(x) = 3(1.1)^x$ and $g(x) = 10 + 5x$. Estimate the first whole number x at which f exceeds g. A calculator may be used. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q1/5

$f(x) = \log_2 x$, and g has $g(1) = 0$, $g(3) = 1$, $g(9) = 2$. Find the average rates of change of f and of g on $1 \le x \le 9$, each to the nearest hundredth. A calculator may be used.

Rate of f: _____

Rate of g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q2/5

True or false?

A function with a larger y-intercept is always larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q3/5

Which grows fastest for large x?

A. $\sqrt{(x - 1)}$

B. $3(1.1)^x$

C. $5x + 10$

D. $\log_2 x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q4/5

Why does $g(x) = \log_3 x$ stay below $f(x) = \log_2 x$ for $x > 1$?

A. Because $3 > 2$ makes every value of $\log_3 x$ larger, so g stays below f only for $0 < x < 1$.

B. Because both equal 0 at $x = 1$, and two graphs that share a point never cross after it.

C. Because $\log_3 x$ is negative for $x > 1$ while $\log_2 x$ is positive there, so g is below f.

D. $\log_3 x = \log_2 x / \log_2 3$, and $\log_2 3 > 1$, so for $x > 1$ (where $\log_2 x > 0$), g is a smaller positive multiple of f.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q5/5

SET 53 • HSF-IF.C.9

FORWARD 2

Build a difference function to compare • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b

Write $d(x) = f(x) - g(x)$ for $f(x) = 3(1.1)^x$ and $g(x) = 5x + 10$.

$d(x) =$ _____

$d(x) = f(x) - g(x)$ is the difference of two functions. What does the sign of $d(x)$ tell you?

A. $d(x) > 0$ means f is increasing and g is decreasing at x.

B. $d(x) > 0$ means $f(x) > g(x)$; $d(x) < 0$ means $g(x) > f(x)$; $d(x) = 0$ means they are equal.

C. $d(x) > 0$ means both functions are positive at x; $d(x) < 0$ means both are negative.

D. The sign of $d(x)$ gives the slope of f: positive means f rises and negative means f falls.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q2/5

True or false?

A zero of $d(x) = f(x) - g(x)$ is an x where $f(x) = g(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q3/5

For $h_1(t) = 12 + 10\sin(\pi t/15)$ and $h_2(t) = 10 + 8\sin(\pi t/20)$, $h_1(0) - h_2(0)$ is

A. 2

B. 22

C. -2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q4/5

Two Ferris wheel heights are $h_1(t) = 12 + 10\sin(\pi t/15)$ and $h_2(t) = 10 + 8\sin(\pi t/20)$. Which is $h_1(t) - h_2(t)$?

A. $2 + 10\sin(\pi t/15) + 8\sin(\pi t/20)$

B. $2 + 2\sin(\pi t/60)$

C. $2 + 10\sin(\pi t/15) - 8\sin(\pi t/20)$

D. $22 + 10\sin(\pi t/15) + 8\sin(\pi t/20)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q5/5

Build models by combining Algebra II function types

MAIN	Algebra II main question	**HSF-BF.A.1.b**
BACKWARD 1	Combine linear and quadratic functions	**HSF-BF.A.1.b**
BACKWARD 2	Multiply and subtract polynomials	**HSA-APR.A.1**
BACKWARD 3	Evaluate exponential and trigonometric expressions	**HSF-IF.A.2**
FORWARD 1	Interpret the parts of a combined model	**HSF-BF.A.1.b**
FORWARD 2	Compose functions in context	**HSF-BF.A.1.c**

SET 54 • HSF-BF.A.1.b

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b

Revenue is $R(x) = x(50 - 0.5x)$ and cost is $C(x) = 200 + 10x$ dollars for x items. Write the profit $P(x) = R(x) - C(x)$ with its terms multiplied out, and find $P(30)$.

$P(x) =$ _____

$P(30) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q1/5

The cost of x posters is C(x) = 1200 + 3x dollars. Write the average cost per poster A(x) = C(x)/x and find A(100).

A(x) = _____

A(100) ($) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q2/5

A town has $P(t) = 5000(1.02)^t$ people and $F(t) = 6000 + 100t$ units of food after t years. Write the food per person $F(t)/P(t)$, and find it at $t = 10$ to the nearest hundredth. A calculator may be used.

Food per person = _____

At t = 10: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q3/5

A spring starts at displacement +10 cm at t = 0. Its amplitude envelope is 10 cm and shrinks by 10% each second, and it oscillates once per second, so $y(t) = A(t)\cos(2\pi t)$. Write the envelope $A(t)$ and find $y(1)$.

$A(t) =$ _____

$y(1) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q4/5

Coffee starts 70°C above a 20°C room, and the difference shrinks 5% each minute. Write its temperature T(t) after t minutes as a sum of a constant and an exponential, and find T(10) to the nearest hundredth. A calculator may be used.

T(t) = _____

T(10) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q5/5

SET 54 • HSF-BF.A.1.b

BACKWARD 1

Combine linear and quadratic functions • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b

$R(x) = 8x$ and $C(x) = 3x + 50$. Write $P = R - C$.

$P(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q1/5

$f(x) = x + 2$ and $g(x) = x - 2$. Write $f \cdot g$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q2/5

True or false?

Profit is revenue minus cost.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q3/5

Average cost is

A. total cost times the number of items

B. total cost divided by the number of items

C. total cost minus total revenue

D. total revenue divided by total cost

$f(x) = 2x$ and $g(x) = x^2 + 1$. Find $(f + g)(3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSF-BF.A.1.b

BACKWARD 2

Multiply and subtract polynomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Expand $x(50 - 0.5x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q1/5

Simplify $(50x - 0.5x^2) - (200 + 10x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q2/5

True or false?

$-(200 + 10x) = -200 + 10x.$

A. True

B. False

$(3x + 1) - (x - 4)$ equals

A. $2x + 5$

B. $4x - 3$

C. $2x - 3$

D. $2x + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q4/5

Expand $2(x^2 - 3x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSF-BF.A.1.b

BACKWARD 3

Evaluate exponential and trigonometric expressions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Evaluate $70(0.95)^{10}$ to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q1/5

Evaluate $5000(1.02)^{10}$ to the nearest whole number. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q2/5

True or false?

$\cos(2\pi) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q3/5

$10(0.9)^2$ equals

A. 9

B. 18

C. 81

D. 8.1

Evaluate $\cos(2\pi \cdot 0.5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q5/5

SET 54 • HSF-BF.A.1.b

FORWARD 1

Interpret the parts of a combined model • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b

Coffee in a 20°C room has temperature $T(t) = 20 + 70(0.95)^t$ °C after t minutes. What do 20 and $70(0.95)^t$ represent?

A. 20 is the room temperature; $70(0.95)^t$ is how far the coffee is above room temperature, which shrinks 5% each minute.

B. 20 is the starting temperature of the coffee; $70(0.95)^t$ is the room temperature, which falls 5% each minute.

C. 20 is the number of minutes the coffee cools; $70(0.95)^t$ is the temperature of the coffee at time t.

D. 20 is the room temperature; $70(0.95)^t$ is the drop in the coffee's temperature during each minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q1/5

A spring's position is $y(t) = 10(0.9)^t \cos(2\pi t)$. Which factor controls the size of the swings and which controls their timing?

A. $10(0.9)^t$ controls the size (it shrinks 10% each second); $\cos(2\pi t)$ controls the timing (one swing per second).

B. $\cos(2\pi t)$ controls the size (it shrinks each second); $10(0.9)^t$ controls the timing (one swing per 0.9 seconds).

C. Both factors control only the size of the swings; the timing is set by the 10 in front.

D. 10 controls the timing (10 swings per second); 0.9 controls the size (it shrinks the swings).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q2/5

True or false?

In $A(x) = (1200 + 3x)/x = 1200/x + 3$, the term $1200/x$ is the fixed cost spread over x posters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q3/5

As t grows, $T(t) = 20 + 70(0.95)^t$ approaches

A. 20

B. 0

C. 90

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q4/5

A town has $P(t) = 5000(1.02)^t$ people and $F(t) = 6000 + 100t$ units of food after t years. Why does the food per person $F(t)/P(t)$ eventually decrease?

A. $F(t)/P(t)$ is constant because both F and P increase every year, so their ratio does not change as time passes.

B. $P(t) = 5000(1.02)^t$ decreases over time, so there are fewer people for the same food.

C. $F(t) = 6000 + 100t$ decreases over time, so there is less food for the same number of people.

D. P grows exponentially (factor 1.02 each year) while F grows linearly, so P eventually grows much faster.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q5/5

SET 54 • HSF-BF.A.1.b

FORWARD 2

Compose functions in context • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

A balloon's radius is $r(t) = 2t$ cm after t seconds and its volume is $V(r) = (4/3)\pi r^3$. Write $V(r(t))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q1/5

A store takes 20% off with d(p) = 0.8p and then adds a $5 fee with f(p) = p + 5. Write f(d(p)) and find it for p = 50.

f(d(p)) = _____

f(d(50)) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q2/5

True or false?

f(g(x)) and g(f(x)) are always equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q3/5

If $f(x) = x^2$ and $g(x) = x + 1$, then $f(g(2)) =$

A. 4

B. 9

C. 3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q4/5

How does composing two functions differ from multiplying them?

A. Composition $f(g(x))$ adds the two outputs $f(x) + g(x)$; multiplication $f(x) \cdot g(x)$ multiplies them.

B. Composition $f(g(x))$ uses the output of g as the input of f; multiplication $f(x) \cdot g(x)$ multiplies the two outputs.

C. Composition $f(g(x))$ multiplies the inputs of f and g; multiplication $f(x) \cdot g(x)$ multiplies the outputs.

D. They are the same operation written two ways: $f(g(x))$ and $f(x) \cdot g(x)$ are always equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q5/5

Identify vertical shifts, vertical stretches, and horizontal shifts of Algebra II functions

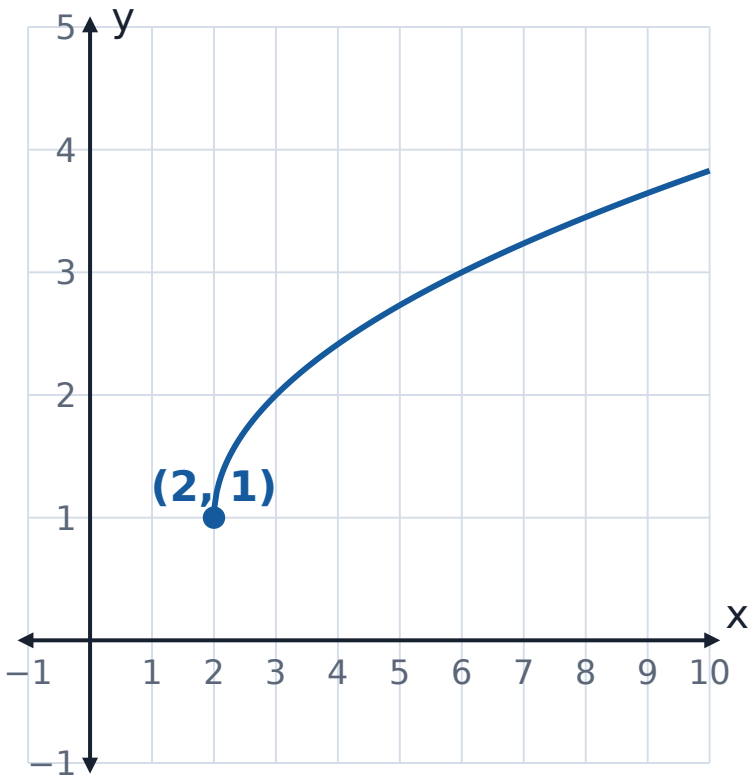
MAIN	Algebra II main question	**HSF-BF.B.3**
BACKWARD 1	Shifts and stretches of linear and quadratic graphs	**HSF-BF.B.3**
BACKWARD 2	Translate points in the plane	**8.G.A.3**
BACKWARD 3	Key points of parent functions	**HSF-IF.C.7.e**
FORWARD 1	Write transformed functions from descriptions and order of transfo..	**HSF-BF.B.3**
FORWARD 2	Read sinusoid parameters as transformations	**HSF-TF.B.5**

SET 55 • HSF-BF.B.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

$f(x) = \sqrt{x}$. The graph of $g(x) = f(x - h) + k$ is shown. Find h and k.



h = _____

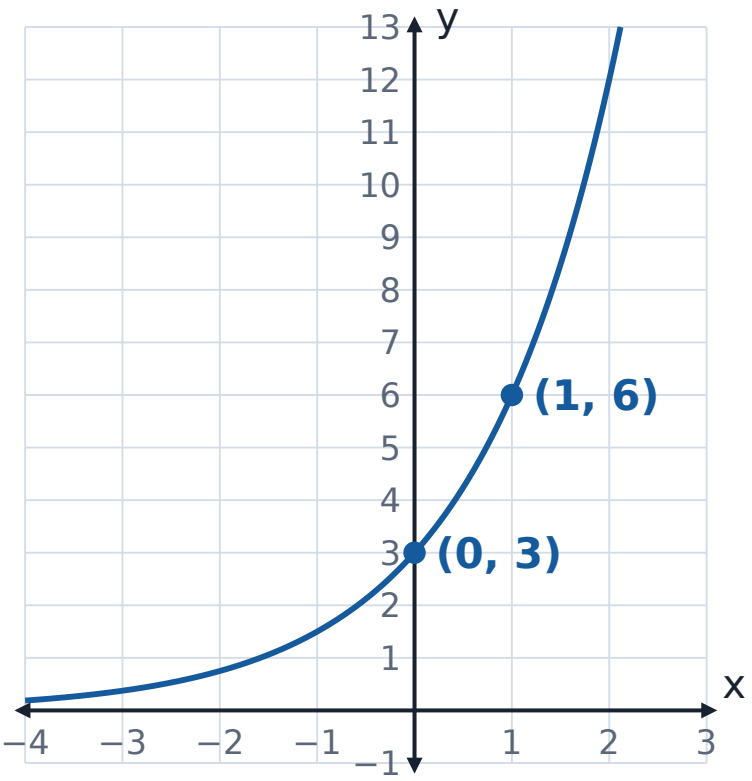
k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q1/5

$f(x) = 2^x$. The graph of $g(x) = a \cdot f(x)$ is shown. Find a.



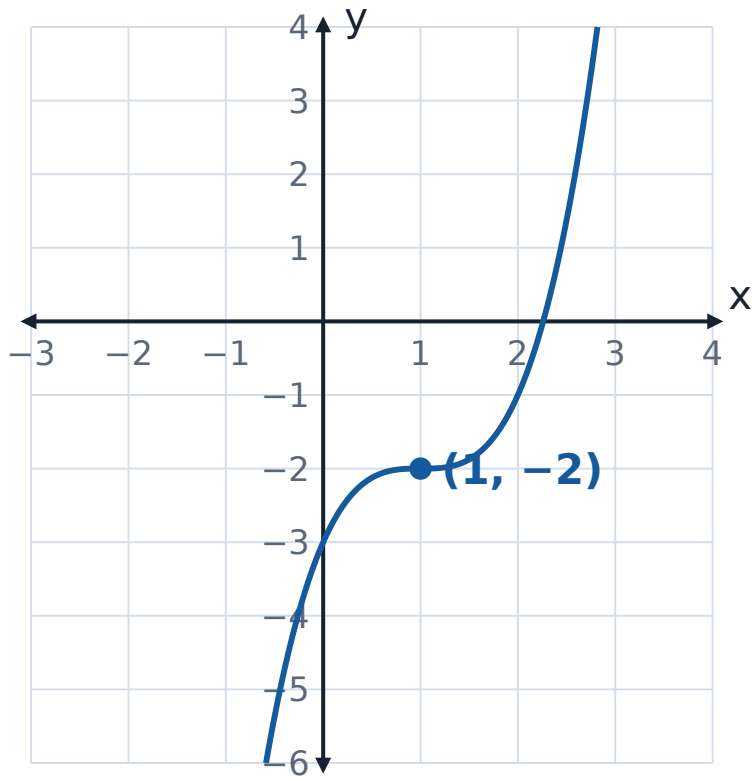
a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q2/5

$f(x) = x^3$. The graph of $g(x) = f(x - h) + k$ is shown. Find h and k.



h = _____

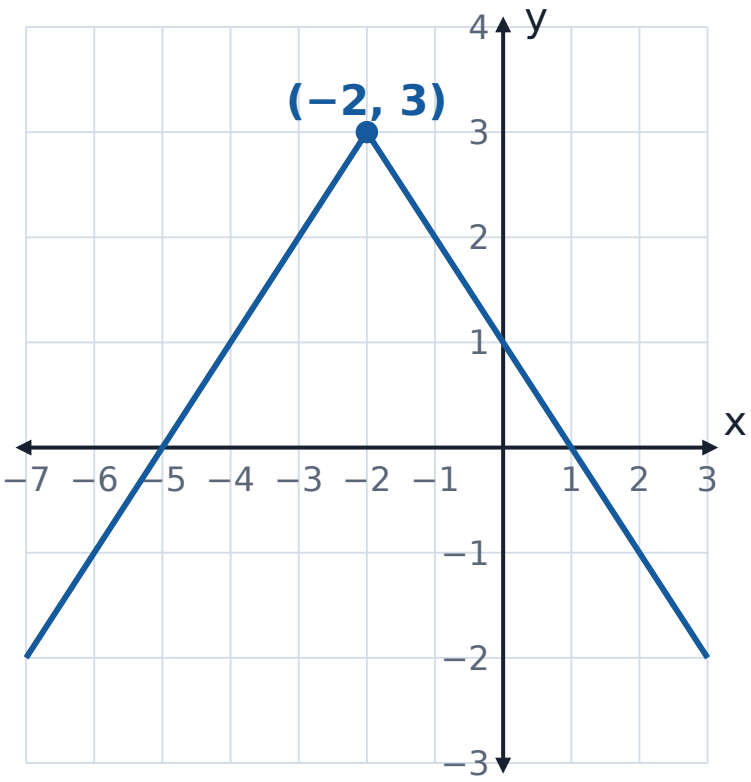
k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q3/5

$f(x) = |x|$. The graph of $g(x) = a \cdot f(x - h) + k$ is shown. Find a, h, and k.



a = _____

h = _____

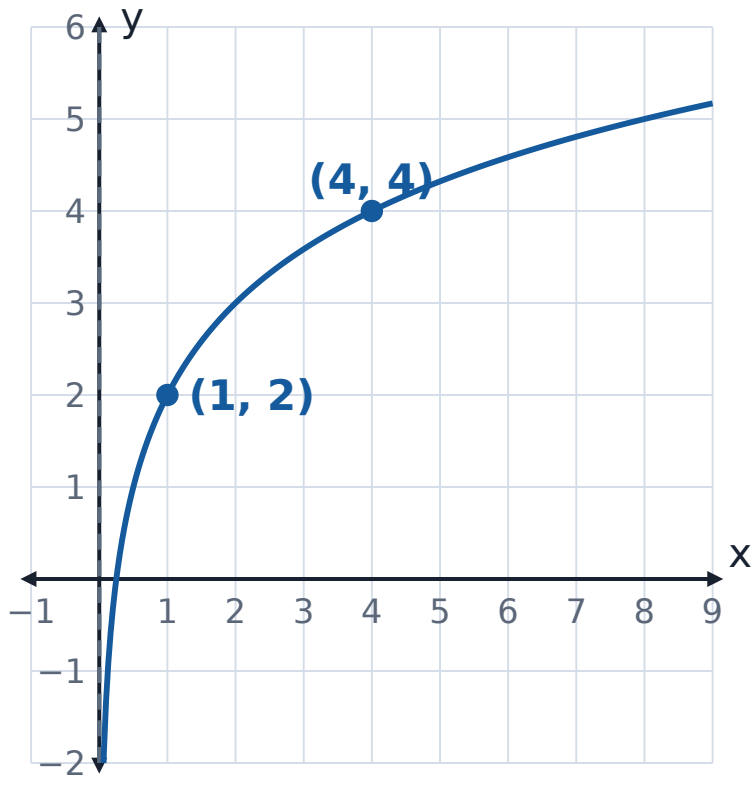
k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q4/5

$f(x) = \log_2 x$. The graph of $g(x) = f(x) + k$ is shown. Find k.



$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q5/5

SET 55 • HSF-BF.B.3

BACKWARD 1

Shifts and stretches of linear and quadratic graphs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How is the graph of $y = (x - 3)^2 + 1$ obtained from the graph of $y = x^2$?

A. Shift left 3 and up 1

B. Shift up 3 and right 1

C. Shift right 3 and up 1

D. Shift right 3 and down 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q1/5

How is the graph of $y = 2x^2$ obtained from the graph of $y = x^2$?

A. Shift up 2

B. Vertical stretch by 2

C. Vertical compression by 1/2

D. Horizontal stretch by 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q2/5

True or false?

$y = f(x + 4)$ is the graph of f shifted right 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q3/5

The vertex of $y = (x + 2)^2 - 5$ is

A. $(2, 5)$

B. $(2, -5)$

C. $(-2, -5)$

D. $(-2, 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q4/5

Write $y = x^2$ shifted left 1 and down 4.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q5/5

SET 55 • HSF-BF.B.3

BACKWARD 2

Translate points in the plane • Grade 8 • Geometry • 8.G.A.3

Translate (0, 0) right 2 and up 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q1/5

Translate (1, 1) left 2 and up 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q2/5

True or false?

Translating (0, 0) down 2 gives (−2, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q3/5

$(x, y) \to (x - 2, y + 3)$ moves $(0, 0)$ to

A. $(-2, 3)$

B. $(-2, -3)$

C. $(2, 3)$

D. $(3, -2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q4/5

Which translation takes the point (0, 0) to (1, −2)?

A. Up 1 and left 2

B. Right 2 and down 1

C. Right 1 and down 2

D. Left 1 and up 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • HSF-BF.B.3

BACKWARD 3

Key points of parent functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

Give the point of $y = \sqrt{x}$ at $x = 0$ and at $x = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q1/5

Give the y-intercept of $y = 2^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q2/5

True or false?

The graph of $y = \log_2 x$ passes through (1, 0).

A. True

B. False

The vertex of $y = |x|$ is

A. (1, 0)

B. (1, 1)

C. (0, 1)

D. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q4/5

Give the center (point of symmetry) of $y = x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q5/5

SET 55 • HSF-BF.B.3

FORWARD 1

Write transformed functions from descriptions and order of transformations • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Write g, the graph of $f(x) = 2^x$ stretched vertically by 3, then shifted down 4.

$g(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q1/5

Which is g, the graph of $f(x) = \sqrt{x}$ shifted down 4 and then stretched vertically by 3?

A. $g(x) = 3\sqrt{(x - 4)}$

B. $g(x) = 3\sqrt{x} - 4$

C. $g(x) = \sqrt{(3x)} - 4$

D. $g(x) = 3\sqrt{x} - 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q2/5

True or false?

A vertical stretch then a shift can give a different graph than the same shift then the stretch.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q3/5

$g(x) = 2f(x) + 1$ moves the point $(3, 4)$ of f to

A. $(6, 5)$

B. $(4, 9)$

C. $(3, 9)$

D. $(3, 10)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q4/5

$g(x) = f(x + 2) - 3$ and f has asymptote $x = 0$. Where is g's vertical asymptote?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q5/5

SET 55 • HSF-BF.B.3

FORWARD 2

Read sinusoid parameters as transformations • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

$y = 3\sin x + 2$ is a transformation of $y = \sin x$. Give the vertical stretch factor, the upward shift, and the maximum value.

Vertical stretch factor: _____

Upward shift: _____

Maximum value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q1/5

The graph of $y = \cos x$ is stretched vertically by 4 and then shifted down 1. Write the result in the form $y = A \cdot \cos x + D$.

A = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q2/5

True or false?

In $y = A \sin x + D$, A is the midline.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q3/5

$y = -2\cos x + 5$ has amplitude and midline

A. 5 and $y = 2$

B. -2 and $y = 5$

C. 2 and $y = -5$

D. 2 and $y = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q4/5

A model is $y = A \sin x + D$ with maximum 9 and minimum 1. Find A ($A > 0$) and D.

$A =$ _____

$D =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q5/5

Horizontal stretches, compressions, and reflections f(kx)

MAIN	Algebra II main question	**HSF-BF.B.3**
BACKWARD 1	Reflections and dilations of points	**8.G.A.3**
BACKWARD 2	Solve kx = c for x	**6.EE.B.7**
BACKWARD 3	Period of y = sin(Bx)	**HSF-TF.B.5**
FORWARD 1	Combine horizontal stretches with shifts	**HSF-BF.B.3**
FORWARD 2	Graph exponential and trigonometric functions with horizontal cha…	**HSF-IF.C.7.e**

SET 56 • HSF-BF.B.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

$f(x) = \sqrt{x}$ passes through (4, 2). For $g(x) = \sqrt{(4x)}$, by what number are the x-coordinates of points of f multiplied, and which point of g has y = 2?

x-coordinates are multiplied by: _____

Point of g with y = 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q1/5

$f(x) = \sin x$ has period 2π. For $g(x) = \sin(3x)$, by what number are the x-coordinates of points of f multiplied, and what is the period of g?

x-coordinates are multiplied by: _____

Period of g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q2/5

$f(x) = 2^x$ passes through (1, 2). For $g(x) = 2^{-x}$, by what number are the x-coordinates of points of f multiplied, and which point of g comes from (1, 2)?

x-coordinates are multiplied by: _____

Point of g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q3/5

f passes through $(-2, 1)$, $(0, 3)$, and $(4, -1)$. List the corresponding points of $g(x) = f(2x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q4/5

f passes through (2, 5). For g(x) = f(−x/2), by what number are the x-coordinates of points of f multiplied, and which point must be on the graph of g?

x-coordinates are multiplied by: _____

Point of g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q5/5

SET 56 • HSF-BF.B.3

BACKWARD 1

Reflections and dilations of points • Grade 8 • Geometry • 8.G.A.3

Reflect (1, 2) over the y-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q1/5

Multiply the x-coordinate of $(4, -1)$ by $1/2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q2/5

True or false?

Reflecting over the y-axis changes the sign of the x-coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q3/5

(x, y) → (x/3, y) applied to (6, 2) gives

A. (18, 2)

B. (6, 2/3)

C. (2, 2)

D. (3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q4/5

Apply $(x, y) \to (-2x, y)$ to $(2, 5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • HSF-BF.B.3

BACKWARD 2

Solve kx = c for x • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve 2x = 4.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q1/5

Solve $-x/2 = 2$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q2/5

True or false?

If 4x = 4, then x = 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q3/5

$3x = 2\pi$ gives $x =$

A. $2\pi/3$

B. $2\pi - 3$

C. $\pi/3$

D. 6π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q4/5

Solve $-x = 1$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • HSF-BF.B.3

BACKWARD 3

Period of y = sin(Bx) • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

Give the period of $y = \sin(2x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q1/5

Give the period of $y = \cos(x/2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q2/5

True or false?

For $y = \sin(Bx)$ with $B > 0$, a larger B gives a longer period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q3/5

$y = \sin(4x)$ has period

A. 8π

B. 4π

C. $\pi/4$

D. $\pi/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q4/5

For $y = \sin(Bx)$ with $B > 0$, which B gives period 6? Give B exactly, in terms of π.

$B =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q5/5

SET 56 • HSF-BF.B.3

FORWARD 1

Combine horizontal stretches with shifts • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Rewrite $g(x) = f(2x - 6)$ in the form $f(k(x - h))$. Give k and h.

$k =$ _____

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q1/5

f passes through (4, 1). Which point is on $g(x) = f(2x - 6)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q2/5

True or false?

f(2x − 6) is f(2x) shifted right 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q3/5

$y = \sqrt{(9x)}$ equals

A. $\sqrt{x} + 9$

B. $9\sqrt{x}$

C. $3\sqrt{x}$

D. $\sqrt{x}/3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q4/5

Write g, the graph of $f(x) = 2^x$ reflected over the y-axis and then shifted right 1.

$g(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q5/5

SET 56 • HSF-BF.B.3

FORWARD 2

Graph exponential and trigonometric functions with horizontal changes • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

For $y = 2^{-x} + 1$, give the y-intercept, the horizontal asymptote, and the y-values at $x = -1$ and at $x = 1$.

y-intercept: _____

Asymptote: $y =$ _____

y at $x = -1$: _____

y at $x = 1$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q1/5

On $0 \le x \le 2\pi$, at which x-values does $y = \cos(2x)$ reach its maximum? List them separated by commas, exactly in terms of π.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q2/5

True or false?

$y = 3^{2x} = 9^{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q3/5

$y = \sin(x/2)$ completes one cycle on

A. $0 \le x \le \pi$

B. $0 \le x \le 2\pi$

C. $0 \le x \le \pi/2$

D. $0 \le x \le 4\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q4/5

Why is $y = 2^{2x}$ both a horizontal compression of $y = 2^x$ and an exponential function with base 4?

A. Because $2^{2x} = 2^x + 2^x$, which doubles each output, and doubling 2 gives the base 4.

B. Replacing x by 2x compresses the graph horizontally by a factor of 1/2, and $2^{2x} = (2^2)^x = 4^x$.

C. Because $2^{2x} = 2 \cdot 2^x$, a horizontal compression by 2, and $2 \cdot 2$ gives the base 4.

D. Replacing x by 2x stretches the graph vertically by a factor of 2, so the base 2 becomes $2 \cdot 2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q5/5

Recognize even and odd functions

MAIN	Algebra II main question	**HSF-BF.B.3**
BACKWARD 1	Signs of powers of negatives	**7.NS.A.2**
BACKWARD 2	Reflect points over the axes and the origin	**6.NS.C.6.b**
BACKWARD 3	Substitute -x into an expression	**HSF-IF.A.2**
FORWARD 1	Reason about sums and products of even and odd functions	**HSF-BF.B.3**
FORWARD 2	Explain trigonometric symmetry with the unit circle	**HSF-TF.A.4**

SET 57 • HSF-BF.B.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Is $f(x) = x^4 - 3x^2$ even, odd, or neither?

A. Even: $f(0) = 0$, and a graph through the origin is even.

B. Even: $f(-x) = (-x)^4 - 3(-x)^2 = x^4 - 3x^2 = f(x)$.

C. Odd: $f(-x) = -x^4 + 3x^2 = -f(x)$, because $-x$ makes each term negative.

D. Neither: $f(1) = -2$ and $f(-1) = 2$, so $f(-1) \neq f(1)$ and f is not even.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q1/5

Is $f(x) = x^3 - x$ even, odd, or neither?

A. Odd: $f(0) = 0$, and every function through the origin is odd.

B. Even: $f(-x) = x^3 - x = f(x)$, because the signs cancel in each term.

C. Odd: $f(-x) = (-x)^3 - (-x) = -x^3 + x = -f(x)$.

D. Neither: $f(-x) = -x^3 - x$, which is neither $f(x)$ nor $-f(x)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q2/5

Is $f(x) = x^2 + x$ even, odd, or neither?

A. Even: $f(0) = 0$, and a graph through the origin is symmetric about the y-axis.

B. Even: x^2 is even, and the term of highest degree decides whether f is even.

C. Odd: x is odd, and adding an odd term to x^2 makes the whole function odd.

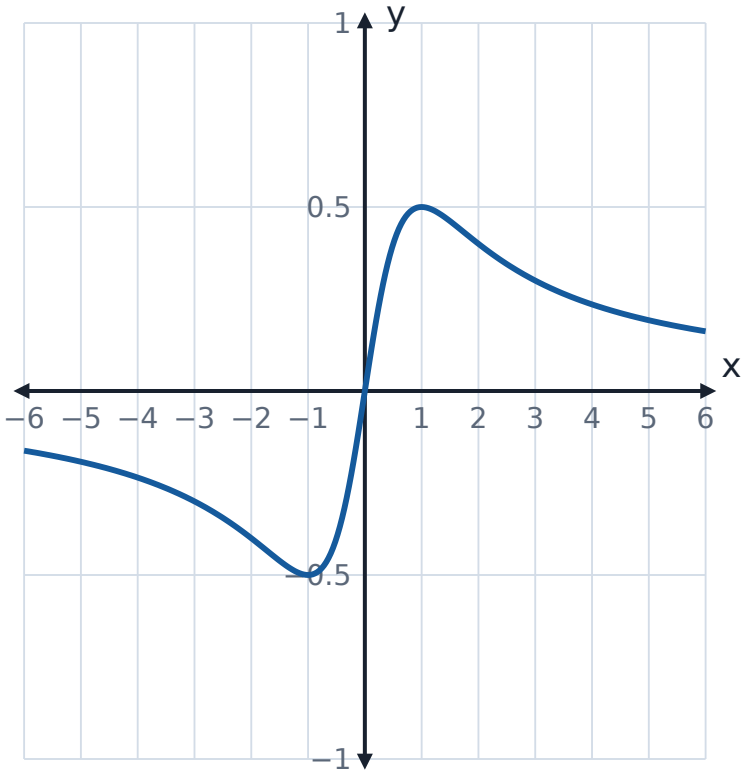
D. Neither: $f(1) = 2$ and $f(-1) = 0$, so $f(-1) \neq f(1)$ and $f(-1) \neq -f(1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q3/5

The graph of $f(x) = x/(x^2 + 1)$ is shown. Is f even, odd, or neither?



A. Even: the graph is symmetric about the y-axis (reflecting it over the y-axis gives the same graph).

B. Odd: the graph passes through the origin, and every graph through the origin is odd.

C. Odd: the graph is symmetric about the origin (rotating it 180° about the origin gives the same graph).

D. Neither: the graph has a peak on one side and a low point on the other, so it has no symmetry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q4/5

Is $f(x) = \cos x + x^2$ even, odd, or neither?

A. Odd: cos is odd.

B. Even: $f(-x) = \cos(-x) + (-x)^2 = \cos x + x^2 = f(x)$.

C. Odd: $f(-x) = -\cos x + x^2$.

D. Neither: cos x and x^2 are different kinds of functions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q5/5

SET 57 • HSF-BF.B.3

BACKWARD 1

Signs of powers of negatives • Grade 7 • The Number System • 7.NS.A.2

Simplify $(-x)^4$ and $(-x)^3$.

$(-x)^4 =$ _____

$(-x)^3 =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q1/5

Simplify $(-x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q2/5

True or false?

$(-x)^5 = x^5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q3/5

$(-2)^2 - (-2)$ equals

A. -6

B. 2

C. 6

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q4/5

Simplify $-(-x)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q5/5

SET 57 • HSF-BF.B.3

BACKWARD 2

Reflect points over the axes and the origin • Grade 6 • The Number System • 6.NS.C.6.b

Reflect (2, 3) over the y-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q1/5

Rotate (2, 3) 180° about the origin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q2/5

True or false?

A graph symmetric about the y-axis contains (−a, b) whenever it contains (a, b).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q3/5

Symmetry about the origin pairs (a, b) with

A. (−a, b)

B. (−a, −b)

C. (a, −b)

D. (b, a)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q4/5

A graph is symmetric about the origin and contains (1, 0.5). Name another point.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q5/5

SET 57 • HSF-BF.B.3

BACKWARD 3

Substitute −x into an expression • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

For $f(x) = x^2 + x$, find $f(-x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q1/5

For $f(x) = x^3 - x$, find $f(-2)$ and $f(2)$.

$f(-2) =$ _____

$f(2) =$ _____

True or false?

For $f(x) = 3x$, $f(-x) = 3x$.

A. True

B. False

For $f(x) = x^4 - 3x^2$, $f(-1)$ is

A. 2

B. 4

C. -2

D. -4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q4/5

For f(x) = 1/x, find f(−x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q5/5

SET 57 • HSF-BF.B.3

FORWARD 1

Reason about sums and products of even and odd functions • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Which argument proves that the sum of two odd functions f and g is odd?

A. x^3 and x^5 are odd and $x^3 + x^5$ is odd, so every sum of two odd functions is odd.

B. $(f + g)(-x) = f(-x) + g(-x) = -f(x) - g(x) = -(f(x) + g(x)) = -(f + g)(x)$.

C. $(f + g)(-x) = f(x) + g(x) = (f + g)(x)$, so the sum is odd.

D. $(f + g)(-x) = -f(-x) - g(-x) = -(f + g)(-x)$, so the sum is odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q1/5

Which argument proves that the product of two odd functions f and g is even?

A. $(fg)(-x) = f(-x)g(-x) = (-f(x))(-g(x)) = f(x)g(x) = (fg)(x)$.

B. $x \cdot x^3 = x^4$ is even, so every product of two odd functions is even.

C. Odd times odd is odd for whole numbers, so the product of two odd functions is odd.

D. $(fg)(-x) = f(-x)g(-x) = -f(x)g(x)$, so the product is odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q2/5

True or false?

The only function that is both even and odd is $f(x) = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q3/5

$f(x) = |x| + x^3$ is

A. even

B. both

C. neither

D. odd

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q4/5

Is $f(x) = x \sin x$ even, odd, or neither?

A. Odd: $f(-x) = (-x) \sin x = -x \sin x = -f(x)$, since only the first factor changes sign.

B. Odd: sin x is odd, and multiplying an odd function by x keeps it odd, so x sin x is odd.

C. Even: x and sin x are both odd, and a product of two odd functions is even.

D. Neither: x is not a trigonometric function, so the rules for even and odd functions do not apply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q5/5

SET 57 • HSF-BF.B.3

FORWARD 2

Explain trigonometric symmetry with the unit circle • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

Why is $\cos(-x) = \cos x$, using the unit circle?

A. The points at angles x and $-x$ are reflections of each other over the x-axis, which keeps the x-coordinate (the cosine).

B. The points at angles x and $-x$ are reflections of each other over the y-axis, which keeps the x-coordinate (the cosine).

C. The points at angles x and $-x$ are the same point on the unit circle, so they have the same x-coordinate.

D. Because cos is never negative, so $\cos(-x)$ and $\cos x$ always have the same value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q1/5

Is tan even, odd, or neither?

A. Odd: $\tan(-x) = \sin(-x)/\cos(-x) = -\sin x/\cos x = -\tan x$.

B. Even: cos is even, so $\tan(-x) = \sin x/\cos(-x) = \sin x/\cos x = \tan x$.

C. Even: $\tan(-x) = \sin x/\cos x = \tan x$.

D. Neither: tan has vertical asymptotes, so it cannot be even or odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q2/5

True or false?

sin is an even function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q3/5

Which is an even function?

A. cos x

B. tan x

C. sin x + cos x

D. sin x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q4/5

Use symmetry to find $\sin(-\pi/6)$ from $\sin(\pi/6) = 1/2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q5/5

Write inverses of radical, rational, cubic, exponential, and logarithmic functions

MAIN	Algebra II main question	**HSF-BF.B.4.a**
BACKWARD 1	Inverses of linear functions	**HSF-BF.B.4.a**
BACKWARD 2	Rearrange formulas with roots, powers, and fractions	**HSA-CED.A.4**
BACKWARD 3	Domain and range	**HSF-IF.B.5**
FORWARD 1	Inverses that are their own inverse, and checking by points	**HSF-BF.B.4.a**
FORWARD 2	Verify inverses by composition	**HSF-BF.B.4.b**

SET 58 • HSF-BF.B.4.a

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

Find the inverse of $f(x) = (x - 1)^3 + 2$ (all real x). It has the form $f^{-1}(x) = \sqrt[3]{(x - a)} + b$; give a and b.

$a =$ _____

$b =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q1/5

Find the inverse of $f(x) = 2x/(x - 3)$, $x \neq 3$, and the value excluded from its domain.

$f^{-1}(x) =$ _____

Excluded value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q2/5

Find the inverse of $f(x) = \sqrt{(x + 4)}$, $x \geq -4$, and its domain.

$f^{-1}(x) =$ _____

Domain of f^{-1}: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q3/5

Which gives the inverse of $f(x) = 5 \cdot 2^x$ and its domain?

A. $f^{-1}(x) = \log_5(x/2),\ x > 0$

B. $f^{-1}(x) = \log_2(x) - 5,\ x > 0$

C. $f^{-1}(x) = \log_2(x/5)$, all real x

D. $f^{-1}(x) = \log_2(x/5),\ x > 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q4/5

Which gives the inverse of $f(x) = \log_3(x - 1)$ and its domain?

A. $f^{-1}(x) = 3^x + 1,\ x > 1$

B. $f^{-1}(x) = 3^x + 1$, all real x

C. $f^{-1}(x) = 3^{x+1}$, all real x

D. $f^{-1}(x) = x^3 + 1$, all real x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q5/5

SET 58 • HSF-BF.B.4.a

BACKWARD 1

Inverses of linear functions • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

Find the inverse of $f(x) = 3x - 6$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q1/5

Solve 2x + 5 = y for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q2/5

True or false?

The inverse undoes the function: $f^{-1}(f(x)) = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q3/5

The inverse of $f(x) = x/4 + 1$ is

A. $4(x - 1)$

B. $(x - 1)/4$

C. $4x - 1$

D. $x/4 - 1$

If $f(2) = 7$, what is $f^{-1}(7)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q5/5

SET 58 • HSF-BF.B.4.a

BACKWARD 2

Rearrange formulas with roots, powers, and fractions • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

Solve $y = \sqrt{(x + 4)}$ for x ($y \geq 0$).

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q1/5

Solve $y = 2x/(x - 3)$ for x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q2/5

True or false?

Solving $y = (x - 1)^3 + 2$ for x gives $x = \sqrt[3]{(y - 2)} + 1$.

A. True

B. False

Solving $y = x^3$ for x gives

A. $y/3$

B. $\sqrt[3]{y}$

C. $\sqrt{y}$

D. y^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q4/5

Solve $y = x/(x + 1)$ for x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • HSF-BF.B.4.a

BACKWARD 3

Domain and range • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

Give the range of $f(x) = \sqrt{(x + 4)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q1/5

Give the range of $f(x) = 5 \cdot 2^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q2/5

True or false?

The range of $f(x) = 2x/(x - 3)$ excludes $y = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q3/5

The domain of $\log_3(x - 1)$ is

A. $x > 1$

B. all reals

C. $x > 0$

D. $x \geq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q4/5

What is the range of $f(x) = \log_3(x - 1)$?

A. $y > 0$

B. $y > 1$

C. All real numbers

D. $y \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q5/5

SET 58 • HSF-BF.B.4.a

FORWARD 1

Inverses that are their own inverse, and checking by points • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

$f(x) = (x + 1)/(x - 1)$, $x \neq 1$. Find $f^{-1}(x)$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q1/5

$f(x) = 5 \cdot 2^x$ passes through (1, 10), and $f^{-1}(x) = \log_2(x/5)$. Check the inverse with this point: find $f^{-1}(10)$.

$f^{-1}(10) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q2/5

True or false?

Every function has an inverse function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q3/5

Which is its own inverse?

A. $f(x) = x^2$

B. $f(x) = x + 1$

C. $f(x) = 2x$

D. $f(x) = 1/x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q4/5

Use the horizontal line test on the graph of $f(x) = x^3 - x$. Does f have an inverse function?

A. No: the horizontal line y = 0 meets the graph three times (x = −1, 0, 1), so f is not one-to-one.

B. Yes: f is a polynomial, and every polynomial passes the horizontal line test, so f has an inverse function.

C. Yes: every cubic passes the horizontal line test, because its graph rises from left to right, so f is one-to-one.

D. No: the vertical line x = 0 meets the graph three times, so f fails the vertical line test and is not one-to-one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q5/5

SET 58 • HSF-BF.B.4.a

FORWARD 2

Verify inverses by composition • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

$f(x) = (x - 1)^3 + 2$ and $g(x) = \sqrt[3]{(x - 2)} + 1$. Which computation of $f(g(x))$ is correct?

A. $f(g(x)) = f(x) \cdot g(x) = ((x - 1)^3 + 2)(\sqrt[3]{(x - 2)} + 1) = x$

B. $f(g(x)) = ((\sqrt[3]{(x - 2)} + 1) - 1)^3 + 1 = (\sqrt[3]{(x - 2)})^3 + 1 = x - 1$

C. $f(g(x)) = ((\sqrt[3]{(x - 2)} + 1) - 1)^3 + 2 = (\sqrt[3]{(x - 2)})^3 + 2 = x - 2 + 2 = x$

D. $f(g(x)) = \sqrt[3]{(((x - 1)^3 + 2) - 2)} + 1 = \sqrt[3]{((x - 1)^3)} + 1 = x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q1/5

$f(x) = (x - 1)^3 + 2$ and $g(x) = \sqrt[3]{(x - 2)} + 1$. Which computation of $g(f(x))$ is correct?

A. $g(f(x)) = \sqrt[3]{(((x - 1)^3 + 2) - 2)} + 1 = \sqrt[3]{((x - 1)^3)} + 1 = x - 1 + 1 = x$

B. $g(f(x)) = g(x) \cdot f(x) = (\sqrt[3]{(x - 2)} + 1)((x - 1)^3 + 2) = x$

C. $g(f(x)) = ((\sqrt[3]{(x - 2)} + 1) - 1)^3 + 2 = (\sqrt[3]{(x - 2)})^3 + 2 = x$

D. $g(f(x)) = \sqrt[3]{(((x - 1)^3 + 2) - 2)} + 2 = \sqrt[3]{((x - 1)^3)} + 2 = x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q2/5

True or false?

To verify inverses, both $f(g(x)) = x$ and $g(f(x)) = x$ must hold on the domains.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q3/5

For $f(x) = \sqrt{(x + 4)}$ and $g(x) = x^2 - 4$ $(x \geq 0)$, $g(f(x)) =$

A. x for all x

B. $|x|$

C. x for $x \geq -4$

D. $x^2 + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q4/5

Why is $g(x) = x^2 - 4$, without a domain restriction, not the inverse of $f(x) = \sqrt{(x + 4)}$?

A. Because $f(g(2)) \neq 2$, so the composition fails even for positive inputs.

B. $f(g(x)) = \sqrt{(x^2)} = |x|$, which is not x when $x < 0$; g is the inverse only on $x \geq 0$ (the range of f).

C. Because $g(x) = x^2 - 4$ is not a function, since it gives two outputs for each input.

D. Because $g(f(x)) = (\sqrt{(x + 4)})^2 - 4$ is undefined for every x, so the composition test cannot be done.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q5/5

Write exact logarithmic solutions of exponential equations

MAIN	Algebra II main question	**HSF-LE.A.4**
BACKWARD 1	Isolate the exponential expression	**HSA-REI.B.3**
BACKWARD 2	Powers of 2, 3, and 10	**8.EE.A.1**
BACKWARD 3	Logarithms as the inverse of exponentials	**HSF-BF.B.4.a**
FORWARD 1	Solve equations with different bases or variables on both sides	**HSF-LE.A.4**
FORWARD 2	Use the inverse relationship to solve logarithmic equations	**HSF-BF.B.5**

SET 59 • HSF-LE.A.4

MAIN

Algebra II main question • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $3 \cdot 2^{5t} = 48$ exactly.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q1/5

Which is the exact solution of $2 \cdot 10^{3t} = 9$?

A. $t = \log(9)/6$

B. $t = (\log 9 - 2)/3$

C. $t = 3 \log 4.5$

D. $t = \log(4.5)/3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q2/5

Which is the exact solution of $5e^{2t} = 30$?

A. $t = (\ln 6) \cdot 2$

B. $t = (\ln 6)/2$

C. $t = \ln 3$

D. $t = \ln 30/10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q3/5

Which is the exact solution of $100(1.05)^t = 300$?

A. $t = 3/1.05$

B. $t = \log 3 - \log 1.05$

C. $t = \log 3/\log 1.05$

D. $t = \log 300/\log 1.05$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q4/5

Which is the exact solution of $4 \cdot 3^{t/2} = 20$?

A. $t = 2 \log_3 5$

B. $t = \log_3 10$

C. $t = 2 \log_3 20$

D. $t = \log_3 5/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q5/5

SET 59 • HSF-LE.A.4

BACKWARD 1

Isolate the exponential expression • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve 3y = 48 for y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q1/5

Solve 5y = 30 for y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q2/5

True or false?

From $2 \cdot 10^{3t} = 9$, we get $10^{3t} = 4.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q3/5

From $4 \cdot 3^{t/2} = 20$, $3^{t/2}$ equals

A. 80

B. 5

C. 1/5

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q4/5

Solve $100y = 300$ for y.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q5/5

SET 59 • HSF-LE.A.4

BACKWARD 2

Powers of 2, 3, and 10 • Grade 8 • Expressions & Equations • 8.EE.A.1

Write 16 as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q1/5

Write 1000 as a power of 10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q2/5

True or false?

$3^4 = 12$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q3/5

$2^{5t} = 2^4$ gives

A. $t = 4$

B. $5t = 4$

C. $5t = 16$

D. $t = 20$

Write 1/8 as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • HSF-LE.A.4

BACKWARD 3

Logarithms as the inverse of exponentials • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

Rewrite $10^x = 4.5$ as a logarithm.

A. $x = 10^{4.5}$

B. $x = 4.5/10$

C. $x = \log 10/4.5$

D. $x = \log 4.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q1/5

Rewrite $e^u = 6$ as a logarithm.

A. $u = \log 6$

B. $u = 6/e$

C. $u = \ln 6$

D. $u = e^6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q2/5

True or false?

$\log_2 8 = 4$ because $2^4 = 8$.

A. True

B. False

$3^y = 5$ means y =

A. $\log_3 5$

B. 5/3

C. $\log_5 3$

D. 3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q4/5

Evaluate $\log_2 32$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q5/5

SET 59 • HSF-LE.A.4

FORWARD 1

Solve equations with different bases or variables on both sides • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $2^x = 3^{x-1}$. Give x to the nearest thousandth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q1/5

Solve $5^{2x} = 7$. Give x to the nearest thousandth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q2/5

True or false?

$\log(ab) = \log a + \log b$ for positive a and b.

A. True

B. False

The exact solution of $e^{x+1} = 10$ is

A. 10/e

B. ln 10/1

C. ln 9

D. ln 10 − 1

Solve $3 \cdot 4^x = 2 \cdot 5^x$. Give x to the nearest thousandth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q5/5

SET 59 • HSF-LE.A.4

FORWARD 2

Use the inverse relationship to solve logarithmic equations • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5

Solve $\log_2(x + 3) = 5$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q1/5

Solve $\log x + \log(x - 3) = 1$. Give every value found after combining the logarithms and solving, and the actual solutions (enter none if there are none).

Values found: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q2/5

True or false?

$\log_b(b^x) = x$ for every real x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q3/5

$\ln(e^7)$ equals

A. $\ln 7$

B. 1

C. 7

D. e^7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q4/5

Solve ln(2x) = 3 exactly.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q5/5

Evaluate logarithmic solutions of exponential models with technology

MAIN	Algebra II main question	**HSF-LE.A.4**
BACKWARD 1	Write exact logarithmic solutions	**HSF-LE.A.4**
BACKWARD 2	Evaluate logarithms with a calculator	**HSF-IF.A.2**
BACKWARD 3	Choose an appropriate level of accuracy	**HSN-Q.A.3**
FORWARD 1	Analyze how parameters change the solution	**HSF-LE.A.4**
FORWARD 2	Rewrite models to show doubling time or half-life	**HSF-IF.C.8.b**

SET 60 • HSF-LE.A.4

MAIN

Algebra II main question • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$2,500 earns 3.5% interest, credited once at the end of each year. Solve $2500(1.035)^t = 4000$ for t (to the nearest hundredth of a year), and give the number of yearly credits after which the balance first reaches at least $4,000. A calculator may be used.

t (years): _____

Number of yearly credits: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q1/5

Carbon-14 has a half-life of 5730 years. A sample has 30 of its original 80 units of carbon-14. Solve $80(0.5)^{t/5730} = 30$ to estimate its age to the nearest year. A calculator may be used.

Age (years): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q2/5

Coffee cools by $T(t) = 20 + 70(0.95)^t$ °C after t minutes. When is it 50°C? Give minutes to the nearest hundredth. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q3/5

A city of 1.2 million grows by $P(t) = 1.2e^{0.015t}$ million people. When does it reach 1.5 million? Give years to the nearest hundredth. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q4/5

A dose leaves $200(0.8)^t$ mg of a drug after t hours. At which whole hour is the amount first below 50 mg? A calculator may be used.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q5/5

SET 60 • HSF-LE.A.4

BACKWARD 1

Write exact logarithmic solutions • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Which is the exact solution of $1.035^t = 1.6$?

A. $t = \log(1.6 - 1.035)$

B. $t = \log 1.6/\log 1.035$

C. $t = \log 1.035/\log 1.6$

D. $t = 1.6/1.035$

Which is the exact solution of $e^{0.015t} = 1.25$?

A. t = ln 1.25/0.015

B. t = 1.25/0.015

C. t = ln(1.25/0.015)

D. t = 0.015 ln 1.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q2/5

True or false?

$0.8^t = 0.25$ gives $t = \log 0.25/\log 0.8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q3/5

$(0.5)^{t/5730} = 0.375$ gives t =

A. $0.375 \cdot 5730$

B. 5730/log 0.375

C. log 0.375/5730

D. 5730 log 0.375/log 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q4/5

Which is the exact solution of $0.95^t = 3/7$?

A. $t = \log(3/7) - \log 0.95$

B. $t = (3/7)/0.95$

C. $t = \log 0.95/\log(3/7)$

D. $t = \log(3/7)/\log 0.95$

SET 60 • HSF-LE.A.4

BACKWARD 2

Evaluate logarithms with a calculator • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Evaluate log 1.6 to four decimal places. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q1/5

Evaluate ln 1.25 to four decimal places. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q2/5

True or false?

log 0.25 is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q3/5

log 1.6/log 1.035 is about

A. 0.20

B. 46.0

C. 1.55

D. 13.66

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q4/5

Evaluate log 0.25/log 0.8 to two decimal places. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • HSF-LE.A.4

BACKWARD 3

Choose an appropriate level of accuracy • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3

Interest is added once a year. A model says a balance reaches its goal at t = 13.66 years. After how many whole years is the goal first reached?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q1/5

Round 8108.16 years to the nearest hundred years.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q2/5

True or false?

A time found from a model should be reported with more decimal places than the data justify.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q3/5

A drug amount first drops below 50 mg at t = 6.21 h. If it is checked every hour, it is first below 50 mg at

A. 6.21 h

B. 6 h

C. 7 h

D. 5 h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q4/5

A population model gives t = 14.88 years. Why might this be reported as "about 15 years"?

A. Because rounding a model's answer up always gives the exact answer for the population.

B. Because 14.88 is not a real number of years, so it must be replaced by a whole number.

C. The model and its data are approximate, so reporting more precision than they justify would be misleading.

D. Because answers about years must always be whole numbers, whatever the model shows.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q5/5

SET 60 • HSF-LE.A.4

FORWARD 1

Analyze how parameters change the solution • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$2,500 is invested at 5% interest, credited once at the end of each year. Using the model $2500(1.05)^t$, find the time t when the model first equals $4,000 (to the nearest hundredth of a year), and the number of yearly credits after which the account first holds at least $4,000. A calculator may be used.

Model time t (years): _____

Number of yearly credits: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q1/5

Money earns 6% interest per year. Using the model $(1.06)^t = 2$, find the doubling time to the nearest hundredth of a year, the Rule of 72 estimate 72/6, and the number of yearly credits after which the money is first at least doubled when interest is credited once a year. A calculator may be used.

Model doubling time (years): _____

Rule of 72 estimate: _____

Number of yearly credits: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F1-Q2**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q2/5

True or false?

Doubling the initial amount in $a(b)^t = d$ halves the time t needed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q3/5

In $200(0.8)^t = 50$, replacing 200 by 400 changes t by

A. doubling it

B. halving it

C. adding $\log 2/\log(1/0.8) \approx 3.11$ hours

D. nothing

At 3.5% yearly interest, why does $2,500 take as long to grow to $4,000 as $5,000 takes to grow to $8,000?

A. Because $5,000 is twice $2,500, and twice the money grows twice as fast, which exactly cancels the doubled target.

B. Both need the growth factor 1.6, and t = log 1.6/log 1.035 depends only on that factor, not on the starting amount.

C. Because both amounts earn the same interest in dollars each year, and both need to gain the same number of dollars.

D. They do not take the same time: the larger amount earns more interest each year, so it reaches its target sooner.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F1-Q5**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q5/5

SET 60 • HSF-LE.A.4

FORWARD 2

Rewrite models to show doubling time or half-life • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

Rewrite $2500(1.035)^t$ in the form $2500 \cdot 2^{t/D}$ and give D, in years to the nearest hundredth. A calculator may be used.

$D \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q1/5

Rewrite $200(0.8)^{t}$ in the form $200(0.5)^{t/H}$ and give H, in hours to the nearest hundredth. A calculator may be used.

H ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q2/5

True or false?

$80(0.5)^{t/5730}$ shows the half-life 5730 directly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q3/5

$1.2e^{0.015t}$ has doubling time about

A. 46.2 years

B. 66.7 years

C. 15 years

D. 0.015 years

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q4/5

A substance decays by $A = A_0(0.5)^{t/h}$, where h is its half-life. Why does this form make it easy to find how much remains after 3 half-lives?

A. After 3 half-lives $t/h = 3$, so $A = A_0(0.5)^3 = A_0/8$.

B. Because the substance is gone after 3 half-lives.

C. Because 3 half-lives leave 3/8 of the amount.

D. Because $A = A_0 - 3h$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q5/5

SET 61 • HSS-ID.A.4 • STATISTICS, INFERENCE, AND DECISIONS

Estimate population percentages with a normal distribution

MAIN	Algebra II main question	**HSS-ID.A.4**
BACKWARD 1	Interpret mean and standard deviation	**HSS-ID.A.2**
BACKWARD 2	Describe the shape of a distribution	**HSS-ID.A.1**
BACKWARD 3	Percents of a whole	**6.RP.A.3.c**
FORWARD 1	Find values from percentiles of a normal model	**HSS-ID.A.4**
FORWARD 2	Use a normal model for a margin of error	**HSS-IC.B.4**

SET 61 • HSS-ID.A.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Adult women's heights are approximately normal with mean 64 in and standard deviation 2.5 in. Estimate the percent between 59 and 69 inches. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q1/5

Adult women's heights are approximately normal with mean 64 in and standard deviation 2.5 in. Estimate the percent of women taller than 66.5 inches. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q2/5

Test scores are approximately normal with mean 500 and standard deviation 100. Estimate the percent of scores between 400 and 700. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q3/5

Battery life is approximately normal with mean 40 h and standard deviation 4 h. Use the table to estimate the percent of batteries lasting less than 46 hours.

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q4

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q4/5

Cereal box weights are approximately normal with mean 16.0 oz and standard deviation 0.2 oz. Use the table to estimate the percent between 15.8 and 16.3 oz.

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q5/5

SET 61 • HSS-ID.A.4

BACKWARD 1

Interpret mean and standard deviation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

A data set has mean 64 and standard deviation 2.5. What value is one standard deviation above the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q1/5

Which value is two standard deviations below a mean of 500 with standard deviation 100?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q2/5

True or false?

A larger standard deviation means the data are more spread out.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q3/5

46 is how many standard deviations above a mean of 40 when the standard deviation is 4?

A. 0.67

B. 4

C. 6

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q4/5

Compute $(15.8 - 16.0)/0.2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • HSS-ID.A.4

BACKWARD 2

Describe the shape of a distribution • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Which describes a histogram that is highest in the middle and falls off evenly on both sides?

A. skewed right

B. uniform

C. bimodal

D. symmetric and mound-shaped

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q1/5

In a symmetric distribution, how do the mean and the median compare?

A. The median is much larger than the mean.

B. The mean is much larger than the median.

C. They are about equal, both at the center.

D. They are at opposite ends of the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-B2-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q2/5

True or false?

A normal curve is symmetric about its mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q3/5

A bell-shaped histogram is called

A. skewed right

B. uniform

C. mound-shaped and symmetric

D. bimodal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q4/5

A continuous distribution is symmetric about its median. What percent of the distribution lies below the median? Give a percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q5/5

SET 61 • HSS-ID.A.4

BACKWARD 3

Percents of a whole • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Find 95% of 2,000.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q1/5

What is 34% + 13.5%?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q2/5

True or false?

Half of 68% is 36%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q3/5

100% − 95% split equally into two tails gives each tail

A. 5%

B. 47.5%

C. 0.5%

D. 2.5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q4/5

Write 0.7745 as a percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q5/5

SET 61 • HSS-ID.A.4

FORWARD 1

Find values from percentiles of a normal model • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Women's heights are approximately normal with mean 64 inches and standard deviation 2.5 inches. Above what height are the tallest 2.5% of women? Give the height in inches. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q1/5

Battery life is approximately normal with mean 40 hours and standard deviation 4 hours. Use the table to find the battery life at the 84th percentile, in hours.

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q2/5

True or false?

In a normal model, the median equals the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q3/5

The 16th percentile of scores with mean 500 and SD 100 is

A. 484

B. 600

C. 340

D. 400

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F1-Q4**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q4/5

A cereal box below 15.7 oz is rejected. Use the table to estimate the percent rejected (mean 16.0, SD 0.2).

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q5/5

SET 61 • HSS-ID.A.4

FORWARD 2

Use a normal model for a margin of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

Simulated sample proportions are approximately normal with mean 0.58 and standard deviation 0.049. Using the 68-95-99.7 rule, within what interval do about 95% of them fall?

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q1/5

Simulated sample proportions from random samples of 100 have standard deviation 0.049. What margin of error does this give for the population proportion? Give a decimal. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q2/5

True or false?

When a simulated statistic is approximately normal and the margin of error is defined as the half-width of a 95% range, the margin of error is about two standard deviations of the simulated statistic.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q3/5

Simulated sample means are approximately normal with standard deviation 0.2 hours. Using a margin of error of 2 standard deviations (the 95% convention), the margin of error is

A. 0.1 hours

B. 0.4 hours

C. 2 hours

D. 0.2 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q4/5

Why can the 68-95-99.7 rule be used to find a 95% range for simulated sample proportions from large random samples?

A. The population values must be normal, even though the sample proportions themselves are not normal.

B. For large enough samples their distribution is approximately normal, so about 95% fall within 2 SD of the mean.

C. Sample proportions are exactly normal for every sample size, so the rule needs no conditions at all.

D. The 68-95-99.7 rule applies to every histogram, whatever its shape, so no check is needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q5/5

SET 62 • HSS-ID.A.4 • STATISTICS, INFERENCE, AND DECISIONS

Recognize data sets for which a normal model is not appropriate

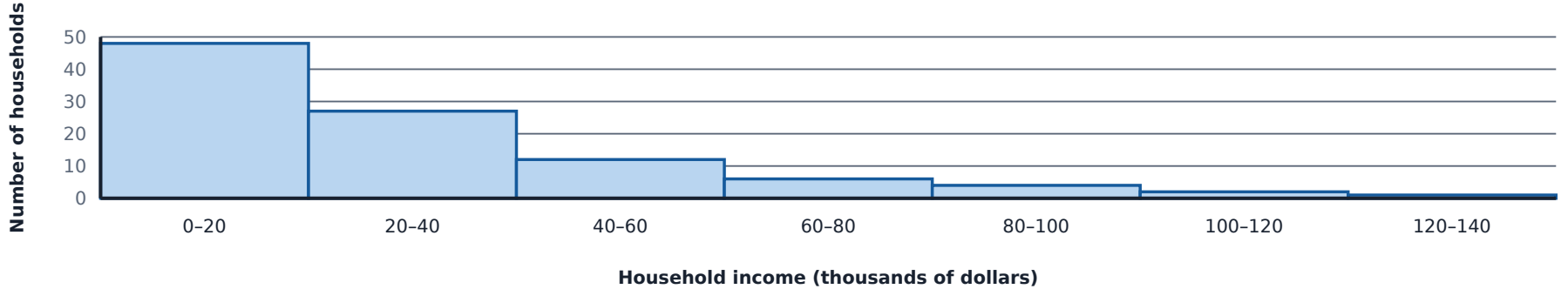
MAIN	Algebra II main question	**HSS-ID.A.4**
BACKWARD 1	Describe skewed, symmetric, bimodal, and uniform shapes	**HSS-ID.A.3**
BACKWARD 2	Measures of center and spread for a data set	**6.SP.A.2**
BACKWARD 3	Groups combined in one data set	**HSS-ID.A.1**
FORWARD 1	Check a normal model with the 68-95-99.7 rule	**HSS-ID.A.4**
FORWARD 2	Decide whether a probability model fits observed data	**HSS-IC.A.2**

SET 62 • HSS-ID.A.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

The histogram shows household incomes in a town. Is a normal model appropriate?



A. No: the distribution is strongly skewed right, not symmetric and mound-shaped.

B. No: it has two separate peaks, so it is bimodal rather than normal.

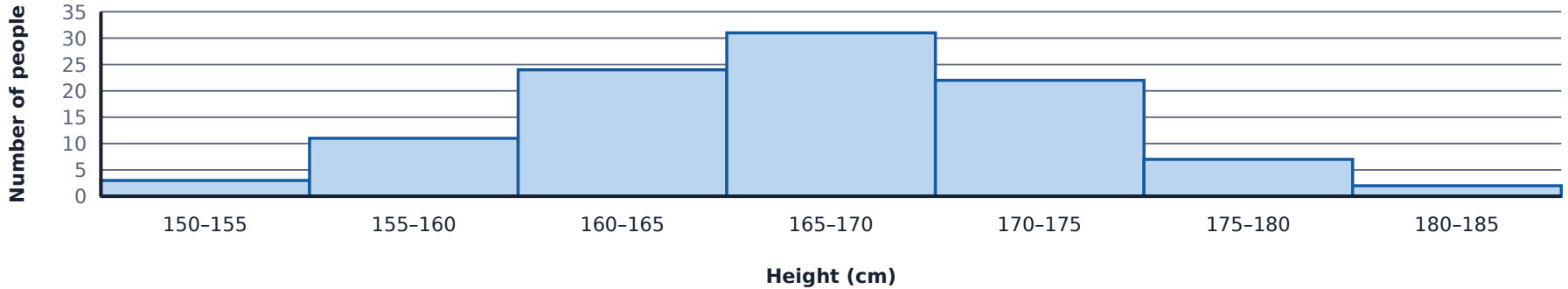
C. Yes: incomes are measured in dollars, and measurements are always normal.

D. Yes: it has a single peak, and any single-peaked distribution is normal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-M-Q1**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q1/5

The histogram shows heights of adults in a club. Is a normal model appropriate?



A. No: it has two separate peaks, one for each group in the club.

B. No: it is strongly skewed right with a long tail to the right.

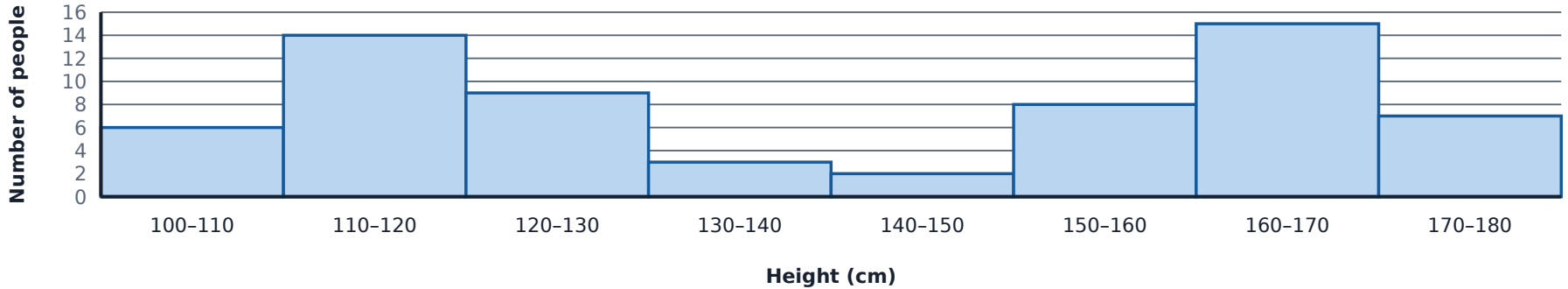
C. Yes: it is roughly symmetric and mound-shaped with a single peak.

D. No: the bars are not exactly equal on both sides of the peak.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-M-Q2**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q2/5

The histogram shows heights of all people at a school event (children and adults). Is a normal model appropriate?



A. No: it is uniform, with all of the bars the same height.

B. No: it has two peaks, from two different groups combined.

C. Yes: heights of people are always approximately normal.

D. Yes: it is roughly symmetric about its middle, so it is normal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-M-Q3**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q3/5

Waiting times at a clinic have mean 4 minutes, standard deviation 5 minutes, and minimum 0. Is a normal model appropriate?

A. No: a normal model requires the standard deviation to be smaller than 1 for any data set.

B. Yes: a normal model can describe any data set once its mean and standard deviation are known.

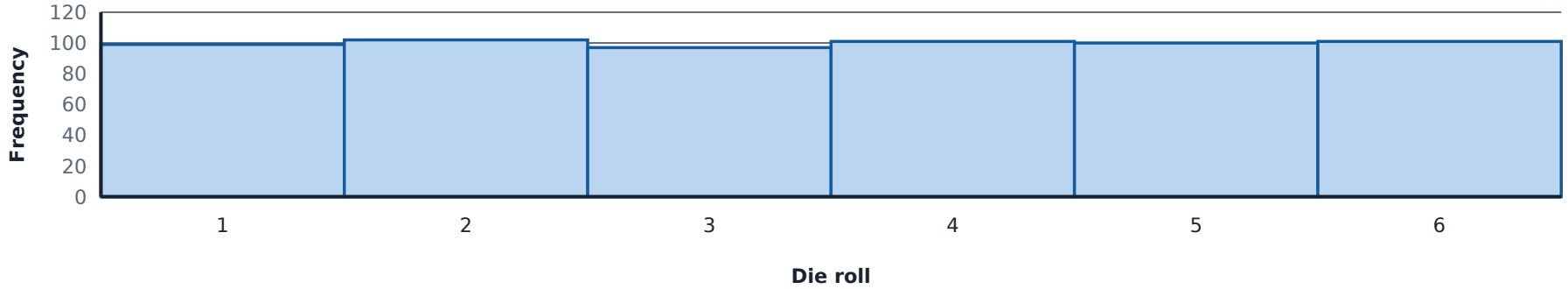
C. Yes: the mean and the standard deviation are both positive, so a normal model fits the times.

D. No: a normal model would put about 16% of times below $4 - 5 = -1$ minute, which is impossible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-M-Q4**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q4/5

The histogram shows the results of 600 rolls of a fair die. Is a normal model appropriate?



A. Yes: it is symmetric about 3.5, so it is normal.

B. No: it is strongly skewed left with a long tail.

C. No: it is roughly uniform (flat), not mound-shaped.

D. Yes: results of dice are always approximately normal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-M-Q5**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q5/5

SET 62 • HSS-ID.A.4

BACKWARD 1

Describe skewed, symmetric, bimodal, and uniform shapes • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

A histogram has a long tail to the right. Which describes its shape?

A. skewed right

B. symmetric

C. uniform

D. skewed left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q1/5

A histogram has two separate peaks. Which describes its shape?

A. symmetric with one peak

B. uniform

C. skewed left

D. bimodal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q2/5

True or false?

In a right-skewed distribution, the mean is usually less than the median.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q3/5

A histogram with bars all about the same height is

A. skewed left

B. uniform

C. bimodal

D. normal

Which measure of center is more resistant to a long tail?

A. the maximum

B. the median

C. the mean

D. both equally

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • HSS-ID.A.4

BACKWARD 2

Measures of center and spread for a data set • Grade 6 • Statistics & Probability • 6.SP.A.2

What does the standard deviation of a data set describe?

A. how spread out the values are around the mean

B. the middle value when the data are in order

C. the number of values in the data set

D. the most common value in the data set

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q1/5

True or false?

A data set has all values at least 0 and mean 4. Some value in the data set can be more than 4 below the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q2/5

True or false?

A distribution can be described by its center, spread, and shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q3/5

Which is a measure of center?

A. median

B. range

C. standard deviation

D. IQR

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q4/5

Give one value two standard deviations below a mean of 4 with SD 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • HSS-ID.A.4

BACKWARD 3

Groups combined in one data set • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Two groups with very different centers are combined into one data set. What will the histogram of the combined data most likely show?

A. a flat shape

B. a long right tail only

C. two peaks

D. one peak exactly in the middle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q1/5

Which data set is most likely to have a histogram with two peaks?

A. heights of the parents only

B. heights of 5-year-olds and their parents together

C. heights of the 5-year-olds only

D. ages of students in one 10th-grade class

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q2/5

True or false?

A histogram of one group can look different from a histogram of two groups combined.

A. True

B. False

A bimodal histogram suggests

A. an error in counting

B. a small standard deviation

C. two groups in the data

D. a normal model

Heights of all people at a school event (young children and adults) form a two-peaked histogram. Which change is most likely to give mound-shaped distributions?

A. Add more children to the combined data set.

B. Round every height in the combined data to the nearest centimeter.

C. Remove the tallest person from the combined data.

D. Make separate histograms for the children and for the adults.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q5/5

SET 62 • HSS-ID.A.4

FORWARD 1

Check a normal model with the 68-95-99.7 rule • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

In 1,000 measurements, 70% are within 1 SD of the mean and 96% are within 2 SDs. Is a normal model reasonable?

A. No: the percents must be exactly 68% and 95%.

B. Yes: any data set with 1,000 values is normal.

C. Yes: 70% and 96% are close to the normal 68% and 95%.

D. No: 96% is more than 95%, so the data cannot be normal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q1/5

A data set has 85% of its values within 1 SD of the mean and 91% within 2 SDs. Is a normal model reasonable?

A. Yes: 91% is close to 95%, so 85% does not matter.

B. Yes: both percents are above 50%.

C. No: a normal model needs 100% within 2 SDs.

D. No: 85% is far above 68% and 91% is below 95%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q2/5

True or false?

If a data set has an extreme outlier, the normal model may still fit the rest but the mean and SD are distorted.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q3/5

About what percent of normal data lies more than 3 SDs from the mean?

A. 5%

B. 3%

C. 0.3%

D. 32%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F1-Q4**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q4/5

A data set of 200 values has 25 values more than 2 SDs from the mean. What percent of the values is that, and about what percent does the 68-95-99.7 rule predict for a normal model?

Observed percent: _____

Normal model percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q5/5

SET 62 • HSS-ID.A.4

FORWARD 2

Decide whether a probability model fits observed data • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A spinner is claimed to land on each of 4 colors equally often. In 400 spins the counts are 98, 105, 97, 100. Are the results consistent with the claim?

A. Yes: each count is close to the expected 100.

B. No: the counts are not all exactly 100.

C. No: 105 proves the spinner favors one color.

D. Yes: any 4 counts adding to 400 are consistent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q1/5

A model says a coin is fair. In 50 flips there were 34 heads. In 200 simulated runs of 50 fair flips, 3 runs had 34 or more heads. Use 5% as the cutoff for unusual. What do you conclude?

A. 1.5% is not unusual, so the data show that the fair-coin model fits.

B. 1.5% < 5%: unusual for a fair coin, so evidence against the model.

C. No simulated run had exactly 34 heads, so a fair coin has probability 0 of giving 34.

D. 1.5% < 5%: this proves the coin is unfair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q2/5

True or false?

A model is rejected only if the observed result would be unusual under the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q3/5

Under a model, a simulated result at least as extreme as the observed one occurs in 34 of 200 runs. Use 5% as the cutoff for unusual. The model is

A. proved true

B. rejected

C. consistent with the data

D. impossible

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q4/5

How does judging a normal model for a data set differ from judging a probability model with a simulation?

A. Both check whether the data form a bell curve; the simulation just repeats the same comparison many more times.

B. Both compare the sample mean with the population mean and ignore the shape of the data and the simulation.

C. The first uses many simulated runs of the model; the second only compares the shape of the histogram with a bell curve.

D. The first compares the shape of the data with a normal curve; the second asks whether an observed result is unusual among simulated results.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q5/5

SET 63 • HSS-IC.A.1 • STATISTICS, INFERENCE, AND DECISIONS

Distinguish population, sample, parameter, and statistic

MAIN	Algebra II main question	**HSS-IC.A.1**
BACKWARD 1	Samples and populations	**7.SP.A.1**
BACKWARD 2	Draw inferences from random samples	**7.SP.A.2**
BACKWARD 3	Compute sample proportions and means	**6.SP.B.5.c**
FORWARD 1	Describe sampling variability	**HSS-IC.A.1**
FORWARD 2	Quantify variability with a margin of error	**HSS-IC.B.4**

SET 63 • HSS-IC.A.1

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

A city surveys 400 randomly chosen residents, and 62% of them support a new park. Which identifies the population, sample, parameter, and statistic correctly?

A. Population: all residents; sample: the 62% who support it; parameter: 400; statistic: 62%.

B. Population: all residents; sample: the 400 surveyed; parameter: percent of all residents in support; statistic: 62%.

C. Population: the 400 surveyed; sample: all residents; parameter: 62%; statistic: percent of all residents in support.

D. Population: all residents; sample: the 400 surveyed; parameter: 62%; statistic: percent of all residents in support.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q1/5

A factory measures 50 randomly selected bolts from a day's production; their mean length is 5.02 cm. Which identifies the parameter and the statistic?

A. Parameter: 5.02 cm, the mean of the 50 bolts; statistic: the mean length of all bolts made that day.

B. Parameter: the number of bolts measured, 50; statistic: the mean length of those bolts, 5.02 cm.

C. Parameter: the mean length of all bolts made that day; statistic: 5.02 cm, the mean of the 50 bolts.

D. Parameter: the 50 bolts that were measured; statistic: all of the bolts that were made that day.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q2/5

A news website posts a poll; 80% of the 2,000 visitors who answered want a later school start. Can this result be generalized to all students?

A. No: it is a voluntary-response sample, so people with strong opinions are over-represented.

B. Yes: 80% is so far above 50% that any bias could not change the overall conclusion.

C. No: a poll can never be used to estimate a population percent, even with random sampling.

D. Yes: 2,000 responses form a large sample, so the result must be accurate for all students.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q3/5

Why does a random sample allow conclusions about the whole population?

A. A random sample always includes every member of the population, so nothing is left out of it.

B. Chance decides who is chosen, so the sample tends to represent the population without systematic bias.

C. A random sample is always larger than a voluntary sample, so its result is always more exact.

D. A random sample always has exactly the same statistic as the population parameter it estimates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q4/5

A school uses the first 30 students arriving in the morning to estimate the students' average hours of sleep. Which statement is correct?

A. It is a random sample, because the times the students arrive vary by chance.

B. It is not random, but 30 students is enough to remove any bias from the estimate.

C. It is a random sample, because no one chose the students by name or by any other rule.

D. It is not a random sample: early arrivers may sleep differently, so the estimate may be biased.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q5/5

SET 63 • HSS-IC.A.1

BACKWARD 1

Samples and populations • Grade 7 • Statistics & Probability • 7.SP.A.1

In statistics, what is a population?

A. the number of people in a city

B. the part of the group that is measured

C. the entire group a study wants to describe

D. a number computed from a sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q1/5

In statistics, what is a sample?

A. the part of the population that is studied

B. the entire group of interest

C. a number describing the whole population

D. a list of every possible outcome

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q2/5

True or false?

A random sample gives every member of the population a chance to be chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q3/5

Surveying every student in a school is

A. a statistic

B. an experiment

C. a sample

D. a census

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q4/5

Why do studies often use a sample instead of the whole population?

A. Samples have no chance variation at all.

B. Samples always give the exact population value.

C. A population cannot be described by a number.

D. Studying everyone may be too costly, slow, or impossible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q5/5

SET 63 • HSS-IC.A.1

BACKWARD 2

Draw inferences from random samples • Grade 7 • Statistics & Probability • 7.SP.A.2

In a random sample of 50 students, 20 walk to school. Estimate how many of 600 students walk.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q1/5

Two random samples of 500 voters from the same city give 40% and 44% support for a plan. What is the best explanation for the difference?

A. The population percent changed between the two samples.

B. Random samples vary by chance, so a 4-point difference is not surprising.

C. One of the two samples must have been chosen incorrectly.

D. Random samples from one population always give the same percent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q2/5

True or false?

Different random samples from the same population usually give slightly different results.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q3/5

From a random sample, 30% of 200 voters prefer a candidate. The best estimate for all voters is

A. 30%

B. 200

C. cannot estimate

D. 60%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q4/5

Why is a larger random sample usually better for estimating a population value?

A. Its statistic varies less from sample to sample, so the estimate is more precise.

B. It always gives exactly the population value, with no chance variation.

C. It makes the population value larger, so the estimate is closer to it.

D. It removes all bias, even when the sample is not chosen at random.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • HSS-IC.A.1

BACKWARD 3

Compute sample proportions and means • Grade 6 • Statistics & Probability • 6.SP.B.5.c

248 of 400 people say yes. What proportion is that?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q1/5

Find the mean of 5.0, 5.1, 4.9, 5.08.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q2/5

True or false?

A proportion can be greater than 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q3/5

1,600 of 2,000 is

A. 16%

B. 1.25%

C. 80%

D. 20%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q4/5

Write 62% as a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q5/5

SET 63 • HSS-IC.A.1

FORWARD 1

Describe sampling variability • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

Two random samples of 400 residents of the same city give 62% and 59% support for a park. Which statement is correct?

A. One sample must be biased, since random samples agree exactly.

B. Each sample estimates a different population percent.

C. The samples differ by chance; both estimate the same population percent.

D. The population percent is exactly their average, 60.5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q1/5

Would random samples of 40 or random samples of 400 from the same population vary more from sample to sample?

A. Samples of 40: small samples are always biased.

B. Samples of 400: more people means more variation.

C. Samples of 40: smaller samples are affected more by chance.

D. They vary the same amount, since both are random.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q2/5

True or false?

A parameter changes from sample to sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q3/5

Which is a statistic?

A. the population size

B. the mean height of 30 sampled students

C. the mean height of all students

D. the population proportion

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F1-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q4/5

Why can a large voluntary-response sample be worse than a small random sample?

A. Voluntary samples are random, but a large one has too many answers to analyze well.

B. Large samples always have a larger margin of error than small random samples.

C. A large size does not remove bias; volunteers are systematically unrepresentative.

D. Small samples have no chance variation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F1-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q5/5

SET 63 • HSS-IC.A.1

FORWARD 2

Quantify variability with a margin of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A city surveys a random sample of 400 residents, and 62% support a new park. Simulated sample proportions (samples of 400 with p = 0.62) have standard deviation 0.024. What is the margin of error? Give a decimal. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q1/5

In a random sample of 400 residents, 62% support a new park, and the margin of error is 4.8 percentage points. Give the interval estimate for the percent of all residents who support the park.

Lower end (%): _____

Upper end (%): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q2/5

True or false?

A margin of error describes chance variation, not bias.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q3/5

If the sample size is multiplied by 4, the margin of error is about

A. unchanged

B. halved

C. doubled

D. divided by 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q4/5

A random sample of 400 residents gives 62% support for a park, with the 95% interval 57.2% to 66.8%. What does the interval say?

A. Exactly 62% of all of the residents of the city support the park.

B. The percent of all residents who support the park is plausibly between 57.2% and 66.8%.

C. 95% of all of the residents of the city support the park.

D. Between 57.2% and 66.8% of the 400 surveyed residents support the park.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q5/5

SET 64 • HSS-IC.A.2 • STATISTICS, INFERENCE, AND DECISIONS

Decide whether a model is consistent with observed results using a simulation

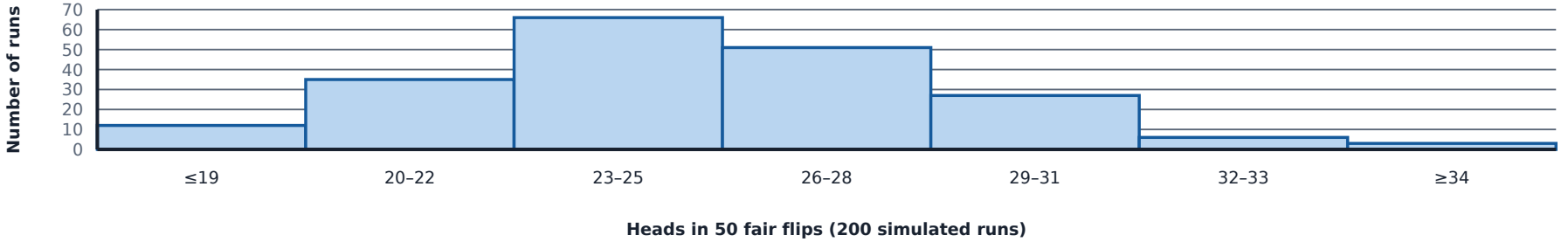
MAIN	Algebra II main question	**HSS-IC.A.2**
BACKWARD 1	Probability as long-run relative frequency	**7.SP.C.6**
BACKWARD 2	Use simulations of compound events	**7.SP.C.8.c**
BACKWARD 3	Relative frequency of a group of outcomes	**6.RP.A.3.c**
FORWARD 1	Design a simulation to test a model	**HSS-IC.A.2**
FORWARD 2	Compare two treatments with a re-randomization simulation	**HSS-IC.B.5**

SET 64 • HSS-IC.A.2

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A coin is claimed to be fair. It landed heads 34 times in 50 flips. The histogram shows 200 simulated runs of 50 flips of a fair coin. Count runs with 34 or more heads. Use 5% as the cutoff for unusual. Which conclusion is justified?



A. 3 of 200 runs (1.5%) had 34 or more heads; 1.5% < 5%, so the data are evidence against a fair coin.

B. 3 of 200 runs (1.5%) had 34 or more heads, so the coin is proved to be unfair.

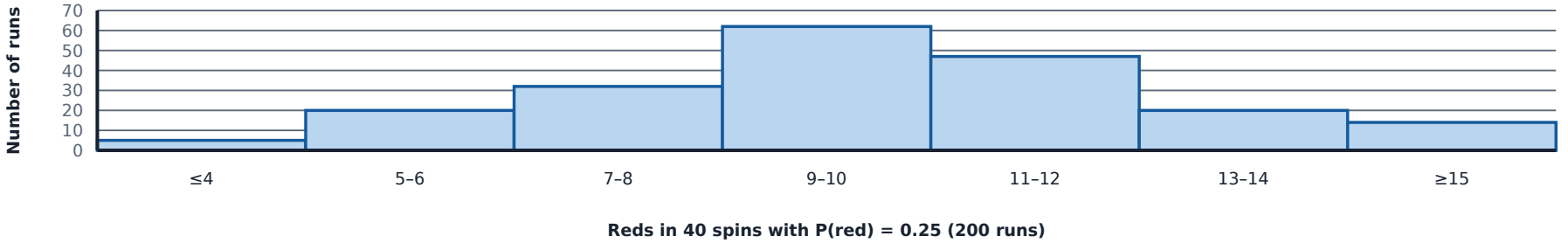
C. A fair coin can never land heads 34 times in 50 flips, so the claim is false.

D. 36 of 200 runs (18%) had 29 or more heads, so 34 heads is not unusual for a fair coin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q1/5

A spinner is claimed to land on red 25% of the time. It landed on red 13 times in 40 spins. The histogram shows 200 simulated runs of 40 spins under the claim. Count runs with 13 or more reds. Use 5% as the cutoff for unusual. Which conclusion is justified?



A. The expected count is 10, and 13 is not 10, so the claim is false.

B. 34 of 200 runs (17%) had 13 or more reds, so the claim is proved true.

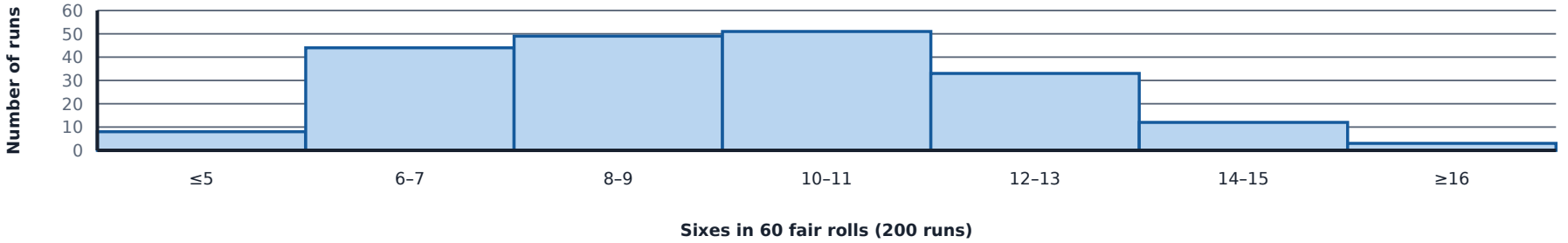
C. Only runs with exactly 13 reds count, which is rare, so the claim is rejected.

D. 34 of 200 runs (17%) had 13 or more reds; 17% > 5%, so the result is consistent with the claim.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q2/5

A die is claimed to be fair. It showed 16 sixes in 60 rolls. The histogram shows 200 simulated runs of 60 rolls of a fair die. Count runs with 16 or more sixes. Use 5% as the cutoff for unusual. Which conclusion is justified?



A. 15 of 200 runs (7.5%) had 14 or more sixes, so 16 sixes is not unusual.

B. 3 of 200 runs (1.5%) had 16 or more sixes; 1.5% < 5%, so the data are evidence against a fair die.

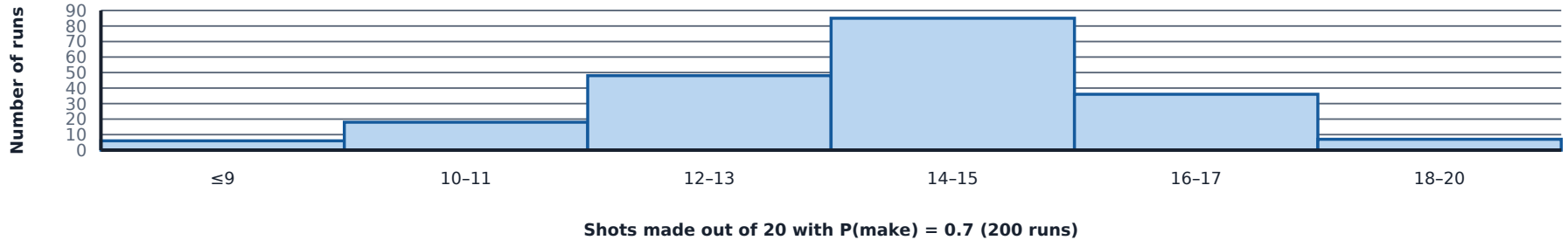
C. 3 of 200 runs (1.5%) had 16 or more sixes, so the die is proved to be unfair.

D. A fair die can never show 16 sixes in 60 rolls, so the claim is false.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q3**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q3/5

A player claims to make 70% of her free throws. She made 11 of 20. The histogram shows 200 simulated runs of 20 shots under the claim. Count runs with 11 or fewer makes. Use 5% as the cutoff for unusual. Which conclusion is justified?



A. She made 11, not the expected 14, so the claim is false.

B. 24 of 200 runs (12%) had 11 or fewer makes, so the 70% claim is proved true.

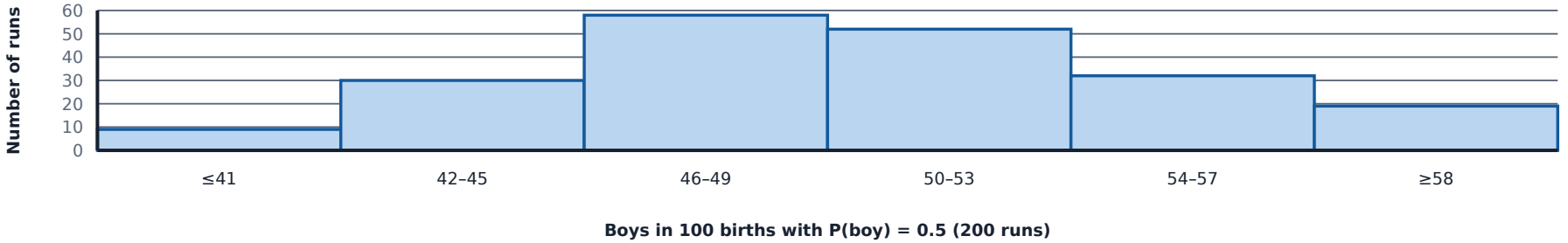
C. 24 of 200 runs (12%) had 11 or fewer makes; 12% > 5%, so the result is consistent with the claim.

D. 6 of 200 runs (3%) had 9 or fewer makes; 3% < 5%, so 11 makes is unusual.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q4/5

A model says boys and girls are equally likely. A hospital had 58 boys in 100 births. The histogram shows 200 simulated runs of 100 births under the model. Count runs with 58 or more boys. Use 5% as the cutoff for unusual. Which conclusion is justified?



A. 19 of 200 runs (9.5%) had 58 or more boys, so the model is proved true.

B. 19 runs had 58 or more boys, and 19 is more than 5, so the result is unusual.

C. There were 58 boys, not the expected 50, so the model is false.

D. 19 of 200 runs (9.5%) had 58 or more boys; 9.5% > 5%, so the result is consistent with the model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q5/5

SET 64 • HSS-IC.A.2

BACKWARD 1

Probability as long-run relative frequency • Grade 7 • Statistics & Probability • 7.SP.C.6

A fair coin is flipped 50 times. About how many heads do you expect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q1/5

A spinner has P(red) = 0.25. About how many reds in 40 spins?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q2/5

True or false?

Getting exactly the expected number every time is likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q3/5

A fair die is rolled 60 times. Expected sixes:

A. 10

B. 16

C. 6

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q4/5

A fair coin is flipped 50 times and lands heads 26 times. Why is this not surprising?

A. 26 is close to the expected 25, and results vary by chance.

B. Every number of heads from 0 to 50 is equally likely.

C. A fair coin always gives exactly 26 heads in 50 flips.

D. 26 is more than half, which proves the coin is fair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q5/5

SET 64 • HSS-IC.A.2

BACKWARD 2

Use simulations of compound events • Grade 7 • Statistics & Probability • 7.SP.C.8.c

Which design simulates 50 flips of a fair coin with random digits 0-9?

A. Use 50 random digits; only the digit 5 = heads.

B. Use 50 random digits; even = heads, odd = tails.

C. Use 50 random digits; 0 = heads, 1-9 = tails.

D. Use 1 random digit and multiply it by 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q1/5

Which design simulates one spin of a spinner with P(red) = 0.25 using two-digit random numbers 00-99?

A. 00-25 = red; 26-99 = not red

B. 25 = red; all other numbers = not red

C. 00-49 = red; 50-99 = not red

D. 00-24 = red; 25-99 = not red

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q2/5

True or false?

Each run of a simulation should copy the real process under the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q3/5

A simulation has 200 runs. Each run represents

A. the final answer

B. one repetition of the whole experiment

C. one histogram bar

D. one flip

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q4/5

Why does a simulation use many runs instead of just one?

A. Because a single run always gives the wrong answer.

B. Because more runs change the model's probabilities.

C. To see the pattern of results the model produces by chance.

D. To make the model match the observed data exactly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q5/5

SET 64 • HSS-IC.A.2

BACKWARD 3

Relative frequency of a group of outcomes • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

3 of 200 runs is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q1/5

6 + 18 runs out of 200 is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q2/5

True or false?

19 out of 200 is 19%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q3/5

34 of 200 is

A. 1.7%

B. 17%

C. 6.8%

D. 34%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q4/5

Is 2% less than 5%? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q5/5

SET 64 • HSS-IC.A.2

FORWARD 1

Design a simulation to test a model • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A player claims to make 70% of her free throws, and she made 11 of 20. Which simulation tests the claim?

A. Digits 0-6 = make; 20 digits per run; repeat many runs; find how often 11 or fewer makes occur.

B. Digits 0-6 = make; 20 digits in a single run; the claim is true if that run has 14 makes.

C. Digits 0-4 = make; 20 digits per run; repeat many runs; find how often exactly 11 makes occur.

D. Digits 0-6 = make; 11 digits per run; repeat many runs; find how often 20 makes occur.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q1/5

A player who claims to make 70% of free throws made 11 of 20. Why should a simulation count runs with 11 or fewer makes, not only runs with exactly 11?

A. Any one exact count can be rare; the question is how often results at least this far below the claim occur.

B. Runs with exactly 11 makes are impossible under the model, so they would never be counted at all.

C. The claim is about making 11 or fewer shots, not about making 70% of her free throws.

D. Counting 11 or fewer always makes the result significant, which is the goal of the simulation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F1-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q2/5

True or false?

A result that happens in 30% of simulated runs is strong evidence against the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q3/5

A common cutoff for "unusual" is

A. about 50% of runs

B. 100% of runs

C. about 5% of runs

D. about 25% of runs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q4/5

A simulation of a model is run 400 times instead of 200 times. What does this change?

A. It changes the observed data that the model is compared with.

B. It changes the probabilities that the model assigns to each outcome.

C. It turns an unusual result into a usual one, because more runs are counted.

D. It gives a more precise estimate of how often a result occurs under the model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q5

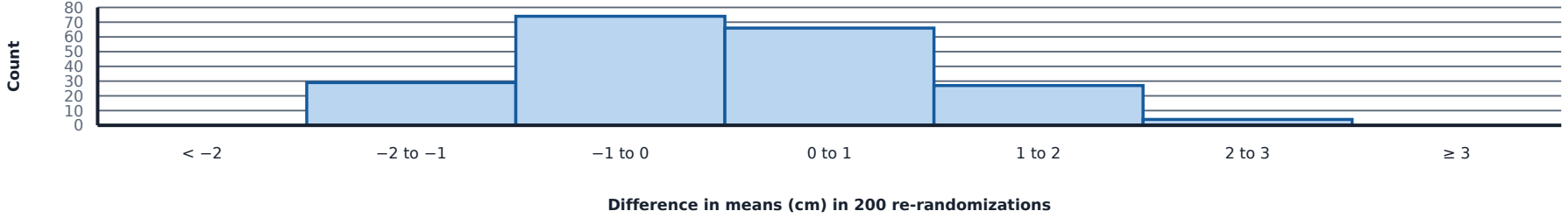
High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q5/5

SET 64 • HSS-IC.A.2

FORWARD 2

Compare two treatments with a re-randomization simulation • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

In an experiment, 20 plants were randomly assigned to fertilizer or no fertilizer, and the fertilized plants grew 2.9 cm more on average. The histogram shows 200 re-randomized differences under the model that fertilizer has no effect; none was 2.9 cm or more. Use 5% as the cutoff for unusual. What do you conclude?



A. 0/200 < 5%: a difference this large is very unlikely by chance, so fertilizer appears to increase growth.

B. The difference is not significant, because no re-randomized difference matched it exactly.

C. 0/200: a difference this large has probability exactly 0, so the fertilizer is proved to work.

D. No conclusion is possible, because the result must first be compared with a fair coin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F2-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q1/5

What model does a re-randomization simulation assume?

A. The two groups were chosen by voluntary response.

B. The observed difference is exactly the true effect.

C. The treatment has no effect, so the group labels can be shuffled.

D. The treatment has the same large effect on every subject.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q2/5

True or false?

Re-randomization is a simulation of a model, like the coin-flip simulation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q3/5

Under the model of no treatment effect, 60 of 200 re-randomized differences were at least as large as the observed one. Use 5% as the cutoff for unusual. The result is

A. significant

B. proof of no effect

C. not statistically significant

D. impossible

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q4/5

How does a re-randomization test for an experiment extend the reasoning of testing a fair-coin claim with a simulation?

A. Both flip a real coin many times and then count how many heads appear in each of the groups of subjects in the study.

B. Both ask how often a model produces a result at least as extreme as the observed one; here the model is "no treatment effect."

C. The coin test uses a probability model, but the re-randomization test does not use any model of the results at all.

D. Both prove the model is true whenever the observed result turns out to be common among the simulated results.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q5/5

SET 65 • HSS-IC.B.3 • STATISTICS, INFERENCE, AND DECISIONS

Identify sample surveys, experiments, and observational studies

MAIN	Algebra II main question	**HSS-IC.B.3**
BACKWARD 1	Statistical questions and data collection	**6.SP.A.1**
BACKWARD 2	Association is not causation	**HSS-ID.C.9**
BACKWARD 3	Random samples	**7.SP.A.1**
FORWARD 1	Identify confounding and design flaws	**HSS-IC.B.3**
FORWARD 2	Analyze the results of an experiment	**HSS-IC.B.5**

SET 65 • HSS-IC.B.3

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Researchers randomly assign 100 volunteers to take a vitamin or a placebo and compare the number of colds. What type of study is this, and what is its purpose?

A. Experiment: to estimate the percent of all people who catch colds.

B. Observational study: to describe colds without any treatment.

C. Experiment: to test whether the vitamin causes fewer colds.

D. Sample survey: to estimate the percent of people who take vitamins.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q1/5

A pollster asks a random sample of 1,000 voters which candidate they prefer. What type of study is this, and what is its purpose?

A. Sample survey: to estimate the preference of all voters.

B. Experiment: to test whether a candidate causes votes.

C. Observational study: to show which candidate causes turnout.

D. Sample survey: to show the poll caused voters' choices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q2/5

Researchers compare the hospital records of smokers and nonsmokers for lung disease. What type of study is this, and can it by itself show causation?

A. Experiment; yes, smoking was the treatment in the study.

B. Observational study; no, the groups were not randomly assigned.

C. Observational study; yes, hospital records are accurate.

D. Sample survey; yes, the sample of records is large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q3**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q3/5

Which study best answers "Does playing music while studying improve test scores?"

A. a census of how many students in the school own headphones

B. an observational study of students who already choose to play music

C. a survey that asks students whether they think music helps them study

D. an experiment: randomly assign students to study with or without music and compare scores

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q4/5

Which study best answers "What percent of students at the school use the library weekly?"

A. a sample survey of a random sample of the school's students

B. a voluntary online poll posted on the library website

C. a randomized experiment that assigns students to use the library

D. an observational study of students in the library today

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q5/5

SET 65 • HSS-IC.B.3

BACKWARD 1

Statistical questions and data collection • Grade 6 • Statistics & Probability • 6.SP.A.1

Is "How many hours do students at our school sleep?" a statistical question?

A. No: it has a single correct answer.

B. Yes: the answers vary from student to student.

C. No: hours of sleep cannot be measured.

D. Yes: it asks about one particular student.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q1/5

Which method of collecting data best answers "What percent of the city's voters favor the new library?"

A. an experiment that assigns voters to read books

B. a survey of people leaving the old library

C. asking the members of the city council

D. a survey of a random sample of the city's voters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q2/5

True or false?

An experiment imposes a treatment on the subjects.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q3/5

Watching and recording without imposing treatments is

A. an experiment

B. an observational study

C. a census

D. a placebo

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q4/5

In an experiment, what is a placebo?

A. a random sample of the population

B. the group that gets the real treatment

C. a fake treatment with no active ingredient

D. a measurement taken before the study

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q5/5

SET 65 • HSS-IC.B.3

BACKWARD 2

Association is not causation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Ice cream sales and sunburns are positively correlated. Which explanation is best supported?

A. Sunburns cause ice cream sales, because a correlation shows the direction of cause.

B. The correlation must be a calculation error, because the variables are unrelated.

C. Ice cream causes sunburn, because the two variables are positively correlated.

D. Hot, sunny weather increases both; the correlation does not show that ice cream causes sunburn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q1/5

"Students with more books at home get higher test scores." Which is a plausible lurking variable?

A. family income

B. the test's page count

C. the student's ID number

D. the color of the book covers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q2/5

True or false?

A strong correlation proves that one variable causes the other.

A. True

B. False

A variable that affects both variables in a study is

A. the placebo

B. the sample

C. a confounding (lurking) variable

D. the response

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q4/5

Which kind of study can show cause and effect?

A. a voluntary-response survey

B. an observational study

C. a randomized experiment

D. a sample survey

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q5/5

SET 65 • HSS-IC.B.3

BACKWARD 3

Random samples • Grade 7 • Statistics & Probability • 7.SP.A.1

Which method gives a simple random sample of 50 students from a school list?

A. Choose 50 students who are standing in the cafeteria line at lunch.

B. Number the students and use a random number generator to pick 50 different numbers.

C. Ask for 50 volunteers by posting a sign-up sheet in the school office.

D. Choose the first 50 names on the list, which is in alphabetical order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q1/5

Is surveying the students in the cafeteria line a random sample of the school?

A. Yes: the students in the line arrived in a random order.

B. No: a random sample must include every student.

C. No: students who are not in the line have no chance to be chosen.

D. Yes: any student could be in the cafeteria line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q2/5

True or false?

A random sample tends to represent the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q3/5

Which sample is random?

A. the teacher's friends among the students

B. students who volunteer when asked in class

C. names drawn from a hat with every student's name

D. the first 50 students to arrive at school

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q4/5

Why does a survey need a random sample?

A. so it needs to ask only 2 people

B. so every person gives the same answer

C. so it can show cause and effect by itself

D. so its results can be generalized to the population

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q5/5

SET 65 • HSS-IC.B.3

FORWARD 1

Identify confounding and design flaws • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Students who choose to take an SAT prep course score higher than students who do not. Which is a confounding variable, with a correct reason?

A. Hair color: it differs between students, so it must confound the result of the study.

B. Motivation: students who choose the course may be more motivated, which also raises scores.

C. The test date: all students take the same test, so the date explains the difference.

D. The course itself: it raises scores, so it is the confounding variable in the study.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F1-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q1/5

Students who choose to take an SAT prep course score higher than students who do not. Which design could show whether the course causes higher scores?

A. Randomly assign volunteers to take the course or not, then compare their scores.

B. Let students choose whether to take the course again, but use a larger sample.

C. Compare the scores of students at two different schools that offer the course.

D. Survey many more students who chose to take the course about their scores.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q2/5

True or false?

An experiment without a control group can still show the treatment caused the result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q3/5

In an experiment, why use a placebo?

A. to make the study observational instead

B. to control for believing one is treated

C. to increase the sample size of the study

D. to make the real treatment work better

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q4/5

Why is an observational study, rather than an experiment, used to study whether smoking causes lung disease in people?

A. Experiments cannot use random assignment.

B. It would be unethical to assign people to smoke.

C. Smokers cannot be counted in an experiment.

D. Observational studies always show causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F1-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q5/5

SET 65 • HSS-IC.B.3

FORWARD 2

Analyze the results of an experiment • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

100 volunteers were randomly assigned to a vitamin or a placebo; the vitamin group had 1.2 fewer colds on average. Under the model that the vitamin has no effect, 40 of 200 re-randomizations gave the vitamin group at least 1.2 fewer colds. Use 5% as the cutoff for unusual. What do you conclude?

A. 40/200 = 20% > 5%: the difference could easily be chance, so it is not significant.

B. 40/200 = 20% > 5%: this proves the vitamin has no effect.

C. 40/200 = 20%: the vitamin prevents 20% of colds in these volunteers.

D. 40/200 = 20% > 5%: the difference is significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q1/5

In a randomized vitamin-versus-placebo experiment, the vitamin group had 1.2 fewer colds on average. Under the model that the vitamin has no effect, 2 of 200 re-randomized differences were that large in the vitamin's favor. Use 5% as the cutoff for unusual. What do you conclude?

A. 2/200 = 1%: this proves the vitamin works for everyone.

B. 2/200 = 1%: the vitamin works for only 1% of people.

C. 2/200 = 1% < 5%: significant, so the vitamin appears to reduce colds.

D. 2/200 = 1% < 5%: the difference is not significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q2/5

True or false?

Random assignment makes it possible to attribute a significant difference to the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q3/5

The re-randomization model assumes

A. the groups are different sizes

B. the treatment has no effect

C. the treatment works

D. the data are normal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q4/5

Why can a significant difference in an observational study not be interpreted as a treatment effect the way it can in a randomized experiment?

A. Observational studies always use smaller samples, so their differences are always due to chance.

B. Observational studies cannot measure any differences between groups, so no difference can be found.

C. A significant difference in an observational study is always the result of a calculation error.

D. Without random assignment, the difference could come from confounding variables rather than the treatment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q5/5

SET 66 • HSS-IC.B.3 • STATISTICS, INFERENCE, AND DECISIONS

Explain random sampling versus random assignment

MAIN	Algebra II main question	**HSS-IC.B.3**
BACKWARD 1	Study types and their purposes	**HSS-IC.B.3**
BACKWARD 2	Choose a random sample	**7.SP.A.1**
BACKWARD 3	Correlation and causation	**HSS-ID.C.9**
FORWARD 1	Improve an experimental design	**HSS-IC.B.3**
FORWARD 2	Use re-randomization to judge a treatment difference	**HSS-IC.B.5**

SET 66 • HSS-IC.B.3

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

A random sample of 500 adults is surveyed about exercise and sleep; adults who exercise more tend to sleep better. Which conclusion is justified?

A. Exercise causes better sleep for all adults, because the sample was chosen at random.

B. The association can be generalized to all adults, but it does not show that exercise causes better sleep.

C. Nothing can be concluded about any adults beyond the 500 who were surveyed.

D. Exercise causes better sleep, but only for the 500 adults who were surveyed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q1/5

60 volunteers are randomly assigned to two diets, and diet A's group loses significantly more weight. Which conclusion is justified?

A. There is evidence that diet A caused more weight loss for these volunteers, but the result cannot be generalized to all people.

B. Diet A is associated with weight loss for all people, but the study cannot show that the diet caused it.

C. Diet A causes more weight loss for all people, because the volunteers were randomly assigned to the two diets.

D. Nothing can be concluded, because the volunteers chose to participate instead of being randomly sampled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q2/5

A random sample of 200 students from a school is randomly assigned to two study methods, and method 1 leads to significantly higher scores. Which conclusion is justified?

A. Method 1 is associated with higher scores, but the study cannot show that it causes them.

B. There is evidence that method 1 causes higher scores, but only for the 200 students in the study.

C. There is evidence that method 1 causes higher scores, and the result can be generalized to the school's students.

D. Method 1 causes higher scores for students at every school, because the result was significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q3**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q3/5

A teacher compares test scores of students in her class who chose to use a review app with those who did not. Which conclusion is justified?

A. The app causes higher scores, but only for the students in this class.

B. The app causes higher scores for all students, because the groups were compared.

C. The result can be generalized to all students, but it does not show causation.

D. Only an association in this class: there was no random assignment and no random sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q4/5

How can 40 subjects be randomly assigned to two groups of 20?

A. Put the first 20 subjects who arrive in group 1 and the other 20 subjects in group 2.

B. Put the 20 tallest subjects in group 1 and the 20 shortest subjects in group 2.

C. Let each subject choose a group, and stop when one of the groups has 20 subjects.

D. Number the subjects 1-40; choose 20 different random numbers for group 1; the rest form group 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q5/5

SET 66 • HSS-IC.B.3

BACKWARD 1

Study types and their purposes • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Which type of study uses random assignment of treatments?

A. an observational study

B. an experiment

C. a sample survey

D. a census

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q1/5

Which type of study is designed to estimate a population value?

A. a case study of one person

B. an experiment

C. a sample survey

D. a placebo

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q2/5

True or false?

An observational study assigns treatments.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q3/5

Random assignment is used to

A. avoid measuring

B. make the sample larger

C. make the groups comparable

D. choose who is in the population

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q4/5

What does random sampling allow you to do?

A. generalize from the sample to the population

B. conclude cause and effect from the data

C. remove all chance variation from results

D. skip measuring the people in the sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q5/5

SET 66 • HSS-IC.B.3

BACKWARD 2

Choose a random sample • Grade 7 • Statistics & Probability • 7.SP.A.1

Which method chooses 20 names at random from a numbered list of 500?

A. Pick 20 different random numbers from 1 to 500 and take those names.

B. Pick 20 names that look like a good mix of the whole list.

C. Take the first 20 names on the numbered list, numbers 1 to 20.

D. Email all 500 and take the first 20 people who reply to the email.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q1/5

Is a sample of volunteers a random sample? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q2/5

True or false?

A sample of volunteers may differ from the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q3/5

Which method gives a random sample?

A. choosing the first 20 people on the list

B. drawing numbered slips from a well-mixed hat

C. asking for volunteers to sign up

D. asking friends to take part

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q4/5

Names are written on slips and drawn from a hat. Why must the slips be well mixed?

A. so the drawing takes longer

B. so the first slip drawn is always the same

C. so every slip has the same chance of being drawn

D. so the hat can hold more slips

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • HSS-IC.B.3

BACKWARD 3

Correlation and causation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Can an association between two variables in observational data prove causation? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q1/5

Students who exercise more tend to sleep better. Which is a plausible lurking variable?

A. the day of the month they were born

B. stress level

C. their locker number

D. the brand of their shoes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q2/5

True or false?

Random assignment controls lurking variables on average.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q3/5

Causation can be concluded from

A. an observational study

B. a correlation alone

C. a randomized experiment

D. a survey

Why does a correlation alone not show which variable causes the other?

A. Correlation can be found only for categories, not for numbers.

B. Correlation is always 0 when one variable causes the other.

C. Correlated variables can never be related to each other.

D. Either variable could affect the other, or a third variable could affect both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q5/5

SET 66 • HSS-IC.B.3

FORWARD 1

Improve an experimental design • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

60 volunteers try a new diet and lose weight. Why is a control group (for example a second, randomly assigned diet) needed to judge the diet?

A. Without a comparison group, the weight loss could be due to other factors, not the diet.

B. A control group makes every volunteer in the study lose more weight on the diet.

C. A control group proves that the new diet works for everyone who tries it.

D. A control group is needed only in studies with more than 100 volunteers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-F1-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q1/5

A drug study blocks subjects by age, then randomly assigns treatments within each age group. Why?

A. so the older subjects all get the drug and the younger the placebo

B. so every subject in each age group receives both treatments

C. so age differences cannot hide or create a treatment effect

D. so the study can avoid random assignment within the groups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q2/5

True or false?

In a double-blind experiment, neither subjects nor evaluators know who gets the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q3/5

60 volunteers were randomly assigned to two diets, and diet A led to more weight loss. Which change would let the result be generalized to all adults?

A. measuring the volunteers more often during the study

B. using a placebo for one group

C. adding more volunteers to each diet

D. randomly selecting subjects from all adults

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q4/5

Why are many subjects better than few in an experiment?

A. More subjects remove the need for random assignment to the groups.

B. More subjects always make the treatment itself work better.

C. More subjects make any observed result statistically significant.

D. Chance differences between groups are smaller, so real effects are easier to detect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q5/5

SET 66 • HSS-IC.B.3

FORWARD 2

Use re-randomization to judge a treatment difference • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

60 volunteers were randomly assigned to two diets; diet A's group lost 2.4 kg more on average. Under the model that the diets have the same effect, 5 of 200 re-randomizations gave a difference of at least 2.4 kg in favor of A. Use 5% as the cutoff for unusual. Is the difference significant?

A. No: 5/200 = 2.5% < 5%.

B. Yes: 5/200 = 2.5% < 5%.

C. No: 5/200 = 25% > 5%.

D. Yes: 5 is more than 2.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q1/5

60 volunteers (not a random sample of adults) were randomly assigned to two diets, and diet A's group lost significantly more weight. Which conclusion is justified?

A. Diet A is associated with weight loss, but the study cannot show that it caused it.

B. There is evidence that diet A caused more weight loss for these volunteers; it may not apply to all adults.

C. Nothing can be concluded, because the volunteers were not a random sample of adults.

D. Diet A causes more weight loss for all adults, because the result was significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q2/5

True or false?

Re-randomization shuffles the group labels as if the treatment had no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q3/5

If 70 of 200 re-randomized differences were at least 2.4 kg, conclude

A. the difference could be due to chance

B. the study was observational

C. diet B works

D. diet A works

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q4/5

Why is a re-randomization test valid only when the original groups were formed by random assignment?

A. Without random assignment the group means cannot be computed, so there is nothing to shuffle in the test.

B. Random assignment makes every difference significant, so the test can only be used after random assignment.

C. Re-randomization needs exactly the same number of subjects as a fair coin needs flips, so it requires random groups.

D. The shuffle copies the random assignment; if the groups were not randomly formed, the shuffle does not model how they arose.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q5/5

SET 67 • HSS-IC.B.4 • STATISTICS, INFERENCE, AND DECISIONS

Estimate a population proportion with a margin of error from a simulation

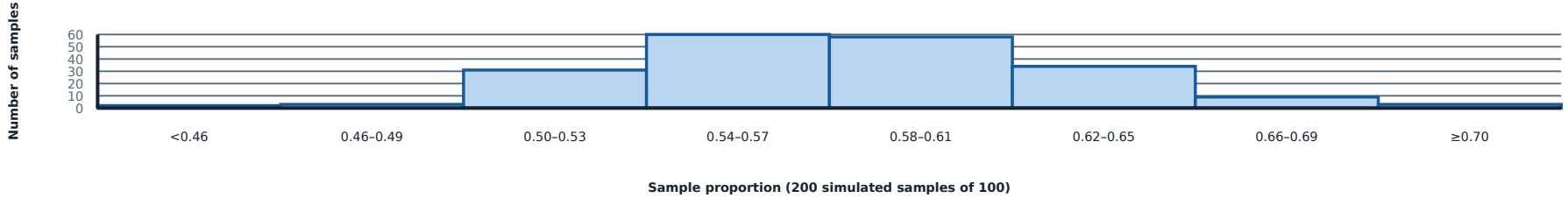
MAIN	Algebra II main question	**HSS-IC.B.4**
BACKWARD 1	Statistics, parameters, and sampling variability	**HSS-IC.A.1**
BACKWARD 2	Proportions and percents	**6.RP.A.3.c**
BACKWARD 3	Standard deviation as typical distance from the mean	**HSS-ID.A.2**
FORWARD 1	Plan a sample size for a desired margin of error	**HSS-IC.B.4**
FORWARD 2	Evaluate reports that use margins of error	**HSS-IC.B.6**

SET 67 • HSS-IC.B.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

In a random sample of 100 students, 58 want a later start time. The histogram shows 200 simulated samples of 100 from a population with p = 0.58; their standard deviation is 0.049. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the margin of error and the interval estimate, as decimals.



Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q1/5

In a random sample of 300 voters, 120 support a measure. Simulated sample proportions (n = 300, p = 0.40) have standard deviation 0.0275. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the estimate, the margin of error, and the interval, as decimals.

Estimate: _____

Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q2/5

In a random sample of 50 shoppers, 35 use a store app. Simulated sample proportions (n = 50, p = 0.70) have standard deviation 0.064. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the margin of error and the interval, as decimals.

Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q3/5

Three random-sample surveys have simulated-proportion standard deviations 0.064 (n = 50), 0.049 (n = 100), and 0.0275 (n = 300). Using margin of error = 2 SD, which survey has the smallest margin of error, and why?

A. n = 100 (margin 0.098): the middle size balances the two kinds of error.

B. All are equal: the margin depends only on the population size.

C. n = 300 (margin 0.055): larger samples give less variable sample proportions.

D. n = 50 (margin 0.128): smaller samples are easier to measure exactly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q4/5

A report from a random sample says "52% of voters support the plan, with a margin of error of 3 percentage points (95% confidence)." Which interpretation is correct?

A. The percent of all voters who support the plan is plausibly between 49% and 55%.

B. Exactly 52% of all voters support the plan, give or take nothing at all.

C. The plan will win, because 52% is more than 50% of the voters.

D. Between 49% and 55% of the sampled voters support the plan.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q5/5

SET 67 • HSS-IC.B.4

BACKWARD 1

Statistics, parameters, and sampling variability • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

In a random sample of 100 students, 58 want a later start time. What is the statistic?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q1/5

In a random sample of 100 students, 58% want a later start time. What parameter does 58% estimate?

A. the margin of error of the survey

B. the percent of all students who want a later start

C. the number of students in the sample

D. the percent of the 100 sampled students who want it

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-B1-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q2/5

True or false?

Different random samples of 100 would give different sample proportions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q3/5

A histogram of 200 simulated sample proportions (samples of 100) shows

A. the population parameter exactly

B. how sample proportions vary

C. the whole population of values

D. one sample of 100 values

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q4/5

Why must a sample be random for its margin of error to be meaningful?

A. A random sample always has a margin of error of 0, so any error is bias.

B. A biased sample has a smaller margin of error than a random one.

C. The margin describes chance variation in random samples; a biased sample can be off by more.

D. The margin depends only on the size of the population, not the sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-B1-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q5/5

SET 67 • HSS-IC.B.4

BACKWARD 2

Proportions and percents • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write 120 out of 300 as a proportion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q1/5

Write 0.58 ± 0.10 as an interval.

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q2/5

True or false?

35 out of 50 is 35%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q3/5

52% ± 3% gives

A. 52% to 55%

B. 49% to 55%

C. 50% to 54%

D. 49% to 52%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q4/5

Write 0.055 as a percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q5/5

SET 67 • HSS-IC.B.4

BACKWARD 3

Standard deviation as typical distance from the mean • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

By the 68-95-99.7 rule, what percent of a normal distribution lies within 2 standard deviations of the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q1/5

If the SD is 0.049, what is 2 SDs?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q2/5

True or false?

A smaller standard deviation means more variability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q3/5

Which set of simulated proportions varies least?

A. SD 0.1

B. SD 0.0275

C. SD 0.049

D. SD 0.064

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q4/5

Compute 2×0.064.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q5/5

SET 67 • HSS-IC.B.4

FORWARD 1

Plan a sample size for a desired margin of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A survey of a random sample of 100 students has a margin of error of 0.10. Quadrupling the sample size halves the margin of error. What is the margin of error with n = 400? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q1/5

About what sample size gives a margin of error of 0.05 when $p \approx 0.5$? (Use margin $\approx 2\sqrt{(p(1-p)/n)}$.)

$n \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q2/5

True or false?

Doubling the sample size halves the margin of error.

A. True

B. False

A random sample of n = 100 people gives a margin of error for a proportion. Which change to the sample size reduces the margin of error the most?

A. use n = 900

B. use n = 200

C. use n = 110

D. take a second sample of 100 and report it alone

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F1-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q4/5

Two random-sample polls of the same population have margins of error 0.05 and 0.025 for a proportion. The margin of error is about proportional to $1/\sqrt{n}$. How many times as large is the second sample as the first?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q5/5

SET 67 • HSS-IC.B.4

FORWARD 2

Evaluate reports that use margins of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A poll shows candidate A at 51% and B at 49%, each with a margin of error of ±4 percentage points. A headline says "A leads." Evaluate.

A. Justified: the margin of error is the same for both, so it cancels and A leads by 2.

B. Not justified: A must have less than 50% support, since 51% − 4% is below 50%.

C. Justified: 51% is greater than 49%, and the poll used a large sample.

D. Not justified: the intervals 47%−55% and 45%−53% overlap, so either could lead.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q1/5

A report gives 58% ± 10% from a random sample of 100 students and claims "a majority of students want a later start." Evaluate.

A. Not clearly justified: the interval 48% to 68% includes values below 50%.

B. Justified: 58% is more than 50%.

C. Justified: the whole interval is above 50%.

D. Not justified: a sample of 100 can never support a claim.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q2/5

True or false?

A margin of error accounts for bias from a voluntary-response sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q3/5

Which claim is best supported by its data?

A. A random sample gives 52% ± 4% support, so exactly 52% support it.

B. A random sample gives 70% ± 3% support, so a majority support the plan.

C. A random sample gives 51% ± 5% support, so a majority support the plan.

D. An online poll gives 60% support, so a majority support the plan.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q4/5

Besides the percent, what should a careful poll report include?

A. the sample size, how the sample was chosen, and the margin of error

B. only the percent, rounded to the nearest whole number

C. only last year's result, so readers can compare

D. the names and answers of the people who were surveyed

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q5/5

SET 68 • HSS-IC.B.4 • STATISTICS, INFERENCE, AND DECISIONS

Estimate a population mean with a margin of error from a simulation

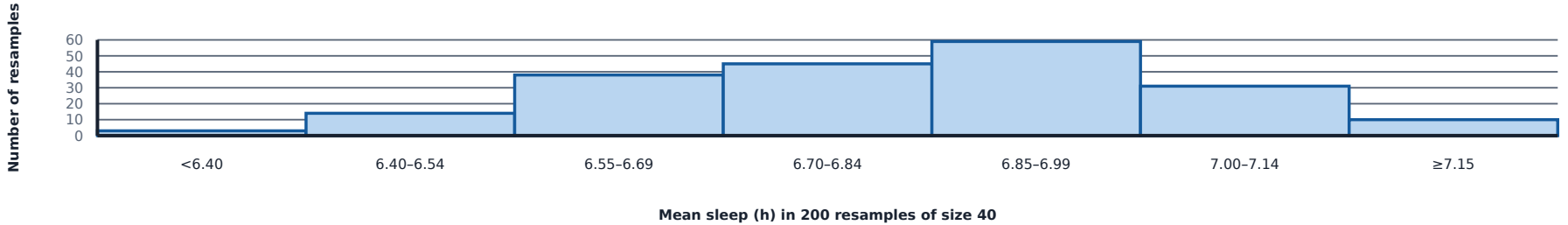
MAIN	Algebra II main question	**HSS-IC.B.4**
BACKWARD 1	Mean of a data set	**6.SP.B.5.c**
BACKWARD 2	What resampling simulates	**HSS-IC.A.1**
BACKWARD 3	Two standard deviations and the 95% range	**HSS-ID.A.2**
FORWARD 1	Use intervals to judge claims about a mean	**HSS-IC.B.4**
FORWARD 2	Compare two group means with re-randomization	**HSS-IC.B.5**

SET 68 • HSS-IC.B.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A random sample of 40 students slept a mean of 6.8 hours. The histogram shows the means of 200 resamples of the data; their standard deviation is 0.20 hours. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the margin of error and the interval, in hours.



Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q1/5

A random sample of 25 apples has mean weight 152 g. Simulated sample means have standard deviation 3.6 g. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the margin of error and the interval, in grams.

Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q2/5

A random sample of 60 commuters has mean commute 24.5 minutes. Simulated sample means have standard deviation 1.4 minutes. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the margin of error and the interval, in minutes.

Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q3/5

A random sample of 40 students gives the 95% interval 6.4 to 7.2 hours for the mean hours of sleep. Which interpretation is correct?

A. Every student at the school sleeps at least 6.4 hours each night.

B. The 40 sampled students each slept between 6.4 and 7.2 hours.

C. The mean sleep of all students is plausibly between 6.4 and 7.2 hours.

D. 95% of all students sleep between 6.4 and 7.2 hours each night.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q4/5

A random sample of 25 apples gives a margin of error of 7.2 g for the mean weight. Assume apple weights vary the same way and the same 2-SD (95%) convention is used, so the margin of error is proportional to $1/\sqrt{n}$. What margin of error do you expect from a random sample of 100 apples? Give grams.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q5/5

SET 68 • HSS-IC.B.4

BACKWARD 1

Mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean of 6, 7, 7.5, 6.5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q1/5

What does the mean of a random sample estimate?

A. the margin of error

B. the population mean

C. the largest value in the population

D. the sample size

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q2/5

True or false?

The mean is not affected by extreme values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q3/5

The mean of 150, 152, 154 is

A. 154

B. 150

C. 152

D. 456

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q4/5

Each of 40 values in a data set increases by 0.5. By how much does the mean increase? Give a number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q5/5

SET 68 • HSS-IC.B.4

BACKWARD 2

What resampling simulates • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

Why can resampling a random sample (drawing with replacement) show how sample means vary?

A. Resampling collects new data from the whole population each time it is done.

B. Resampling always gives the true population mean as the center.

C. Resampling removes all chance variation from the sample means.

D. The resamples act like new samples from a population that looks like the sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q1/5

In a histogram of 200 resample means, what does one bar's height show?

A. the chance the population mean is in that interval

B. the mean of the original sample

C. how many of the 200 resample means fell in that interval

D. the number of values in each resample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q2/5

True or false?

Sample means vary less than individual values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q3/5

A random sample of 40 students slept a mean of 6.8 hours. The parameter being estimated is

A. 6.8

B. the mean sleep of all students

C. the 40 students' mean

D. the resample SD

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q4/5

A random sample of 40 students had a mean of 6.8 hours of sleep. Is 6.8 hours a parameter or a statistic?

A. a statistic: it describes the sample of 40

B. a parameter: it describes all students

C. a statistic: it describes all students

D. a parameter: it describes the sample of 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q5/5

SET 68 • HSS-IC.B.4

BACKWARD 3

Two standard deviations and the 95% range • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Compute $6.8 - 2(0.20)$ and $6.8 + 2(0.20)$.

$6.8 - 2(0.20) =$ _____

$6.8 + 2(0.20) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q1/5

Compute 2×1.4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q2/5

True or false?

About 95% of a mound-shaped distribution is within 2 SDs of its mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q3/5

152 ± 7.2 gives

A. 144.8 to 159.2

B. 145 to 159

C. 152 to 159.2

D. 144.8 to 152

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q4/5

Is 7.5 inside the interval from 6.4 to 7.2? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q5/5

SET 68 • HSS-IC.B.4

FORWARD 1

Use intervals to judge claims about a mean • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A random sample gives the 95% interval 6.4 to 7.2 hours for students' mean sleep. A health guide says students average 8 hours. Does the interval support this?

A. Yes: some of the students in the sample sleep 8 hours or more.

B. No: 8 is outside the interval of plausible values for the mean.

C. Yes: 8 is close to the upper end of the interval, 7.2 hours.

D. No: the interval proves the mean is exactly its center, 6.8 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F1-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q1/5

A random sample gives the 95% interval 144.8 g to 159.2 g for the mean apple weight. A grower claims the apples average 155 g. Is the claim consistent with the interval?

A. No: the mean must be exactly 152 g.

B. Yes: the claim is proved true because 155 is inside it.

C. No: 155 g is not the center of the interval.

D. Yes: 155 g is inside the interval, so it is a plausible mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F1-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q2/5

True or false?

A value inside a 95% interval estimate for a mean is proved to be the true mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q3/5

A random sample of 60 commuters gives an interval for the mean commute time. Which change would most likely narrow the interval?

A. sampling 240 commuters instead of 60

B. using the median

C. rounding the data

D. sampling 30 commuters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F1-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q4/5

A random sample of students at one school gives the interval 6.4 to 7.2 hours for their mean sleep. Why should the interval not be applied to students at a different school?

A. The students at the other school must have a higher mean sleep time.

B. An interval for a mean can only describe a single student's sleep.

C. The interval is too narrow to be applied to any group of students.

D. The sample came only from this school, so it estimates only this school's mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q5/5

SET 68 • HSS-IC.B.4

FORWARD 2

Compare two group means with re-randomization • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

In an experiment, 40 students were randomly assigned to use a study app or not (20 each). The app group scored 6 points more on average. Under the model that the app has no effect, 9 of 200 re-randomizations gave the app group at least 6 more points. Using a one-sided count and a 5% cutoff, is the difference significant?

A. No: 9/200 = 4.5% > 5%.

B. Yes: 9/200 = 4.5% < 5%.

C. No: 9 is more than 6.

D. Yes: 9/200 = 45%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q1/5

Students were randomly assigned to a treatment group (study app) or a control group (no app), and the difference in mean scores was re-randomized 200 times. What null model do the re-randomizations represent?

A. The app has no effect on scores, so the group labels are interchangeable.

B. The app raises every score by exactly 6 points.

C. The students were chosen by voluntary response.

D. The control group scores lower in every re-randomization.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q2/5

True or false?

An interval for one mean and a re-randomization test for two means both use simulated variability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q3/5

In an app experiment, the app group scored 6 points more on average. Under the model of no app effect, 30 of 200 re-randomizations gave the app group at least 6 more points. Use 5% as the cutoff for unusual. The conclusion is

A. the difference could be chance (15% > 5%)

B. the app works (15% < 50%)

C. the app is proved to have no effect

D. the app lowers scores

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q4/5

How is the margin of error for one mean related to judging a difference between two means?

A. A difference between two means never varies by chance, so it does not need any measure of chance variation.

B. The margin of error for one mean is always equal to the difference between the two group means.

C. Both ask how much a statistic varies by chance; a difference is meaningful only if it is large compared with that variability.

D. They are unrelated: one uses the data, and the other uses only the number of subjects in each group.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q5/5

SET 69 • HSS-IC.B.5 • STATISTICS, INFERENCE, AND DECISIONS

Compare two treatments with a re-randomization simulation

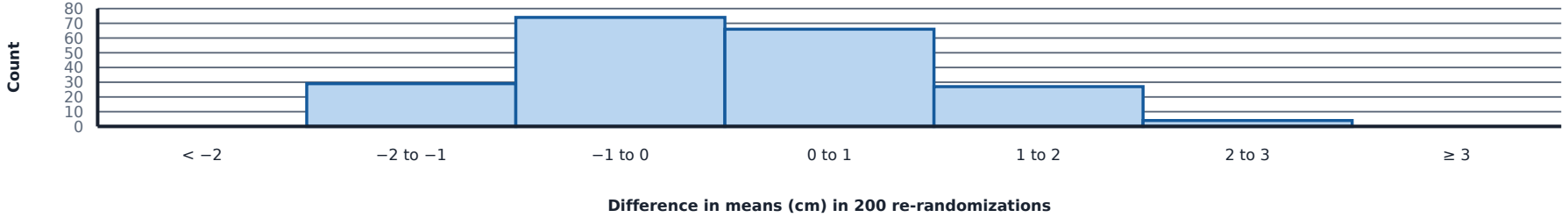
MAIN	Algebra II main question	**HSS-IC.B.5**
BACKWARD 1	Compare the means of two groups	**7.SP.B.4**
BACKWARD 2	Random assignment in experiments	**HSS-IC.B.3**
BACKWARD 3	Probability as relative frequency in simulations	**7.SP.C.6**
FORWARD 1	Two-sided comparisons and design choices	**HSS-IC.B.5**
FORWARD 2	Evaluate reports of experiments	**HSS-IC.B.6**

SET 69 • HSS-IC.B.5

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

10 plants got fertilizer and 10 did not (randomly assigned). Fertilized plants grew 2.9 cm more on average. The histogram shows 200 re-randomized differences under the model that fertilizer has no effect. Use a one-sided count (2.9 cm or more) and a 5% cutoff. Which conclusion is justified?



A. Most re-randomized differences were near 0, so the observed difference is not significant.

B. 4 of 200 (2%) were 2 cm or more, so the observed difference of 2.9 cm is not significant.

C. 0 of 200 were 2.9 cm or more; 0% < 5%, so the difference is significant: fertilizer appears to increase growth.

D. 0 of 200 were 2.9 cm or more, so this difference has probability exactly 0 and fertilizer is proved to work.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q1/5

In a randomized caffeine experiment, the caffeine group's mean reaction time was 0.04 s faster. Under the model that caffeine has no effect, 62 of 200 re-randomizations gave the caffeine group a mean at least 0.04 s faster. Using a one-sided count and a 5% cutoff, is the difference significant?

A. No: 62/200 = 31% > 5%, so the difference could easily be chance.

B. Yes: 62/200 = 31% is more than 5%, so the difference is significant.

C. No: 31% > 5%, and this proves that caffeine has no effect.

D. Yes: 0.04 s is less than 5% of a second, so it is significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-M-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q2/5

Students were randomly assigned to use a study app or not, and the app group scored 6 points higher on average. Under the model that the app has no effect, 9 of 200 re-randomizations gave the app group at least 6 more points. Using a one-sided count and a 5% cutoff, is the difference significant?

A. Yes: 9/200 = 45%, which is above 5%.

B. No: 9 is more than 6, so the app has no effect.

C. No: 9/200 = 4.5%, which is above 5%.

D. Yes: 9/200 = 4.5%, which is below 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-M-Q3**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q3/5

What does a re-randomization simulation assume, and how is it carried out?

A. It assumes no treatment effect; each group's values are re-sorted from smallest to largest, and the two sorted lists are compared once to find the difference.

B. It assumes the treatment works; the observed difference is doubled and halved many times, and the results are graphed to show how large the effect could be.

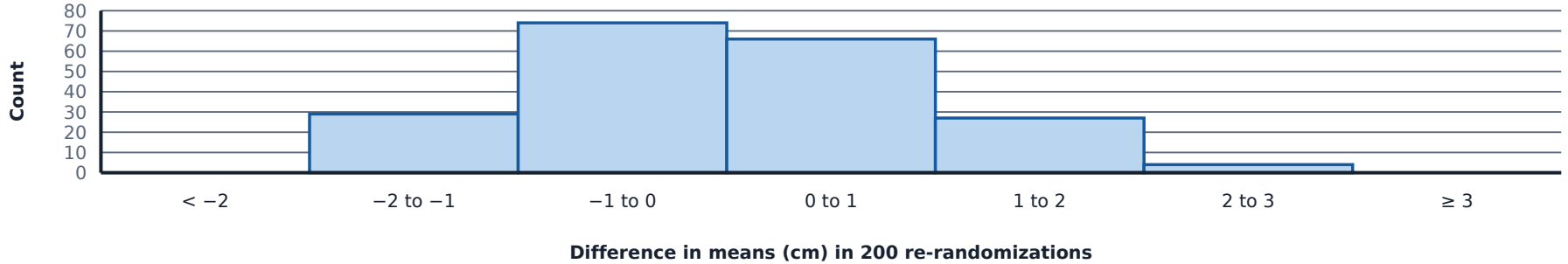
C. It assumes no treatment effect; all responses are pooled and randomly re-split into groups of the original sizes, and the difference in means is recorded many times.

D. It assumes the treatment works; new subjects are recruited for each run, and the difference in their group means is recorded many times to estimate the true effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q4/5

Using the histogram of 200 re-randomized differences, how many differences were 2 cm or more, and would an observed difference of 2 cm be significant using a one-sided count and a 5% cutoff?



A. 0 of 200; a difference of 2 cm has probability 0 under the model.

B. 4 of 200 (2%); 2% < 5%, so yes, it would be significant.

C. 31 of 200 (15.5%); 15.5% > 5%, so no, it would not be significant.

D. 4 of 200 (2%); 2% < 5%, so no, it would not be significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-M-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q5/5

SET 69 • HSS-IC.B.5

BACKWARD 1

Compare the means of two groups • Grade 7 • Statistics & Probability • 7.SP.B.4

Group A has mean 25.4 cm and group B has mean 22.5 cm. Find the difference A − B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q1/5

Group A: 4, 6, 8. Group B: 3, 5, 4. Find the difference in means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q2/5

True or false?

A difference in sample means could happen even if the treatments have the same effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q3/5

Mean A = 0.52 s and mean B = 0.48 s. A − B is

A. −0.04 s

B. 0.4 s

C. 1.00 s

D. 0.04 s

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q4/5

Why is a difference between two group means compared with the variability in the groups?

A. Any difference between the group means proves a treatment effect.

B. The variability in the groups gives the exact size of the effect.

C. A difference is meaningful only if it is large relative to the natural variation.

D. Variability matters only when the two group means are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • HSS-IC.B.5

BACKWARD 2

Random assignment in experiments • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

In a fertilizer experiment, 20 plants were randomly assigned to fertilizer or no fertilizer. Why were they randomly assigned?

A. to make the groups comparable, so a difference can be due to the fertilizer

B. so that no measurements of growth are needed afterward

C. so the result can be generalized to all plants of every kind, everywhere

D. to make sure every plant grows exactly the same amount

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-B2-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q1/5

How can 20 plants be randomly assigned to two groups of 10?

A. Put the 10 tallest plants in the fertilizer group.

B. Let the gardener pick 10 healthy-looking plants for fertilizer.

C. Put the 10 plants nearest the window in the fertilizer group.

D. Number the plants 1-20 and use random numbers to choose 10 for fertilizer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q2/5

True or false?

Without random assignment, a re-randomization test does not justify a cause-and-effect conclusion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q3/5

Random assignment is a feature of

A. censuses

B. experiments

C. observational studies

D. surveys

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q4/5

A randomized experiment gives a significant difference. Which conclusion does it allow?

A. the difference is due to chance alone, not the treatment

B. there is evidence the treatment caused the difference for these subjects

C. the treatment causes the same difference in everyone everywhere

D. the subjects were a random sample of everyone in the population

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q5/5

SET 69 • HSS-IC.B.5

BACKWARD 3

Probability as relative frequency in simulations • Grade 7 • Statistics & Probability • 7.SP.C.6

62 of 200 is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q1/5

9 of 200 is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q2/5

True or false?

An event that happens in 40% of simulated runs is unusual.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q3/5

0 of 200 runs suggests the observed result is

A. very unlikely under the model

B. certain

C. impossible

D. typical

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q4/5

Is 4/200 less than 5%? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q5/5

SET 69 • HSS-IC.B.5

FORWARD 1

Two-sided comparisons and design choices • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

In a caffeine experiment, the observed difference in mean reaction time was 0.04 s. The question is whether caffeine changes reaction time in either direction. Which re-randomized differences should be counted?

A. only those with difference ≥ 0.04 s

B. those with |difference| ≥ 0.04 s, in either direction

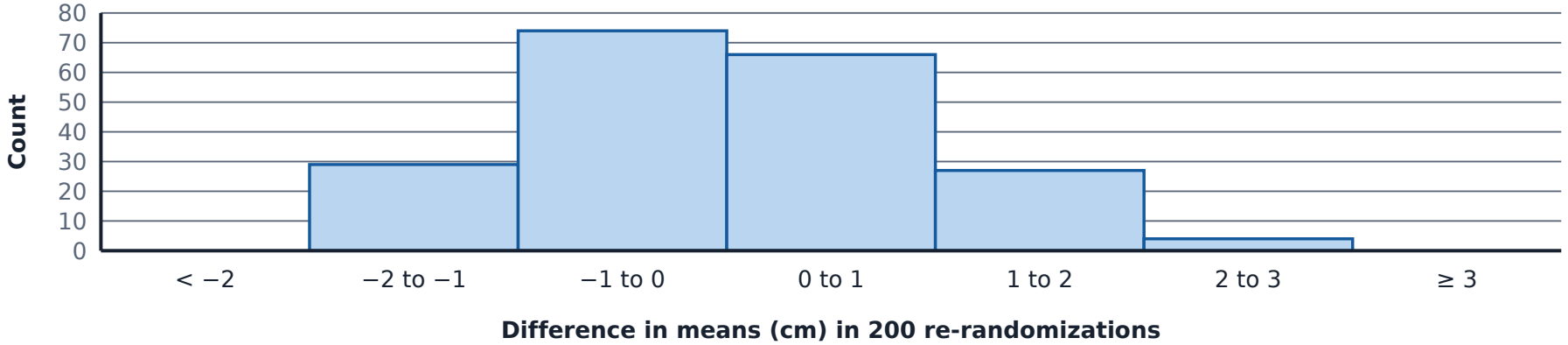
C. only those with |difference| ≤ 0.04 s

D. only those with difference exactly 0.04 s

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F1-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q1/5

The histogram shows 200 re-randomized differences in mean growth (cm). How many differences were 2 cm or more, or less than −2 cm?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q2/5

True or false?

Using more plants in each group usually makes the re-randomized differences less spread out.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q3/5

A small observed difference can be significant when

A. the groups have very little natural variation

B. the data are rounded to whole numbers

C. the groups were not randomly assigned

D. the sample in each group is small

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q4/5

Why should the cutoff for significance (such as 5%) be chosen before seeing the data?

A. The cutoff changes the data that are collected in the study.

B. The cutoff must equal the observed proportion of the data.

C. Choosing it afterward could bias the conclusion toward the result one wants.

D. Cutoffs chosen in advance are always 50%, the fairest choice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q5/5

SET 69 • HSS-IC.B.5

FORWARD 2

Evaluate reports of experiments • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A company says its supplement "improved memory scores by 3 points" in a study of 8 people with no control group. Which evaluation is correct?

A. Strong evidence: an increase of 3 points is a real improvement in memory for these people.

B. Weak evidence, but only because an increase of 3 points is too small a number to matter.

C. Strong evidence: every one of the 8 people in the study took the supplement as directed.

D. Weak evidence: with no control group and only 8 people, the change could be due to other factors or chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F2-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q1/5

A report says a new teaching method "significantly" raised scores in a randomized experiment. Under the no-effect model, 2 of 200 re-randomized differences were at least as large as the observed difference. Use 5% as the cutoff for unusual. Evaluate the report.

A. Supported: the assignment was random, and 2/200 = 1% < 5%.

B. Supported: the method is proved to work for all students.

C. Not supported: re-randomization cannot be used for experiments.

D. Not supported: 2/200 = 1% is too small to matter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q2/5

True or false?

"Statistically significant" means the effect is large and important.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q3/5

Which report is most convincing?

A. a study of 500 users with no comparison group or random assignment

B. testimonials from many satisfied customers

C. a randomized experiment with a control group and a rare result

D. an online poll of 10,000 users of the product

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F2-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q4/5

Which information should a report of an experiment include?

A. only the observed difference between the two groups

B. the assignment method, group sizes, observed difference, and how unusual it is

C. only the conclusion that the treatment works for everyone

D. only the names of the treatments, the date of the study, and who paid for it

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q5/5

SET 70 • HSS-IC.B.6 • STATISTICS, INFERENCE, AND DECISIONS

Evaluate claims in reports based on data

MAIN	Algebra II main question	**HSS-IC.B.6**
BACKWARD 1	Random samples and bias	**HSS-IC.A.1**
BACKWARD 2	Study design and causation	**HSS-IC.B.3**
BACKWARD 3	Read scales on graphs	**3.MD.B.3**
FORWARD 1	Rewrite claims so that they follow from the data	**HSS-IC.B.6**
FORWARD 2	Analyze decisions based on test results	**HSS-MD.B.7**

SET 70 • HSS-IC.B.6

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

An ad says "4 out of 5 dentists recommend our toothpaste." Only 5 dentists, chosen by the company, were asked. Evaluate the claim.

A. Weak: the sample is tiny and not random, so it cannot represent all dentists.

B. Strong: dentists are experts, so any group of them is representative.

C. Weak: the claim would only be fine if all 5 of the dentists agreed.

D. Strong: 80% is a large majority of the dentists who were asked.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q1/5

A report says "Students who eat breakfast earn higher grades, so breakfast causes higher grades." The data came from a survey. Evaluate.

A. Justified: the survey found that students who eat breakfast have higher grades.

B. Not justified: breakfast and grades cannot be associated in a survey.

C. Not justified: a survey shows association, not causation; lurking variables could explain it.

D. Justified: the survey was large enough to show cause and effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-M-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q2/5

A bar graph shows sales of 50 and 52 thousand dollars, with the vertical axis starting at 48 thousand, so one bar looks twice as tall. Evaluate the impression the graph gives.

A. Fair: starting the axis at 48 thousand makes the graph easier to read.

B. Fair: the tops of the bars show the correct values of 50 and 52 thousand.

C. Misleading: the taller bar should be drawn more than twice as tall.

D. Misleading: the truncated axis makes a 4% difference look like a 100% difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q3/5

A TV show asks viewers to text their vote; 90% of 10,000 texts favor a policy. The show claims most citizens support it. Evaluate.

A. Justified: 90% is so far above 50% that the method of collecting the votes does not matter.

B. Not justified: texters are a voluntary response, not a random sample.

C. Justified: 10,000 texts form a very large sample of the citizens.

D. Not justified: 10,000 texts is too few to represent a whole country.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q4/5

A poll shows candidate A at 51% and B at 49%, each with a margin of error of ±4 percentage points. A headline says "A is winning." Evaluate.

A. Not justified: the intervals 47%−55% and 45%−53% overlap, so the lead is within the margin of error.

B. Not justified: B is actually winning, because the margin favors the trailing candidate.

C. Justified: the margin of error is the same for both candidates, so it cancels out.

D. Justified: A has more support than B in the poll, so A is winning the race.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-M-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q5/5

SET 70 • HSS-IC.B.6

BACKWARD 1

Random samples and bias • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

Why is a voluntary-response sample usually biased?

A. Every member of the population responds to it.

B. Voluntary samples are always too small to use.

C. The sample is chosen with random numbers instead.

D. People with strong opinions are more likely to respond.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q1/5

What does a random sample allow?

A. concluding cause and effect from the data

B. skipping the margin of error in the report

C. generalizing from the sample to the population

D. eliminating all chance variation from results

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q2/5

True or false?

A very large sample cannot be biased.

A. True

B. False

Which sample is most likely biased?

A. customers who chose to write reviews

B. a random sample of 1,000

C. randomly chosen customers

D. every customer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q4/5

Is "5 dentists chosen by the company" a random sample? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q5/5

SET 70 • HSS-IC.B.6

BACKWARD 2

Study design and causation • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

A survey records students' breakfast habits and grades. What type of study is it?

A. an observational study

B. a randomized experiment

C. a placebo-controlled experiment

D. a simulation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q1/5

Which design could test whether eating breakfast causes higher grades?

A. a voluntary online poll about eating breakfast and grades

B. a larger survey of students' breakfast habits and grades

C. a randomized experiment that assigns students to eat breakfast or not

D. comparing grades of students who already eat breakfast or not

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q2/5

True or false?

Only randomized experiments can establish cause and effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q3/5

A confounding variable is

A. the response variable that is measured in the study

B. a variable related to both the explanatory and response variables

C. the number of subjects in the sample of the study

D. the margin of error of the estimate

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q4/5

Students who eat breakfast tend to earn higher grades. Which is a plausible confounding variable?

A. the students' locker numbers

B. the amount of sleep students get

C. the color of the cereal box

D. the day the survey was printed

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • HSS-IC.B.6

BACKWARD 3

Read scales on graphs • Grade 3 • Measurement & Data • 3.MD.B.3

A bar graph's vertical axis starts at 48. One bar reaches 50 and another reaches 52. What values do the two bars show?

First bar: _____

Second bar: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q1/5

What percent larger is 52 than 50?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q2/5

True or false?

An axis that does not start at 0 can make differences look larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q3/5

On a graph whose axis starts at 48, a bar of 52 looks how many times as tall as a bar of 50?

A. the same

B. 1.04 times

C. 2 times

D. 4 times

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q4/5

A bar graph of sales of 50 and 52 thousand dollars has a vertical axis starting at 48 thousand. How should it be redrawn to be fair?

A. Make the bars wider.

B. Remove the axis labels.

C. Start the axis at 51 to show the difference better.

D. Start the vertical axis at 0, or clearly mark the axis break.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q5/5

SET 70 • HSS-IC.B.6

FORWARD 1

Rewrite claims so that they follow from the data • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A survey found that students who eat breakfast earn higher grades. Which claim is supported by the survey?

A. Higher grades cause students to eat breakfast.

B. Eating breakfast causes higher grades.

C. Students who eat breakfast tend to have higher grades.

D. Skipping breakfast causes lower grades.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q1/5

A poll shows 51% for A and 49% for B, each with a margin of error of ±4 percentage points. Which headline is accurate?

A. "The race is too close to call"

B. "B is winning"

C. "A is winning"

D. "A has a certain 2-point lead"

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q2/5

True or false?

Saying "is associated with" instead of "causes" is appropriate for an observational study.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q3/5

Which conclusion is justified by a random sample of 1,000 voters showing 62% ± 3% support?

A. the sample was biased

B. most voters support the measure

C. exactly 62% support it

D. the measure will pass by 24 points

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q4/5

An ad says "4 out of 5 dentists recommend our toothpaste," based on 5 dentists chosen by the company. What would make the claim more convincing?

A. printing the claim in larger letters on the box

B. a large random sample of dentists, with the question wording reported

C. asking the same 5 dentists again a year later

D. choosing 5 different dentists who already like the brand

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q5/5

SET 70 • HSS-IC.B.6

FORWARD 2

Analyze decisions based on test results • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A screening test detects 95% of people who have a disease (sensitivity 95%) and gives a false positive for 5% of healthy people. In a group of 10,000 people, 1% have the disease. How many true positives are there, and how many positive results in all?

True positives: _____

All positives: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q1/5

A screening test has sensitivity 95% and a 5% false-positive rate. In a group of 10,000 people where 1% have the disease, there are 95 true positives and 495 false positives. A report says "a positive test means you almost surely have the disease." Evaluate.

A. Correct: only 5% of positive results are false.

B. Misleading: 495/590 ≈ 84% of positives have the disease.

C. Correct: the test detects 95% of cases.

D. Misleading: only 95/590 ≈ 16% of positives have the disease.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q2**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q2/5

True or false?

The probability of having the disease given a positive result can be low even with an accurate test.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q3/5

A screening test has sensitivity 95% and a 5% false-positive rate. If 1% of a population has the disease, P(disease | positive) is about

A. 0.99

B. 0.05

C. 0.95

D. 0.16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q4/5

For a screening test, P(disease | positive) ≈ 0.16. Which follow-up decision makes the most sense after a positive result?

A. Begin treatment at once, since the test detects 95% of cases.

B. Ignore the result, since it is probably a false positive.

C. Repeat the same test and treat only if it is negative.

D. Confirm with a second, more accurate test before treatment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q5/5

SET 71 • HSS-MD.B.6 • STATISTICS, INFERENCE, AND DECISIONS

Design and judge fair decision procedures with probabilities

MAIN	Algebra II main question	**HSS-MD.B.6**
BACKWARD 1	Probabilities of equally likely outcomes	**7.SP.C.7.a**
BACKWARD 2	Sample spaces for compound events	**7.SP.C.8.b**
BACKWARD 3	Probability between 0 and 1 and redoing outcomes	**7.SP.C.5**
FORWARD 1	Design fair procedures and count expected trials	**HSS-MD.B.6**
FORWARD 2	Analyze strategies and decisions with probability	**HSS-MD.B.7**

SET 71 • HSS-MD.B.6

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

Three friends choose a driver by flipping a fair coin twice: HH = Ana, HT = Ben, TH = Cy, and TT = flip twice again. Is this fair?

A. No: TT is ignored, so the three probabilities add to 3/4 instead of 1.

B. Yes: each friend has probability 1/4, the same as each of the four outcomes.

C. Yes: HH, HT, and TH are equally likely and TT is redone, so each friend has probability 1/3.

D. No: Ana has probability 1/2, because HH has two heads and the others have only one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q1**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q1/5

Four people use a fair die: 1 = A, 2 = B, 3 = C, 4 = D, and 5 or 6 means roll again. Is this fair?

A. No: D is less likely, because 4 is next to 5 and 6.

B. Yes: each person has probability 1/4 after rerolls.

C. Yes: each person has probability 1/6.

D. No: each person has 1/6, and 1/3 of the time is wasted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q2**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q2/5

A well-mixed hat holds 3 slips with Ana's name and 2 with Ben's. One slip is drawn at random to choose who goes first. Is this fair?

A. No: P(Ana) = 3/5 and P(Ben) = 2/5.

B. Yes: the slips are well mixed.

C. Yes: each name is drawn or not, so each has 1/2.

D. No: P(Ana) = 2/5 and P(Ben) = 3/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q3**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q3/5

A random number generator gives a whole number 1-100, each equally likely. Numbers 1-33 choose A, 34-66 choose B, 67-99 choose C, and 100 means generate again. Is this fair?

A. Yes: each has probability 33/100, and the leftover 1/100 does not matter.

B. Yes: each person has 33 equally likely numbers, and 100 is redone, so each has 1/3.

C. No: A has 33 numbers but C has 34 numbers, so C is a little more likely.

D. No: 100 is wasted, so the probabilities are 0.33 each and do not add to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q4**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q4/5

Two fair dice are rolled; if the sum is even, player 1 wins, otherwise player 2 wins. Is this fair?

A. Yes: 18 of the 36 equally likely outcomes have an even sum, so each player has 1/2.

B. Yes: there are 11 possible sums, and about half of the 11 sums are even.

C. No: there are 6 even sums (2, 4, ..., 12) and only 5 odd sums, so player 1 wins more.

D. No: a sum of 7 is the most likely sum, so player 2 wins more often than player 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q5**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q5/5

SET 71 • HSS-MD.B.6

BACKWARD 1

Probabilities of equally likely outcomes • Grade 7 • Statistics & Probability • 7.SP.C.7.a

A fair die is rolled. What is P(3)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q1/5

A bag has 3 red and 2 blue slips. What is P(red)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q2/5

True or false?

Each face of a fair die has the same probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q3/5

Two fair coins are flipped, one after the other. P(first is heads and second is tails) is

A. 1/2

B. 1/4

C. 1/8

D. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q4/5

A spinner has 4 equal sections. What is P(one section)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q5/5

SET 71 • HSS-MD.B.6

BACKWARD 2

Sample spaces for compound events • Grade 7 • Statistics & Probability • 7.SP.C.8.b

A fair coin is flipped twice. Which list is the sample space of equally likely outcomes?

A. H, T

B. HH, HT, TT

C. two heads, one head, no heads

D. HH, HT, TH, TT

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q1/5

How many outcomes does rolling two dice have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q2/5

True or false?

When a fair coin is flipped twice, the 3 outcomes "two heads," "one head," and "no heads" are equally likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q3/5

How many outcomes for two dice give an even sum?

A. 21

B. 12

C. 18

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q4/5

List the outcomes of rolling two dice with a sum of 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q5/5

SET 71 • HSS-MD.B.6

BACKWARD 3

Probability between 0 and 1 and redoing outcomes • Grade 7 • Statistics & Probability • 7.SP.C.5

A fair die is rolled; a roll of 5 or 6 is ignored and the die is rolled again until it shows 1-4. What are the probabilities of the final results 1, 2, 3, and 4?

A. Each becomes 1/5, because one reroll is added.

B. 1 becomes more likely than 4, since it is first.

C. Each stays 1/6, because the die is still fair.

D. They stay equally likely, and each becomes 1/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q1/5

A fair die is rolled, and any 5 or 6 is rerolled until the result is 1-4. What is P(final result is 1)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q2/5

True or false?

Probabilities of all outcomes in a sample space add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q3/5

An event that can never happen has probability

A. 1

B. -1

C. 1/2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q4/5

Three equally likely outcomes: what is each probability?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q5/5

SET 71 • HSS-MD.B.6

FORWARD 1

Design fair procedures and count expected trials • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

Which procedure chooses one of 5 people fairly using a fair coin?

A. Flip 5 times; the person whose number equals the count of heads is chosen, and 0 heads means flip again.

B. Flip twice (4 equally likely outcomes); one outcome each for persons 1-4, and person 5 also gets HH.

C. Flip 3 times (8 equally likely outcomes); assign 5 outcomes to the people and flip again on the other 3.

D. Flip 3 times (8 equally likely outcomes); assign 5 outcomes to people and give the other 3 to person 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F1-Q1**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q1/5

Three friends choose a driver by flipping a coin twice: HH, HT, and TH each pick a friend, and TT means flip again. What is the probability that a choice is made on the first try?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions HSS-MD.B.6 • Set 71 • FORWARD 1 • Q2/5

True or false?

A fair procedure must give each person an equal chance, even if it may take several tries.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q3/5

To choose among 6 people fairly with one die, you

A. roll twice and add

B. assign one face to each person

C. assign 1-3 to person A

D. reroll on 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions HSS-MD.B.6 • Set 71 • FORWARD 1 • Q4/5

A teacher picks a student from a class of 28 using a random number generator that gives whole numbers 1-30, each equally likely. Which rule is fair?

A. Assign numbers 1-28 to the students, and generate again on 29 or 30.

B. Assign numbers 1-28; 29 and 30 both choose student 1.

C. Assign numbers 1-28; 29 and 30 choose students 1 and 2.

D. Generate one number and divide it by 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F1-Q5**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions HSS-MD.B.6 • Set 71 • FORWARD 1 • Q5/5

SET 71 • HSS-MD.B.6

FORWARD 2

Analyze strategies and decisions with probability • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A game costs $1. You roll a fair die and win $4 if you roll a 6, and nothing otherwise. Find the expected net gain per game, in dollars, as a fraction.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q1/5

A game costs $1. You roll a fair die and win a prize if you roll a 6, and nothing otherwise. With a $4 prize, what is the expected net gain in dollars? What prize makes the expected net gain 0?

Expected net gain ($): _____

Fair prize ($): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q2/5

True or false?

A game can be fair in who is chosen but still have a negative expected value for the player.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q3/5

With P(win) = 1/4 and prize \$8 for a \$2 ticket, the expected net gain is

A. \$2

B. −\$2

C. \$0

D. \$6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q4/5

What is the difference between a fair choice procedure and a fair game?

A. A fair game always has a prize; a fair choice procedure never uses probability at all.

B. A fair choice procedure has expected net gain 0; a fair game gives everyone equal probability.

C. A fair choice procedure gives everyone equal probability; a fair game has expected net gain 0.

D. They mean the same thing: both require each player to have probability 1/2 of winning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q5/5

SET 72 • HSS-MD.B.7 • STATISTICS, INFERENCE, AND DECISIONS

Analyze decisions and strategies with probabilities

MAIN	Algebra II main question	**HSS-MD.B.7**
BACKWARD 1	Probability of compound independent events	**7.SP.C.8.a**
BACKWARD 2	Two-way tables of counts	**8.SP.A.4**
BACKWARD 3	Weighted averages of outcomes	**6.RP.A.3.c**
FORWARD 1	Investigate how changing conditions changes the decision	**HSS-MD.B.7**
FORWARD 2	Find expected values of random variables	**HSS-MD.A.2**

SET 72 • HSS-MD.B.7

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A disease affects 1% of 10,000 people. A test detects 95% of people who have it and gives a false positive for 5% of healthy people. Find the number of true positives, the number of false positives, and P(disease | positive) as a decimal rounded to the nearest thousandth.

True positives: _____

False positives: _____

P(disease | positive): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q1**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q1/5

A shipment is accepted if 2 randomly tested items are both non-defective. Each item is defective with probability 0.1, independently. What is the probability the shipment is accepted? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q2/5

A $30 warranty covers a repair that costs $200 and is needed with probability 0.1 (otherwise there is no cost). Using expected cost as the decision rule, which is correct?

A. Expected repair cost $0.1 \times 200 = \$20 < \$30$, so the warranty is worth buying.

B. Expected repair cost $0.9 \times 200 = \$180 > \$30$, so the warranty is worth buying.

C. Expected repair cost $200 - 30 = \$170 > \$30$, so the warranty is worth buying.

D. Expected repair cost $0.1 \times 200 = \$20 < \$30$, so the warranty is not worth buying.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q3**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q3/5

Option A pays $10 with probability 0.5 and $0 otherwise; option B pays $50 with probability 0.1 and $0 otherwise. Compare the expected values and the risks.

A. Both have expected value $5; A is riskier, because it pays less.

B. A has expected value $10 and B has $5, so A is better.

C. Both have expected value $5; B is riskier, because it usually pays $0.

D. A has expected value $5 and B has $50, so B is better.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q4**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q4/5

A test gives +1 point for a correct answer, −1/4 point for a wrong answer, and 0 for a blank, on 4-choice questions. Using expected score as the decision rule, should a student with no idea of the answer guess randomly or leave it blank?

A. Leave it blank: the expected score of guessing is (1/4)(1) + (3/4)(−1) = −1/2.

B. Guess: the expected score is (1/4)(1) + (3/4)(−1/4) = 1/16, more than 0 for a blank.

C. Leave it blank: a wrong answer loses points, so guessing always lowers the score.

D. Guess: the expected score is (1/4)(1) = 1/4, because wrong answers do not count.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q5**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q5/5

SET 72 • HSS-MD.B.7

BACKWARD 1

Probability of compound independent events • Grade 7 • Statistics & Probability • 7.SP.C.8.a

Each item is non-defective with probability 0.9, independently of the others. Find the probability that two items are both non-defective. Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q1/5

Two fair coins: P(both heads)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q2/5

True or false?

For independent events, $P(A \text{ and } B) = P(A) \cdot P(B)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q3/5

Each item is non-defective with probability 0.9, independently of the others. The probability that three items are all non-defective is

A. 0.9

B. 0.27

C. 0.729

D. 2.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q4/5

Each item is defective with probability 0.1, independently of the others. Find the probability that at least one of two items is defective. Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q5/5

SET 72 • HSS-MD.B.7

BACKWARD 2

Two-way tables of counts • Grade 8 • Statistics & Probability • 8.SP.A.4

Of 10,000 people, 1% have a disease. How many have it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q1/5

Of 9,900 healthy people, 5% test positive. How many?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q2/5

True or false?

A two-way table can show test results for people with and without a disease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q3/5

If 95 sick and 495 healthy people test positive, the total positives are

A. 10,000

B. 590

C. 95

D. 495

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q4/5

What fraction of positives are sick (95 of 590)? Give a decimal to the nearest thousandth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q5/5

SET 72 • HSS-MD.B.7

BACKWARD 3

Weighted averages of outcomes • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Find 10% of $200.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q1/5

Compute 0.5 × 10 and 0.1 × 50.

0.5 × 10 = _____

0.1 × 50 = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q2/5

True or false?

$1/4 \times 1 + 3/4 \times (-1/4) = 1/4.$

A. True

B. False

$0.2 \times 30 + 0.8 \times 0$ equals

A. 30

B. 24

C. 0.2

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q4/5

Compute $0.9 \times 0 + 0.1 \times 200$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q5/5

SET 72 • HSS-MD.B.7

FORWARD 1

Investigate how changing conditions changes the decision • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A test detects 95% of people who have a disease and gives a false positive for 5% of healthy people. In a group of 10,000 where 10% have the disease, find P(disease | positive). Give a decimal rounded to the nearest thousandth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q1/5

A $30 warranty covers a repair that costs $200; the repair is needed with probability 0.2 and otherwise there is no cost. Using expected cost as the decision rule, is the warranty worth buying?

A. Yes: the repair cost of $200 is more than $30.

B. No: the repair probably will not happen, since $0.2 < 0.5$.

C. Yes: the expected repair cost is $0.2 \times 200 = \$40$, more than $30.

D. No: the expected repair cost is $40, which is more than $30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F1-Q2**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q2/5

True or false?

A positive result is more convincing when the disease is more common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q3/5

A test gives +1 point for a correct answer and −1/4 point for a wrong answer on 4-choice questions. A student eliminates one wrong choice and guesses randomly among the 3 remaining choices. What is the expected score for the guess?

A. (1/3)(1) + (2/3)(−1/4) = 1/6

B. (1/3)(1) + (2/3)(−1) = −1/3

C. (1/4)(1) + (3/4)(−1/4) = 1/16

D. (1/3)(1) = 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F1-Q4**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q4/5

A shipment is accepted only if all n tested items are non-defective. Each item is defective with probability 0.1, independently. What is the smallest whole number n for which the probability of acceptance is below 0.5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q5/5

SET 72 • HSS-MD.B.7

FORWARD 2

Find expected values of random variables • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

A random variable X takes values 0, 1, 2 with probabilities 0.25, 0.5, 0.25. Find E(X).

E(X) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q1/5

A raffle sells 500 tickets at $2 each; one ticket wins a $400 prize. Find the expected net gain of one ticket, in dollars, as a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q2/5

True or false?

The expected value is the long-run average of many repetitions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q3/5

For a fair die, E(roll) is

A. 3

B. 21

C. 3.5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q4**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q4/5

A $30 warranty covers a $200 repair that is needed with probability 0.1 (otherwise no cost). Why is deciding whether to buy it an expected-value calculation?

A. The repair cost is a random variable (0 or 200) with expected value $20, compared with the certain $30 price.

B. The repair will cost exactly $20 for certain, so the warranty price is compared with $20.

C. The expected repair cost is 0.1 + 200 = $200.10, which is compared with the $30 price.

D. The warranty price is a random variable whose expected value is $200, compared with the $30 price.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q5**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q5/5

SET 73 • HSN-CN.A.3 • ADVANCED EXTENSION (+)

Use conjugates to find moduli and quotients of complex numbers

MAIN	Algebra II main question	**HSN-CN.A.3**
BACKWARD 1	Multiply complex numbers	**HSN-CN.A.2**
BACKWARD 2	Distance from the origin	**8.G.B.8**
BACKWARD 3	Powers of i	**HSN-CN.A.1**
FORWARD 1	Prove properties of conjugates and moduli	**HSN-CN.A.3**
FORWARD 2	Write complex numbers in polar form	**HSN-CN.B.4**

SET 73 • HSN-CN.A.3

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

Give the conjugate of $3 - 4i$ and find $|3 - 4i|$.

Conjugate: _____

$|3 - 4i|$ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q1/5

Write (2 + 3i)/(1 − i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q2

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q2/5

Write (5 + i)/(2 + 3i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q3/5

Find $|-5 + 12i|$ and compute $(-5 + 12i)(-5 - 12i)$.

$|-5 + 12i| =$ _____

Product: _____

Write 1/(4 + 3i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q5/5

SET 73 • HSN-CN.A.3

BACKWARD 1

Multiply complex numbers • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply $(2 + 3i)(1 + i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q1/5

Multiply $(3 - 4i)(3 + 4i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q2/5

True or false?

$(a + bi)(a - bi) = a^2 + b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q3/5

$(5 + i)(2 - 3i)$ equals

A. $13 + 13i$

B. $7 - 13i$

C. $10 - 3i$

D. $13 - 13i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q4/5

Multiply $(1 - i)(1 + i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q5/5

SET 73 • HSN-CN.A.3

BACKWARD 2

Distance from the origin • Grade 8 • Geometry • 8.G.B.8

Find the distance from (0, 0) to (3, −4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q1/5

Find the distance from (0, 0) to (−5, 12).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q2/5

True or false?

The distance from (0, 0) to (a, b) is $\sqrt{(a^2 + b^2)}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q3/5

The distance from (0, 0) to (1, 1) is

A. 0

B. 1

C. 2

D. $\sqrt{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q4/5

Find the distance from (0, 0) to (4, 3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q5/5

SET 73 • HSN-CN.A.3

BACKWARD 3

Powers of i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Simplify i^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q1/5

Simplify $-12i \cdot i$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q2/5

True or false?

$-3i^2 = -3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q3/5

Which is real?

A. $3i$

B. $(4 + 3i)^2$

C. $(4 + 3i)(4 - 3i)$

D. $4 + 3i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q4/5

Write 13/13 − 13i/13 in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q5/5

SET 73 • HSN-CN.A.3

FORWARD 1

Prove properties of conjugates and moduli • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

Which argument proves $z \cdot \bar{z} = |z|^2$ for $z = a + bi$ (a and b real)?

A. $(a + bi)(a - bi) = a^2 + b^2i^2$, which is $|z|^2$ because $|z| = \sqrt{(a^2 + b^2i^2)}$.

B. $(a + bi)(a - bi) = a^2 - b^2$, and $|z|^2 = a^2 - b^2$.

C. $(a + bi)(a - bi) = a^2 - b^2i^2 = a^2 + b^2$, and $|z|^2 = a^2 + b^2$.

D. For $z = 3 + 4i$: $(3 + 4i)(3 - 4i) = 25 = 5^2$, so the statement holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q1**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q1/5

Let $z = 3 + 4i$ and $w = 1 + i$. Find zw, $|zw|$, and $|z||w|$, giving the moduli exactly in simplest radical form.

$zw =$ _____

$|zw| =$ _____

$|z||w| =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q2/5

True or false?

The conjugate of a real number is itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q3/5

If $z = 2 - 5i$, then $z + \bar{z}$ is

A. $4 - 10i$

B. 0

C. $-10i$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q4/5

Use $\bar{z}/|z|^2$ to write $1/z$ for $z = 1 + 2i$ in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q5/5

SET 73 • HSN-CN.A.3

FORWARD 2

Write complex numbers in polar form • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4

Write $1 + i$ in polar form $r(\cos\theta + i\sin\theta)$ with $0 \le \theta < 2\pi$. Give r and θ exactly.

$r =$ _____

$\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q1/5

Write $2(\cos(\pi/2) + i \sin(\pi/2))$ in rectangular form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q2/5

True or false?

The modulus of a complex number is the r in its polar form.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q3/5

The polar form of -3 is

A. $3(\cos \pi + i \sin \pi)$

B. $-3(\cos \pi + i \sin \pi)$

C. $3(\cos 0 + i \sin 0)$

D. $3(\cos(\pi/2) + i \sin(\pi/2))$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q4/5

In the complex plane, what do the modulus and the angle of a complex number represent?

A. The modulus is the sum of the parts a + b; the angle is the ratio of the parts b/a.

B. The modulus is the distance from 1; the angle is measured from the positive imaginary axis.

C. The modulus is the real part; the angle is the imaginary part of the number written as a + bi.

D. The modulus is the distance from 0; the angle is the direction measured from the positive real axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.d • ADVANCED EXTENSION (+)

Graph rational functions with zeros and asymptotes

MAIN	Algebra II main question	**HSF-IF.C.7.d**
BACKWARD 1	Rewrite a rational expression by inspection	**HSA-APR.D.6**
BACKWARD 2	Domain and excluded values	**HSF-IF.B.5**
BACKWARD 3	Sign of a quotient on intervals	**7.NS.A.2**
FORWARD 1	Graph rational functions with slant asymptotes	**HSF-IF.C.7.d**
FORWARD 2	Interpret a rational model's features in context	**HSF-IF.B.4**

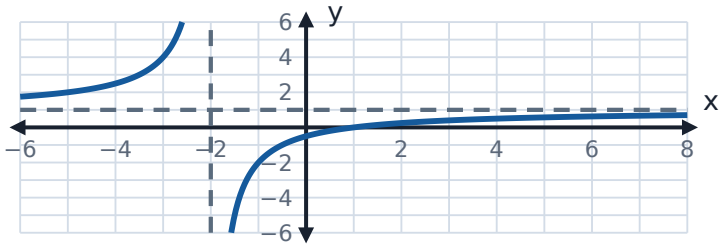
SET 74 • HSF-IF.C.7.d

MAIN

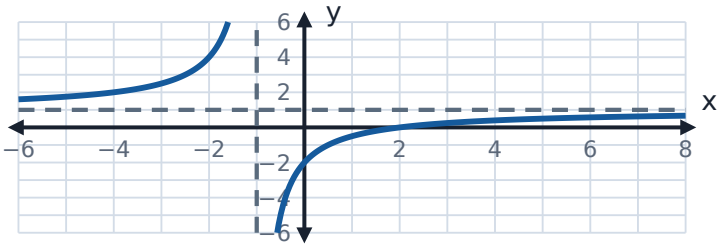
Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Which graph shows y = (x + 1)/(x − 2), with its asymptotes dashed?

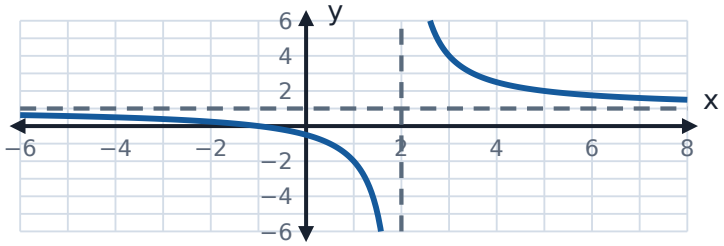
A



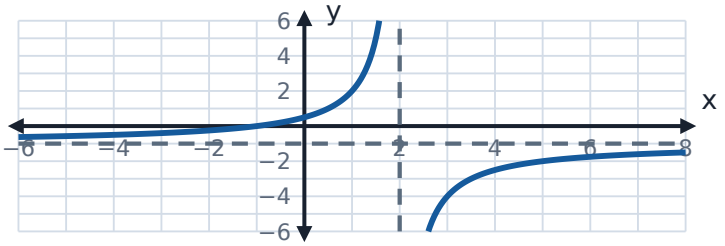
B



C



D



A. Graph A

B. Graph B

C. Graph C

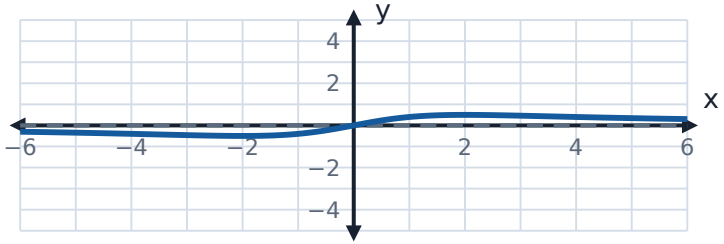
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q1**

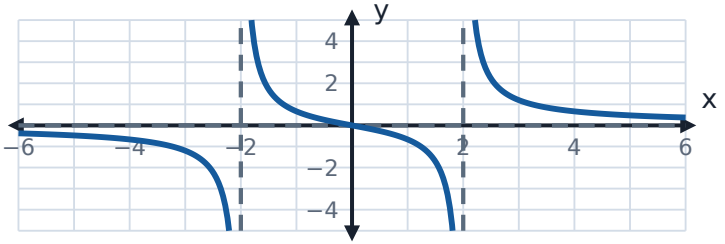
High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q1/5

Which graph shows $y = 2x/(x^2 - 4)$, with its asymptotes dashed?

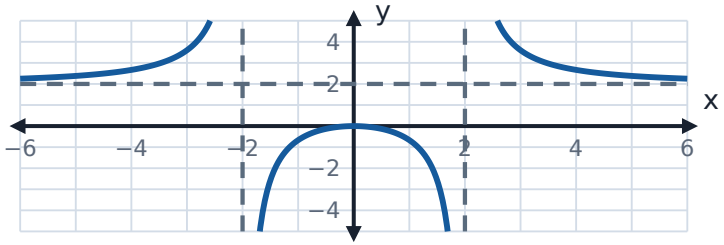
A



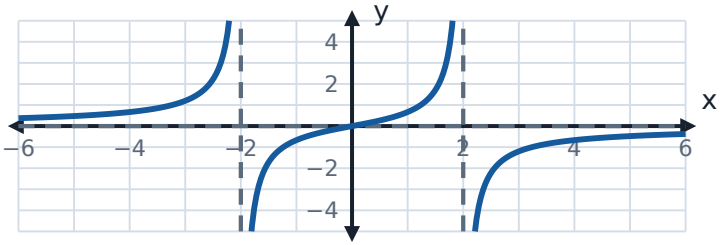
B



C



D



A. Graph A

B. Graph B

C. Graph C

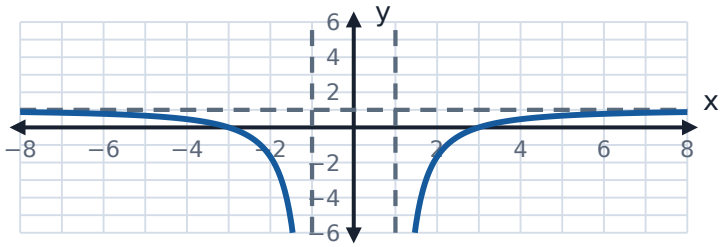
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q2**

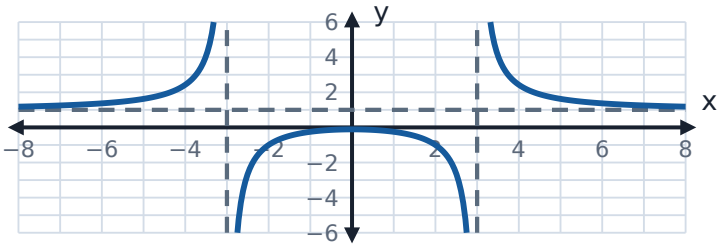
High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q2/5

Which graph shows $y = (x^2 - 1)/(x^2 - 9)$, with its asymptotes dashed?

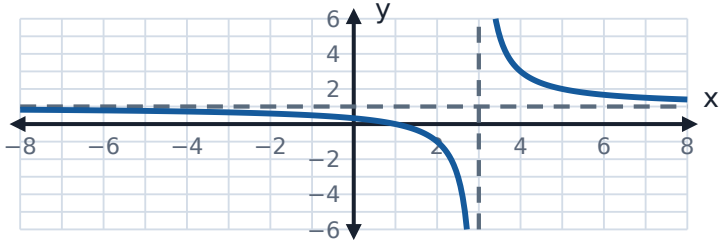
A



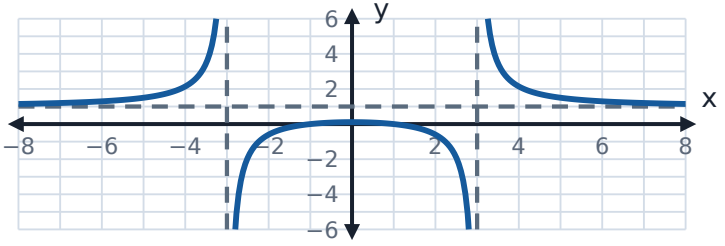
B



C



D



A. Graph A

B. Graph B

C. Graph C

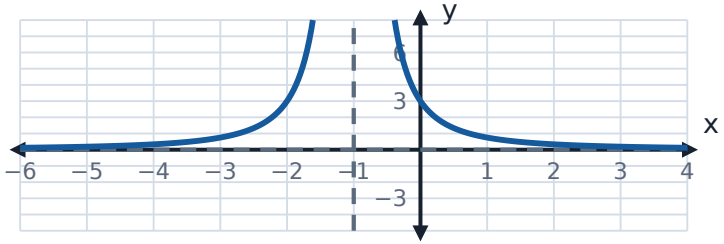
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q3**

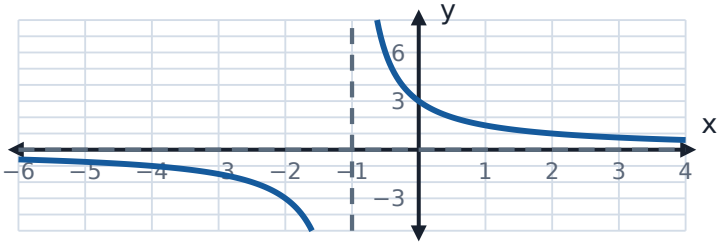
High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q3/5

Which graph shows $y = 3/(x + 1)^2$, with its asymptotes dashed?

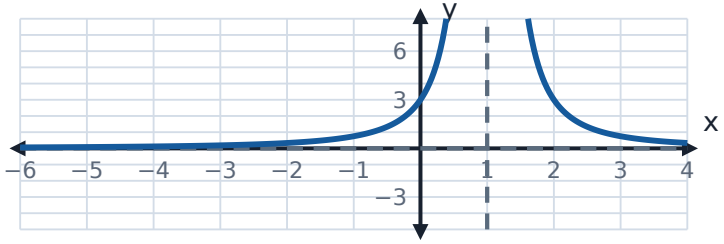
A



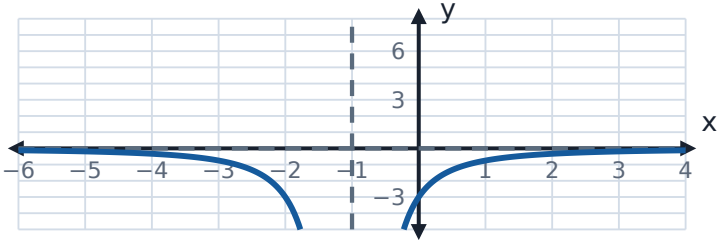
B



C



D



A. Graph A

B. Graph B

C. Graph C

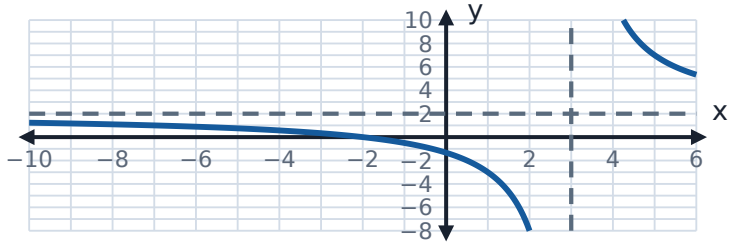
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q4**

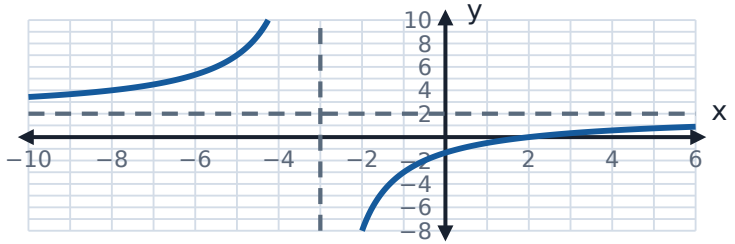
High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q4/5

Which graph shows $y = (2x - 4)/(x + 3)$, with its asymptotes dashed?

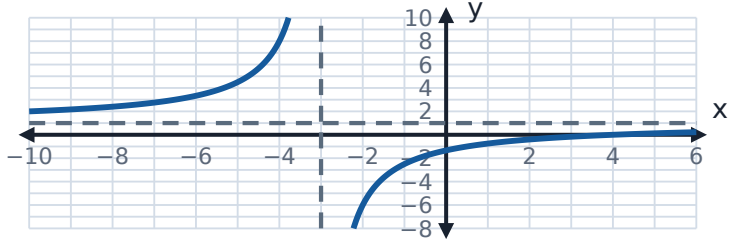
A



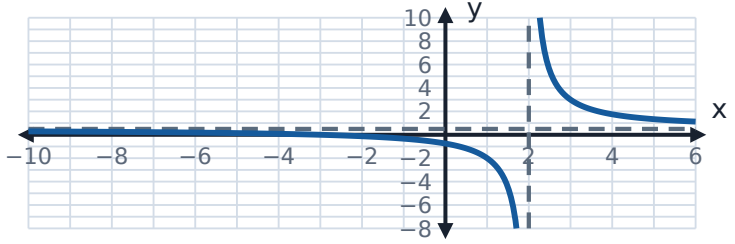
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q5/5

SET 74 • HSF-IF.C.7.d

BACKWARD 1

Rewrite a rational expression by inspection • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Rewrite $(x + 1)/(x - 2)$ as $1 + r/(x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q1/5

Rewrite $(2x - 4)/(x + 3)$ as $2 + r/(x + 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q2/5

True or false?

The constant q in $q + r/(x - a)$ is the horizontal asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q3/5

$(x^2 - 1)/(x^2 - 9)$ equals

A. $1 + 8/(x^2 - 9)$

B. $1 - 8/(x^2 - 9)$

C. $1 + 1/(x^2 - 9)$

D. $1/9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-B1-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q4/5

Factor the denominator of $2x/(x^2 - 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q5/5

SET 74 • HSF-IF.C.7.d

BACKWARD 2

Domain and excluded values • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

Give the domain of $(x + 1)/(x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q1/5

Give the domain of $3/(x + 1)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q2/5

True or false?

A rational function is undefined where its denominator is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q3/5

$2x/(x^2 - 4)$ is undefined at

A. $x = \pm 2$

B. $x = 4$

C. $x = 2$ only

D. $x = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q4/5

List the excluded values of $(x^2 - 1)/(x^2 - 9)$, separated by commas.

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.d

BACKWARD 3

Sign of a quotient on intervals • Grade 7 • The Number System • 7.NS.A.2

Is $(x + 1)/(x - 2)$ positive or negative at $x = 0$? Answer Positive or Negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q1/5

Is $2x/(x^2 - 4)$ positive or negative at $x = 3$? Answer Positive or Negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q2/5

True or false?

A quotient of two negative numbers is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q3/5

At $x = 2.1$, $(x + 1)/(x - 2)$ is

A. large and negative

B. undefined

C. zero

D. large and positive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q4/5

True or false?

$3/(x + 1)^2$ is negative for some value of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q5/5

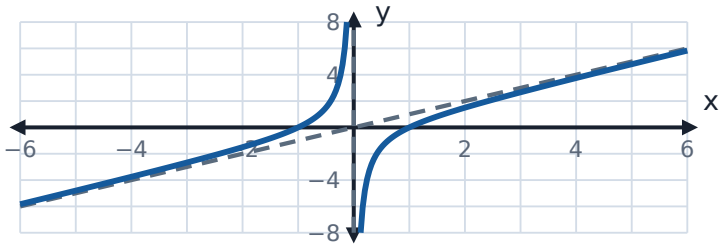
SET 74 • HSF-IF.C.7.d

FORWARD 1

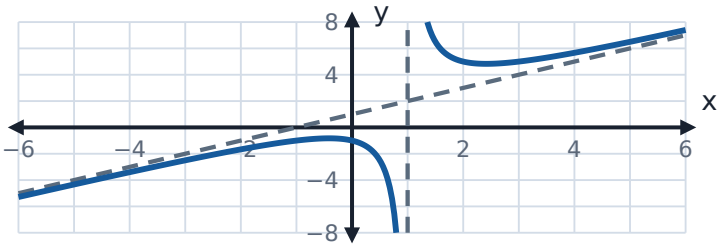
Graph rational functions with slant asymptotes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Which graph shows $y = (x^2 + 1)/x$, with its asymptotes dashed?

A



B



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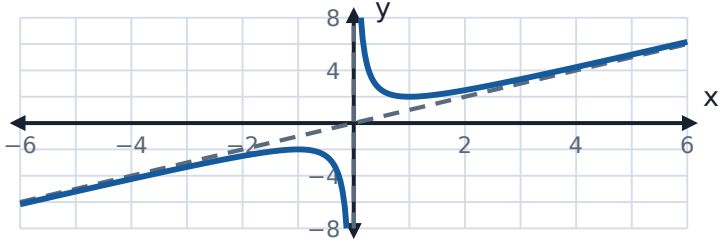
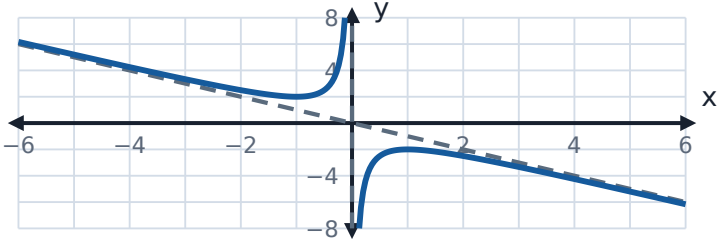


D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q1/5

Find the slant asymptote of $y = (x^2 - x - 1)/(x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q2/5

True or false?

A rational function can cross its horizontal asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q3/5

$y = (3x^2 + 1)/(x^2 + 4)$ has horizontal asymptote

A. none

B. $y = 3$

C. $y = 0$

D. $y = 1/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q4/5

Which rational function has vertical asymptote $x = 1$, horizontal asymptote $y = 2$, and a zero at $x = 3$?

A. $f(x) = (x - 3)/(2(x - 1))$

B. $f(x) = 2(x - 3)/(x - 1)$

C. $f(x) = 2(x - 1)/(x - 3)$

D. $f(x) = 2(x + 3)/(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q5/5

SET 74 • HSF-IF.C.7.d

FORWARD 2

Interpret a rational model's features in context • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

The average cost per item is $A(x) = (500 + 4x)/x$ dollars for x items. What do the vertical asymptote $x = 0$ and the horizontal asymptote $y = 4$ mean in context?

A. $x = 0$: the fixed cost of production is \$0; $y = 4$: no more than 4 items can be made at this average cost per item.

B. $x = 0$: the average cost grows without bound as the number of items nears 0; $y = 4$: the average cost approaches \$4 per item as x grows.

C. $x = 0$: the first item is free to make; $y = 4$: the total cost of all of the items approaches \$4 as the number of items grows.

D. $x = 0$: the average cost is exactly \$500 for very few items; $y = 4$: the average cost is never more than \$4 per item.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q1/5

A drug concentration is $C(t) = 20t/(t^2 + 4)$ for $t \geq 0$ hours. What does its horizontal asymptote mean?

A. $y = 0$: the concentration is 0 at every time after $t = 2$.

B. $y = 20$: the concentration approaches 20 as time goes on.

C. $y = 0$: the drug never enters the blood.

D. $y = 0$: the concentration approaches 0 as time goes on.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q2/5

True or false?

For a context, only the part of a rational graph in the context domain is meaningful.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q3/5

In $A(x) = (500 + 4x)/x$, as x grows the average cost

A. increases

B. is constant

C. approaches 0

D. decreases toward 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q4/5

The average cost per item is A(x) = (500 + 4x)/x dollars for x items. Find the number of items for which the average cost is $5.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q5/5

SET 75 • HSF-IF.C.7.d • ADVANCED EXTENSION (+)

Distinguish holes from vertical asymptotes

MAIN	Algebra II main question	**HSF-IF.C.7.d**
BACKWARD 1	Factor numerators and denominators	**HSA-SSE.A.2**
BACKWARD 2	Simplify rational expressions and keep exclusions	**HSA-APR.D.7**
BACKWARD 3	Evaluate the simplified expression	**HSF-IF.A.2**
FORWARD 1	Construct and graph rational functions with holes	**HSF-IF.C.7.d**
FORWARD 2	Relate holes to extraneous solutions	**HSA-REI.A.2**

SET 75 • HSF-IF.C.7.d

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

$f(x) = (x^2 - 4)/(x - 2)$ has exactly one hole. Give the hole as a point (x, y), and give every vertical asymptote as an x-value (enter none if there is none).

Hole: _____

Vertical asymptote: x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q1/5

$f(x) = (x - 3)/(x^2 - 9)$ has exactly one hole. Give the hole as a point (x, y), and give every vertical asymptote as an x-value (enter none if there is none).

Hole: _____

Vertical asymptote: x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q2/5

$f(x) = (x^2 + x - 6)/(x^2 - x - 2)$ has exactly one hole. Give the hole as a point (x, y), and give every vertical asymptote as an x-value (enter none if there is none).

Hole: _____

Vertical asymptote: x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q3/5

$f(x) = (x + 1)/(x^2 - 1)$ has exactly one hole. Give the hole as a point (x, y), and give every vertical asymptote as an x-value (enter none if there is none).

Hole: _____

Vertical asymptote: x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q4/5

$f(x) = x(x - 1)/(x(x + 2))$ has exactly one hole. Give the hole as a point (x, y), and give every vertical asymptote as an x-value (enter none if there is none).

Hole: _____

Vertical asymptote: x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q5/5

SET 75 • HSF-IF.C.7.d

BACKWARD 1

Factor numerators and denominators • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 + x - 6$ and $x^2 - x - 2$.

$x^2 + x - 6 =$ _____

$x^2 - x - 2 =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q1/5

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q2/5

True or false?

$x^2 - 1 = (x - 1)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q3/5

$x^2 - 4$ equals

A. $(x - 4)(x + 1)$

B. $(x - 2)^2$

C. $(x - 2)(x + 2)$

D. $x(x - 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q4/5

Factor $x^2 + 2x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q5/5

SET 75 • HSF-IF.C.7.d

BACKWARD 2

Simplify rational expressions and keep exclusions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Simplify $(x - 3)/((x - 3)(x + 3))$ and list the excluded values of the original expression, separated by commas.

Simplified: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q1/5

Simplify $(x + 1)/((x - 1)(x + 1))$ and list the excluded values of the original expression, separated by commas.

Simplified: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q2/5

True or false?

After simplifying, the original excluded values are still excluded.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q3/5

$x(x - 1)/(x(x + 2))$ simplifies to

A. $-1/2$

B. $(x - 1)/(x + 2)$ for all x

C. $x - 1$

D. $(x - 1)/(x + 2)$, $x \neq 0, -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-B2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q4/5

Simplify $(x^2 - 4)/(x - 2)$ and give the excluded value of the original expression.

Simplified: _____

Excluded value: $x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q5/5

SET 75 • HSF-IF.C.7.d

BACKWARD 3

Evaluate the simplified expression • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Evaluate $1/(x + 3)$ at $x = 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q1/5

Evaluate $(x + 3)/(x + 1)$ at $x = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q2/5

True or false?

The y-coordinate of a hole is the value of the simplified expression there.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q3/5

$(x - 1)/(x + 2)$ at $x = 0$ is

A. 1/2

B. $-1/2$

C. undefined

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q4/5

Evaluate $1/(x - 1)$ at $x = -1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q5/5

SET 75 • HSF-IF.C.7.d

FORWARD 1

Construct and graph rational functions with holes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Which rational function has a hole at $x = 1$, a vertical asymptote $x = -3$, and horizontal asymptote $y = 0$?

A. $f(x) = (x - 1)/((x - 1)(x + 3))$

B. $f(x) = (x - 1)(x + 3)/(x - 1)$

C. $f(x) = (x - 1)/(x + 3)$

D. $f(x) = (x + 3)/((x - 1)(x + 3))$

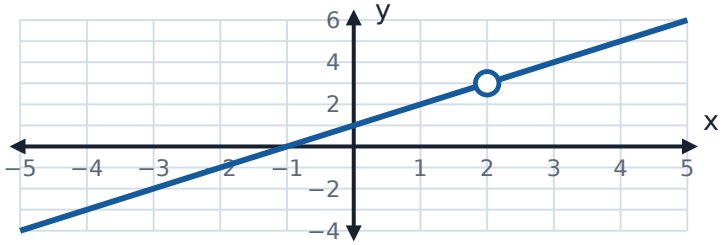
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q1

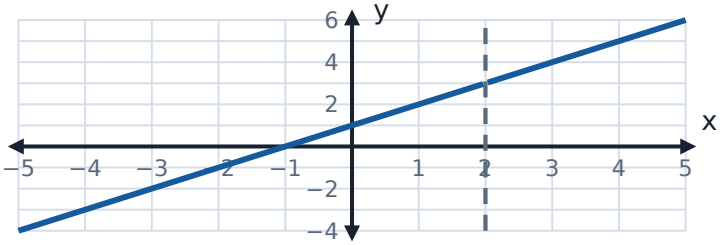
High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q1/5

Which graph shows $y = (x^2 - x - 2)/(x - 2)$?

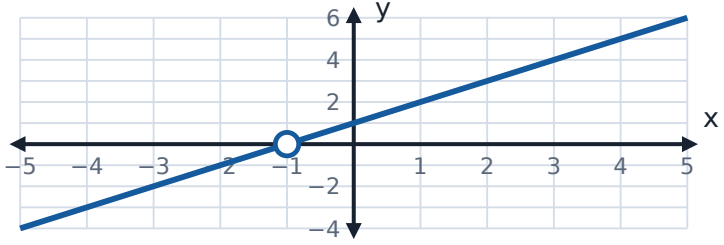
A



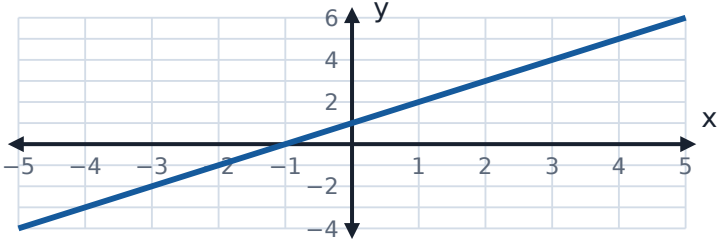
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q2/5

True or false?

A rational function can have both a hole and a vertical asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q3/5

$f(x) = (x - 4)^2/(x - 4)$ has

A. no special features

B. a hole at (4, 4)

C. a vertical asymptote $x = 4$

D. a hole at (4, 0)

For $f(x) = (x - 3)/(x^2 - 9)$, why is $x = -3$ a vertical asymptote but $x = 3$ a hole?

A. Negative excluded values always give asymptotes, and positive excluded values always give holes in a graph.

B. Both $x - 3$ and $x + 3$ cancel, so $x = -3$ and $x = 3$ are both holes, and the asymptote comes from x^2 alone.

C. After canceling $x - 3$, the factor $x + 3$ remains in the denominator, but $x - 3$ does not; near 3 the values approach 1/6.

D. The numerator is 0 at $x = -3$, so the graph has a vertical asymptote at -3, and a hole is at the other zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-F1-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q5/5

SET 75 • HSF-IF.C.7.d

FORWARD 2

Relate holes to extraneous solutions • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $(x^2 - 4)/(x - 2) = 4$.

A. $x = 4$, because the hole of the graph is at the height $y = 4$.

B. $x = 2$, because it satisfies the simplified equation $x + 2 = 4$.

C. Every $x \neq 2$, because the function equals 4 at every point of its domain.

D. No solution: $x + 2 = 4$ gives $x = 2$, which is excluded (the hole at (2, 4)).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q1/5

Solve $(x - 3)/(x^2 - 9) = 1/6$.

A. $x = 3$ or $x = -3$: both make $x^2 - 9 = 0$, so both solve it.

B. $x = -3$: then $1/(x + 3) = 1/6$ after simplifying the left side.

C. No solution: $1/(x + 3) = 1/6$ gives $x = 3$, which is excluded.

D. $x = 3$: it satisfies $1/(x + 3) = 1/6$, the simplified equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q2/5

True or false?

An extraneous solution of f(x) = c can occur exactly at a hole whose height is c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q3/5

$(x + 1)/(x^2 - 1) = -1/2$ has

A. no solution

B. $x = -1$

C. $x = 0$

D. $x = 1$

Solve $(x^2 + x - 6)/(x^2 - x - 2) = 2$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q5/5

SET 76 • HSF-BF.A.1.c • ADVANCED EXTENSION (+)

Compose functions and interpret compositions with domains

MAIN	Algebra II main question	**HSF-BF.A.1.c**
BACKWARD 1	Build combined models	**HSF-BF.A.1.b**
BACKWARD 2	Evaluate a function at an expression	**HSF-IF.A.2**
BACKWARD 3	Domains of radical, rational, and log functions	**HSF-IF.B.5**
FORWARD 1	Decompose functions into compositions	**HSF-BF.A.1.c**
FORWARD 2	Verify inverses by composition	**HSF-BF.B.4.b**

SET 76 • HSF-BF.A.1.c

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

$f(x) = \sqrt{x}$ and $g(x) = x - 4$. Which gives $f(g(x))$ and $g(f(x))$ with the domain of each?

A. $f(g(x)) = \sqrt{x - 4}$, $x \geq 0$; $g(f(x)) = \sqrt{(x - 4)}$, $x \geq 4$

B. $f(g(x)) = \sqrt{(x - 4)}$, $x \geq 4$; $g(f(x)) = \sqrt{x} - 4$, $x \geq 0$

C. $f(g(x)) = \sqrt{(x - 4)}$, all real x; $g(f(x)) = \sqrt{x} - 4$, all real x

D. $f(g(x)) = \sqrt{(x - 4)}$, $x \geq 0$; $g(f(x)) = \sqrt{x} - 4$, $x \geq 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q1/5

The temperature is $T(h) = 20 - 6.5h$ °C at height h km, and a balloon's height is $h(t) = 0.3t$ km after t minutes. Write $T(h(t)) = a + bt$, and find the temperature at $t = 10$.

$a =$ _____

$b =$ _____

Temperature at $t = 10$ (°C): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q2/5

$f(x) = 1/(x - 1)$ and $g(x) = x^2$. Write $f(g(x))$ and list the real numbers excluded from its domain, separated by commas. Use ^ for exponents.

$f(g(x)) =$ _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q3/5

A dollar amount d converts to euros by e(d) = 0.9d, and a store's price with tax in euros is p(e) = 1.2e. Then p(e(d)) = kd. Find k and the price in euros for d = 50.

k = _____

p(e(50)) (euros): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q4/5

$f(x) = \log_2 x$ and $g(x) = x^2 - 9$. Which gives $f(g(x))$ and its domain?

A. $f(g(x)) = \log_2(x^2 - 9)$; domain $x < -3$ or $x > 3$

B. $f(g(x)) = (\log_2 x)^2 - 9$; domain $x > 0$

C. $f(g(x)) = \log_2(x^2) - 9$; domain $x \neq 0$

D. $f(g(x)) = \log_2(x^2 - 9)$; domain $x > 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q5/5

SET 76 • HSF-BF.A.1.c

BACKWARD 1

Build combined models • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b

$f(x) = x^2$ and $g(x) = x + 1$. Find $f(x) + g(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q1/5

$f(x) = x^2$ and $g(x) = x + 1$. Find $f(x) \cdot g(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q2/5

True or false?

f(g(x)) is a different combination from f(x) · g(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q3/5

For $f(x) = 2x$ and $g(x) = x + 3$, $f(x) - g(x)$ is

A. $x - 3$

B. $2x^2 + 6x$

C. $x + 3$

D. $3x + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q4/5

Which operation combines T and h in T(h(t))?

A. division

B. addition

C. composition

D. multiplication

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q5/5

SET 76 • HSF-BF.A.1.c

BACKWARD 2

Evaluate a function at an expression • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

$f(x) = \sqrt{x}$. Find $f(x - 4)$.

A. $\sqrt{x} - \sqrt{4}$

B. $\sqrt{(x)} \cdot (-4)$

C. $\sqrt{x} - 4$

D. $\sqrt{(x - 4)}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q1/5

$T(h) = 20 - 6.5h$. Find $T(0.3t)$ in the form $a + bt$. Give a and b.

$a =$ _____

$b =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q2/5

True or false?

If $f(x) = 1/(x - 1)$, then $f(x^2) = 1/(x^2 - 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q3/5

If $g(x) = x^2 - 9$, then $g(3)$ is

A. -3

B. -6

C. 9

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q4/5

p(e) = 1.2e. Find p(45).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q5/5

SET 76 • HSF-BF.A.1.c

BACKWARD 3

Domains of radical, rational, and log functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

Give the domain of $\sqrt{(x - 4)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q1/5

Give the domain of $\log_2 u$ in terms of u.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q2/5

True or false?

$1/(x^2 - 1)$ is undefined only at $x = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q3/5

$x^2 - 9 > 0$ when

A. $x < -3$ or $x > 3$

B. $-3 < x < 3$

C. all x

D. $x > 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q4/5

Give the domain of $\sqrt{x} - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q5/5

SET 76 • HSF-BF.A.1.c

FORWARD 1

Decompose functions into compositions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

Which pair writes $h(x) = (3x + 1)^5$ as $f(g(x))$?

A. $g(x) = 3x + 1$, $f(x) = x^5$

B. $g(x) = 3x + 1$, $f(x) = 5x$

C. $g(x) = 3x$, $f(x) = x^5 + 1$

D. $g(x) = x^5$, $f(x) = 3x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q1/5

Which pair writes $h(x) = 2^{x^2}$ as $f(g(x))$?

A. $g(x) = 2x$, $f(x) = x^2$

B. $g(x) = x^2$, $f(x) = 2^x$

C. $g(x) = x^2$, $f(x) = 2x$

D. $g(x) = 2^x$, $f(x) = x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q2/5

True or false?

$h(x) = (x + 1)^2$ can be written as $f(g(x))$ in more than one way, for example with $g(x) = x + 1$ and $f(x) = x^2$, or with $g(x) = x^2 + 2x$ and $f(x) = x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q3/5

If $f(x) = x + 1$, $g(x) = 2x$, $k(x) = x^2$, then $f(g(k(3)))$ is

A. 37

B. 19

C. 64

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-F1-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q4/5

Which context describes a composition V(r(t))?

A. The time t depends on the volume V, and the volume depends on the radius r.

B. The volume V of a balloon depends on its radius r, and the radius depends on time t.

C. The volume V of a balloon is multiplied by its radius r at each time t.

D. The volume V and the radius r of a balloon are added together at each time t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q5/5

SET 76 • HSF-BF.A.1.c

FORWARD 2

Verify inverses by composition • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

Which work shows that $f(x) = 2x + 3$ and $g(x) = (x - 3)/2$ are inverses?

A. $f(0) = 3$ and $g(0) = -3/2$, which are opposite in sign

B. $f(x) \cdot g(x) = (2x + 3)(x - 3)/2 = x$

C. $f(g(x)) = 2((x - 3)/2) + 3 = x + 6$ and $g(f(x)) = x$

D. $f(g(x)) = 2((x - 3)/2) + 3 = x$ and $g(f(x)) = ((2x + 3) - 3)/2 = x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q1/5

Compute $f(g(x))$ for $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q2/5

True or false?

If f(g(x)) = x for all x in g's domain and g(f(x)) = x for all x in f's domain, then f and g are inverses.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q3/5

For $f(x) = x + 5$ and $g(x) = x - 5$, $f(g(2))$ is

A. 2

B. -3

C. 12

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-F2-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q4/5

Why is composition the natural test for inverse functions?

A. Composition adds the two functions, and inverses must add to 0 for every input x.

B. Composition reverses the graph across the x-axis, which is where an inverse must lie.

C. An inverse undoes the function, so applying one after the other must return the original input.

D. Two functions are inverses exactly when their product f(x) · g(x) is equal to 1 for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q5/5

SET 77 • HSF-BF.B.4.b • ADVANCED EXTENSION (+)

Verify inverses by composition on the stated domains

MAIN	Algebra II main question	**HSF-BF.B.4.b**
BACKWARD 1	Find inverses algebraically	**HSF-BF.B.4.a**
BACKWARD 2	Compose functions	**HSF-BF.A.1.c**
BACKWARD 3	Square roots of squares	**HSN-RN.A.2**
FORWARD 1	Show that proposed inverses fail	**HSF-BF.B.4.b**
FORWARD 2	Read inverse values from tables and graphs	**HSF-BF.B.4.c**

SET 77 • HSF-BF.B.4.b

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

Which work verifies by composition that $f(x) = 2x + 3$ and $g(x) = (x - 3)/2$ are inverses?

A. $f(x) \cdot g(x) = (2x + 3)(x - 3)/2 = 1$, so f and g are inverses.

B. $f(g(x)) = 2 \cdot (x - 3)/2 + 3 = x$ and $g(f(x)) = ((2x + 3) - 3)/2 = x$, for all real x.

C. $f(1) = 5$ and $g(5) = 1$, so f and g are inverses.

D. $f(g(x)) = 2 \cdot (x - 3)/2 + 3 = x$ for all real x, and one composition is enough by itself.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q1/5

Which work verifies by composition that $f(x) = x^3 - 1$ and $g(x) = \sqrt[3]{(x + 1)}$ are inverses?

A. $f(g(x)) = (\sqrt[3]{(x + 1)})^3 - 1 = x$, but only for $x \geq 0$, so f and g are inverses only when $x \geq 0$.

B. $g(f(x)) = \sqrt[3]{(x^3 - 1)} + 1 = x$, so f and g are inverses.

C. $f(2) = 7$ and $g(7) = 2$, so f and g are inverses.

D. $f(g(x)) = (\sqrt[3]{(x + 1)})^3 - 1 = x$ and $g(f(x)) = \sqrt[3]{((x^3 - 1) + 1)} = \sqrt[3]{(x^3)} = x$, for all real x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q2/5

Which work verifies by composition that $f(x) = 2^x$ and $g(x) = \log_2 x$ are inverses, with the correct domains?

A. $f(g(x)) = 2 \cdot \log_2 x = x$ for $x > 0$.

B. $f(g(x)) = 2^{\log_2 x} = x$ and $g(f(x)) = \log_2(2^x) = x$, both for every real number x.

C. $f(3) = 8$ and $g(8) = 3$, so f and g are inverses for all x.

D. $f(g(x)) = 2^{\log_2 x} = x$ for $x > 0$, and $g(f(x)) = \log_2(2^x) = x$ for all real x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q3/5

Which work verifies by composition that $f(x) = 1/(x - 2)$ $(x \neq 2)$ and $g(x) = 1/x + 2$ $(x \neq 0)$ are inverses?

A. $f(3) = 1$ and $g(1) = 3$, so f and g are inverses.

B. $f(g(x)) = 1/((1/x + 2) - 2) = 1/(1/x) = x$, so f and g are inverses for all real x with no excluded values needed.

C. $f(g(x)) = 1/((1/x + 2) - 2) = 1/(1/x) = x$ for $x \neq 0$, and $g(f(x)) = 1/(1/(x - 2)) + 2 = (x - 2) + 2 = x$ for $x \neq 2$.

D. $g(f(x)) = 1/(x - 2) + 2 = (2x - 3)/(x - 2)$, so they are not inverses.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q4/5

$f(x) = x^2$ with domain $x \geq 0$ and $g(x) = \sqrt{x}$. Which work verifies that they are inverses and explains why the domain restriction is needed?

A. The restriction is needed because $\sqrt{x}$ is undefined for $x > 0$.

B. $f(g(x)) = (\sqrt{x})^2 = x$ for $x \geq 0$, and $g(f(x)) = \sqrt{(x^2)} = |x| = x$ because $x \geq 0$. Without the restriction, $\sqrt{(x^2)} = |x| \neq x$ for $x < 0$.

C. $f(g(x)) = (\sqrt{x})^2 = x$ and $g(f(x)) = \sqrt{(x^2)} = x$ for every real number x, so the two functions are inverses and no domain restriction is needed.

D. $g(f(x)) = \sqrt{(x^2)} = \pm x$, and the restriction chooses the + sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q5/5

SET 77 • HSF-BF.B.4.b

BACKWARD 1

Find inverses algebraically • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

Find the inverse of $f(x) = 2x + 3$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q1/5

Find the inverse of $f(x) = x^3 - 1$.

A. $f^{-1}(x) = \sqrt[3]{(x + 1)}$

B. $f^{-1}(x) = \sqrt[3]{x} + 1$

C. $f^{-1}(x) = \sqrt[3]{(x - 1)}$

D. $f^{-1}(x) = 1/(x^3 - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q2/5

True or false?

The inverse of 2^x is $\log_2 x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q3/5

The inverse of $f(x) = 1/(x - 2)$ is

A. $1/x + 2$

B. $1/(x + 2)$

C. $2 - 1/x$

D. $x - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q4/5

Find the inverse of $f(x) = x^2$ for $x \geq 0$.

A. $f^{-1}(x) = \pm\sqrt{x}$

B. $f^{-1}(x) = \sqrt{x}$

C. $f^{-1}(x) = x/2$

D. $f^{-1}(x) = 1/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q5/5

SET 77 • HSF-BF.B.4.b

BACKWARD 2

Compose functions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

$f(x) = 2x + 3$ and $g(x) = (x - 3)/2$. Compute $f(g(5))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q1/5

$f(x) = 2x + 3$ and $g(x) = (x - 3)/2$. Compute $g(f(5))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q2/5

True or false?

f(g(x)) means apply g first, then f.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q3/5

For $f(x) = x^3$ and $g(x) = x + 1$, $f(g(x))$ is

A. $(x + 1)^3$

B. $x^3 + 1$

C. $x + 1$

D. $3x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q4/5

Compute $\log_2(2^5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q5/5

SET 77 • HSF-BF.B.4.b

BACKWARD 3

Square roots of squares • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

Evaluate $\sqrt{((-3)^2)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q1/5

Evaluate $(\sqrt{7})^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q2/5

True or false?

$\sqrt{(x^2)} = x$ for every real x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q3/5

$\sqrt{(x^2)}$ equals

A. $|x|$

B. x^2

C. $-x$

D. x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q4/5

For which x is $\sqrt{(x^2)} = x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q5/5

SET 77 • HSF-BF.B.4.b

FORWARD 1

Show that proposed inverses fail • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

Which work shows by composition that $f(x) = 3x - 1$ and $g(x) = (x - 1)/3$ are not inverses?

A. $f(0) = -1$ and $g(0) = -1/3$ are different, so they are not inverses

B. $f(g(x)) = 3((x - 1)/3) - 1 = x$, so they are inverses

C. $g(f(x)) = ((3x - 1) - 1)/3 = x$, so they are inverses

D. $f(g(x)) = 3((x - 1)/3) - 1 = x - 2$, which is not x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q1/5

$f(x) = x^2$ (all real x) and $g(x) = \sqrt{x}$. Which composition fails to equal x, and for which x?

A. $g(f(x)) = \sqrt{(x^2)} = |x|$, which is not x for $x < 0$

B. $f(g(x)) = (\sqrt{x})^2$ fails for every $x > 0$

C. $g(f(x)) = x$ for every real x

D. $f(g(x)) = |x|$, which is not x for $x < 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q2/5

True or false?

Checking f(g(x)) = x alone is always enough.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q3/5

For $f(x) = e^x$ and $g(x) = \ln x$, $g(f(x)) =$

A. $|x|$

B. x only for $x > 0$

C. x for all real x

D. $e^{\ln x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q4/5

Fix the domain of $f(x) = (x - 1)^2$ so that $g(x) = 1 + \sqrt{x}$ is its inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q5/5

SET 77 • HSF-BF.B.4.b

FORWARD 2

Read inverse values from tables and graphs • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c

A table gives f(0) = 1, f(1) = 3, f(2) = 7, f(3) = 15. Find $f^{-1}(7)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q1/5

A table gives $f(0) = 1$, $f(1) = 3$, $f(2) = 7$, $f(3) = 15$. Find $f^{-1}(f^{-1}(15))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q2/5

True or false?

If (a, b) is on the graph of f, then (b, a) is on the graph of f^{-1}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q3/5

If $f(4) = 9$, then $f^{-1}(9)$ is

A. 1/4

B. 4

C. 13

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q4/5

A table gives $f(1) = 3$ and $f(2) = 7$. How do these values give $f(f^{-1}(7)) = 7$?

A. $f^{-1}(7) = 1/7$ because an inverse is a reciprocal, and $f(1/7) = 7$, so $f(f^{-1}(7)) = 7$.

B. $f^{-1}(7) = 7 - 3 = 4$ by subtracting the outputs, and $f(4) = 7$ by the pattern.

C. $f^{-1}(7) = 2$ because $f(2) = 7$, and then $f(2) = 7$, so $f(f^{-1}(7)) = 7$.

D. $f^{-1}(7) = 3$ because $f(1) = 3$, and then $f(3) = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q5/5

SET 78 • HSF-BF.B.4.c • ADVANCED EXTENSION (+)

Read values of an inverse function from graphs and tables

MAIN	Algebra II main question	**HSF-BF.B.4.c**
BACKWARD 1	Inverse functions swap inputs and outputs	**HSF-BF.B.4.a**
BACKWARD 2	Read points from graphs and tables	**8.F.A.1**
BACKWARD 3	One-to-one functions	**HSF-BF.B.4.b**
FORWARD 1	Estimate inverse values between data points and sketch inverse...	**HSF-BF.B.4.c**
FORWARD 2	Restrict the domain to make a function invertible	**HSF-BF.B.4.d**

SET 78 • HSF-BF.B.4.c

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c

The table shows f. Find $f^{-1}(15)$.

x	0	1	2	3	4
f(x)	1	3	7	15	31

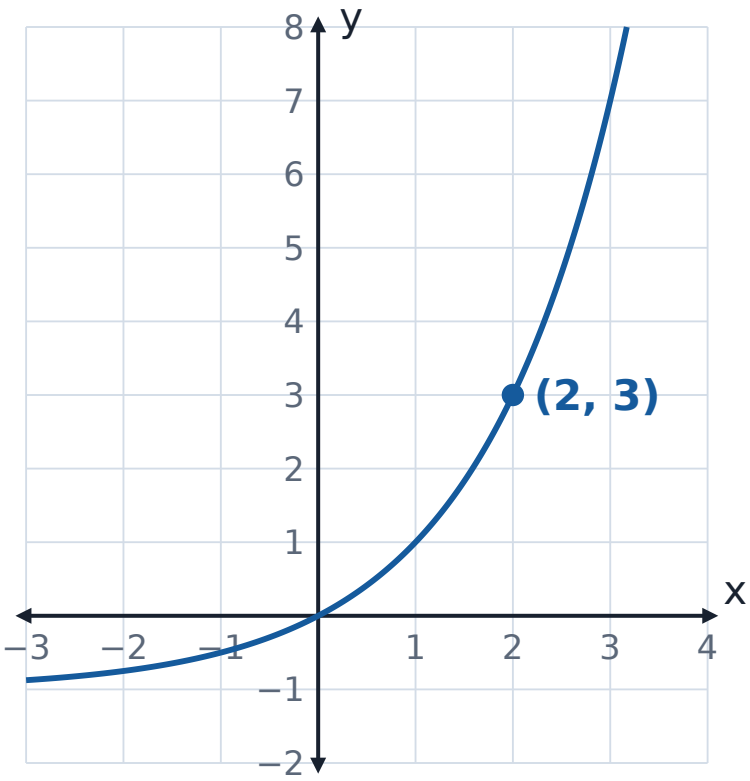
$f^{-1}(15) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q1/5

The graph of $f(x) = 2^x - 1$ is shown. Use it to find $f^{-1}(3)$.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q2/5

The table shows a decreasing function g. Find $g^{-1}(10)$ and $g^{-1}(g(2))$.

x	1	2	3	4	5
g(x)	40	25	16	10	4

$g^{-1}(10) =$ _____

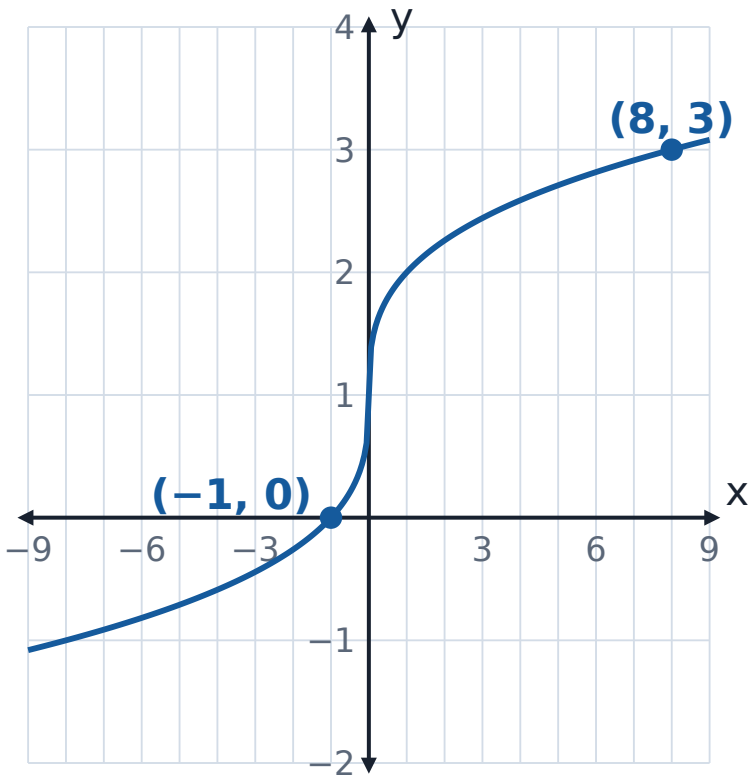
$g^{-1}(g(2)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q3/5

The graph of $f(x) = \sqrt[3]{x} + 1$ is shown. Use it to find $f^{-1}(0)$ and $f^{-1}(3)$.



$f^{-1}(0) =$ _____

$f^{-1}(3) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q4/5

The table gives a plane's altitude A (m) t minutes after takeoff. Find $A^{-1}(1200)$ and interpret it.

t (min)	0	1	2	3	4	5
A (m)	0	250	550	880	1200	1500

A. $A^{-1}(1200) = 4$: the plane climbs 1200 m every minute.

B. $A^{-1}(1200) = 1500$: the altitude one minute after reaching 1200 m.

C. $A^{-1}(1200) = 4$: the plane reaches 1200 m four minutes after takeoff.

D. $A^{-1}(1200) = 1/1200$: the reciprocal of the altitude.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q5/5

SET 78 • HSF-BF.B.4.c

BACKWARD 1

Inverse functions swap inputs and outputs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

If $f(3) = 15$, what is $f^{-1}(15)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q1/5

Write the inverse of $f(x) = 2^x - 1$.

A. $f^{-1}(x) = \log_2 x + 1$

B. $f^{-1}(x) = 2^{-x} + 1$

C. $f^{-1}(x) = \log_2(x - 1)$

D. $f^{-1}(x) = \log_2(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q2/5

True or false?

$f^{-1}(y)$ is the input of f that gives the output y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q3/5

If (5, 4) is on the graph of g, which point is on g^{-1}?

A. (5, 4)

B. (1/5, 1/4)

C. (−5, −4)

D. (4, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q4/5

Use the formula to find $f^{-1}(3)$ for $f(x) = 2^x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q5/5

SET 78 • HSF-BF.B.4.c

BACKWARD 2

Read points from graphs and tables • Grade 8 • Functions • 8.F.A.1

A table gives f(0) = 1, f(1) = 3, f(2) = 7, f(3) = 15. What is f(2)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q1/5

A plane's altitude A (m) at t = 0, 1, 2, 3, 4 minutes is 0, 250, 550, 880, 1200. What is A(3)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q2/5

True or false?

A function gives exactly one output for each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q3/5

$g(1) = 40$, $g(2) = 25$, $g(3) = 16$, $g(4) = 10$. Which input of g gives 16?

A. 16

B. 3

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q4/5

A graph passes through (8, 3). What is the output at input 8?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q5/5

SET 78 • HSF-BF.B.4.c

BACKWARD 3

One-to-one functions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

Why does a strictly increasing function have an inverse function?

A. Its graph passes through the origin, so it can be reflected over $y = x$.

B. Different inputs give different outputs, so each output comes from exactly one input.

C. Its outputs are always positive, so each output can be undone.

D. Its domain is all real numbers, so every input has an output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q1/5

Does a function with f(1) = 4 and f(3) = 4 have an inverse function? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q2/5

True or false?

A decreasing function can have an inverse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q3/5

Which table has an inverse function?

A. x: 1, 2, 3; y: 5, 7, 5

B. x: 1, 2, 3; y: 5, 7, 9

C. x: 1, 2, 3; y: 0, 1, 0

D. x: 1, 2, 3; y: 2, 2, 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S78-B3-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q4/5

A table gives $f(2) = 7$ and $f(3) = 15$. Use the table to find $f^{-1}(15)$, then $f(f^{-1}(15))$.

$f^{-1}(15) =$ _____

$f(f^{-1}(15)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q5/5

SET 78 • HSF-BF.B.4.c

FORWARD 1

Estimate inverse values between data points and sketch inverse graphs • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c

A plane's altitude is 880 m at t = 3 min and 1200 m at t = 4 min. Assume it climbs at a constant rate between these times. Estimate the time when it reaches 1000 m, to the nearest tenth of a minute.

t ≈ _____

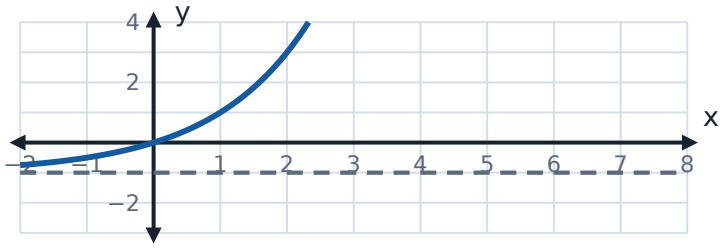
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q1

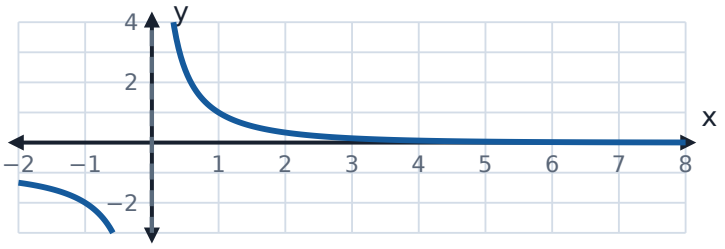
High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q1/5

Which graph shows the inverse of $f(x) = 2^x - 1$?

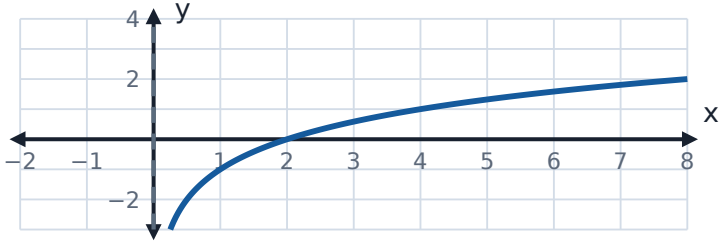
A



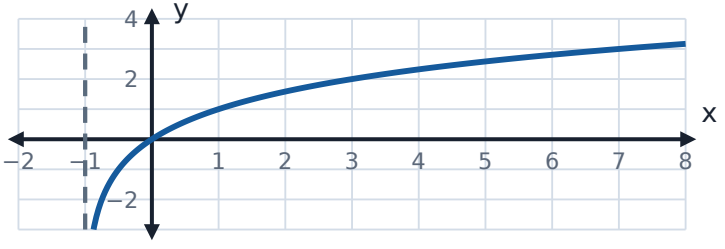
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S78-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q2/5

True or false?

The graph of f^{-1} is the reflection of the graph of f across $y = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q3/5

A decreasing function has $g(1) = 40$, $g(2) = 25$, $g(3) = 16$, $g(4) = 10$. Estimate $g^{-1}(20)$.

A. between 1 and 2

B. between 2 and 3

C. between 4 and 5

D. between 3 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q4/5

A table gives $f(0) = 1$, $f(1) = 3$, $f(2) = 7$, $f(3) = 15$. Find $(f^{-1})^{-1}(2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q5/5

SET 78 • HSF-BF.B.4.c

FORWARD 2

Restrict the domain to make a function invertible • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

$f(x) = (x - 3)^2$ is not invertible on all real numbers. Which restriction and inverse are correct?

A. $x \geq 3;\ f^{-1}(x) = 3 + \sqrt{x}$

B. $x \geq 3;\ f^{-1}(x) = 3 - \sqrt{x}$

C. $x \geq 0;\ f^{-1}(x) = \sqrt{x} + 3$

D. $x \leq 3;\ f^{-1}(x) = 3 + \sqrt{x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q1/5

$h(x) = x^2$ has $h(-2) = 4$, $h(0) = 0$, and $h(2) = 4$. Why is $h^{-1}(4)$ not defined, and how can the domain be restricted to fix it?

A. 4 is the output of two inputs (−2 and 2); restrict h to $x \geq 0$ (or to $x \leq 0$).

B. 0 has no inverse, so remove $x = 0$ from the domain of h.

C. 4 is the output of two inputs (−2 and 2); restrict h to $-2 \leq x \leq 2$.

D. $h^{-1}(4) = 1/4$, so it is defined, and no restriction is needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q2/5

True or false?

Restricting the domain can turn a non-invertible function into an invertible one.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q3/5

$f(x) = x^2$ restricted to $x \leq 0$ has inverse

A. $\sqrt{x}$

B. x^2

C. $-\sqrt{x}$

D. $1/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q4/5

$f(x) = (x - 3)^2$ has range $y \geq 0$. When its domain is restricted to make it invertible, why should the restricted function still take every output value $y \geq 0$?

A. So the restricted function has more outputs than the original function, which makes the inverse larger.

B. So the inverse is defined for every output of the original function; for example, $x \geq 3$ still gives every $y \geq 0$.

C. Because the inverse function must be identical to the original function on the restricted domain.

D. Because a restricted function must always have exactly the same domain as the original function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q5/5

SET 79 • HSF-BF.B.4.d • ADVANCED EXTENSION (+)

Restrict a domain to make a function invertible and write the inverse

MAIN	Algebra II main question	**HSF-BF.B.4.d**
BACKWARD 1	Find the vertex of a quadratic	**HSF-IF.C.8.a**
BACKWARD 2	Write inverses of one-to-one functions	**HSF-BF.B.4.a**
BACKWARD 3	Domains and ranges	**HSF-IF.B.5**
FORWARD 1	Compare different restrictions	**HSF-BF.B.4.d**
FORWARD 2	Restrict trigonometric functions to define inverses	**HSF-TF.B.6**

SET 79 • HSF-BF.B.4.d

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

Restrict $f(x) = (x - 3)^2$ to $x \geq 3$. Which is f^{-1} with its domain?

A. $f^{-1}(x) = \sqrt{x} + 3,\ x \geq 3$

B. $f^{-1}(x) = \sqrt{(x + 3)},\ x \geq -3$

C. $f^{-1}(x) = \sqrt{x} - 3,\ x \geq 0$

D. $f^{-1}(x) = \sqrt{x} + 3,\ x \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q1/5

Restrict $f(x) = x^2 + 1$ to $x \le 0$. Which is f^{-1} with its domain?

A. $f^{-1}(x) = \sqrt{(x - 1)},\ x \ge 1$

B. $f^{-1}(x) = -\sqrt{(x + 1)},\ x \ge -1$

C. $f^{-1}(x) = -\sqrt{(x - 1)},\ x \ge 1$

D. $f^{-1}(x) = -\sqrt{(x - 1)},\ x \le 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q2/5

Restrict $f(x) = |x + 2|$ to $x \geq -2$. Which is f^{-1} with its domain?

A. $f^{-1}(x) = x + 2,\ x \geq -2$

B. $f^{-1}(x) = x - 2,\ x \geq 0$

C. $f^{-1}(x) = -x - 2,\ x \geq 0$

D. $f^{-1}(x) = x - 2,\ x \geq -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q3/5

Restrict $f(x) = (x + 1)^2 - 4$ to $x \geq -1$. Which is f^{-1} with its domain?

A. $f^{-1}(x) = -1 + \sqrt{(x + 4)}, x \geq -4$

B. $f^{-1}(x) = -1 - \sqrt{(x + 4)}, x \geq -4$

C. $f^{-1}(x) = -1 + \sqrt{(x + 4)}, x \geq -1$

D. $f^{-1}(x) = 1 + \sqrt{(x - 4)}, x \geq 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q4/5

Restrict $f(x) = x^4$ to $x \geq 0$. Which is f^{-1} with its domain?

A. $f^{-1}(x) = x/4,\ x \geq 0$

B. $f^{-1}(x) = \pm\sqrt[4]{x},\ x \geq 0$

C. $f^{-1}(x) = \sqrt[4]{x},\ x \geq 0$

D. $f^{-1}(x) = 1/x^4,\ x > 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q5/5

SET 79 • HSF-BF.B.4.d

BACKWARD 1

Find the vertex of a quadratic • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a

Give the vertex of $y = (x + 1)^2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q1/5

Give the vertex of $y = x^2 + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q2/5

True or false?

A parabola is increasing on one side of its vertex and decreasing on the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q3/5

The vertex of $y = (x - 3)^2$ is

A. (0, 9)

B. (0, 3)

C. (−3, 0)

D. (3, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q4/5

On which side of the vertex is $y = (x - 3)^2$ increasing?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q5/5

SET 79 • HSF-BF.B.4.d

BACKWARD 2

Write inverses of one-to-one functions • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

Solve $y = (x - 3)^2$ for x when $x \geq 3$.

A. $x = 3 - \sqrt{y}$

B. $x = \sqrt{(y + 3)}$

C. $x = \sqrt{y} - 3$

D. $x = 3 + \sqrt{y}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q1/5

Solve $y = x^2 + 1$ for x when $x \leq 0$.

A. $x = \sqrt{(y - 1)}$

B. $x = -\sqrt{y} - 1$

C. $x = -\sqrt{(y + 1)}$

D. $x = -\sqrt{(y - 1)}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q2/5

True or false?

When $x \leq 0$, the solution of $x^2 = 4$ is $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q3/5

Solving $y = x + 2$ $(x \geq -2)$ for x gives

A. $2 - y$

B. $y + 2$

C. $y - 2$

D. $-y - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q4/5

Solve $y = x^4$ for x when $x \geq 0$. Give x as a power of y, using ^ for exponents.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q5/5

SET 79 • HSF-BF.B.4.d

BACKWARD 3

Domains and ranges • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

Give the range of $f(x) = (x - 3)^2$ for $x \geq 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q1/5

Give the range of $f(x) = (x + 1)^2 - 4$ for $x \geq -1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q2/5

True or false?

The domain of the inverse is the range of the restricted function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q3/5

The range of $x^2 + 1$ for $x \leq 0$ is

A. $y \geq 0$

B. $y \geq 1$

C. all reals

D. $y \leq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q4/5

Give the range of $|x + 2|$ for $x \geq -2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q5/5

SET 79 • HSF-BF.B.4.d

FORWARD 1

Compare different restrictions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

Restrict $f(x) = (x - 3)^2$ to $x \leq 3$. Which is the inverse with its domain?

A. $f^{-1}(x) = 3 - \sqrt{x},\ x \leq 3$

B. $f^{-1}(x) = 3 + \sqrt{x},\ x \geq 0$

C. $f^{-1}(x) = 3 - \sqrt{x},\ x \geq 0$

D. $f^{-1}(x) = \sqrt{x} - 3,\ x \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q1/5

A ball's height is $h(t) = -5(t - 2)^2 + 20$ for $0 \le t \le 4$. Which restriction and inverse give the time as a function of height on the way down?

A. $2 \le t \le 4;\ t = 2 + \sqrt{((20 - h)/5)}$

B. $2 \le t \le 4;\ t = 2 - \sqrt{((20 - h)/5)}$

C. $0 \le t \le 4;\ t = 2 + \sqrt{((20 - h)/5)}$

D. $0 \le t \le 2;\ t = 2 - \sqrt{((20 - h)/5)}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q2/5

True or false?

A function can have more than one valid restriction that makes it invertible.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q3/5

Which restriction does not make $f(x) = x^2$ invertible?

A. $x \geq 0$

B. $x \leq 0$

C. $-1 \leq x \leq 1$

D. $1 \leq x \leq 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q4/5

Why is the restriction $x \geq 0$ natural for $f(x) = x^2$ in the area formula $A = s^2$?

A. x^2 is always increasing.

B. Side lengths are nonnegative.

C. Squares have four sides.

D. Areas can be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q5/5

SET 79 • HSF-BF.B.4.d

FORWARD 2

Restrict trigonometric functions to define inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

Why is sin x not invertible on all real numbers?

A. It repeats values (for example sin 0 = sin π = 0), so it is not one-to-one.

B. Its outputs are larger than 1 for large x, outside the range of an inverse.

C. It is not defined for negative x, so an inverse would be incomplete.

D. It is an odd function, and odd functions never have an inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q1/5

What is the standard restricted domain of sin x, and why does it work?

A. $0 \le x \le \pi$: sin is increasing there and takes every value from -1 to 1 once.

B. $-\pi/2 \le x \le \pi/2$: sin is decreasing there and takes every value from 0 to 1 once.

C. $0 \le x \le 2\pi$: sin takes every value from -1 to 1 there, which is all that is needed.

D. $-\pi/2 \le x \le \pi/2$: sin is increasing there and takes every value from -1 to 1 once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q2/5

True or false?

cos x is restricted to $0 \leq x \leq \pi$ to define arccos.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q3/5

arcsin(1) equals

A. 0

B. 3π/2

C. π/2

D. π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q4/5

Why would $0 \le x \le \pi$ not work as a restricted domain for an inverse of sine?

A. sin is not defined at $x = \pi$, so the interval is not closed.

B. sin is negative everywhere on that interval, so it misses positive values.

C. sin increases then decreases there, so values repeat ($\sin(\pi/6) = \sin(5\pi/6)$).

D. sin is increasing on that whole interval, so it is one-to-one there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q5/5

SET 80 • HSF-BF.B.5 • ADVANCED EXTENSION (+)

Use the inverse relationship of exponents and logarithms to solve equations

MAIN	Algebra II main question	**HSF-BF.B.5**
BACKWARD 1	Write exact logarithmic solutions of exponential equations	**HSF-LE.A.4**
BACKWARD 2	Check for extraneous solutions	**HSA-REI.A.2**
BACKWARD 3	Domains of logarithmic expressions	**HSF-IF.B.5**
FORWARD 1	Use log properties and change of base in equations	**HSF-BF.B.5**
FORWARD 2	Check logarithmic solutions graphically	**HSA-REI.D.11**

SET 80 • HSF-BF.B.5

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5

Solve $\log_3 x = 4$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q1/5

Solve $2\log_5(x - 1) = 4$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q2/5

Solve $\ln(x + 2) = 1$ exactly.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q3/5

Solve $\log_2 x + \log_2(x - 2) = 3$. The equation $x(x - 2) = 8$ gives two candidates. Give the actual solution and the extraneous value.

Solution: $x =$ _____

Extraneous value: $x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q4/5

Solve $\log x - \log(x - 9) = 1$ (base 10) and check.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q5/5

SET 80 • HSF-BF.B.5

BACKWARD 1

Write exact logarithmic solutions of exponential equations • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $3^x = 81$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q1/5

Solve $e^x = 5$ exactly.

A. $x = 5/e$

B. $x = \ln 5$

C. $x = e^5$

D. $x = \log 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q2/5

True or false?

$\log_b y = x$ means $b^x = y$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q3/5

$5^2 = 25$ written as a logarithm is

A. $\log_5 2 = 25$

B. $\log_2 25 = 5$

C. $\log_{25} 5 = 2$

D. $\log_5 25 = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q4/5

Solve $10^x = 1000$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q5/5

SET 80 • HSF-BF.B.5

BACKWARD 2

Check for extraneous solutions • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Check $x = -2$ in $\log_2 x + \log_2(x - 2) = 3$.

A. $\log_2(-2) = 1$, so the left side is 3 and $x = -2$ is a solution.

B. $\log_2(-2) = -1$, so $x = -2$ is a solution.

C. $\log_2(-2)$ is undefined, so $x = -2$ is not a solution.

D. $(-2)(-4) = 8 = 2^3$, so $x = -2$ is a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q1/5

Check $x = 4$ in $\log_2 x + \log_2(x - 2) = 3$: find the value of the left side, then say whether $x = 4$ is a solution.

Left side: _____

Solution? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q2/5

True or false?

Combining logs can introduce solutions outside the original domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q3/5

Which value is excluded from log(x − 9)?

A. x = 20

B. x = 100

C. x = 5

D. x = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q4/5

Solve $x(x - 2) = 8$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q5/5

SET 80 • HSF-BF.B.5

BACKWARD 3

Domains of logarithmic expressions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

Give the domain of $\log_5(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q1/5

Give the domain of ln(x + 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q2/5

True or false?

$\log 0 = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q3/5

The domain of $\log_2 x + \log_2(x - 2)$ is

A. $x > 2$

B. $x \neq 2$

C. $x > 0$

D. all reals

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q4/5

Give the domain of $\log x - \log(x - 9)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q5/5

SET 80 • HSF-BF.B.5

FORWARD 1

Use log properties and change of base in equations • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5

Solve $\log_2(x + 6) - \log_2 x = 2$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q1/5

Solve $\log_4 x = \log_2 3$. (Hint: change of base.)

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q2/5

True or false?

$\log_2(x + 6) - \log_2 x = \log_2((x + 6)/x)$ for $x > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q3/5

$\log(x + 1) + \log(x - 1) = \log 8$ has solution

A. no solution

B. $x = 8$

C. $x = 3$

D. $x = -3$ and 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q4/5

Why does $\log_2(x - 5) = \log_2(1 - x)$ have no solution?

A. Both sides are defined only if $x > 5$ and $x < 1$ at the same time, which is impossible.

B. Setting $x - 5 = 1 - x$ gives $x = 3$, which is a solution, so the equation does have one.

C. The equation has a solution only when x is negative.

D. Logarithms with base 2 can never be equal to each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q5/5

SET 80 • HSF-BF.B.5

FORWARD 2

Check logarithmic solutions graphically • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

How do the graphs of $y = \log_2 x + \log_2(x - 2)$ and $y = 3$ show that the only solution is $x = 4$?

A. They intersect only at $x = 4$; the left graph exists only for $x > 2$, so there is no point at $x = -2$.

B. They intersect at $x = 4$ and at $x = -2$, so both values are solutions of the equation.

C. They never intersect, so $x = 4$ must come from the algebra only.

D. They intersect at $y = 4$, which is the solution of the equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q1/5

Use a graph or table to approximate the solution of $\ln x = 2 - x$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q2/5

True or false?

An extraneous solution of a log equation corresponds to no intersection point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q3/5

The graphs of $y = \log x$ and $y = 1$ meet at

A. $x = e$

B. $x = 10$

C. $x = 1$

D. $x = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q4/5

Why can $\ln x = 2 - x$ not be solved exactly with logarithm rules?

A. $\ln x$ is undefined for every x, so the equation has no solution at all.

B. Logarithm rules apply only to equations with base-10 logarithms.

C. The equation has infinitely many solutions, so no single exact value exists.

D. x appears both inside a logarithm and outside it, so no algebraic step isolates x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q5/5

SET 81 • HSF-TF.A.3 • ADVANCED EXTENSION (+)

Derive values from special triangles and relate π − x, π + x, 2π − x to x

MAIN	Algebra II main question	**HSF-TF.A.3**
BACKWARD 1	Special right triangle side ratios	**HSG-SRT.C.6**
BACKWARD 2	Unit-circle values in all quadrants	**HSF-TF.A.2**
BACKWARD 3	The Pythagorean Theorem	**8.G.B.7**
FORWARD 1	Derive more special values and the reflection identities	**HSF-TF.A.3**
FORWARD 2	Explain symmetry and periodicity	**HSF-TF.A.4**

SET 81 • HSF-TF.A.3

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3

A 30°−60°−90° triangle with hypotenuse 1 is half of an equilateral triangle with side 1. Which derivation of sin(π/3) and cos(π/3) is correct?

A. The short leg is 1/2 and the long leg is $\sqrt{(1 - 1/4)} = \sqrt{3}/2$. The angle π/3 is opposite the long leg, so $\sin(\pi/3) = \sqrt{3}/2$ and $\cos(\pi/3) = 1/2$.

B. The long leg is $\sqrt{(1 + 1/4)} = \sqrt{5}/2$, so $\sin(\pi/3) = \sqrt{5}/2$.

C. The short leg is 1/2 and the long leg is $\sqrt{(1 - 1/4)} = \sqrt{3}/2$. The angle π/3 is opposite the short leg, so $\sin(\pi/3) = 1/2$ and $\cos(\pi/3) = \sqrt{3}/2$.

D. The legs are 1/2 and $1 - 1/2 = 1/2$, so $\sin(\pi/3) = \cos(\pi/3) = 1/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q1/5

Which derivation of $\tan(\pi/4)$ from a $45°-45°-90°$ triangle is correct?

A. The hypotenuse is twice a leg, so $\tan(\pi/4) = 1/2$.

B. $\tan(\pi/4) = \sin(\pi/4) + \cos(\pi/4) = \sqrt{2}$.

C. The legs are 1 and $\sqrt{2}$, so $\tan(\pi/4)$ = opposite/adjacent = $1/\sqrt{2} = \sqrt{2}/2$.

D. The two legs are equal, so $\tan(\pi/4)$ = opposite/adjacent = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q2/5

Which uses $\sin(\pi - x) = \sin x$ to find $\sin(5\pi/6)$ exactly, with a correct unit-circle explanation?

A. $\sin(5\pi/6) = -\sin(\pi/6) = -1/2$: the points for x and $\pi - x$ are reflections over the x-axis, so their y-coordinates are opposite.

B. $\sin(5\pi/6) = \sin(\pi/6) = 1/2$: the points for x and $\pi - x$ are reflections over the y-axis, so they have the same y-coordinate.

C. $\sin(5\pi/6) = \cos(\pi/6) = \sqrt{3}/2$: the point for $\pi - x$ is a quarter turn from the point for x, so y becomes x.

D. $\sin(5\pi/6) = \sin(\pi/6) = 1/2$: the points for x and $\pi - x$ are reflections over the x-axis, so they have the same y-coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-M-Q3**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q3/5

Use $\cos(\pi + x) = -\cos x$ to find $\cos(7\pi/6)$ exactly.

$\cos(7\pi/6) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q4/5

Use $\tan(2\pi - x) = -\tan x$ to find $\tan(5\pi/3)$ exactly.

$\tan(5\pi/3) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q5/5

SET 81 • HSF-TF.A.3

BACKWARD 1

Special right triangle side ratios • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

A 30°−60°−90° triangle has hypotenuse 2. Give the lengths of its legs exactly.

Short leg: _____

Long leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q1/5

Give the hypotenuse of a 45°−45°−90° triangle with legs 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q2/5

True or false?

In a 30°−60°−90° triangle, the shortest side is half the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q3/5

cos 30° equals

A. $\sqrt{3}$

B. $\sqrt{2}/2$

C. $1/2$

D. $\sqrt{3}/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q4/5

Give tan 60°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q5/5

SET 81 • HSF-TF.A.3

BACKWARD 2

Unit-circle values in all quadrants • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Give $\cos(5\pi/6)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q1/5

Give $\sin(7\pi/6)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q2/5

True or false?

sin is positive in quadrant II.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q3/5

Which quadrant contains 5π/3?

A. II

B. I

C. IV

D. III

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q4/5

Give $\tan(\pi/3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q5/5

SET 81 • HSF-TF.A.3

BACKWARD 3

The Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

A right triangle has hypotenuse 1 and leg 1/2. Find the other leg.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q1/5

A right isosceles triangle has hypotenuse 1. Find each leg.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q2/5

True or false?

$(1/2)^2 + (\sqrt{3}/2)^2 = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q3/5

Legs 1 and 1 give hypotenuse

A. 2

B. $\sqrt{3}$

C. $\sqrt{2}$

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q4/5

Find x if $x^2 + x^2 = 1$ ($x > 0$).

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q5/5

SET 81 • HSF-TF.A.3

FORWARD 1

Derive more special values and the reflection identities • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3

Which derivation of $\tan(\pi/6)$ from a special triangle is correct?

A. The opposite side is half the hypotenuse, so $\tan(\pi/6) = 1/2$.

B. In the $30°-60°-90°$ triangle with hypotenuse 1, $\pi/6$ is opposite the leg $1/2$ and adjacent to the leg $\sqrt{3}/2$, so $\tan(\pi/6) = (1/2)/(\sqrt{3}/2) = 1/\sqrt{3} = \sqrt{3}/3$.

C. In the $30°-60°-90°$ triangle with hypotenuse 1, $\pi/6$ is opposite the leg $\sqrt{3}/2$ and adjacent to the leg $1/2$, so $\tan(\pi/6) = (\sqrt{3}/2)/(1/2) = \sqrt{3}$ exactly.

D. $\tan(\pi/6) = (1/2)/(\sqrt{3}/2) = \sqrt{3}/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F1-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q1/5

Which unit-circle argument explains why $\cos(\pi - x) = -\cos x$?

A. The points for x and $\pi - x$ are reflections of each other over the y-axis, so their x-coordinates (the cosines) are opposites.

B. $\cos(\pi - \pi/3) = -1/2 = -\cos(\pi/3)$, so the rule always holds.

C. $\cos(\pi - x) = \cos \pi - \cos x = -1 - \cos x$.

D. The points for x and $\pi - x$ are reflections of each other over the x-axis, so their x-coordinates (the cosines) are opposites.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q2/5

True or false?

$\sin(\pi + x) = -\sin x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q3/5

$\cos(2\pi - x)$ equals

A. $\cos x$

B. $-\cos x$

C. $\sin x$

D. $-\sin x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q4/5

Use $\sin(2\pi - x) = -\sin x$ to find $\sin(11\pi/6)$ exactly.

$\sin(11\pi/6) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q5/5

SET 81 • HSF-TF.A.3

FORWARD 2

Explain symmetry and periodicity • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

Which unit-circle reasoning explains why $\sin(-x) = -\sin x$?

A. The points for x and −x are opposite points, so their x-coordinates are equal.

B. The points for x and −x are reflections over the y-axis, so their y-coordinates are opposites.

C. The points for x and −x are reflections over the x-axis, so their y-coordinates are opposites.

D. The points for x and −x are the same point, so their y-coordinates are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q1/5

Is cos an even or odd function, and why?

A. Even: the points for x and −x have the same y-coordinate, so cos(−x) = cos x.

B. Even: the points for x and −x have the same x-coordinate, so cos(−x) = cos x.

C. Odd: the points for x and −x have opposite x-coordinates, so cos(−x) = −cos x.

D. Neither: cos(−x) is not related to cos x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q2/5

True or false?

$\sin(x + 2\pi) = \sin x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q3/5

$\tan(-\pi/4)$ equals

A. undefined

B. -1

C. 1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q4/5

Using $2\pi - x = -x + 2\pi$, which reasoning explains why $\tan(2\pi - x) = -\tan x$?

A. $\tan(-x + 2\pi) = \tan(-x)$ by periodicity, and tan is odd, so $\tan(-x) = -\tan x$.

B. $\tan(-x + 2\pi) = \tan x$ by periodicity, and tan is even.

C. $\tan(-x + 2\pi) = \tan(-x) + \tan(2\pi) = -\tan x + 1$.

D. $\tan(2\pi - x) = 2\pi - \tan x$, because tan distributes over subtraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q5/5

SET 82 • HSF-TF.A.4 • ADVANCED EXTENSION (+)

Explain the symmetry and periodicity of trigonometric functions with the unit circle

MAIN	Algebra II main question	**HSF-TF.A.4**
BACKWARD 1	Explain the unit-circle definition	**HSF-TF.A.2**
BACKWARD 2	Even and odd functions	**HSF-BF.B.3**
BACKWARD 3	Reflect points over the axes	**8.G.A.3**
FORWARD 1	Use periodicity and symmetry to evaluate and simplify	**HSF-TF.A.4**
FORWARD 2	Restrict trigonometric functions so they have inverses	**HSF-TF.B.6**

SET 82 • HSF-TF.A.4

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

Which unit-circle explanation shows why $\sin(-x) = -\sin x$?

A. The points for x and $-x$ are (a, b) and $(-a, b)$, reflections of each other over the y-axis, so their sines are opposites.

B. $\sin(-\pi/6) = -1/2 = -\sin(\pi/6)$, so the rule always holds.

C. The points for x and $-x$ are (a, b) and $(a, -b)$, reflections over the x-axis, so the sines b and $-b$ are opposites.

D. A negative sign can always be moved outside a function, so $\sin(-x) = -\sin x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q1/5

Which unit-circle explanation shows why $\cos(x + 2\pi) = \cos x$?

A. Adding 2π moves the point to the opposite side of the unit circle, and opposite points on the circle have the same x-coordinate, so the cosine is the same.

B. $\cos(x + 2\pi) = \cos x + \cos 2\pi = \cos x + 1$.

C. Adding 2π travels one full turn around the unit circle and ends at the same point, so the x-coordinate (the cosine) is the same.

D. $\cos(0 + 2\pi) = 1 = \cos 0$, so the rule always holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S82-M-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q2/5

Which unit-circle explanation shows why $\tan(x + \pi) = \tan x$?

A. Adding π gives the point $(a, -b)$, a reflection over the x-axis, so $\tan(x + \pi) = -b/a = -\tan x$, which is not $\tan x$.

B. Adding π gives the opposite point $(-a, -b)$ of (a, b), and $\tan = (-b)/(-a) = b/a$, the same ratio.

C. Adding π travels a full turn, so the point is the same.

D. $\tan(\pi/4 + \pi) = 1 = \tan(\pi/4)$, so the rule always holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q3/5

Use symmetry to evaluate $\cos(-2\pi/3)$.

A. $1/2$, because cos is even: $\cos(-2\pi/3) = \cos(\pi/3) = 1/2$

B. $-1/2$, because cos is even: $\cos(-2\pi/3) = \cos(2\pi/3) = -1/2$

C. $1/2$, because cos is odd: $\cos(-2\pi/3) = -\cos(2\pi/3) = 1/2$

D. $-\sqrt{3}/2$, because $\cos(-2\pi/3) = -\sin(2\pi/3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S82-M-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q4/5

Is $f(x) = \sin x \cos x$ even, odd, or neither?

A. Even: a product of two functions is always even

B. Odd: $f(-x) = \sin(-x)\cos(-x) = (-\sin x)(\cos x) = -f(x)$

C. Even: $f(-x) = \sin(-x)\cos(-x) = \sin x \cos x = f(x)$

D. Neither: sin is odd but cos is even

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q5/5

SET 82 • HSF-TF.A.4

BACKWARD 1

Explain the unit-circle definition • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Which point on the unit circle represents angle x?

A. (cos x, x)

B. (sin x, cos x)

C. (x, sin x)

D. (cos x, sin x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q1/5

In which direction from (1, 0) does a negative angle go on the unit circle? Answer Clockwise or Counterclockwise.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q2/5

True or false?

$\tan x = \sin x / \cos x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q3/5

The point for angle π is

A. $(0, -1)$

B. $(-1, 0)$

C. $(0, 1)$

D. $(1, 0)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q4/5

Give $\cos(2\pi/3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q5/5

SET 82 • HSF-TF.A.4

BACKWARD 2

Even and odd functions • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

Which is the definition of an even function?

A. $f(x + 2\pi) = f(x)$ for every x in the domain

B. $f(-x) = f(x)$ for every x in the domain

C. $f(-x) = -f(x)$ for every x in the domain

D. $f(x) > 0$ for every x in the domain

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q1/5

Which is the definition of an odd function?

A. $f(-x) = f(x)$ for every x in the domain

B. $f(x) < 0$ for every x in the domain

C. $f(x + \pi) = -f(x)$ for every x in the domain

D. $f(-x) = -f(x)$ for every x in the domain

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q2/5

True or false?

An odd function's graph is symmetric about the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q3/5

x^3 is

A. odd

B. neither

C. even

D. both

Is the product of an odd function and an even function odd or even? Answer Odd or Even.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q5/5

SET 82 • HSF-TF.A.4

BACKWARD 3

Reflect points over the axes • Grade 8 • Geometry • 8.G.A.3

Reflect (a, b) over the x-axis.

A. (b, a)

B. (−a, b)

C. (a, −b)

D. (−a, −b)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q1/5

Rotate (a, b) by 180° about the origin.

A. (−a, −b)

B. (a, −b)

C. (−a, b)

D. (−b, a)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q2/5

True or false?

Reflecting over the x-axis keeps the y-coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q3/5

Reflecting (0.6, 0.8) over the x-axis gives

A. (0.8, 0.6)

B. (−0.6, 0.8)

C. (−0.6, −0.8)

D. (0.6, −0.8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q4/5

Rotate the point $(\sqrt{3}/2, 1/2)$ by 180° about the origin.

A. $(\sqrt{3}/2, -1/2)$

B. $(-\sqrt{3}/2, -1/2)$

C. $(-1/2, \sqrt{3}/2)$

D. $(-\sqrt{3}/2, 1/2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q5/5

SET 82 • HSF-TF.A.4

FORWARD 1

Use periodicity and symmetry to evaluate and simplify • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

Find $\sin(17\pi/6)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q1/5

Find $\tan(-11\pi/4)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q2/5

True or false?

$\cos(x + \pi) = -\cos x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q3/5

$\sin(x + \pi) + \sin x$ equals

A. 1

B. $\sin 2x$

C. $2 \sin x$

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q4/5

Simplify $\cos(-x) + \cos(x + 2\pi)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q5/5

SET 82 • HSF-TF.A.4

FORWARD 2

Restrict trigonometric functions so they have inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

How does periodicity prevent sin x from having an inverse on all real numbers?

A. Each value appears only once in every period, so periodicity changes nothing.

B. Periodicity makes sin increasing everywhere, so it has no turning points.

C. Each value repeats every 2π, so an output does not determine its input.

D. Periodic functions are undefined at multiples of 2π, which breaks the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q1/5

What is the restricted domain used for tan x, and why?

A. $-\pi/2 < x < \pi/2$: tan increases there and takes every real value once.

B. $0 < x < \pi$: tan is defined and increasing everywhere there.

C. $-\pi/2 \leq x \leq \pi/2$: tan is defined at both endpoints.

D. $0 \leq x < 2\pi$: tan takes every real value there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q2/5

True or false?

arccos gives values between 0 and π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q3/5

arctan(−1) equals

A. 7π/4

B. 3π/4

C. −π/4

D. π/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q4/5

Why is $\arcsin(-1/2) = -\arcsin(1/2)$?

A. $\arcsin(-1/2) = 7\pi/6$ because $\sin(7\pi/6) = -1/2$, which is $-\arcsin(1/2)$.

B. sin is even on $[-\pi/2, \pi/2]$, so $\sin(-\pi/6) = 1/2$, giving $\arcsin(-1/2) = \pi/6$.

C. A minus sign can always be moved outside any function, so $\arcsin(-a) = -\arcsin(a)$.

D. sin is odd on $[-\pi/2, \pi/2]$, so $\sin(-\pi/6) = -1/2$, giving $\arcsin(-1/2) = -\pi/6 = -\arcsin(1/2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q5/5

SET 83 • HSF-TF.B.6 • ADVANCED EXTENSION (+)

Choose and justify domain restrictions that make trigonometric functions invertible

MAIN	Algebra II main question	**HSF-TF.B.6**
BACKWARD 1	Restrict a domain to make a function invertible	**HSF-BF.B.4.d**
BACKWARD 2	Exact values on the unit circle	**HSF-TF.A.2**
BACKWARD 3	Periodicity of trigonometric functions	**HSF-TF.A.4**
FORWARD 1	Compose trigonometric functions with their inverses	**HSF-TF.B.6**
FORWARD 2	Solve trigonometric equations with inverses	**HSF-TF.B.7**

SET 83 • HSF-TF.B.6

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

Why does sin x have no inverse on all real numbers, and on which restricted domain does it have one?

A. sin repeats values, so it is not one-to-one; on $0 \le x \le \pi$ it is increasing and takes each value in $[-1, 1]$ once.

B. sin is not defined for all real numbers; on $0 \le x \le \pi$ it is defined and takes each value in $[-1, 1]$ once.

C. sin repeats values, so it is not one-to-one; on $-\pi/2 \le x \le \pi/2$ it is increasing and takes each value in $[-1, 1]$ once.

D. sin has outputs greater than 1 for large x; on $-\pi \le x \le \pi$ it takes each value in $[-1, 1]$ exactly once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q1/5

What is the standard restricted domain of cos x, and why does it work?

A. $0 \le x \le 2\pi$: cos takes every value from -1 to 1 there, so it is enough.

B. $0 \le x \le \pi$: cos decreases from 1 to -1 there, taking each value once.

C. $-\pi/2 \le x \le \pi/2$: cos increases from -1 to 1 there, taking each value once.

D. $-\pi \le x \le 0$: cos decreases from 1 to -1 there, taking each value once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q2/5

What is the standard restricted domain of tan x, and why does it work?

A. $-\pi/2 < x < \pi/2$: tan increases from $-\infty$ to ∞ there, taking each real value once.

B. $0 < x < \pi$: tan is defined and increasing at every point of that interval.

C. $0 \le x < \pi/2$: tan increases there and takes each real value exactly once.

D. $-\pi/2 \le x \le \pi/2$: tan increases there, including its values at both endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q3/5

What is arcsin(1/2), and why is it not 5π/6?

A. π/6 and 5π/6: arcsin gives every angle in [0, 2π) whose sine is 1/2.

B. π/3: arcsin(1/2) is the angle whose cosine is 1/2, not 5π/6.

C. 5π/6: arcsin gives the largest angle in [0, 2π) whose sine is 1/2.

D. π/6: arcsin outputs only angles in [−π/2, π/2], and 5π/6 is outside that interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q4/5

Evaluate $\arccos(-1/2)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q5/5

SET 83 • HSF-TF.B.6

BACKWARD 1

Restrict a domain to make a function invertible • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

On which domain is $f(x) = x^2$ invertible?

A. all real numbers

B. $x \geq -2$

C. $x \geq 0$

D. $-1 \leq x \leq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q1/5

Why must a function be strictly increasing or strictly decreasing on an interval to be invertible there?

A. Then the function is defined at every real number, inside and outside the interval.

B. Then the function has no zeros on the interval, so it never crosses the x-axis.

C. Then the outputs are all positive on the interval, so each can be undone.

D. Then no output repeats on the interval, so each output comes from exactly one input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q2/5

True or false?

A function that is always increasing on an interval is one-to-one there.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q3/5

$f(x) = (x - 1)^2$ is one-to-one on

A. $-1 \leq x \leq 3$

B. all reals

C. $x \geq 1$

D. $x \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q4/5

Which is the inverse of $f(x) = x^2$ on $x \geq 0$?

A. $-\sqrt{x}$

B. $\sqrt{x}$

C. $x/2$

D. $1/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q5/5

SET 83 • HSF-TF.B.6

BACKWARD 2

Exact values on the unit circle • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Give $\sin(\pi/6)$ and $\sin(5\pi/6)$ exactly.

$\sin(\pi/6) =$ _____

$\sin(5\pi/6) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q1/5

Give $\cos(2\pi/3)$ and $\cos(4\pi/3)$ exactly.

$\cos(2\pi/3) =$ _____

$\cos(4\pi/3) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q2/5

True or false?

$\tan(\pi/4) = \tan(3\pi/4)$.

A. True

B. False

Which angle in [0, π] has cos = −1/2?

A. 2π/3

B. π/3

C. 4π/3

D. 5π/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q4/5

Give $\sin(-\pi/6)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q5/5

SET 83 • HSF-TF.B.6

BACKWARD 3

Periodicity of trigonometric functions • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

What is the period of sin x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q1/5

What is the period of tan x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q2/5

True or false?

A periodic function cannot be one-to-one on its whole domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q3/5

$\sin x = 0$ for

A. $x = \pi/2 + k\pi$

B. no x

C. $x = 2k\pi$ only

D. $x = k\pi$

Which two angles both have cosine 1?

A. 0 and 2π

B. 0 and π

C. π and 2π

D. π/2 and 3π/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q5/5

SET 83 • HSF-TF.B.6

FORWARD 1

Compose trigonometric functions with their inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

Evaluate $\arcsin(\sin(5\pi/6))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q1/5

Evaluate sin(arcsin(0.3)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q2/5

True or false?

arccos(cos x) = x for every real x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q3/5

arctan(tan(3π/4)) equals

A. $-\pi/4$

B. $-3\pi/4$

C. $3\pi/4$

D. $\pi/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q4/5

What are the domain and range of arccos?

A. all real numbers; range $[0, \pi]$

B. domain $[0, \pi]$; range $[-1, 1]$

C. domain $[-1, 1]$; range $[-\pi/2, \pi/2]$

D. domain $[-1, 1]$; range $[0, \pi]$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q5/5

SET 83 • HSF-TF.B.6

FORWARD 2

Solve trigonometric equations with inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

Solve $\sin x = 0.4$ on $0 \leq x < 2\pi$. Give both solutions to the nearest hundredth, separated by a comma. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q1/5

Solve $\cos x = 0.4$ on $0 \leq x < 2\pi$. Give both solutions to the nearest hundredth, separated by a comma. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q2/5

True or false?

For $0 < c < 1$, arcsin c gives only one of the two solutions of $\sin x = c$ on $[0, 2\pi)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q3/5

The second solution of $\sin x = c$ in $[0, 2\pi)$ (with $0 < c < 1$) is

A. $\pi + \arcsin c$

B. $-\arcsin c$

C. $\pi - \arcsin c$

D. $2\pi - \arcsin c$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q4/5

For $-1 < c < 1$, how is the restricted range of arccos used to find both solutions of $\cos x = c$ on $[0, 2\pi)$?

A. arccos c gives the solution in $(0, \pi)$; the other is $2\pi -$ arccos c, by symmetry over the x-axis.

B. arccos c gives both solutions at once, because its range is the whole interval $[0, 2\pi)$.

C. arccos c gives the solution in $(0, \pi)$; the other is $\pi -$ arccos c, by symmetry over the y-axis.

D. arccos c gives the solution in $(-\pi/2, \pi/2)$; the other is arccos c + π, by periodicity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q5/5

SET 84 • HSF-TF.B.7 • ADVANCED EXTENSION (+)

Solve trigonometric equations in models with inverse functions

MAIN	Algebra II main question	**HSF-TF.B.7**
BACKWARD 1	Evaluate inverse trigonometric functions on restricted domains	**HSF-TF.B.6**
BACKWARD 2	Parameters of sinusoidal models	**HSF-TF.B.5**
BACKWARD 3	Symmetry of unit-circle solutions	**HSF-TF.A.2**
FORWARD 1	Find all real solutions and solve equations that factor	**HSF-TF.B.7**
FORWARD 2	Solve equations with sum and double-angle identities	**HSF-TF.C.9**

SET 84 • HSF-TF.B.7

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

Solve $2\sin x - 1 = 0$ for $0 \leq x < 2\pi$. Give exact values.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q1/5

Solve $3\cos x = -2$ for $0 \leq x < 2\pi$. Give both solutions to the nearest hundredth, separated by a comma. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q2/5

A Ferris wheel rider's height is $h(t) = 12 + 10\sin(\pi t/15)$ m after t seconds. Find every time in $0 \le t < 30$ when $h = 17$ m, in seconds, separated by commas.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q3/5

Harbor depth is $d(t) = 5 + 4\cos(\pi t/6)$ ft after t hours. Find every time in $0 \le t < 12$ when $d = 7$ ft, in hours, separated by commas.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q4/5

Solve $\tan x = 3$ for $0 \leq x < 2\pi$. Give both solutions to the nearest hundredth, separated by a comma. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q5/5

SET 84 • HSF-TF.B.7

BACKWARD 1

Evaluate inverse trigonometric functions on restricted domains • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

Evaluate arcsin(1/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q1/5

Evaluate arccos(1/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q2/5

True or false?

arctan 3 is between 0 and π/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q3/5

arccos(−2/3) is in quadrant

A. II

B. III

C. I

D. IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q4/5

Evaluate arctan 3 to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q5/5

SET 84 • HSF-TF.B.7

BACKWARD 2

Parameters of sinusoidal models • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

In $h(t) = 12 + 10\sin(\pi t/15)$, what is the period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q1/5

A rider's height is $h(t) = 12 + 10\sin(\pi t/15)$ m. Setting $h(t) = 17$ gives an equation $\sin(\pi t/15) = c$. Find c.

$c =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q2/5

True or false?

$d(t) = 5 + 4\cos(\pi t/6)$ has a maximum of 5.

A. True

B. False

Isolating the cosine in $5 + 4\cos(\pi t/6) = 7$ gives

A. $\cos(\pi t/6) = 1/2$

B. $\cos(\pi t/6) = 3$

C. $\cos(\pi t/6) = 2$

D. $\cos(\pi t/6) = 7/4$

What is the midline of d(t)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q5/5

SET 84 • HSF-TF.B.7

BACKWARD 3

Symmetry of unit-circle solutions • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Give both angles in $[0, 2\pi)$ with $\sin = 1/2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q1/5

Give both angles in [0, 2π) with cos = 1/2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q2/5

True or false?

$\tan x = c$ has two solutions in $[0, 2\pi)$ that differ by π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q3/5

For $-1 < c < 1$, $\cos x = c$ has a solution a in $(0, \pi)$. The other solution in $[0, 2\pi)$ is

A. $2\pi - a$

B. $\pi + a$

C. $-a$

D. $\pi - a$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q4/5

Give both angles in [0, 2π) with tan = 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q5/5

SET 84 • HSF-TF.B.7

FORWARD 1

Find all real solutions and solve equations that factor • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

Which gives all real solutions of $2\sin x - 1 = 0$?

A. $x = \pi/3 + 2k\pi$ or $x = 2\pi/3 + 2k\pi$, k an integer

B. $x = \pi/6 + k\pi$, k an integer

C. $x = \pi/6 + 2k\pi$ or $x = 5\pi/6 + 2k\pi$, k an integer

D. $x = \pi/6 + 2k\pi$, k an integer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q1/5

Solve $2\sin^2 x - \sin x = 0$ for $0 \leq x < 2\pi$. List all solutions exactly, separated by commas.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q2/5

True or false?

$\cos x = 2$ has no real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q3/5

How many solutions does $\sin(2x) = 1/2$ have in $0 \leq x < 2\pi$?

A. 2

B. 4

C. 8

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F1-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q4/5

Solve $\cos^2 x = 1/4$ for $0 \leq x < 2\pi$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q5/5

SET 84 • HSF-TF.B.7

FORWARD 2

Solve equations with sum and double-angle identities • High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9

Use $\sin 2x = 2\sin x \cos x$ to solve $\sin 2x = \cos x$ for $0 \leq x < 2\pi$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q1/5

Use $\cos 2x = 1 - 2\sin^2 x$ to solve $\cos 2x = \sin x$ for $0 \leq x < 2\pi$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q2/5

True or false?

$\sin(x + \pi/2) = \cos x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q3/5

cos 2x equals

A. $1 + 2\sin^2 x$

B. $\cos^2 x - \sin^2 x$

C. $\cos^2 x + \sin^2 x$

D. $2\cos x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q4**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q4/5

Why are identities useful for solving equations that contain different trigonometric functions?

A. They rewrite the equation in one function (or let it factor), so inverse functions can be applied.

B. They turn every equation into a linear equation that has exactly one solution.

C. They remove the need to check the solutions in the original equation.

D. They change the period of each function to 2π, so the graphs match.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q5/5